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(54) **CO-MANIPULATION SURGICAL SYSTEM
HAVING MULTIPLE OPERATIONAL MODES
FOR USE WITH SURGICAL INSTRUMENTS
FOR PERFORMING SURGERY**

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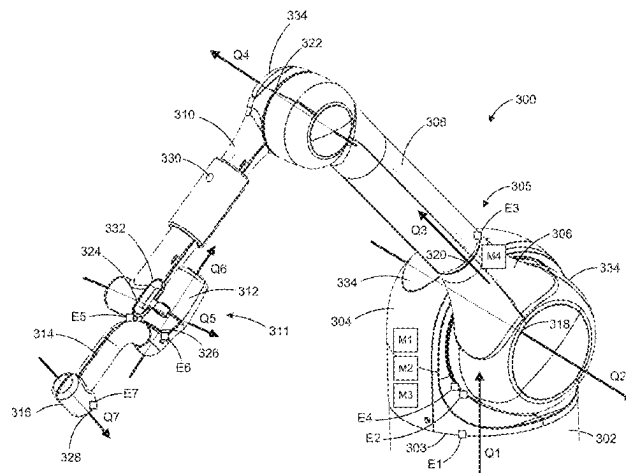
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(57) **ABSTRACT**

Co-manipulation robotic systems are described herein that may be used for assisting with laparoscopic surgical procedures. The co-manipulation robotic systems allow a surgeon to use commercially-available surgical tools while providing benefits associated with surgical robotics. Advantageously, the surgical tools may be seamlessly coupled to the robot arms using a disposable coupler while the reusable portions of the robot arm remain in a sterile drape. Further, the co-manipulation robotic system may operate in multiple modes to enhance usability and safety, while allowing the

(Continued)



surgeon to position the instrument directly with the instrument handle and further maintain the desired position of the instrument using the robot arm.

20 Claims, 61 Drawing Sheets

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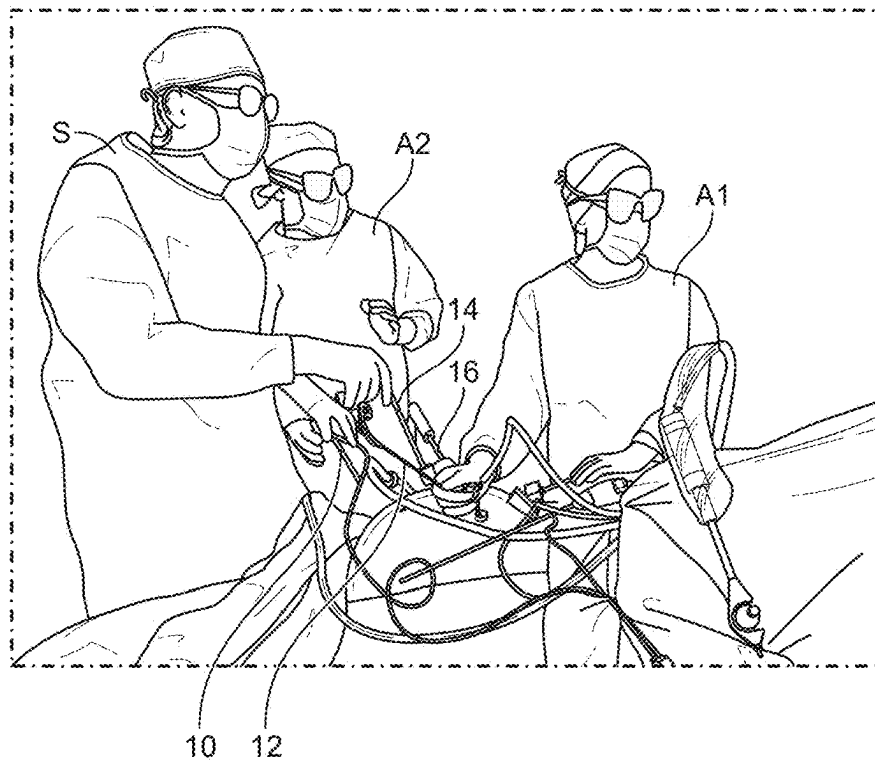


FIG. 1A

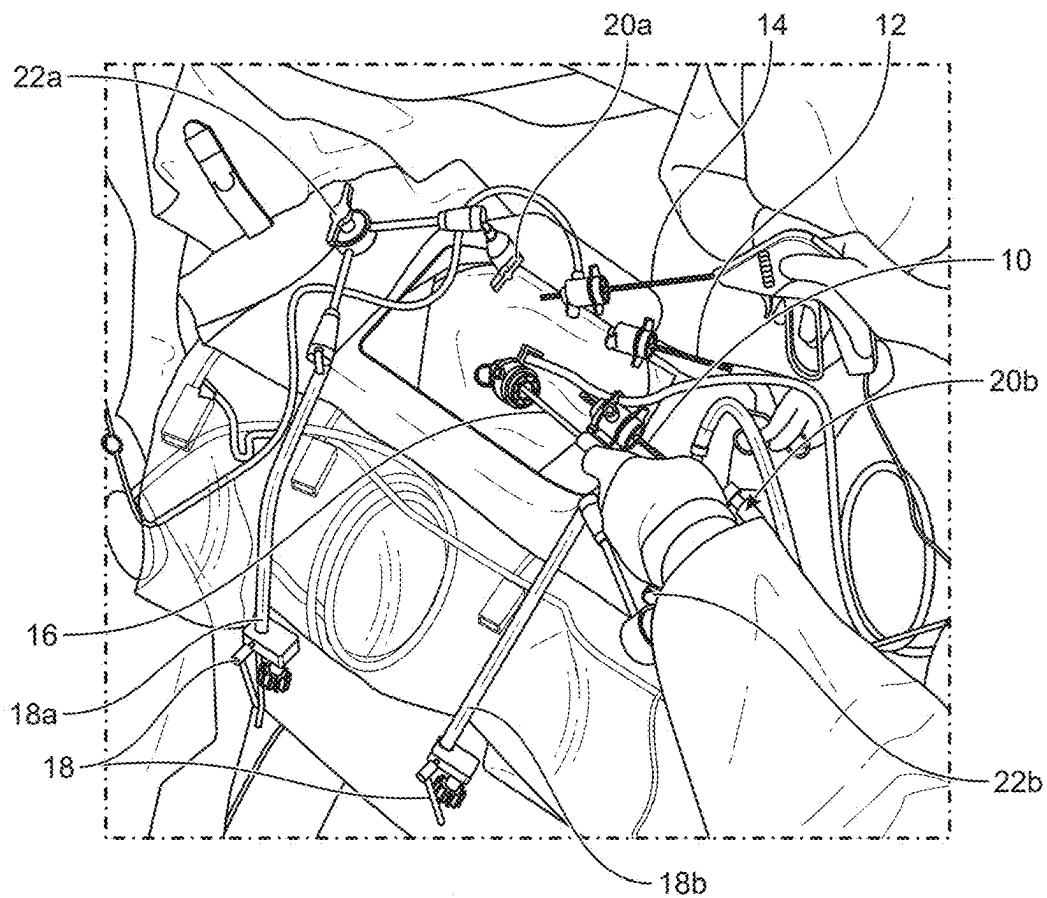


FIG. 1B

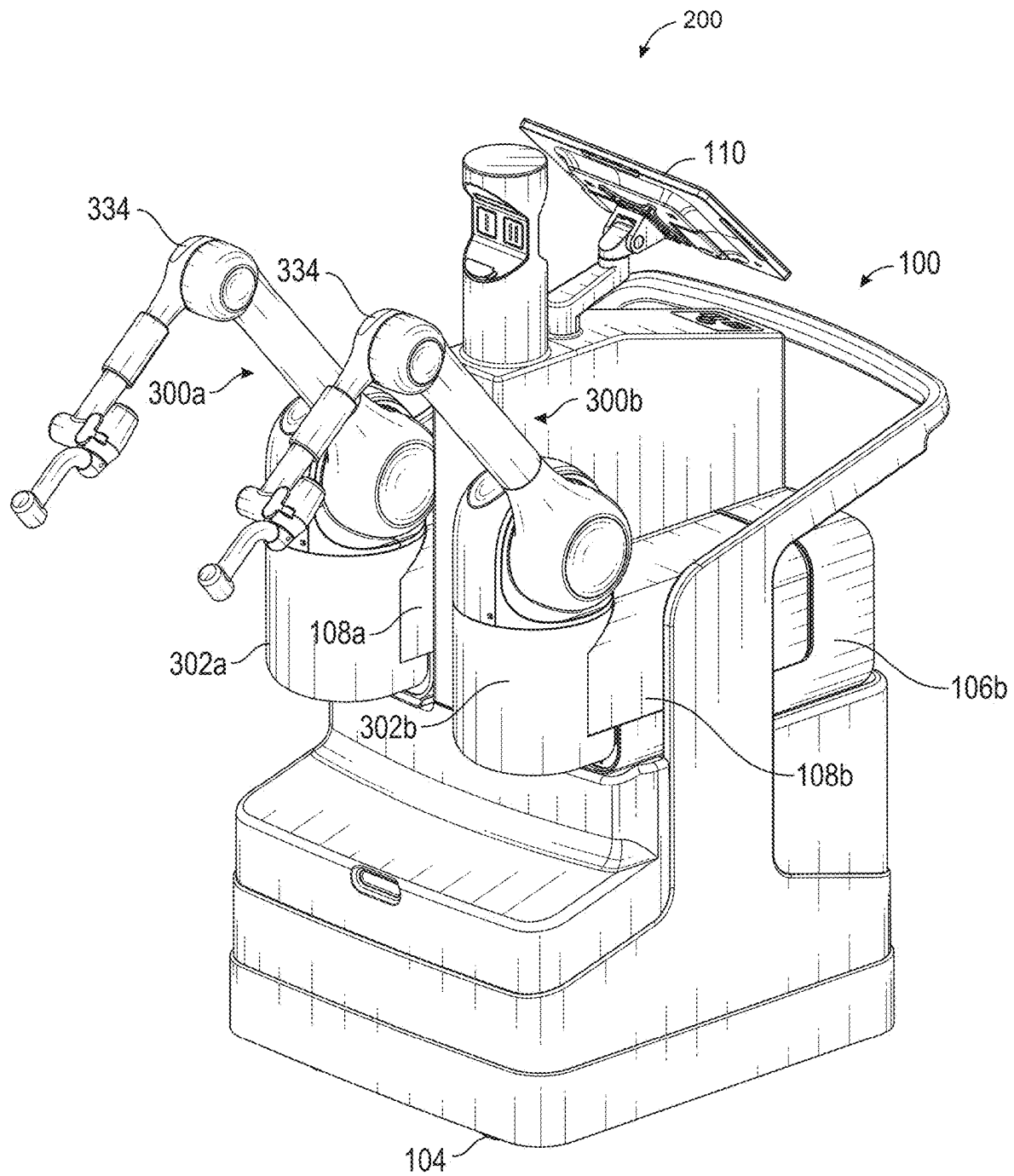


FIG. 2

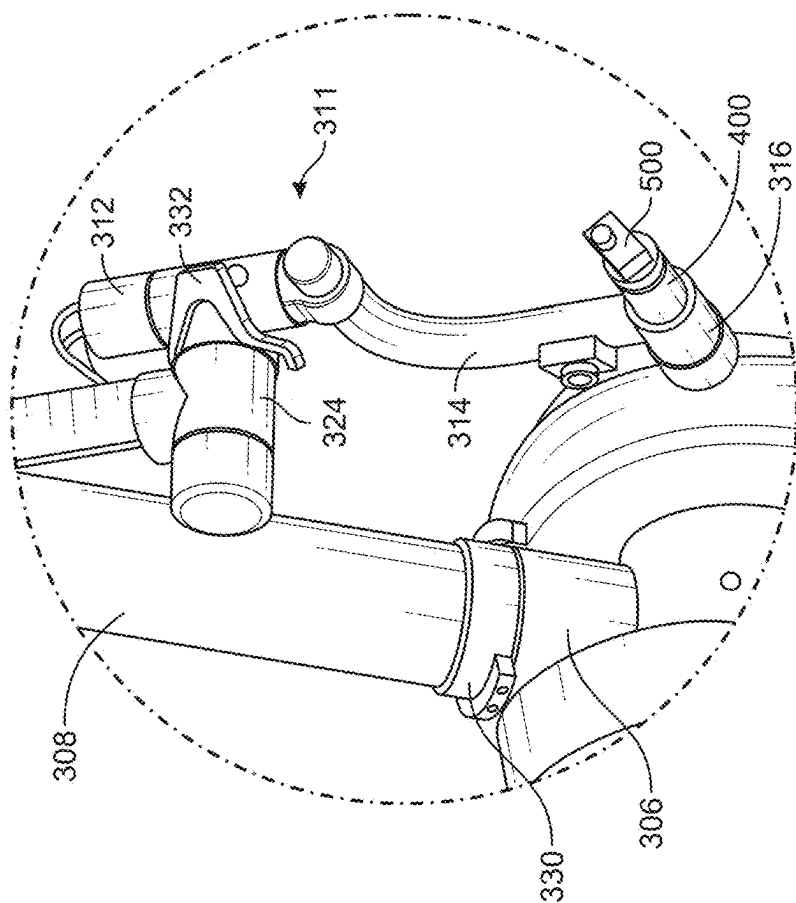


FIG. 4B

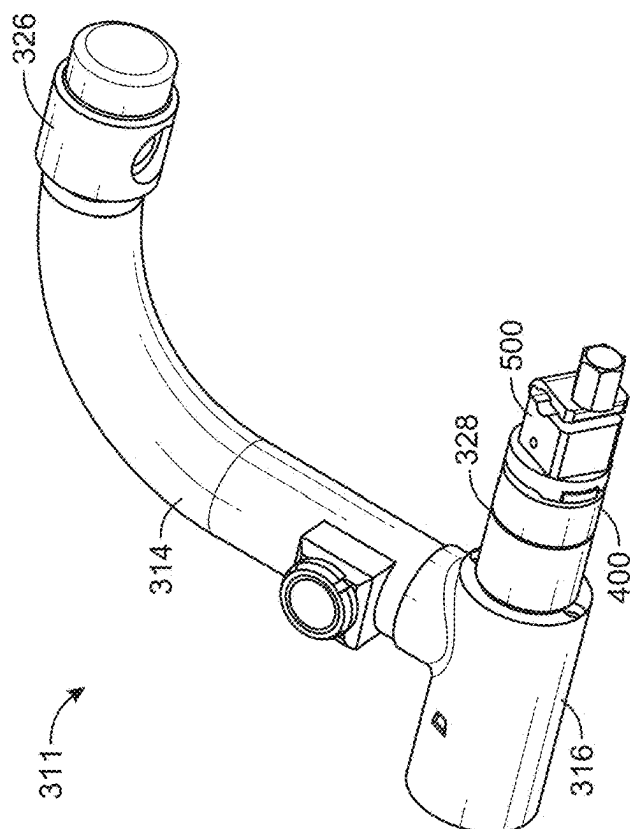


FIG. 4A

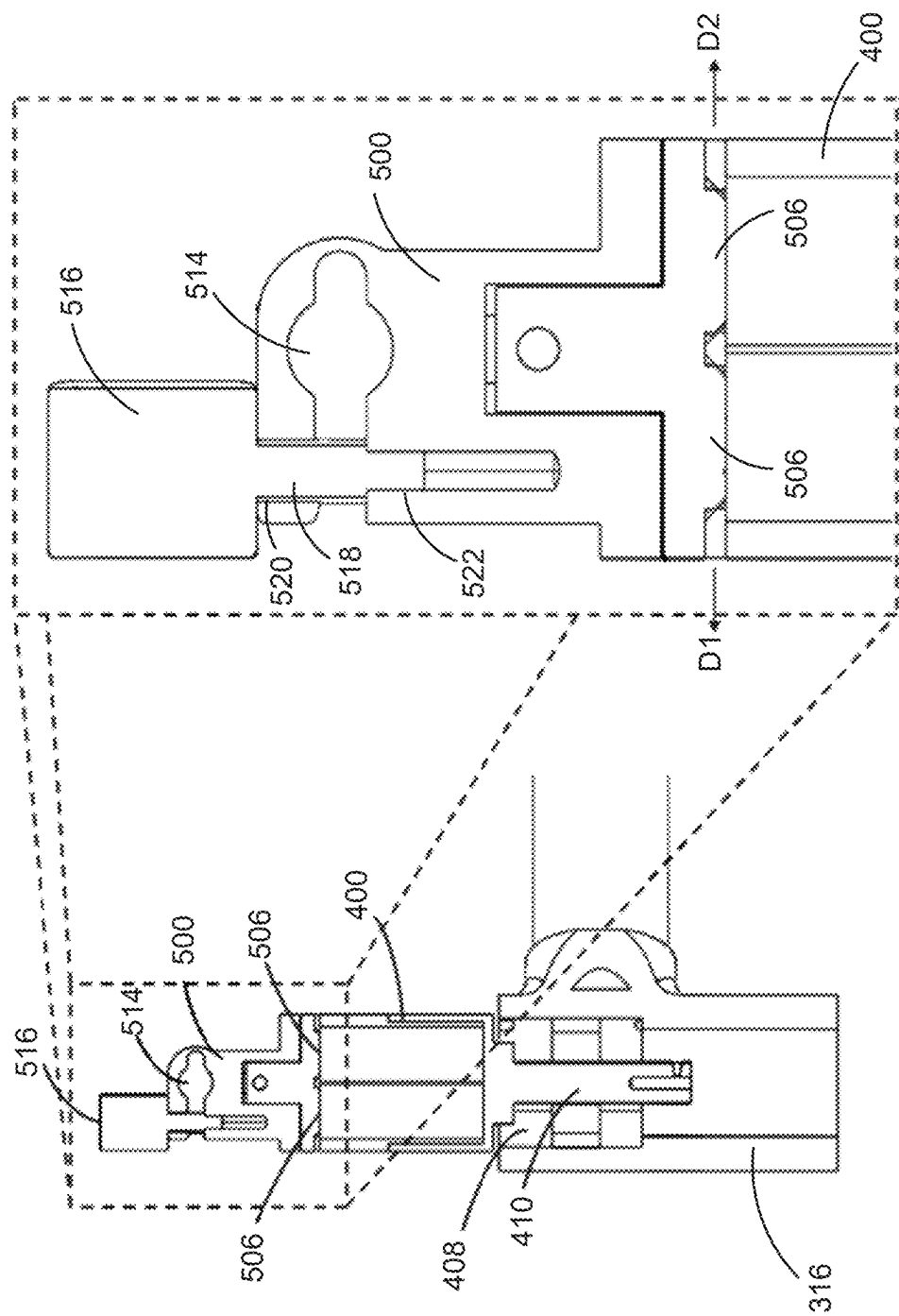


FIG. 4C

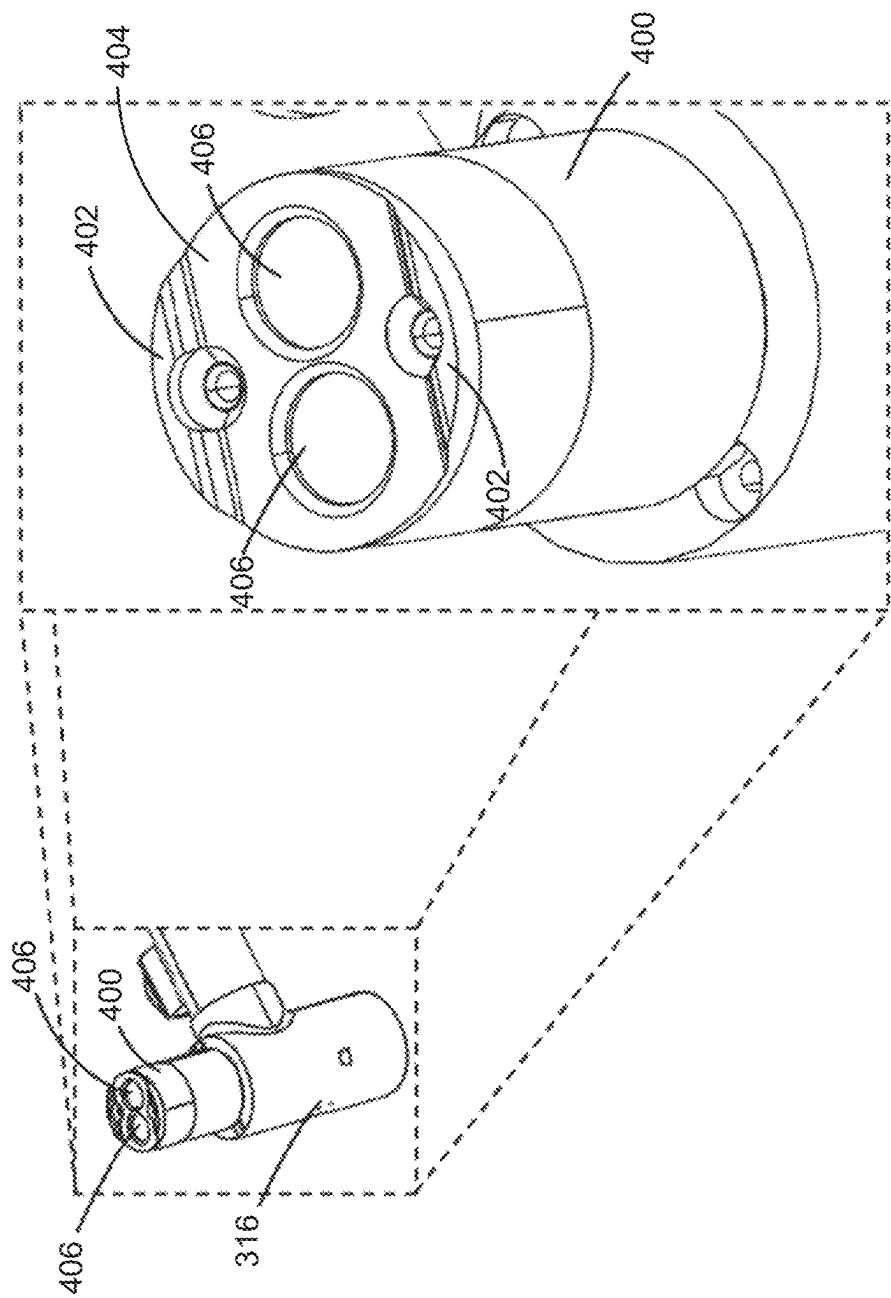


FIG. 4D

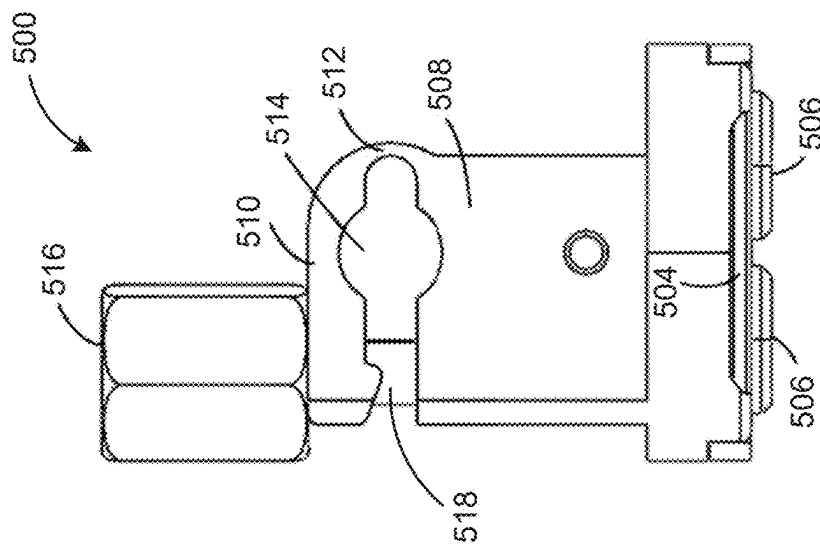


FIG. 5B

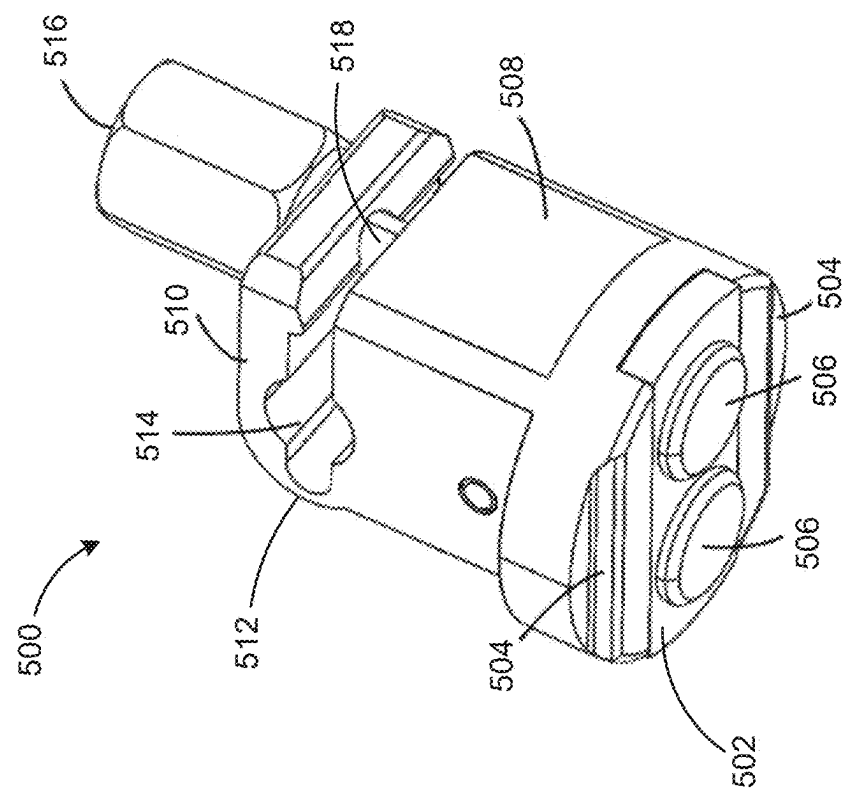


FIG. 5A

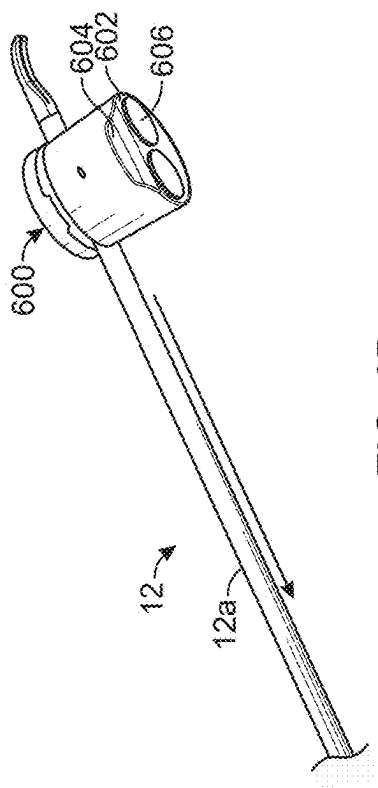


FIG. 6B

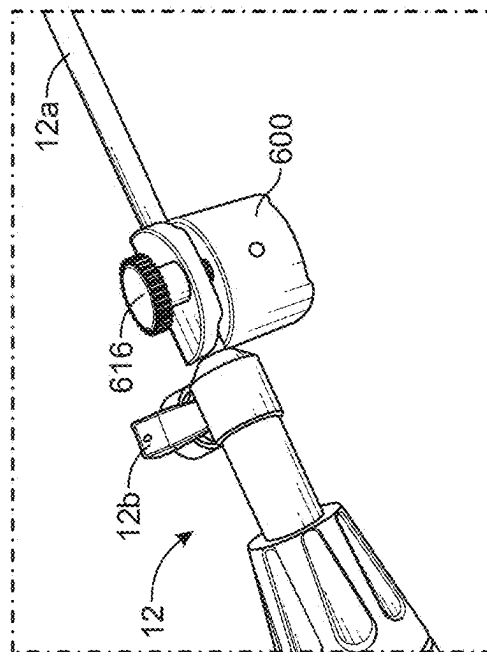


FIG. 6D

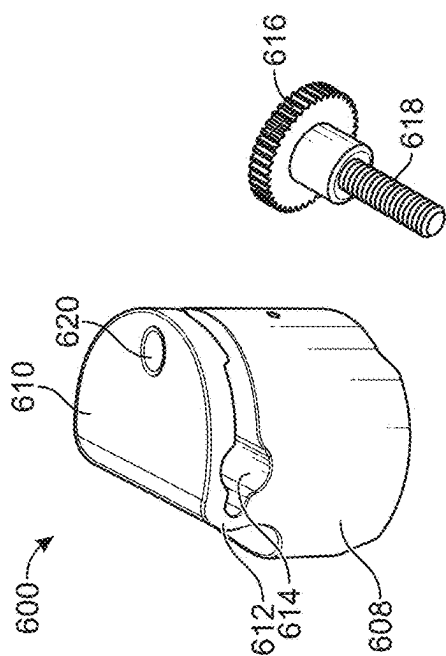


FIG. 6A

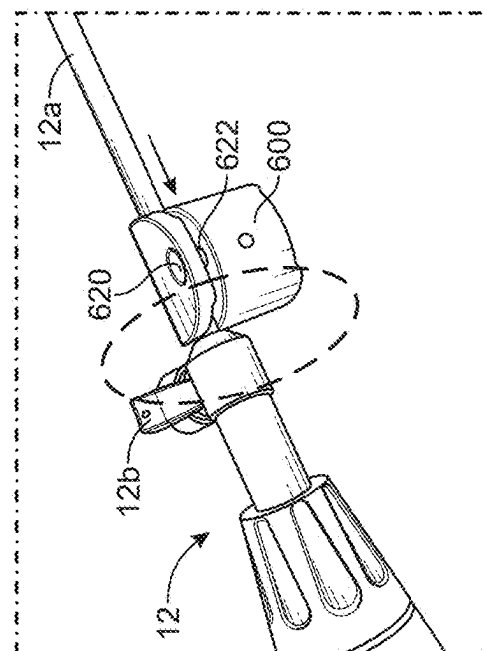


FIG. 6C

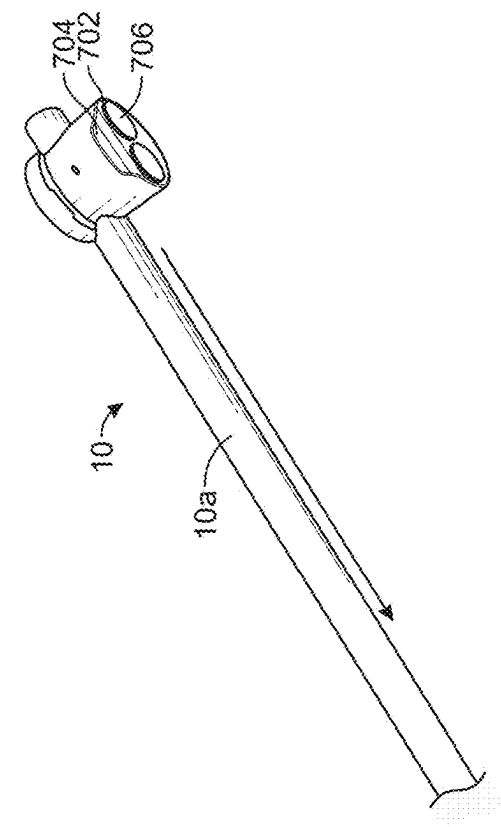


FIG. 7B

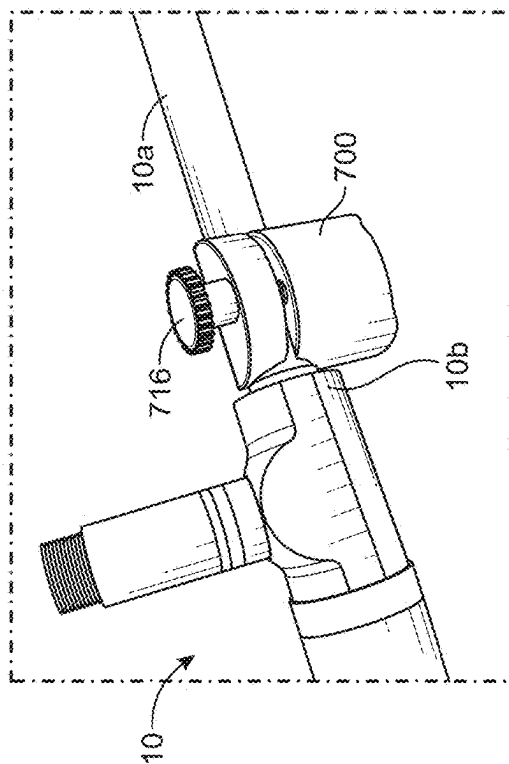


FIG. 7D

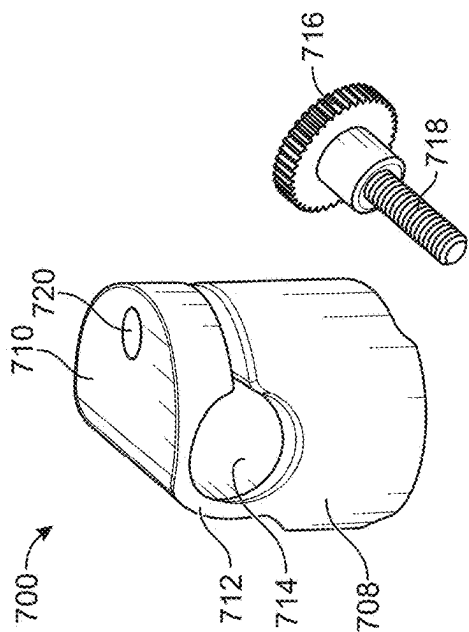


FIG. 7A

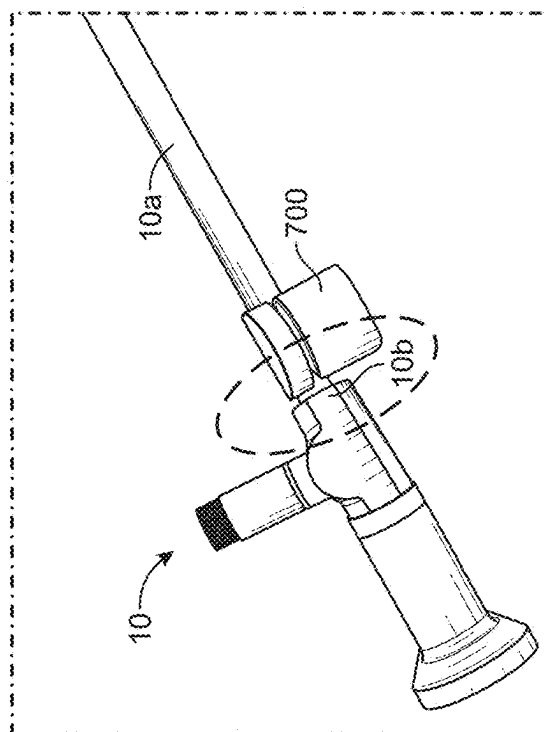


FIG. 7C

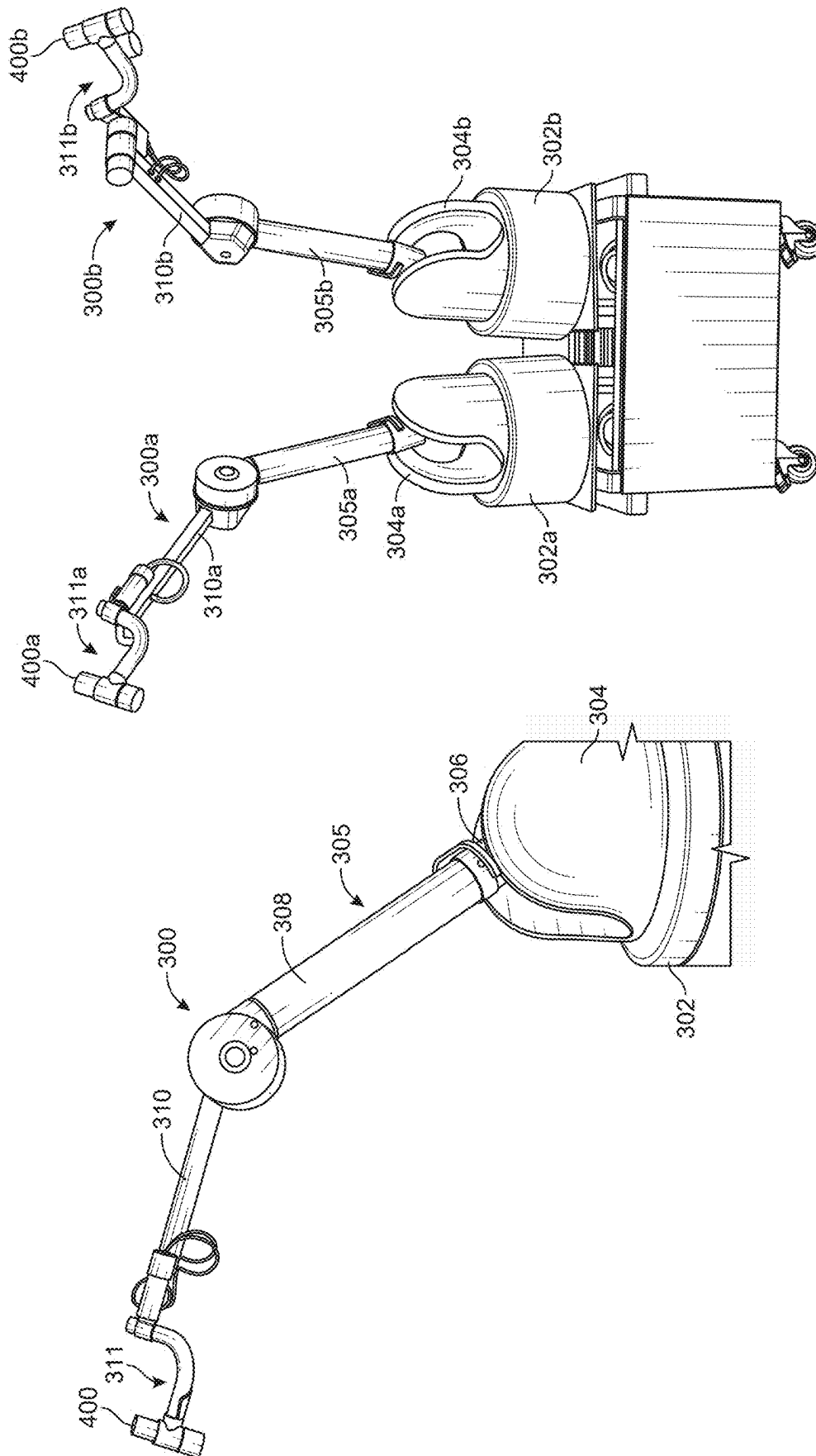
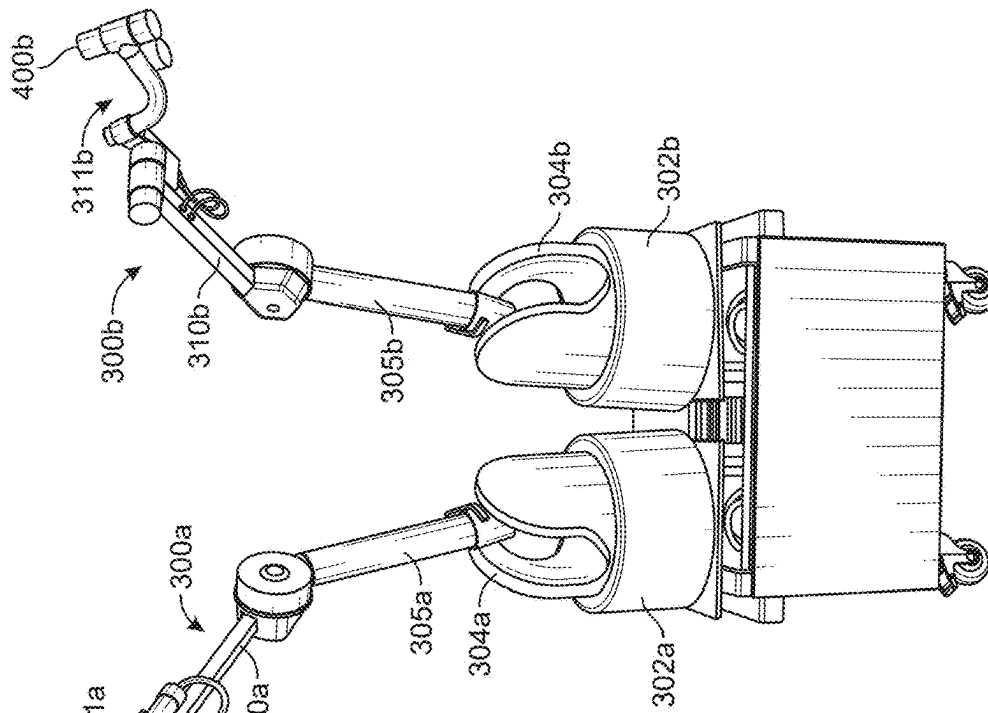


FIG. 8A



உள்ளுரை

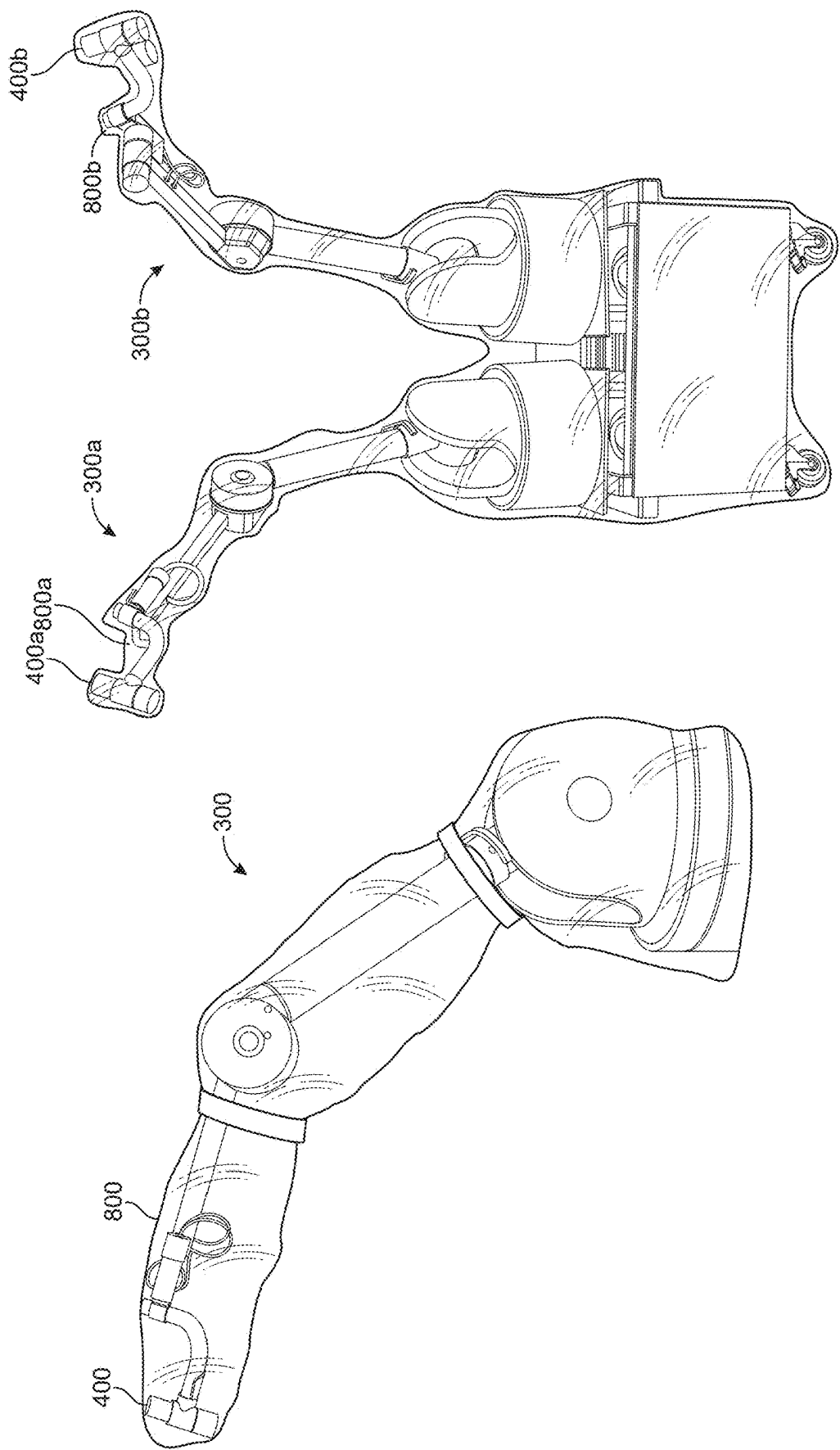


FIG. 9A

FIG. 9B

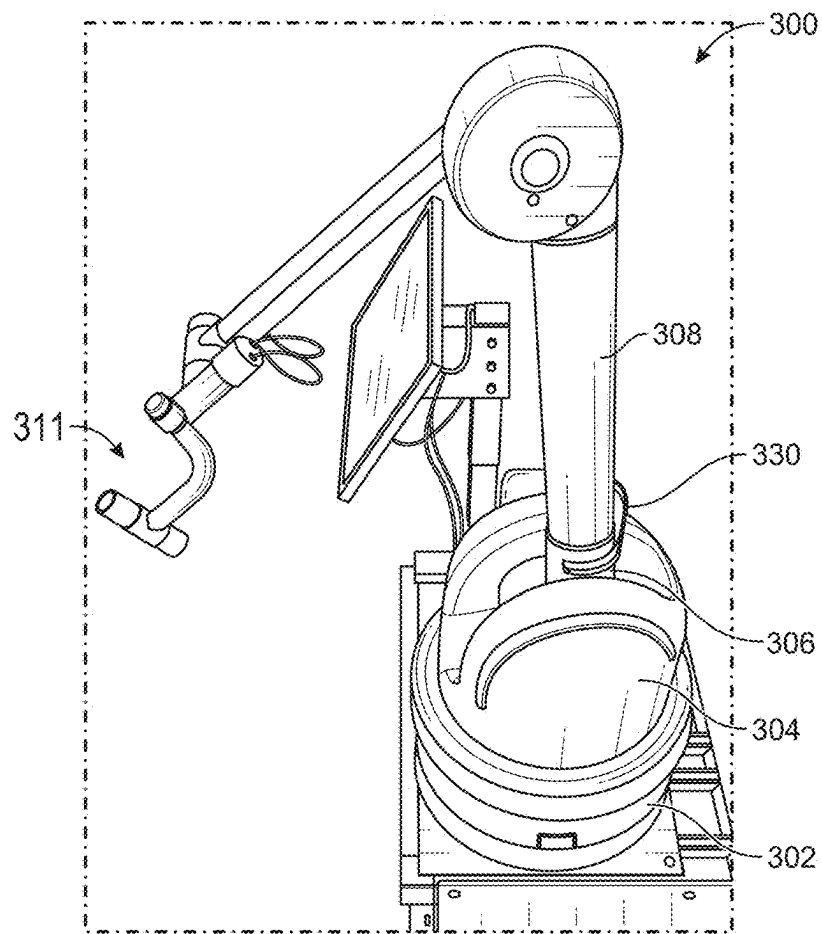


FIG. 10A

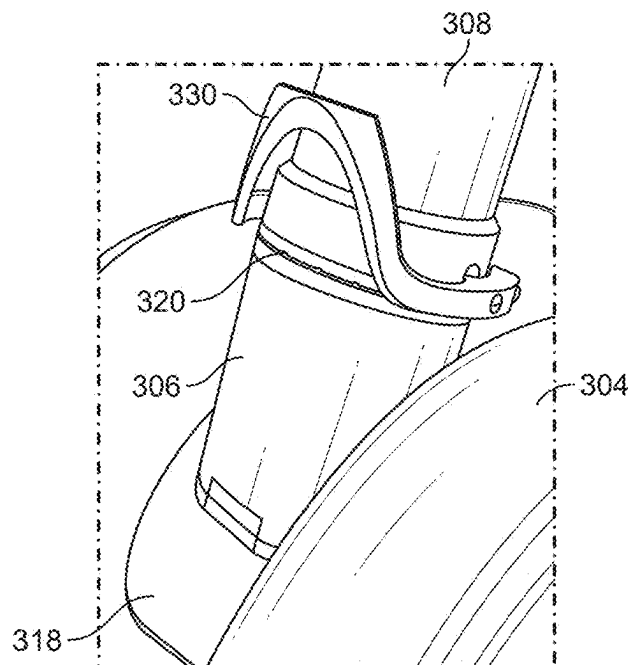


FIG. 10B

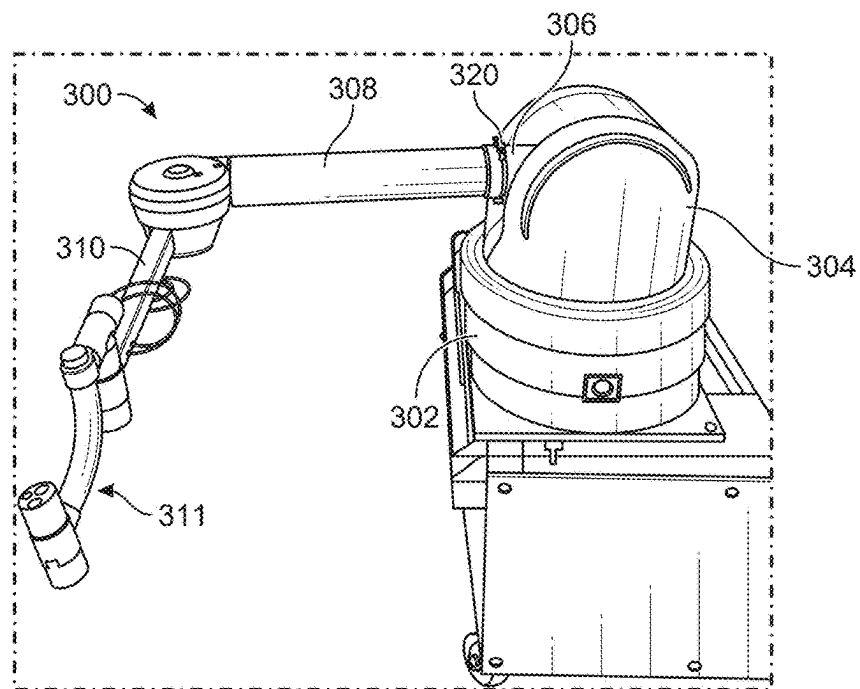


FIG. 10C

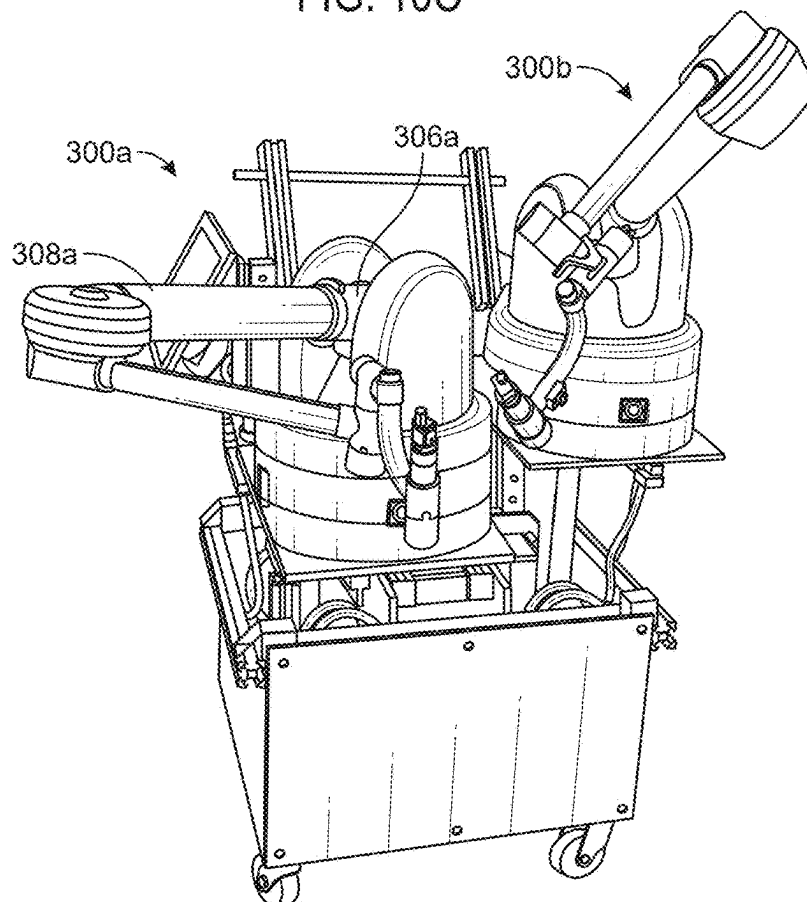
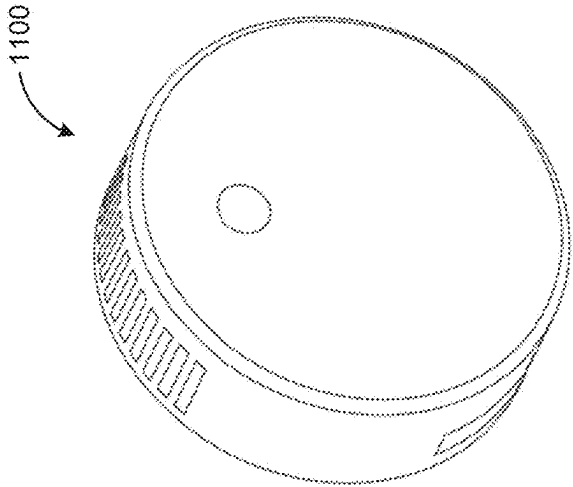
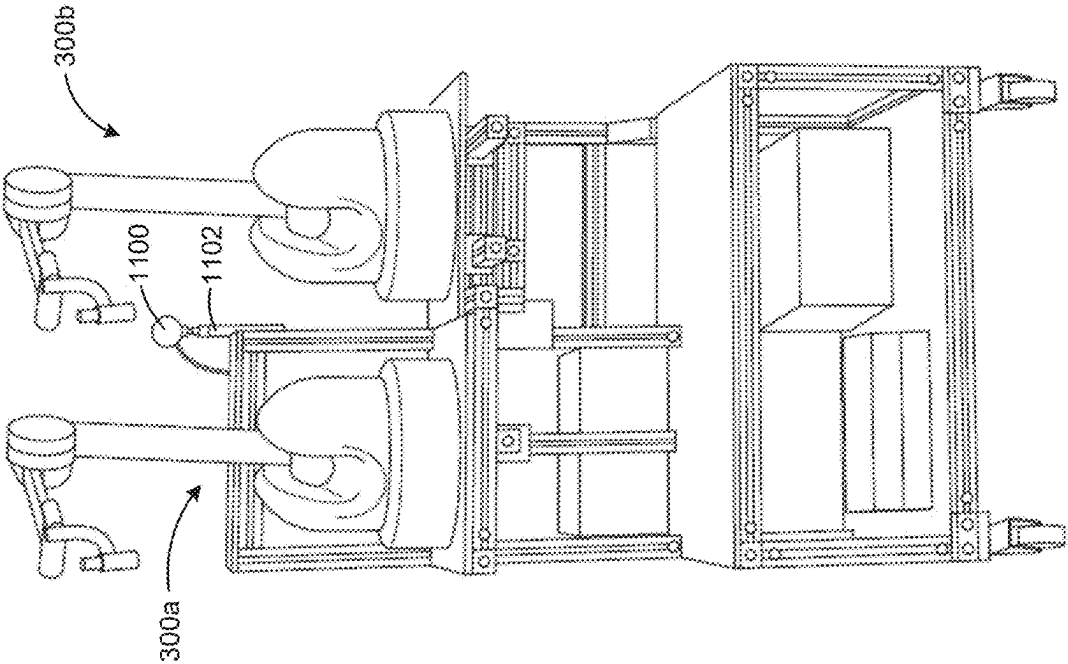


FIG. 10D



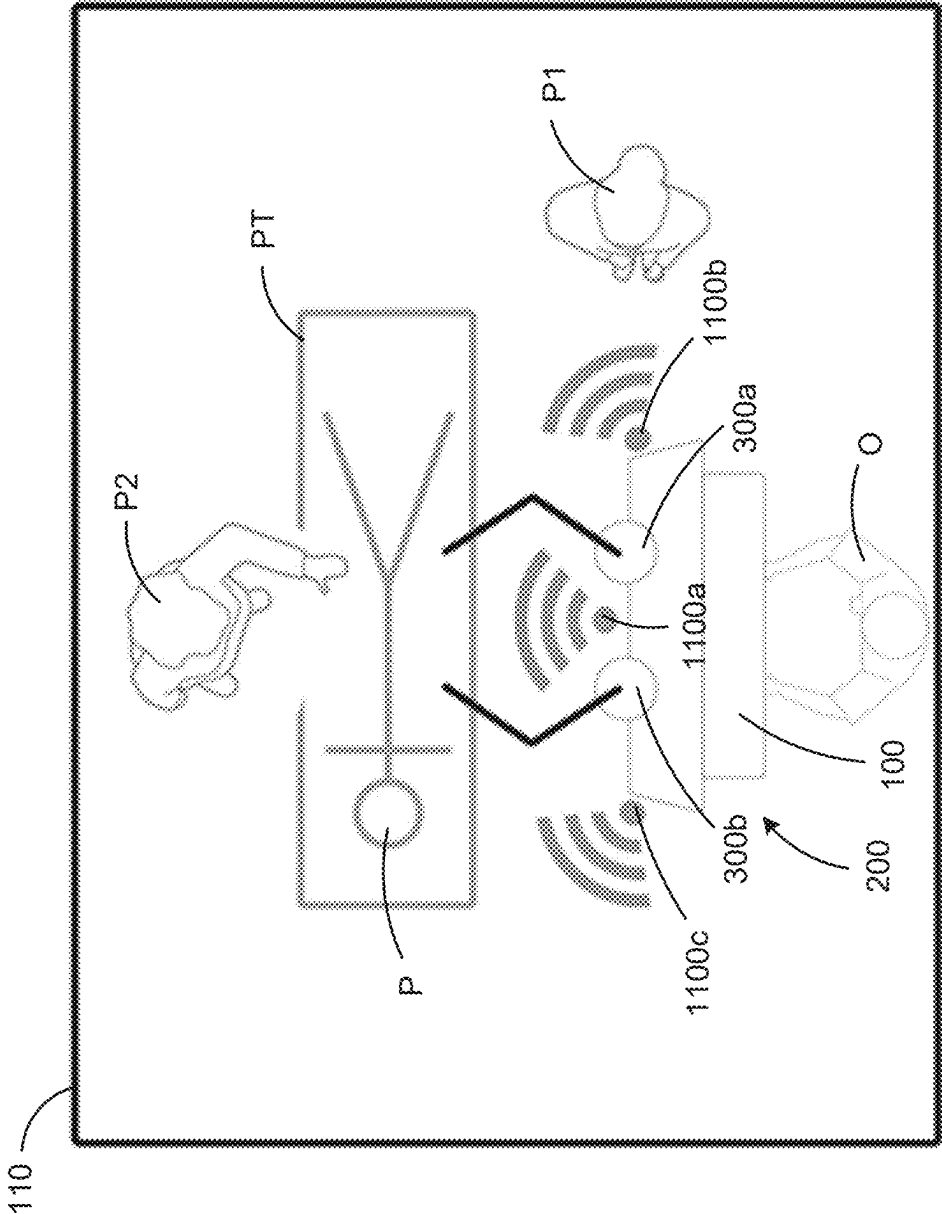


FIG. 11C

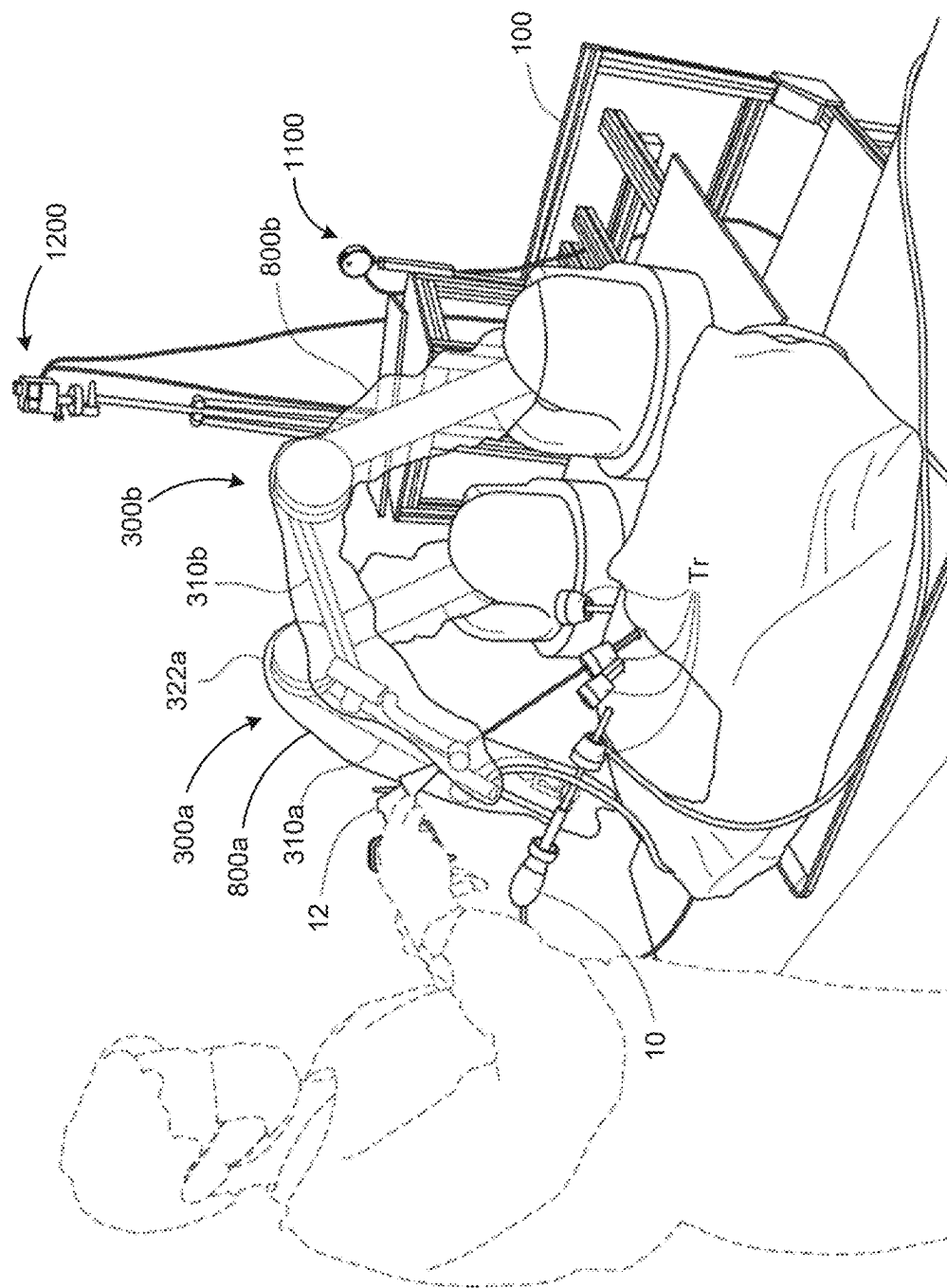


FIG. 12

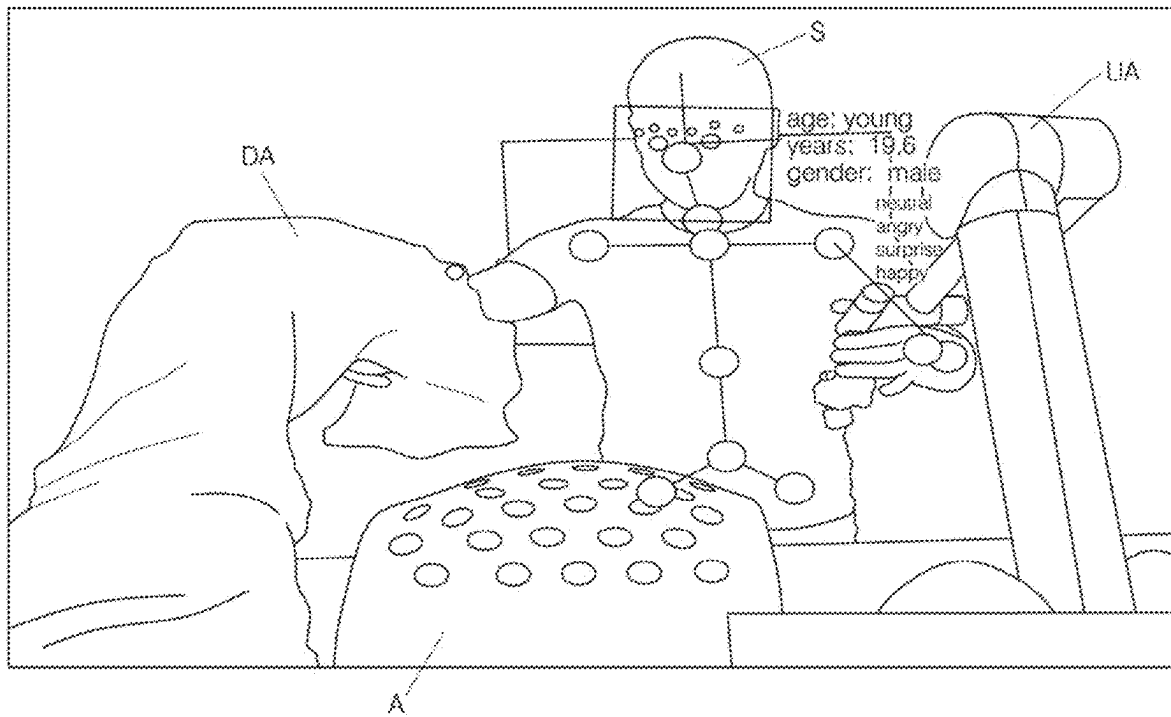


FIG. 13A

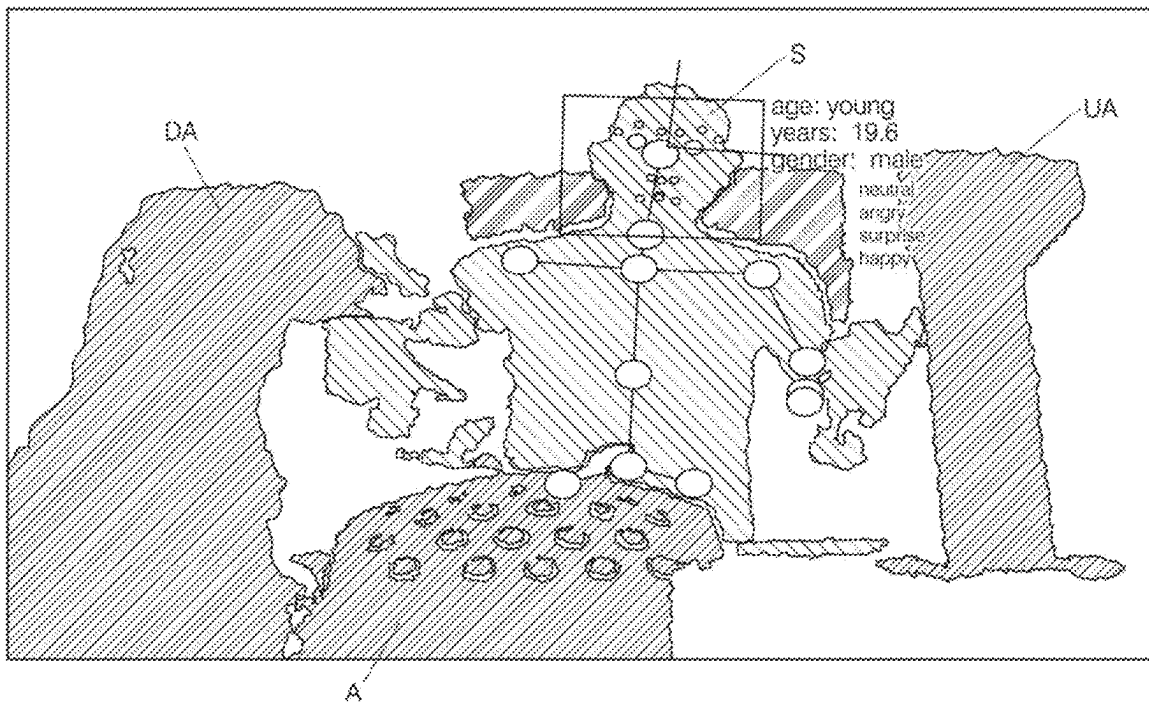
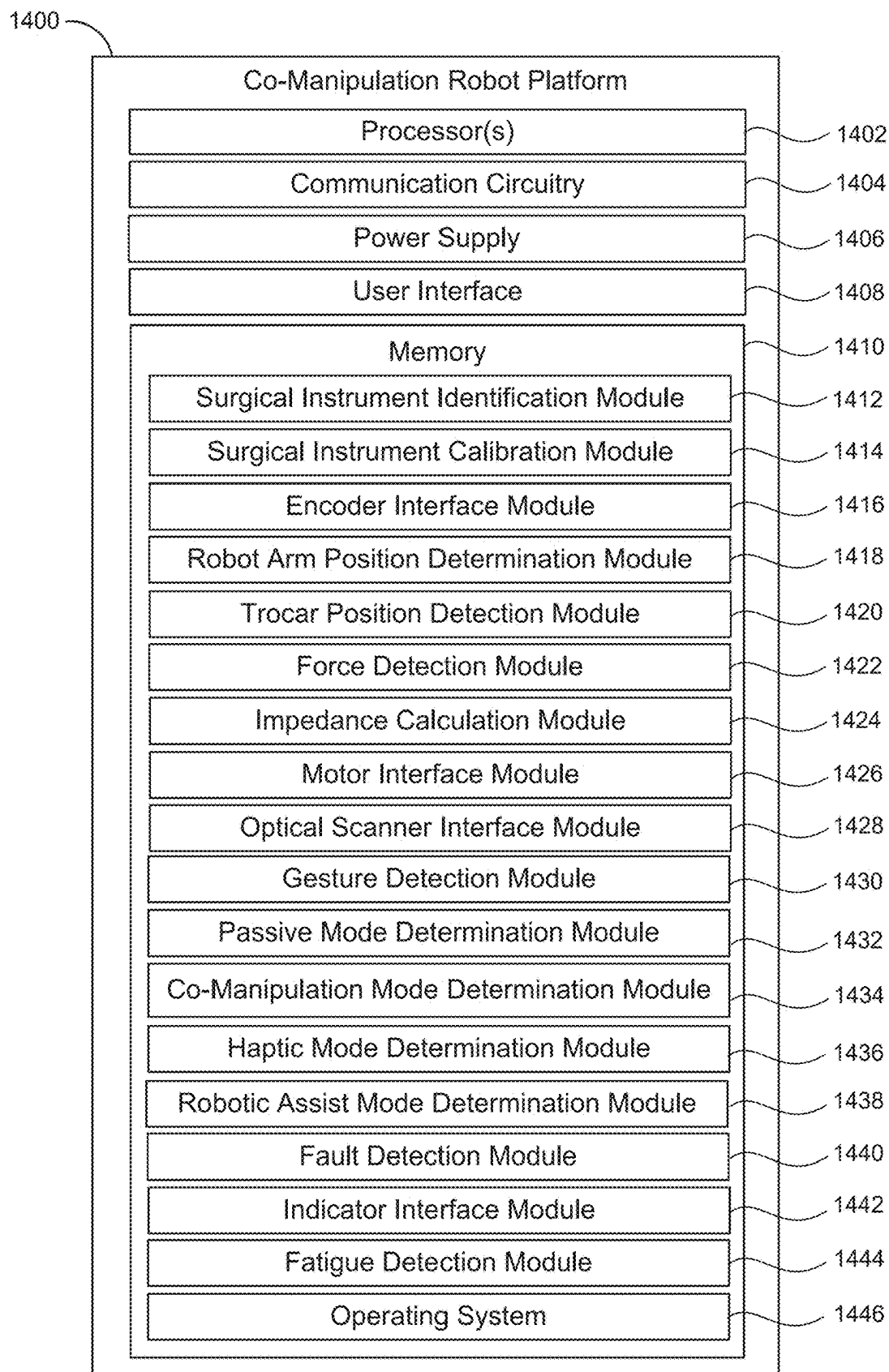


FIG. 13B



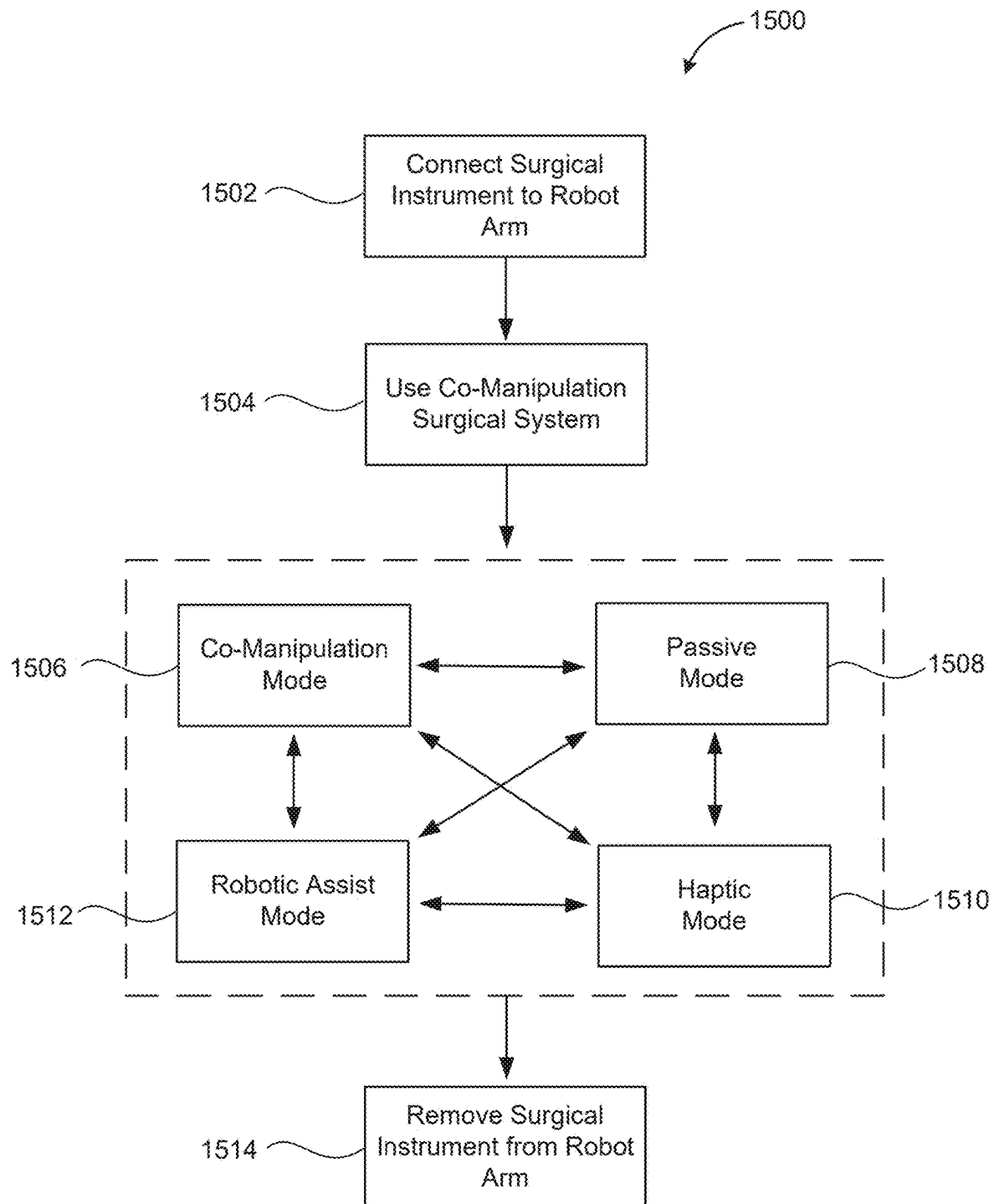


FIG. 15

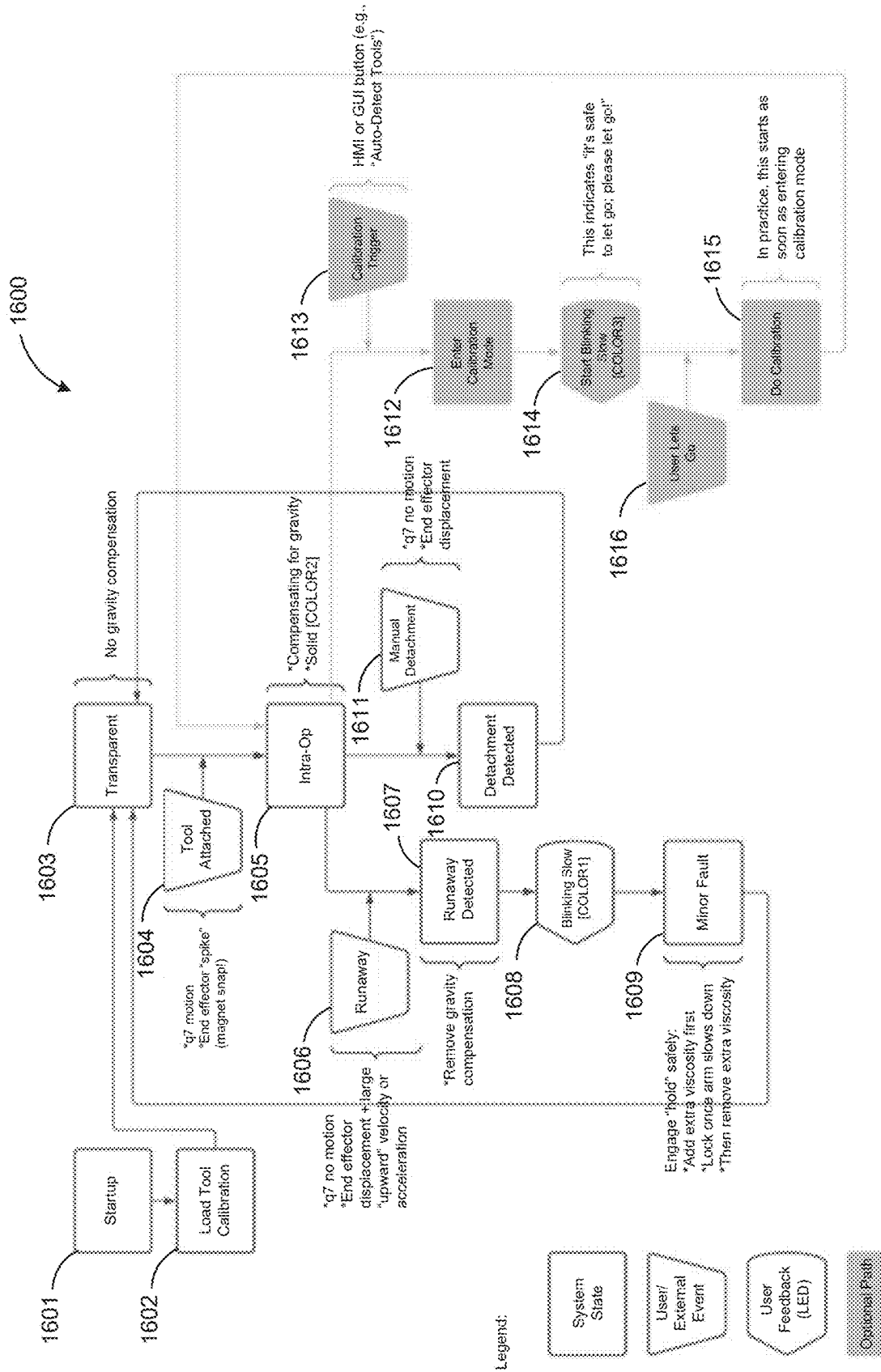


FIG. 16

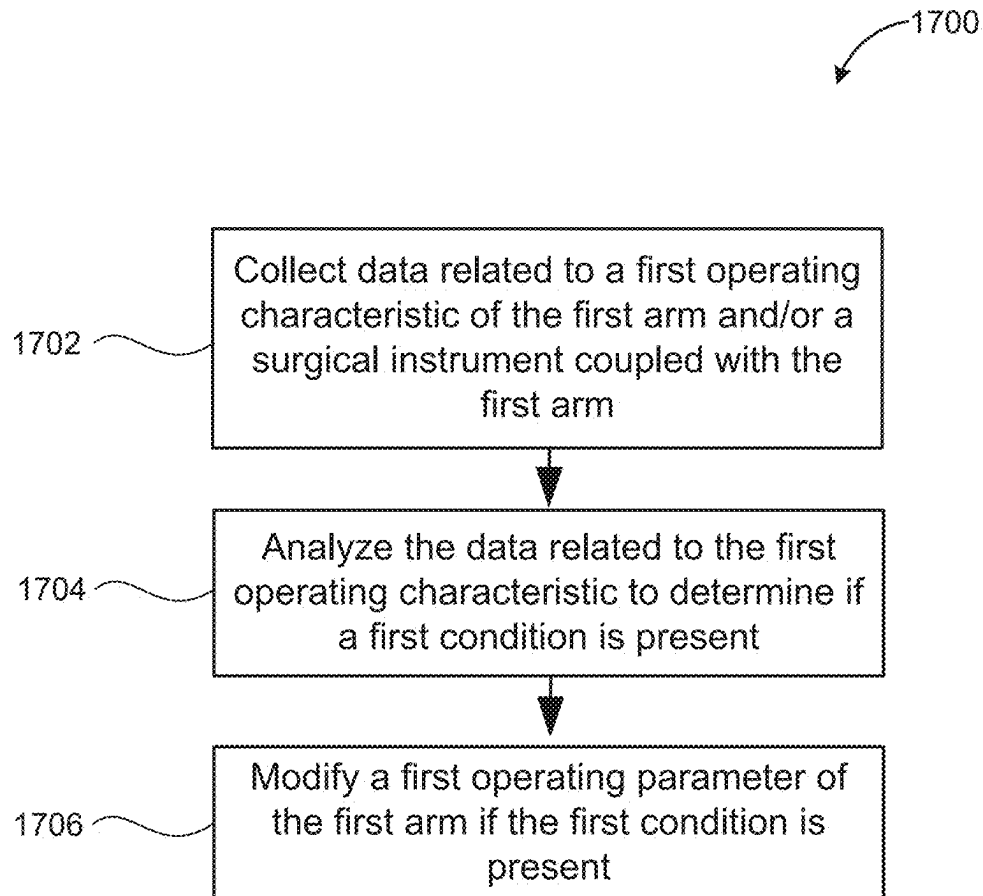


FIG. 17

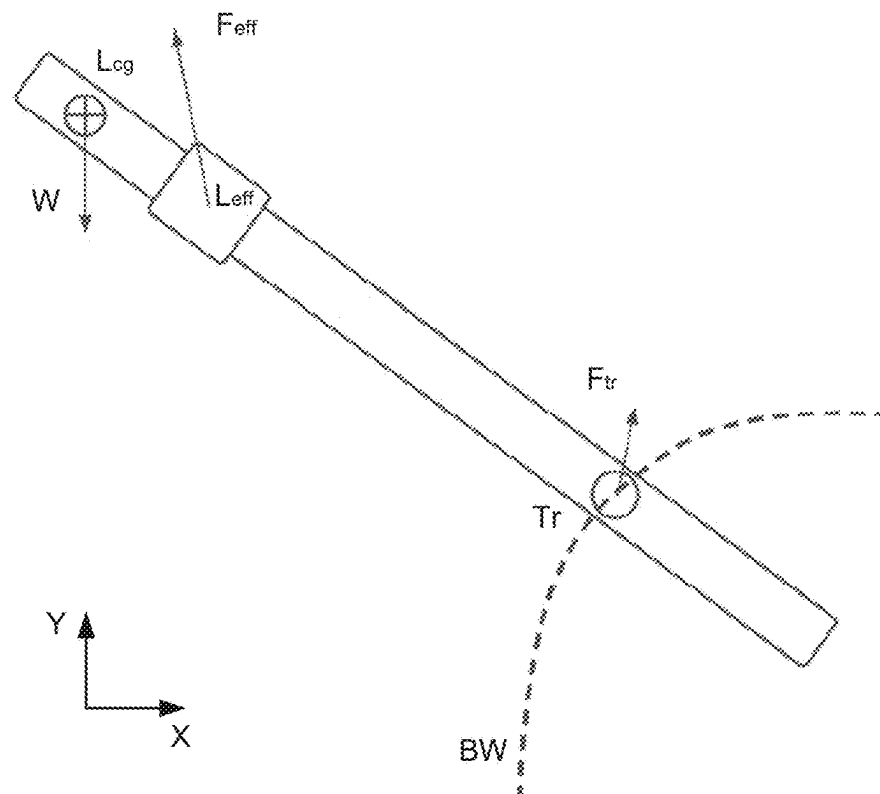


FIG. 18A

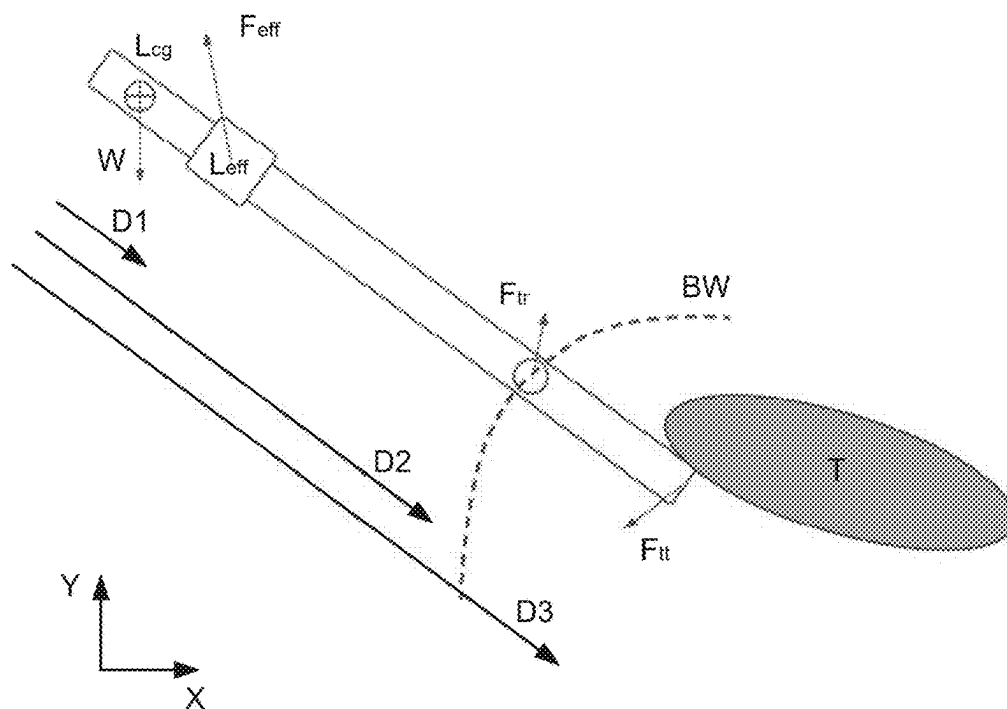


FIG. 18B

Instrument Type	Optional Predefined Threshold Release Force	Optional Viscosity Level	Optional Threshold Dwell Time	Optional Dwell Speed
Scope (e.g., without limitation, 5mm or 10mm)	Medium (e.g., without limitation, 7 N, or approximately 7 N)	Variable*	Low (0.25 second)	Medium (5 mm/second or approximately 5 mm/second)
Tissue manipulation instruments (e.g., without limitation, grasper/retractor)	Variable due to potential for external forces on surgical instrument from tissue (e.g., without limitation, 7 N or approximately 7 N to 15 N or approximately 15 N)	Low (0 N)	Low (0.25 second)	Medium (5 mm/second or approximately 5 mm/second)
Suturing instruments (e.g., without limitation, needle driver)	Medium (e.g., without limitation, 7 N, or approximately 7 N)	Variable*	Low (0.25 second)	Low (1 mm/second or approximately 1 mm/second)
High force instruments (e.g., without limitation, stapler/clip applier)	High (e.g., without limitation, 10 N, or approximately 10 N)	Variable*	Low (0.25 second)	Low (1 mm/second or approximately 1 mm/second)

FIG. 19

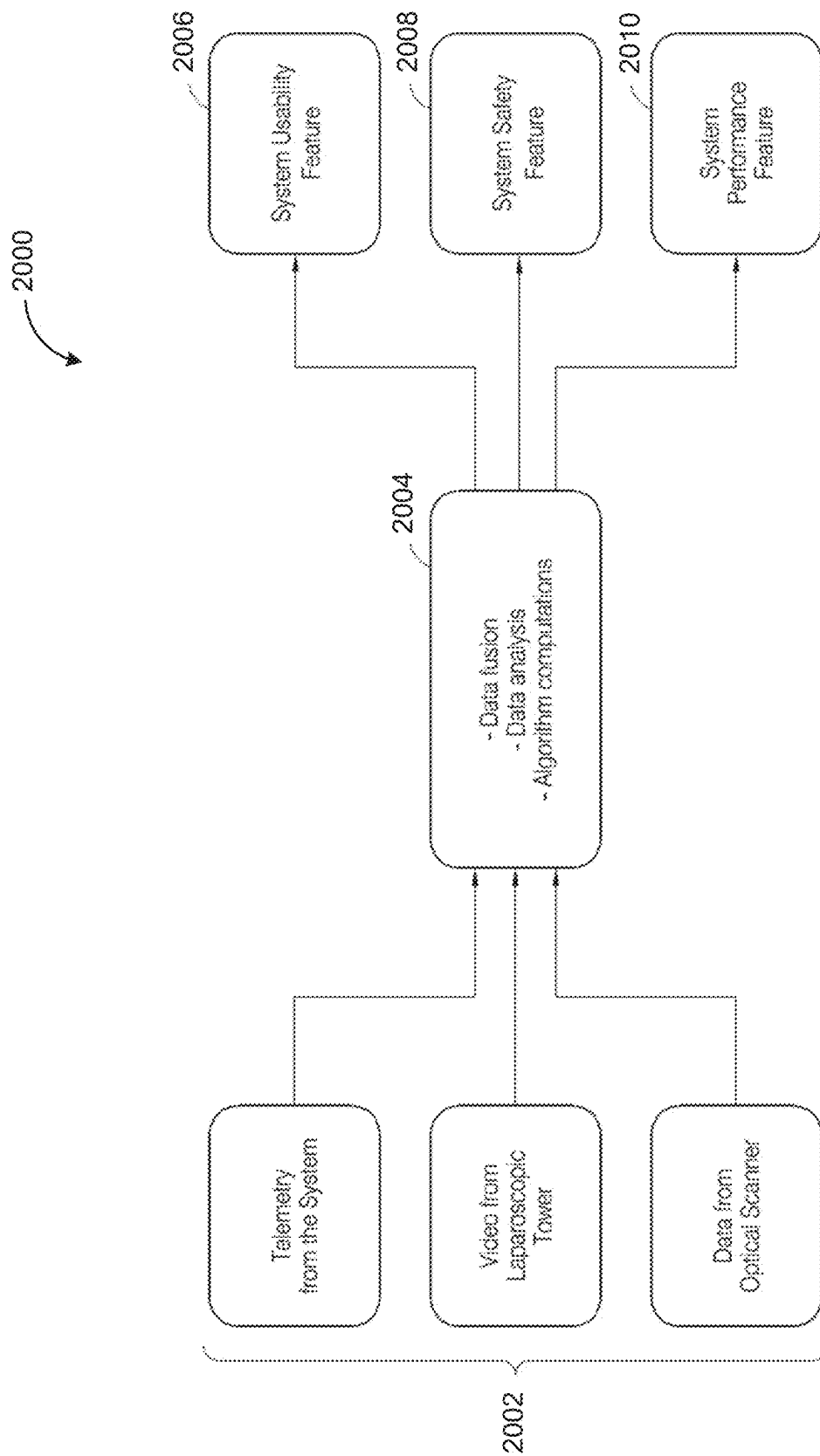


FIG. 20

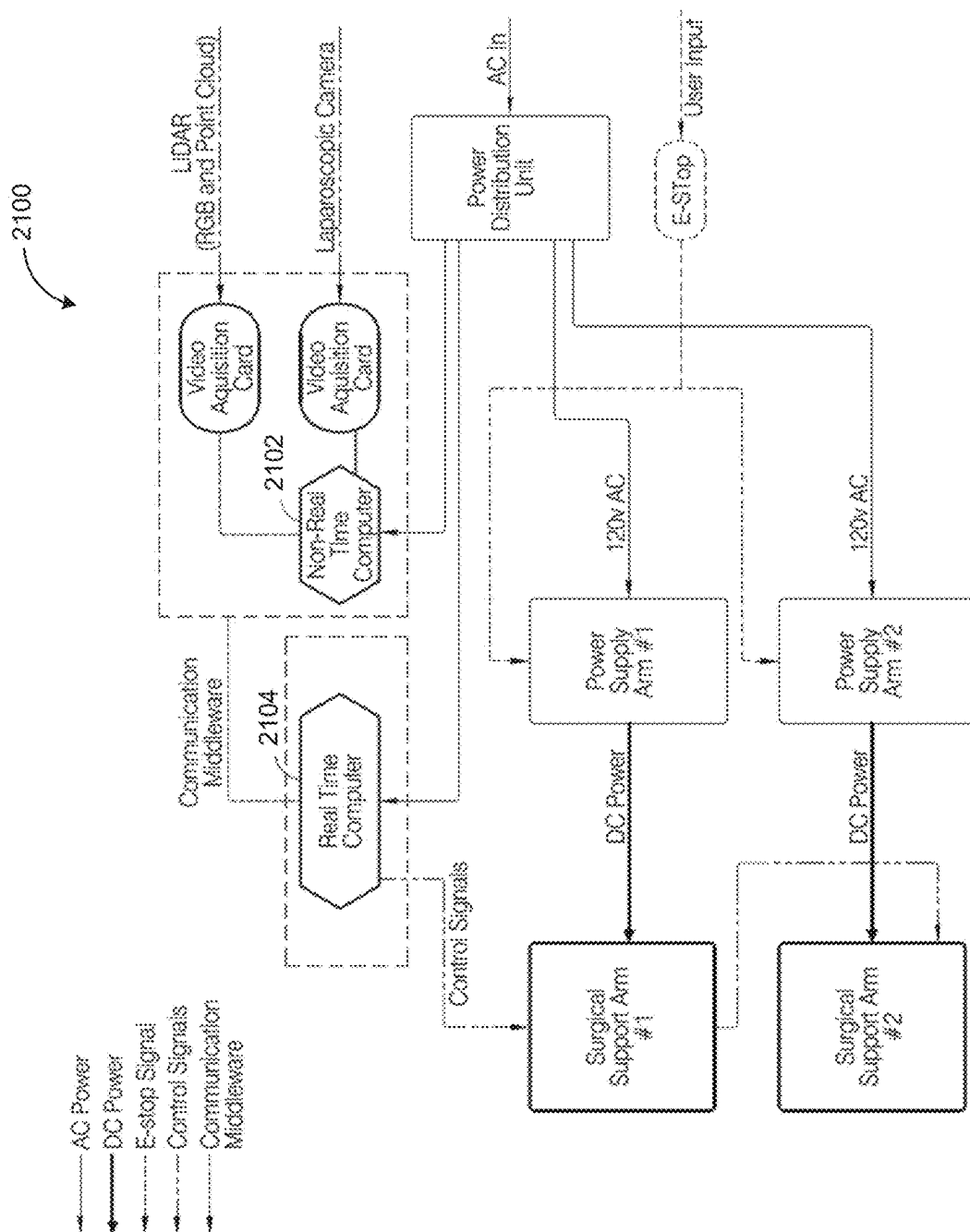


FIG. 21

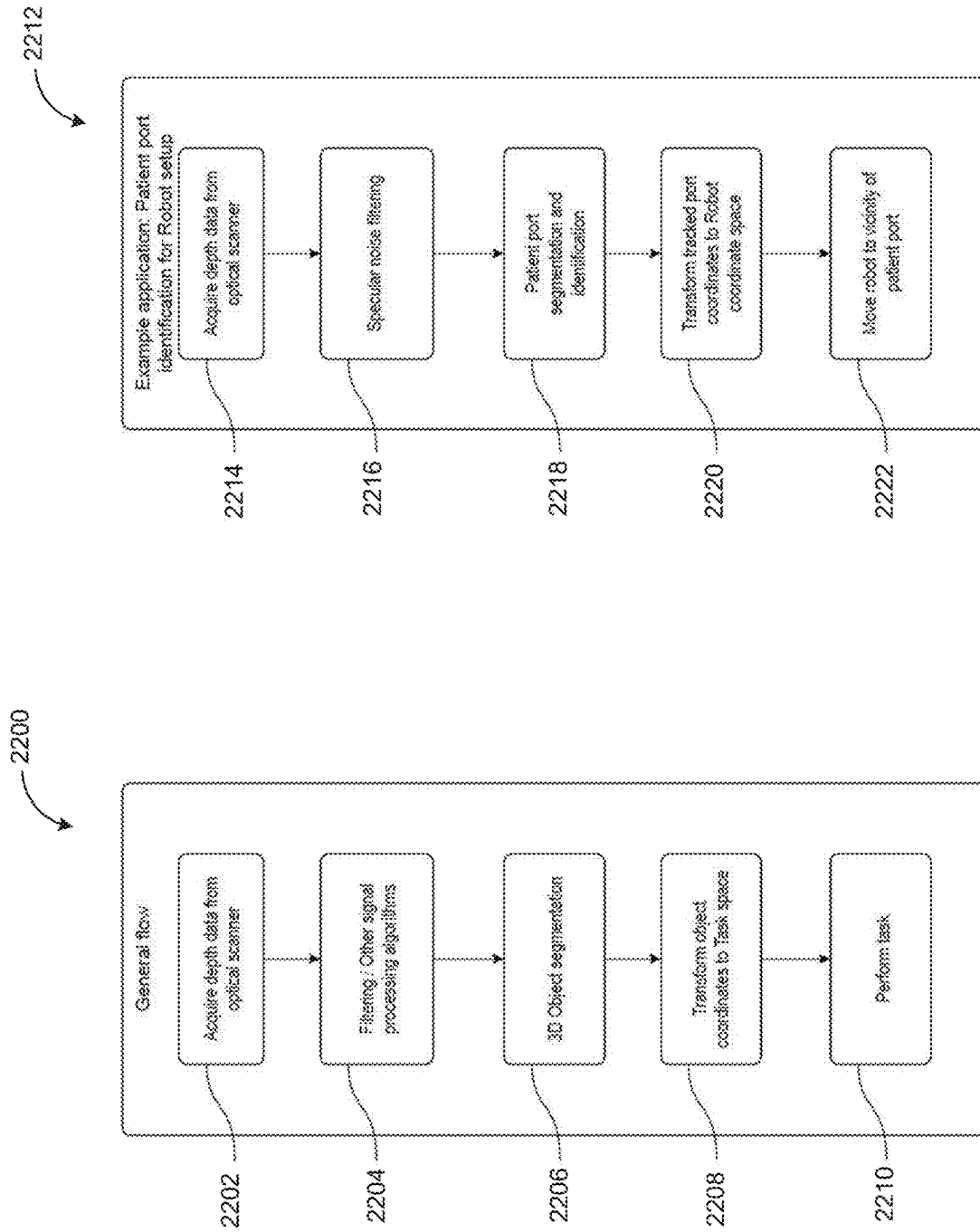


FIG. 22

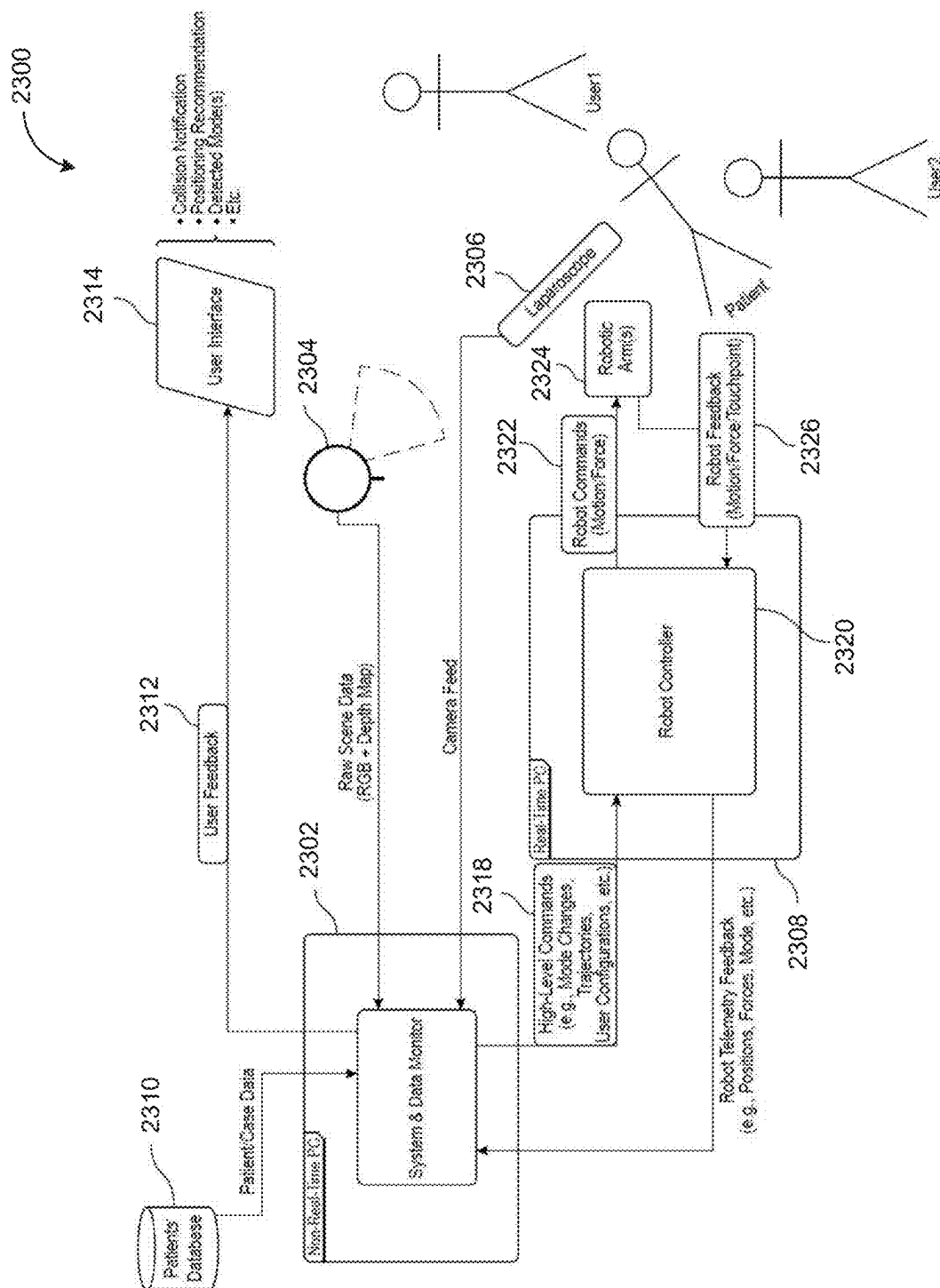


FIG. 23

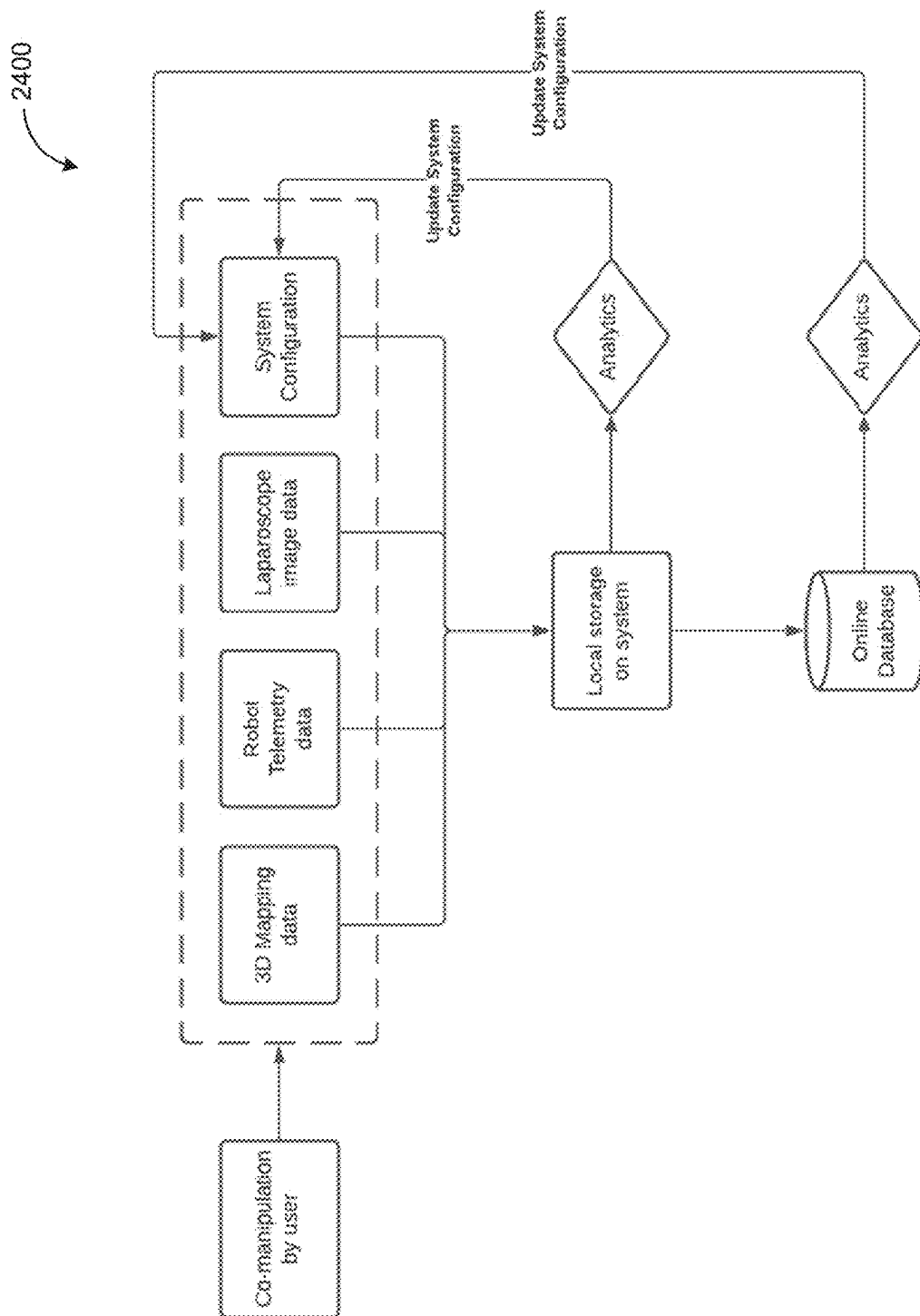


FIG. 24

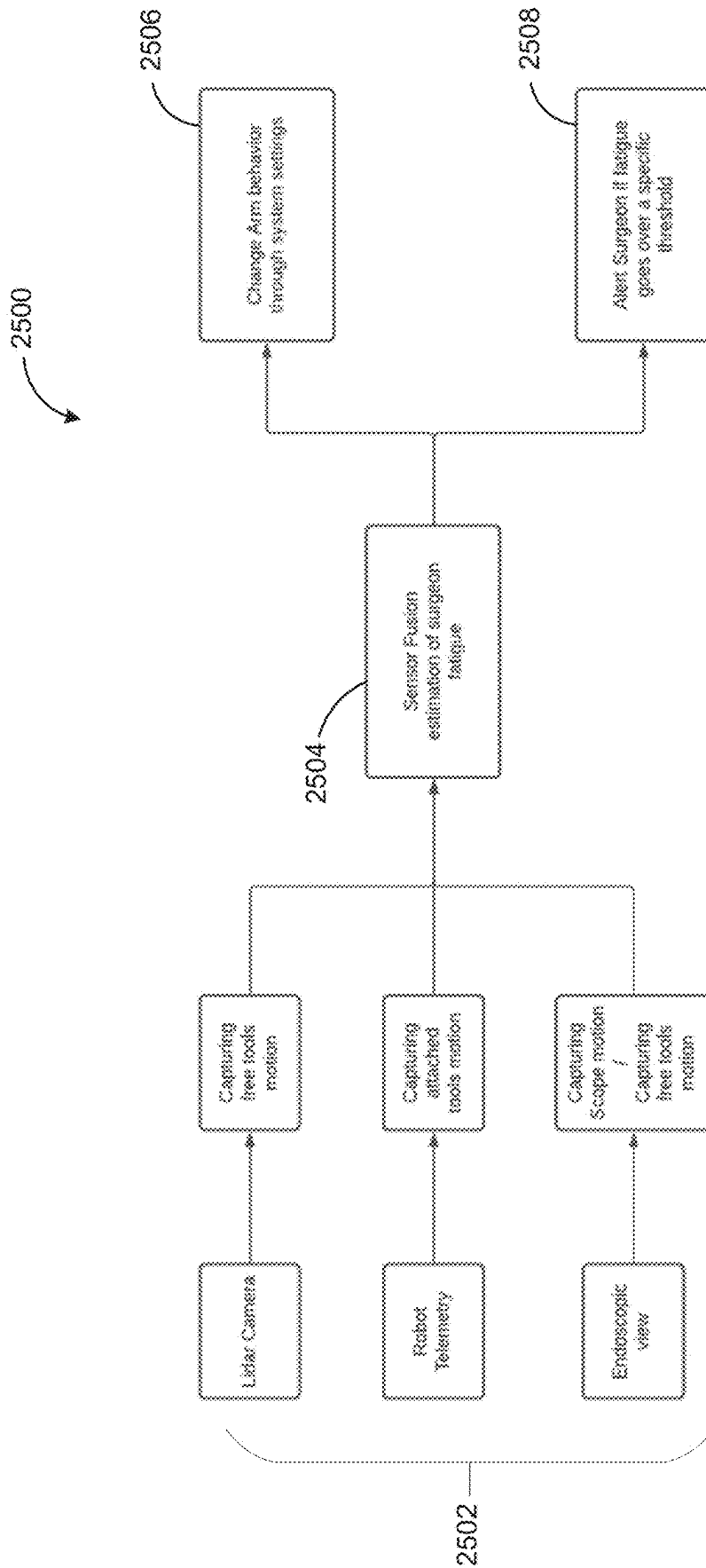


FIG. 25

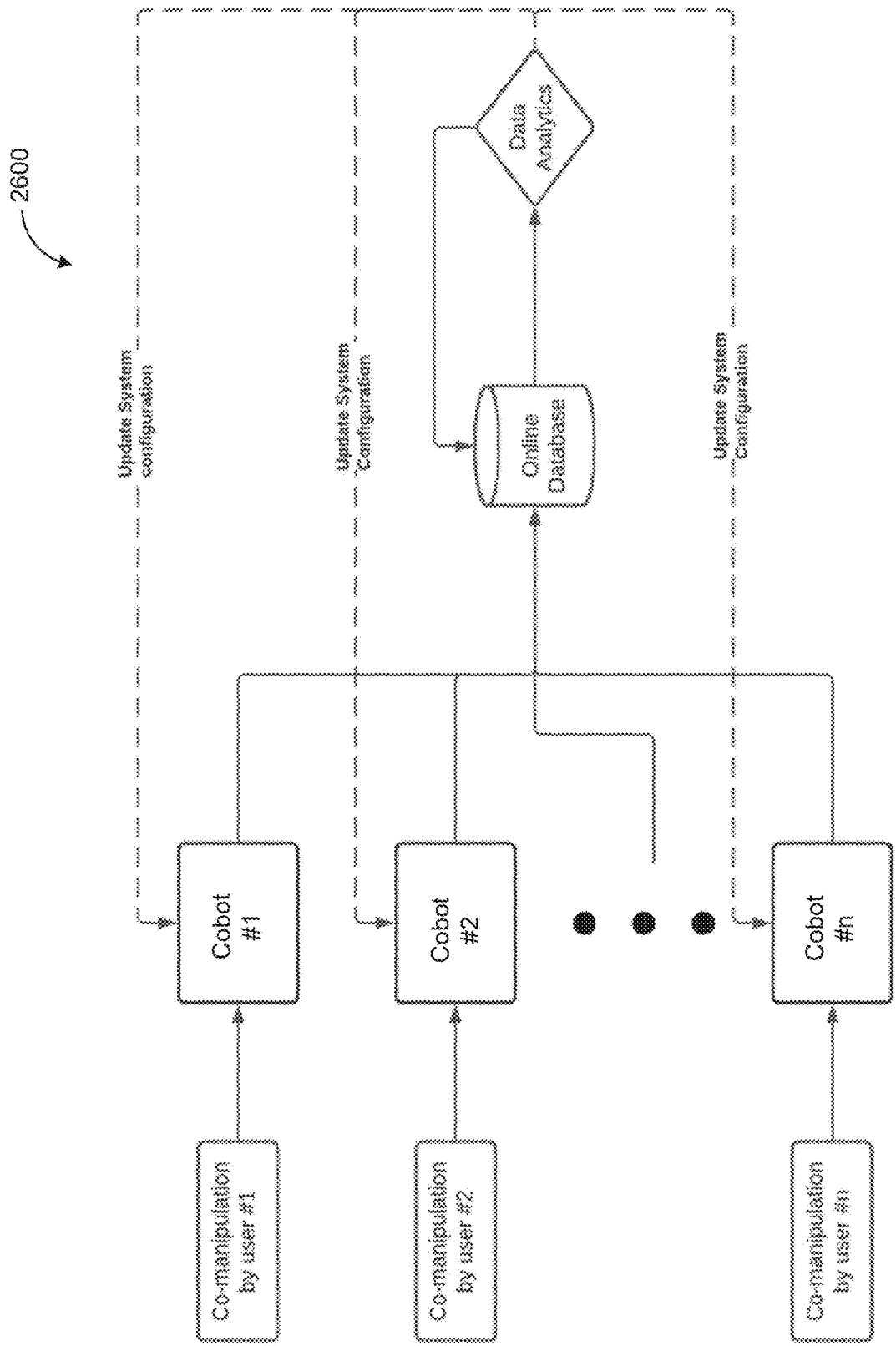


FIG. 26

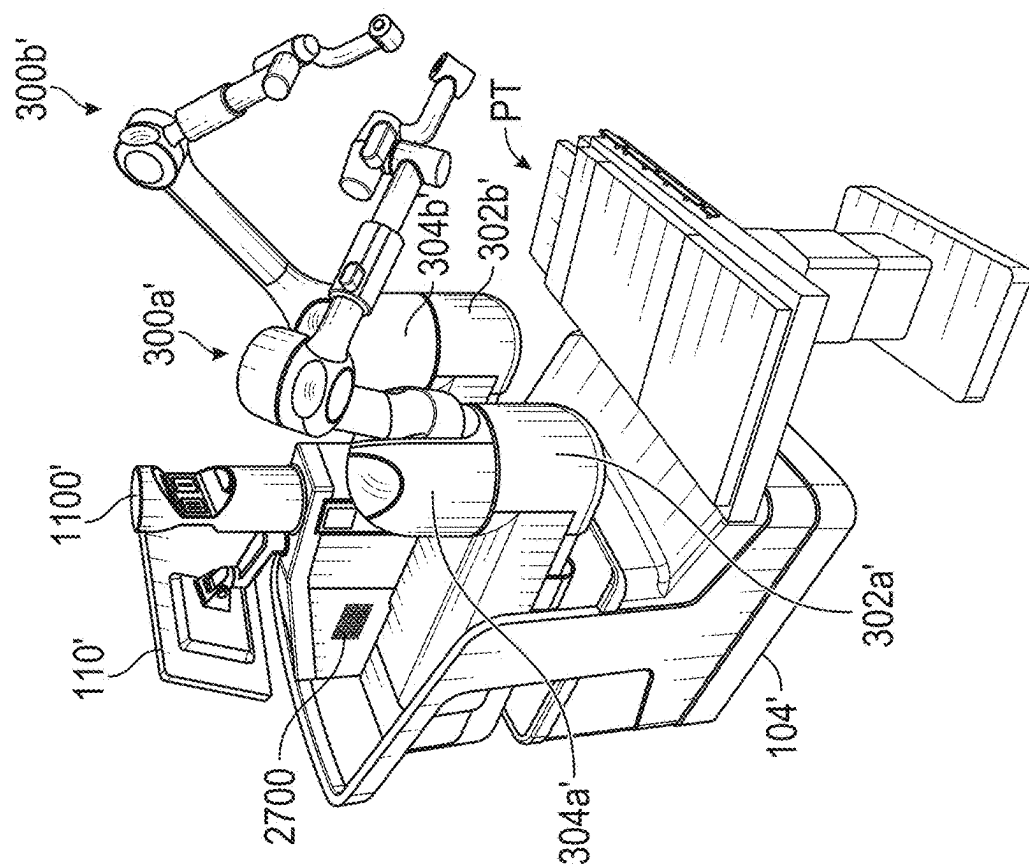


FIG. 27B

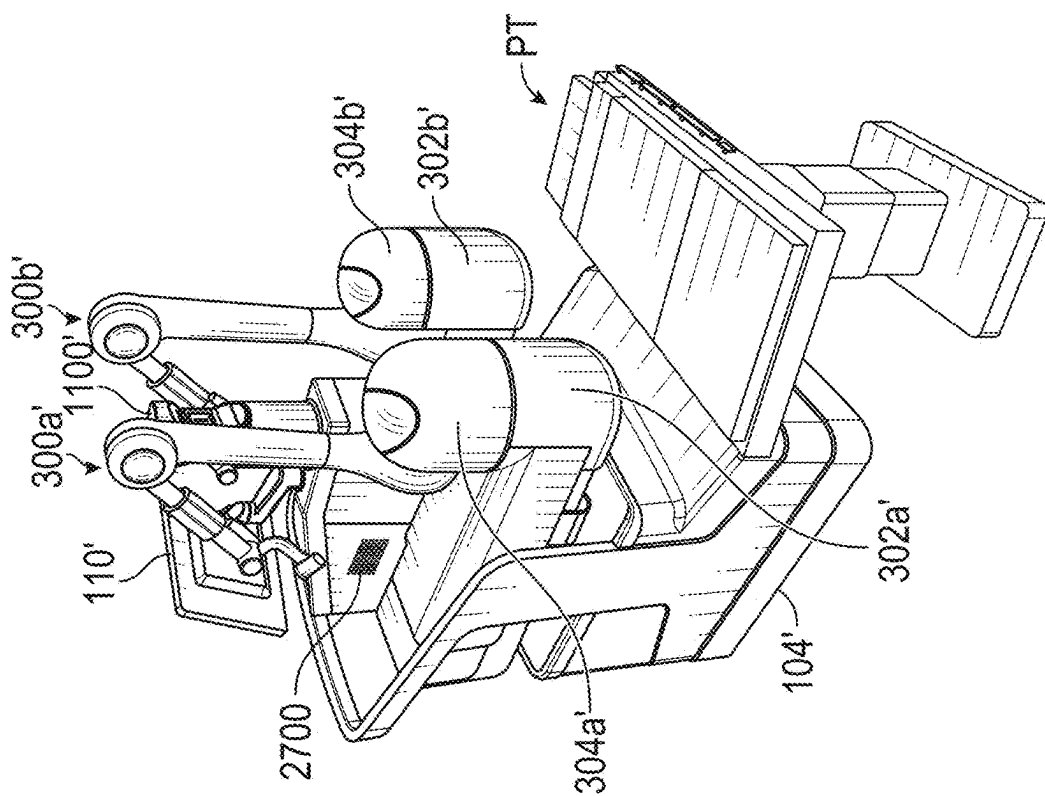


FIG. 27A

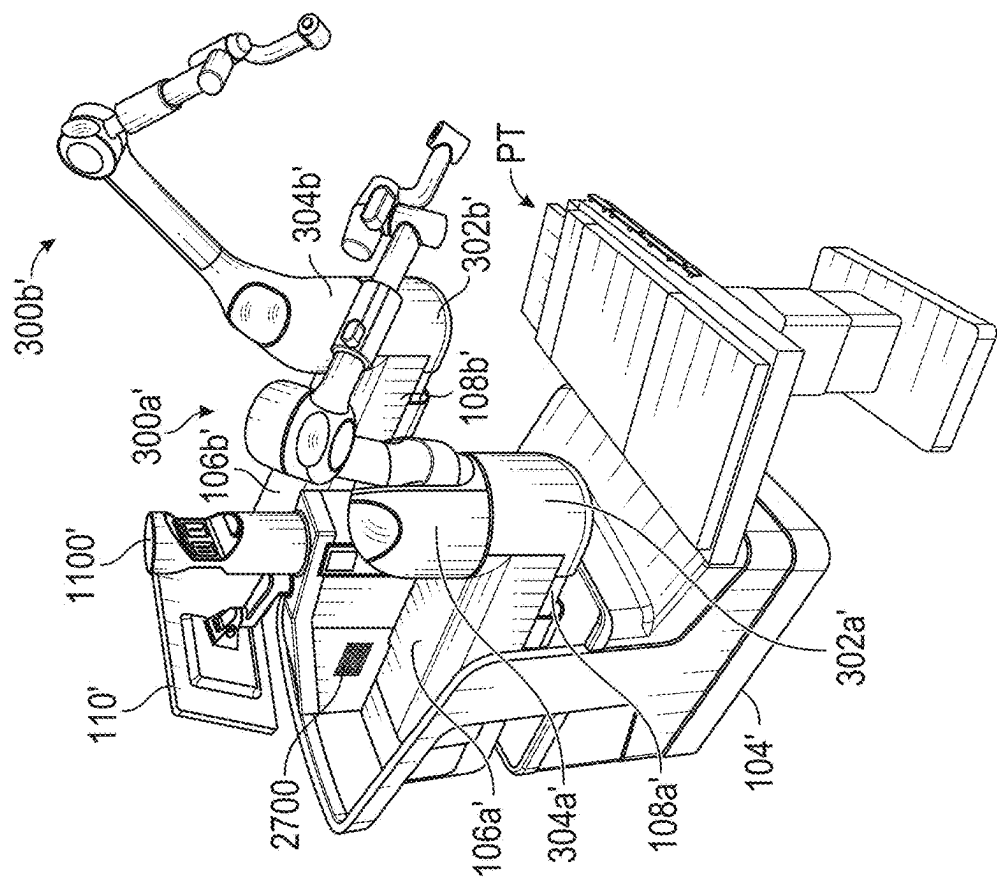


FIG. 27D

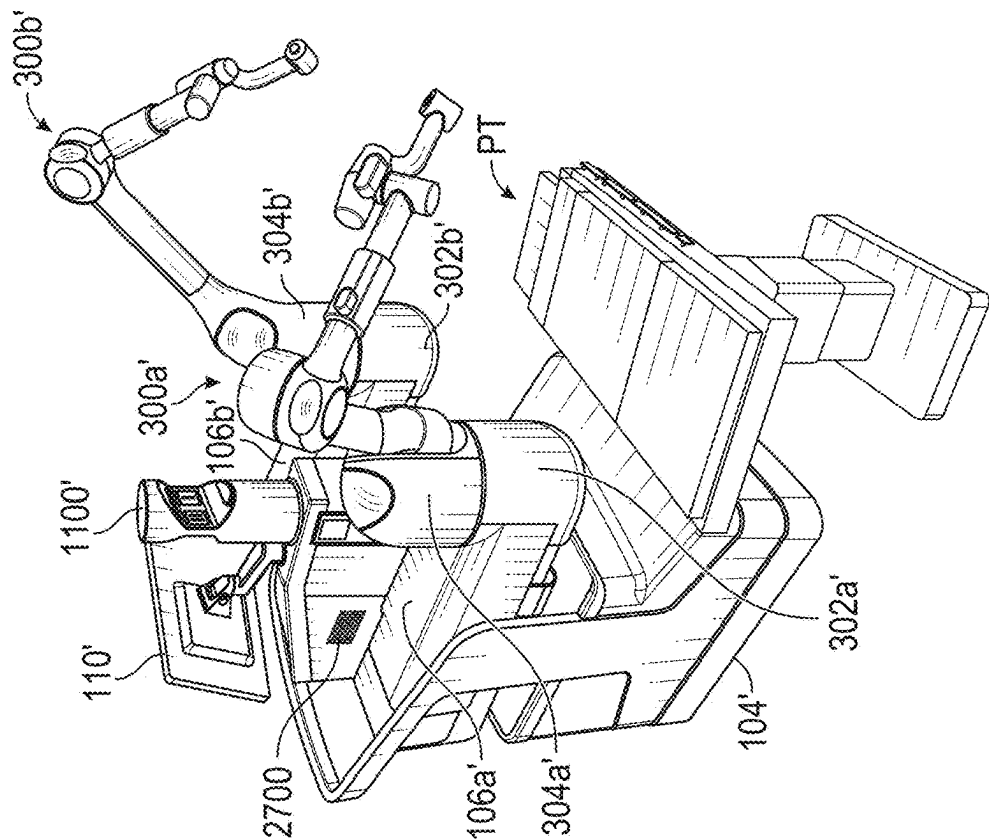


FIG. 27C

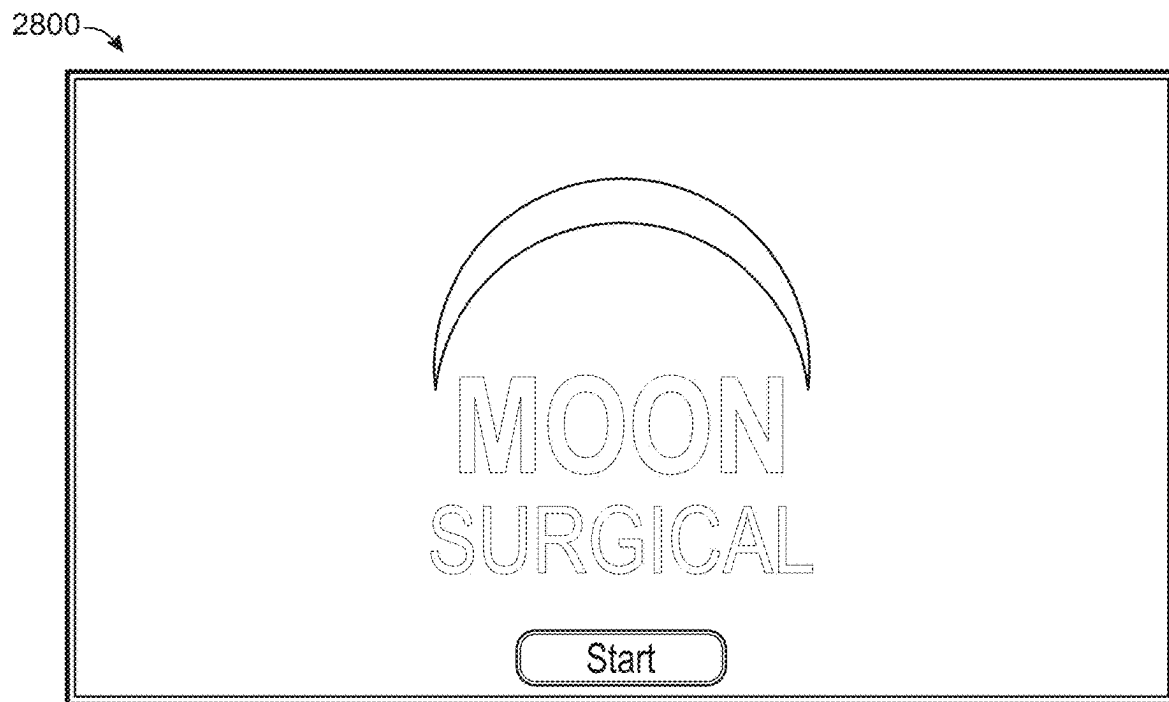


FIG. 28A

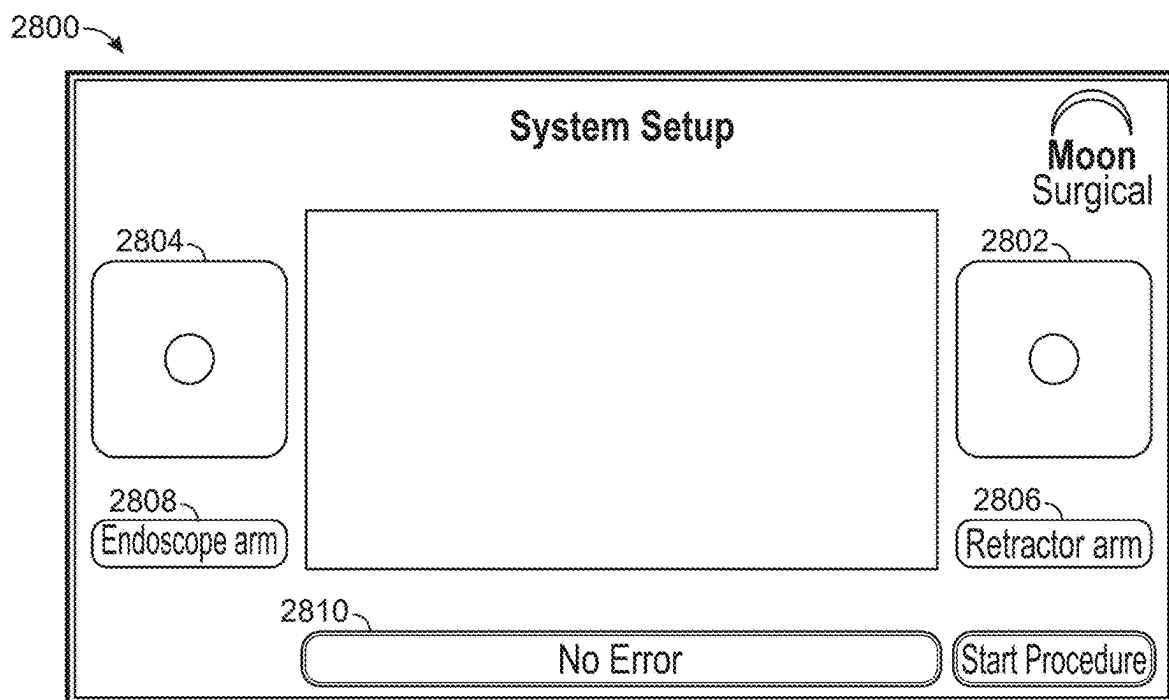


FIG. 28B

2800

Maintenance

Moon Surgical

Tool name: Scope

Tool length (mm): 500

Tip to coupler (mm): 310

Tip to CoG (mm): 385

Mass (g): 700

Unlocking Force (N): 7.0

Auto Calibration

Raw Calibration

Unlocking Force Low

Unlocking Force High

Tool name: Grasper1

Tool length (mm): 420

Tip to coupler (mm): 340

Tip to CoG (mm): 366

Mass (g): 136

Unlocking Force (N): 7.0

Auto Calibration

Raw Calibration

Unlocking Force Low

Unlocking Force High

Endoscope arm

Retractor arm

2808

2806

2810

No Error

Resume

FIG. 28C

2800

Procedure

Moon Surgical

Scope

Trocar Force: 6.9 N

Trocar Force: 0.0 N

Lock

2808

Endoscope arm

Grasper1

Trocar Force: 12.2 N

Tooltip Force: 10.9 N

Lock

2806

Endoscope arm

2810

End Procedure

No Error

Setup

FIG. 28D

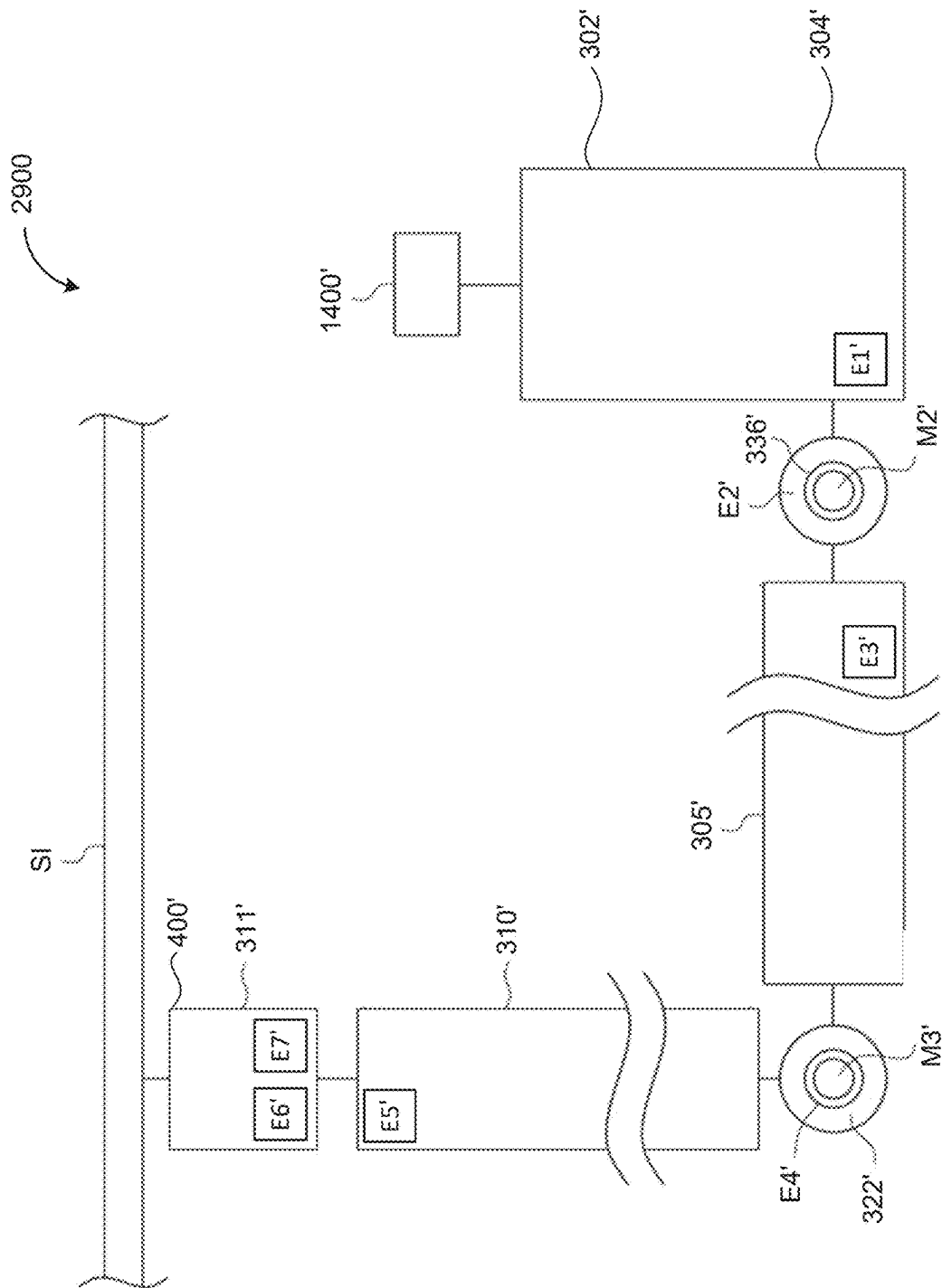


FIG. 29

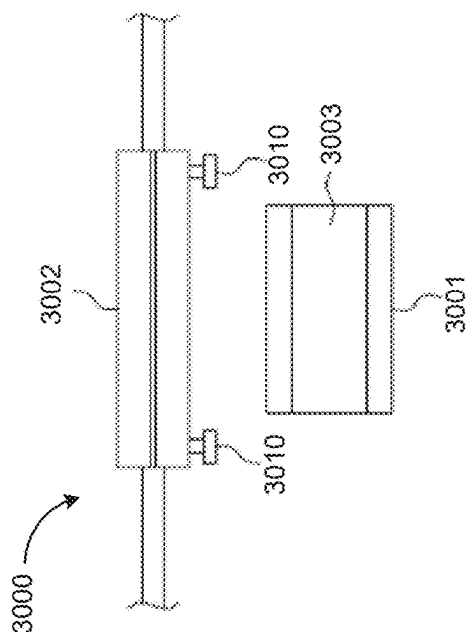


FIG. 30A

FIG. 30B

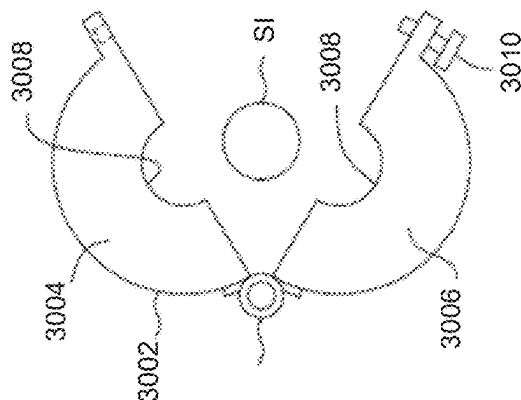


FIG. 30C

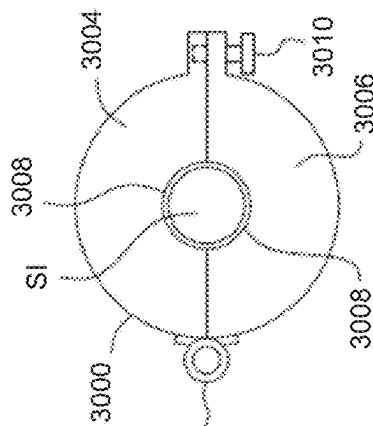
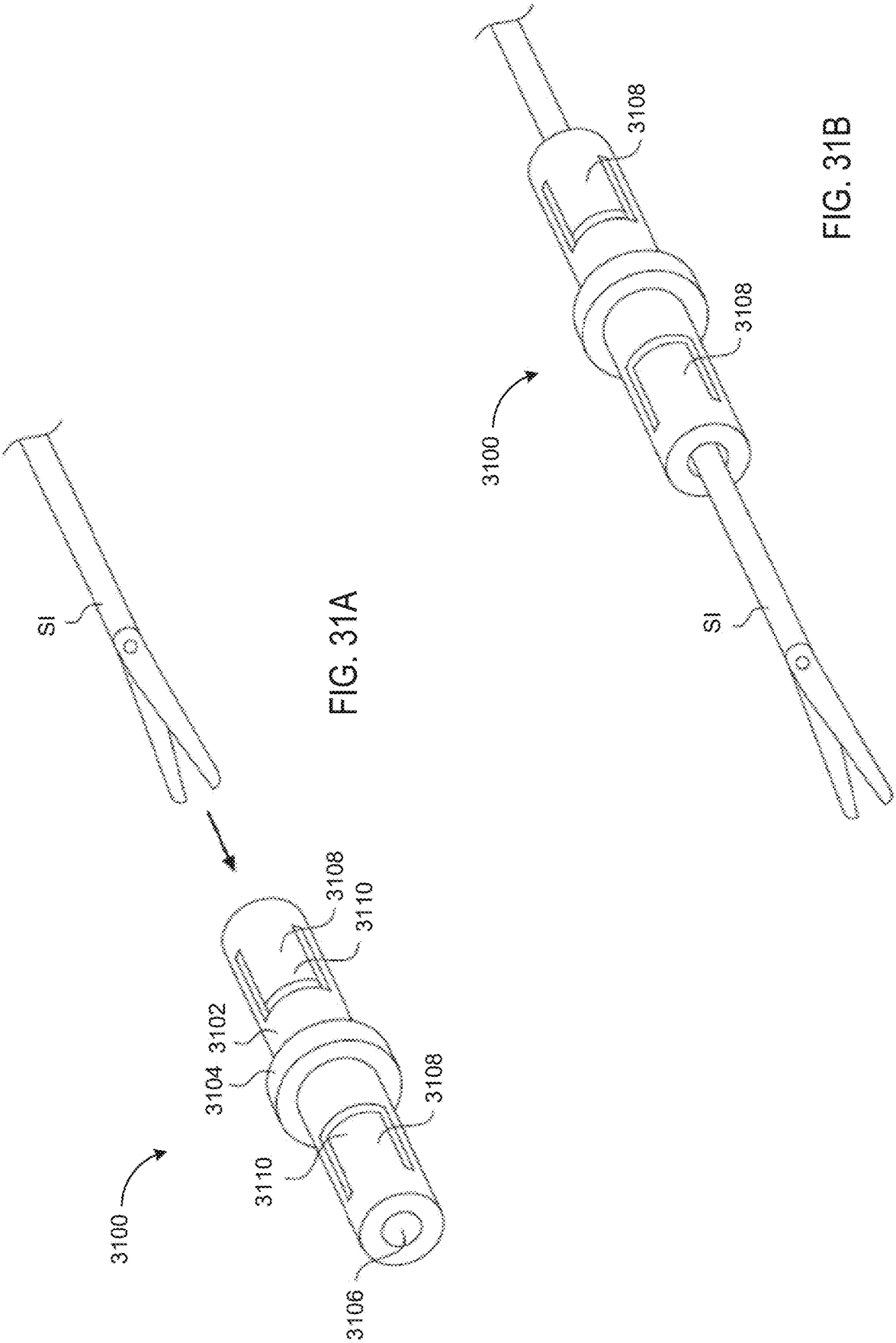


FIG. 30D



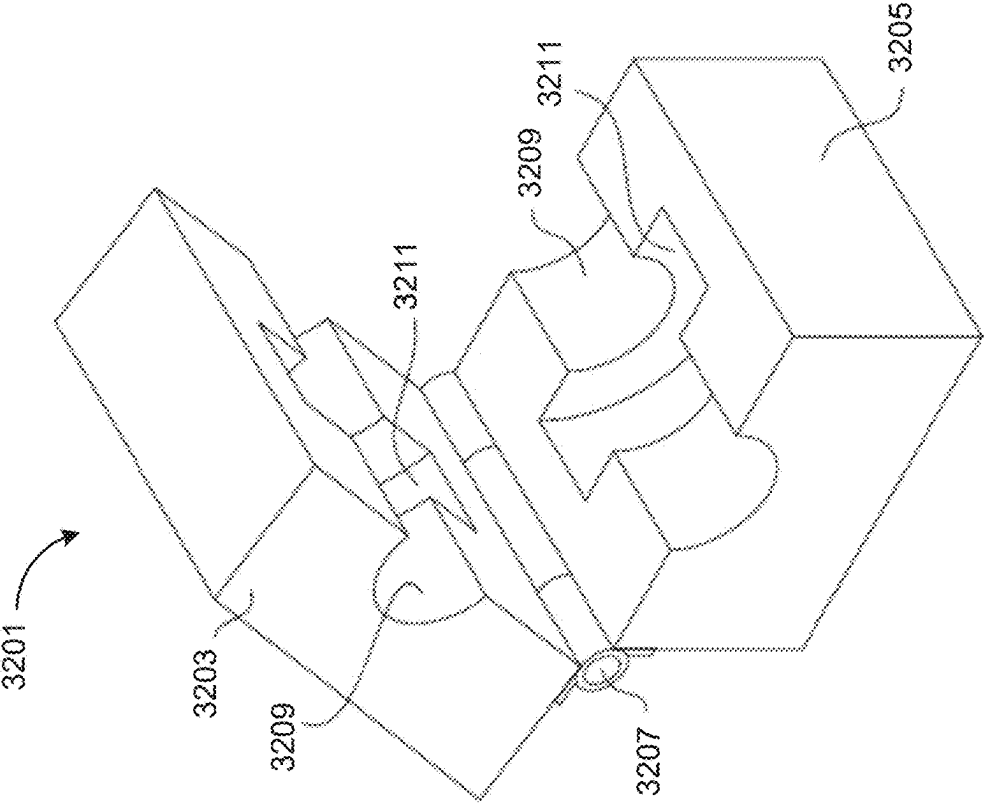


FIG. 31D

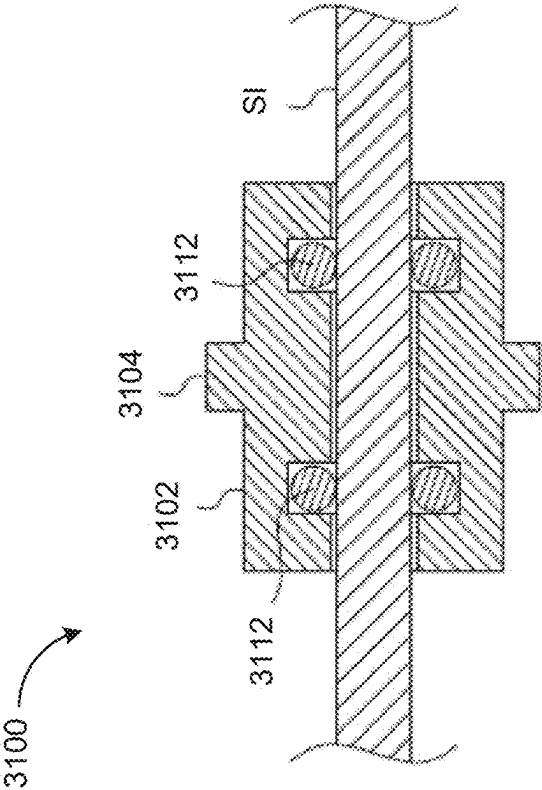


FIG. 31C

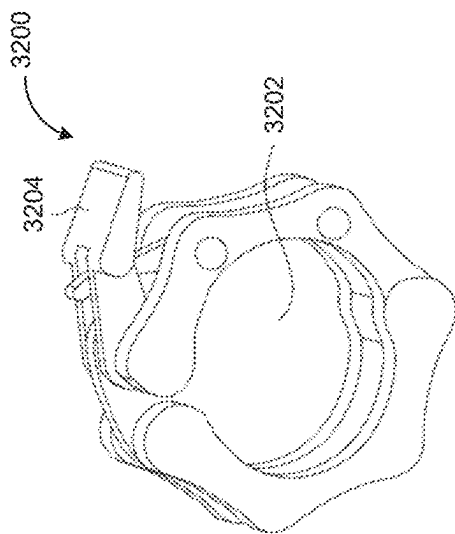


FIG. 32A

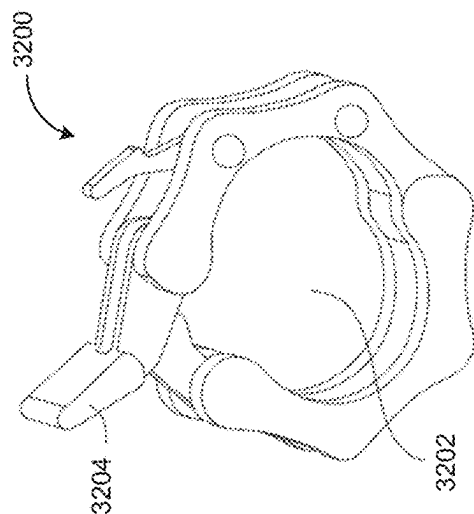


FIG. 32B

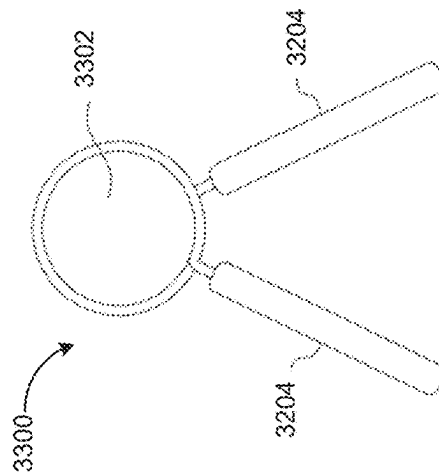


FIG. 33A

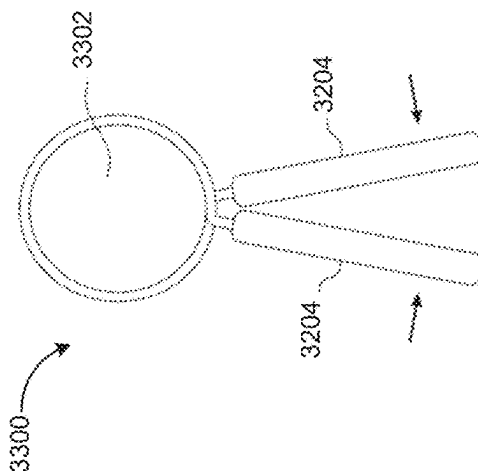


FIG. 33B

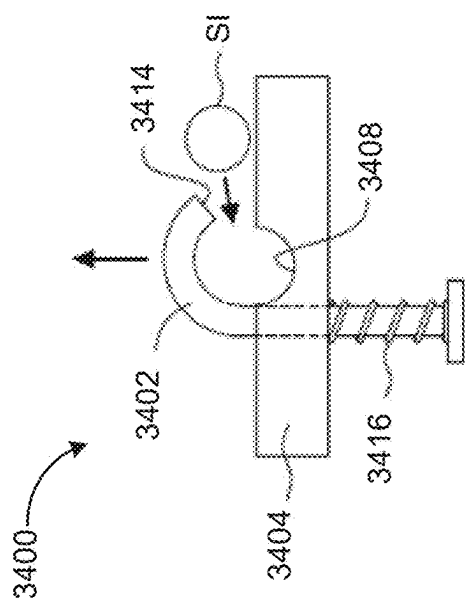


FIG. 34B

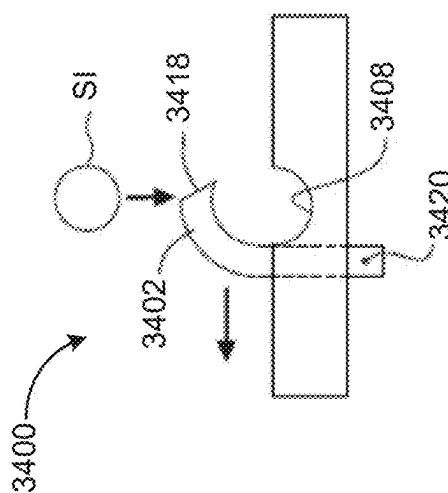


FIG. 34C

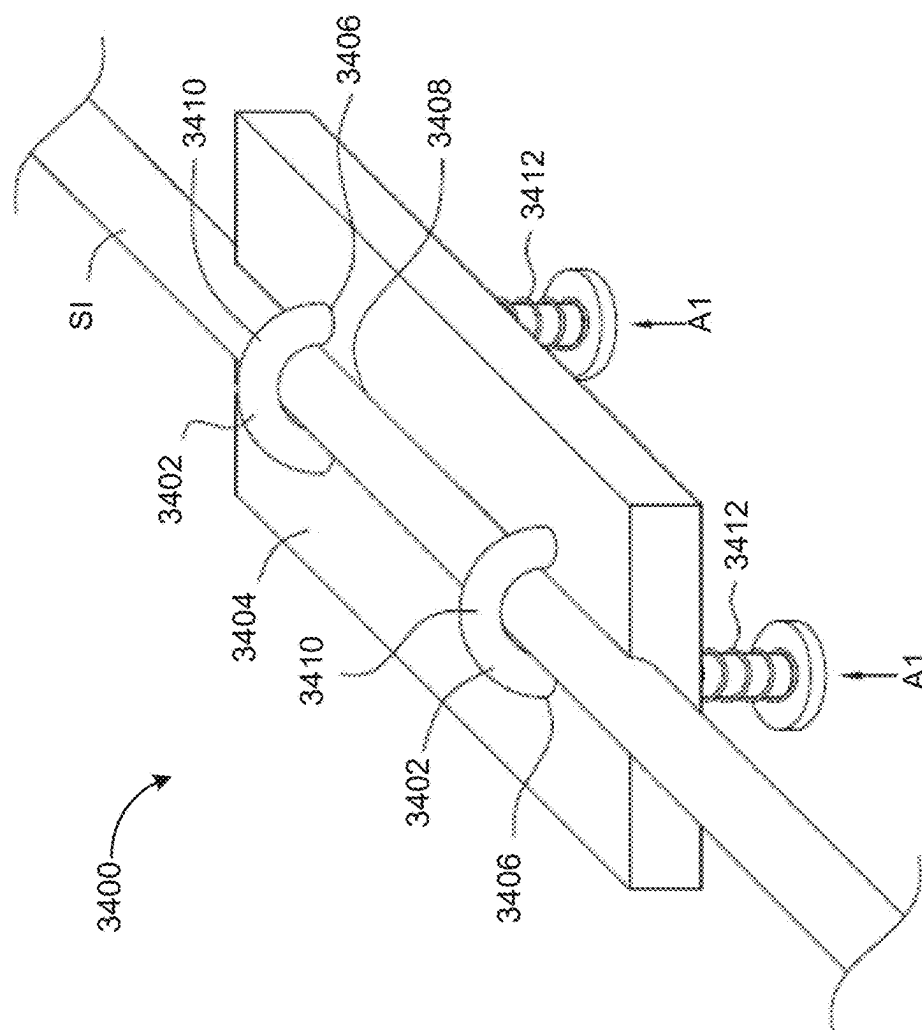


FIG. 34A

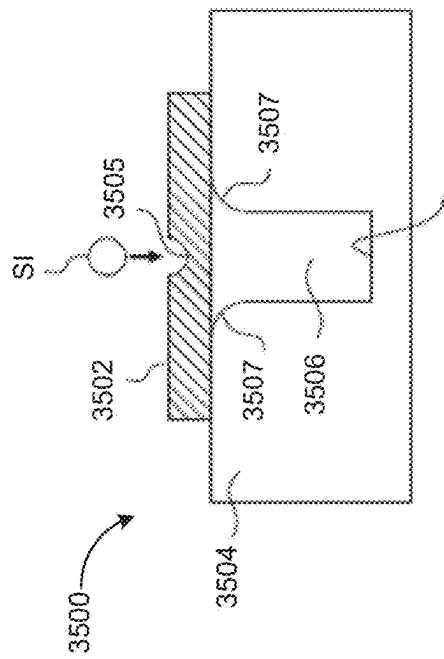


FIG. 35A

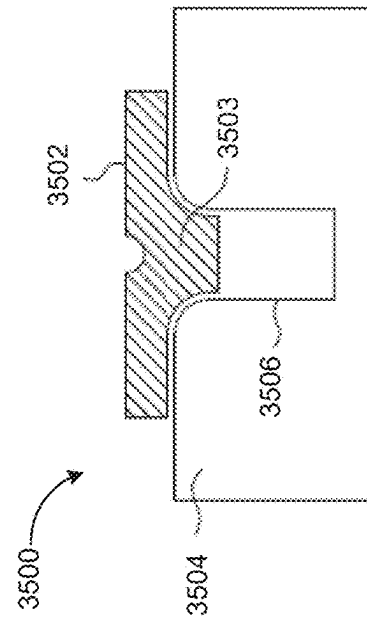
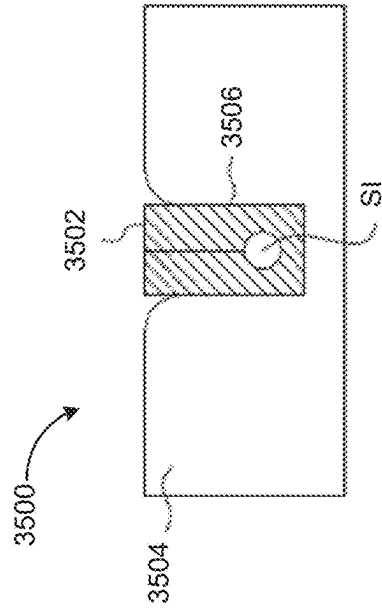
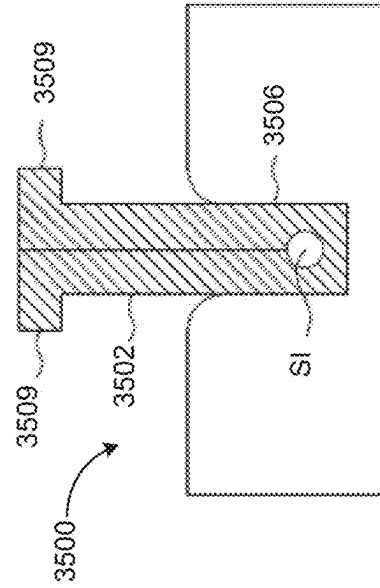


FIG. 35C



FILE 35B



EG 350

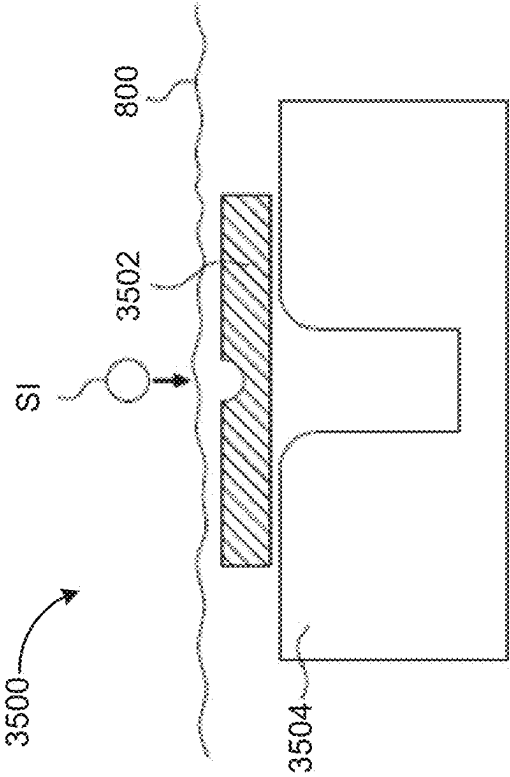


FIG. 35E

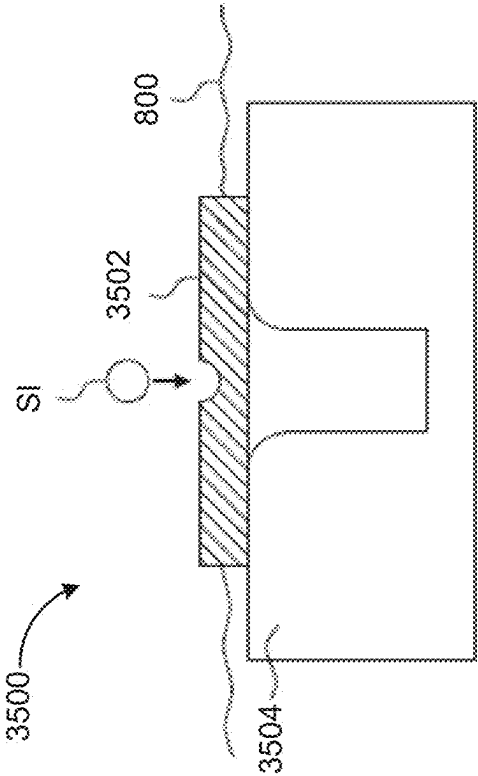
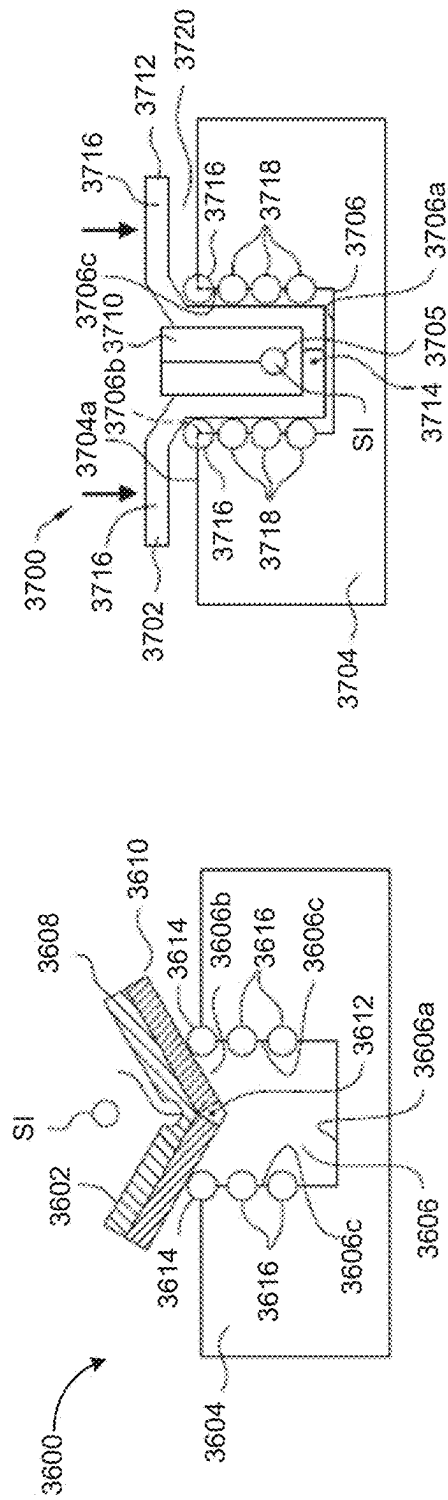


FIG. 35F



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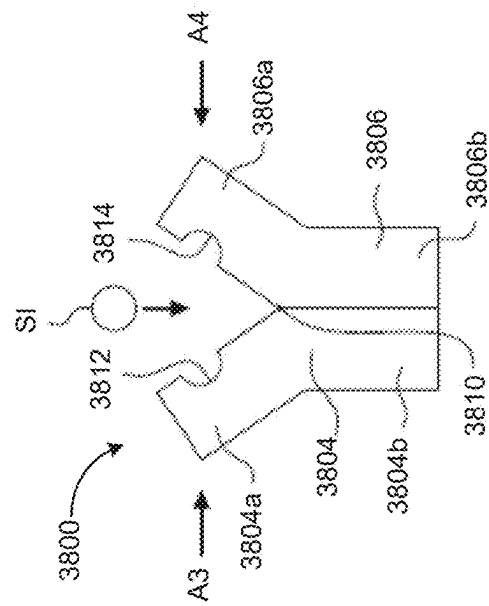
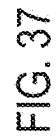
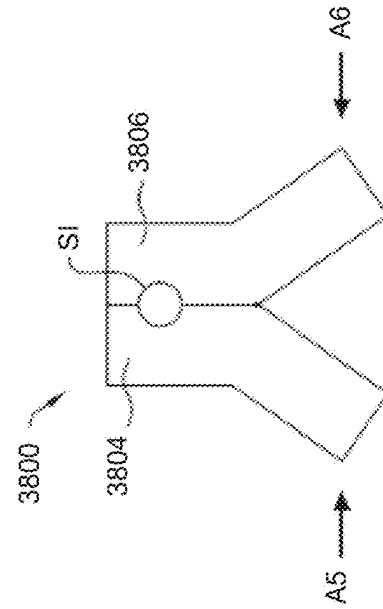


FIG. 38A

[illegible]

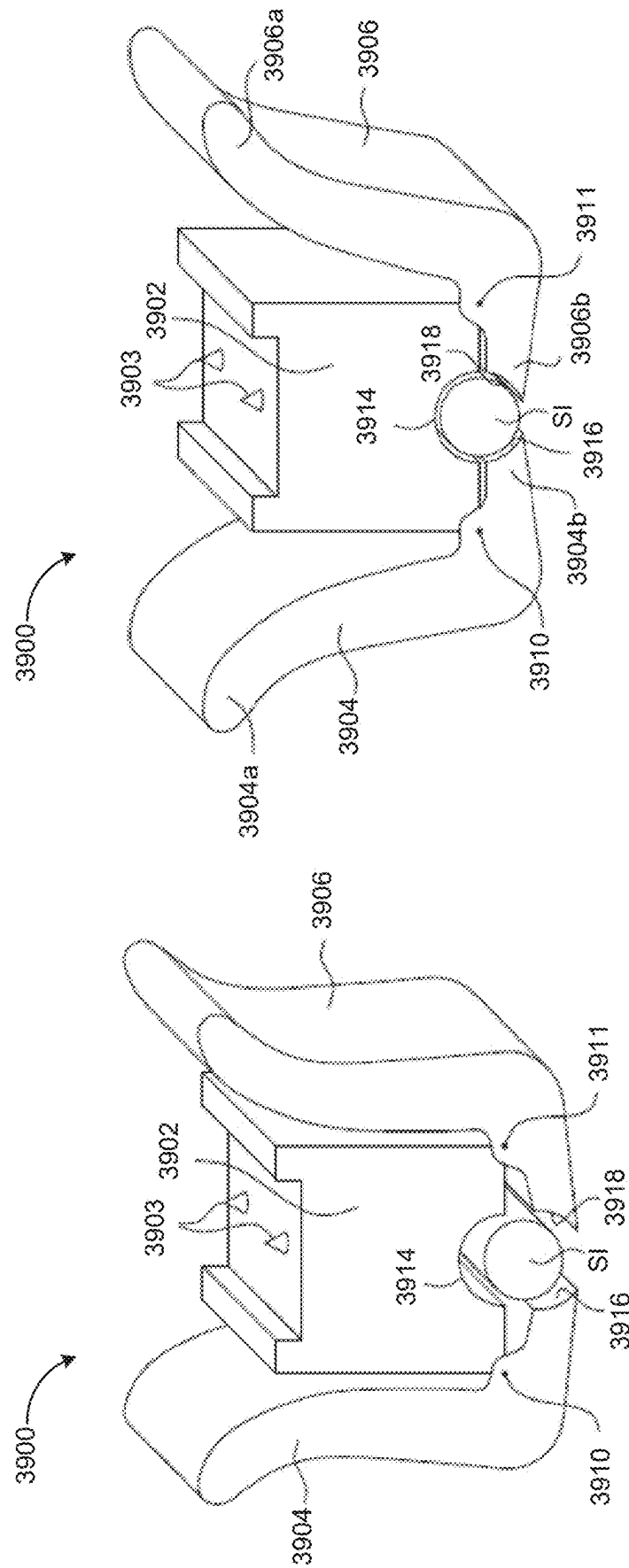
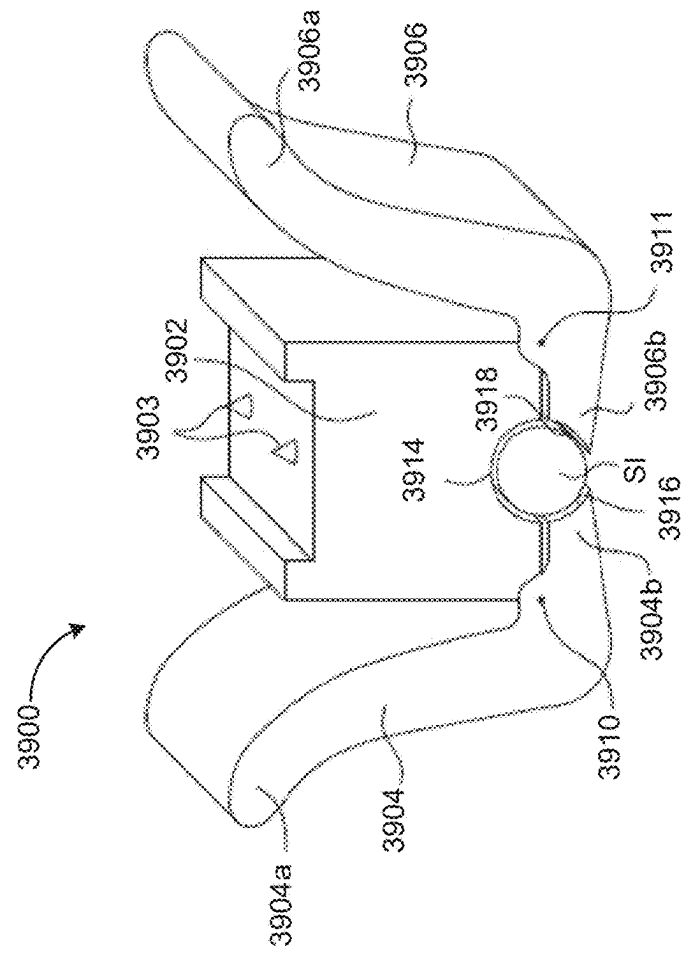


FIG. 39A



FILED

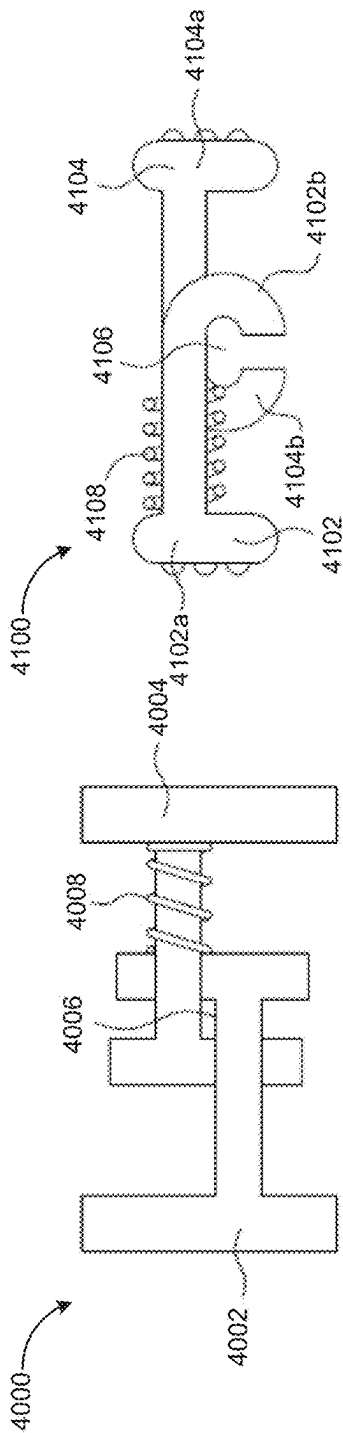


FIG. 41

FIG. 40

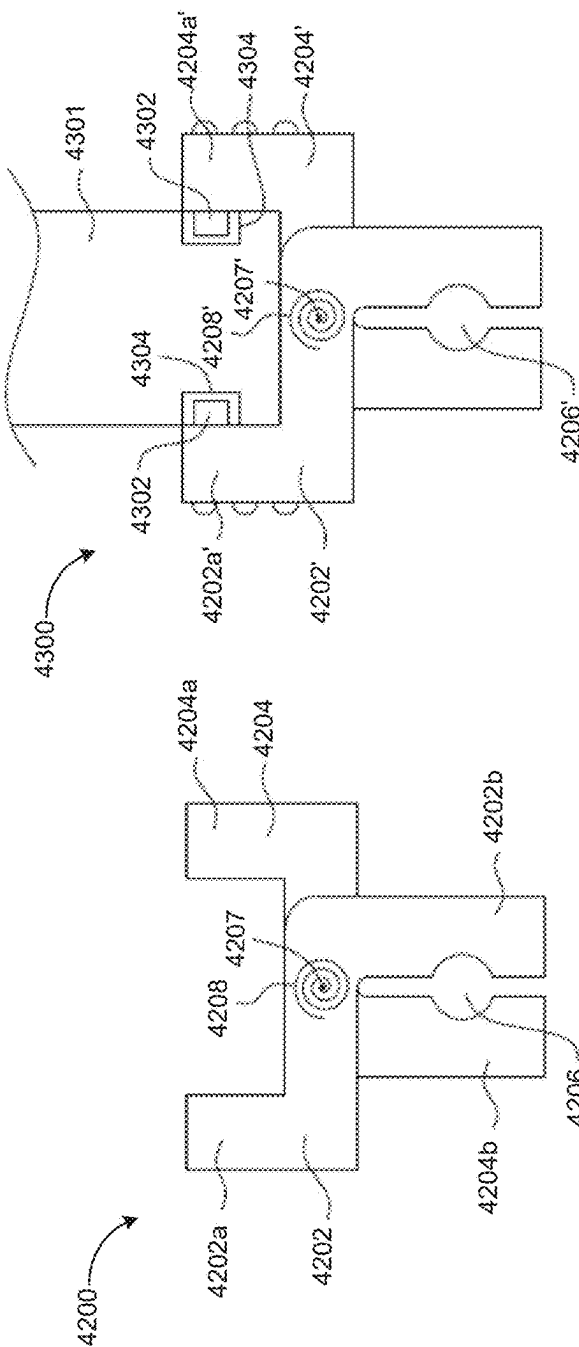


FIG. 43

FIG. 42

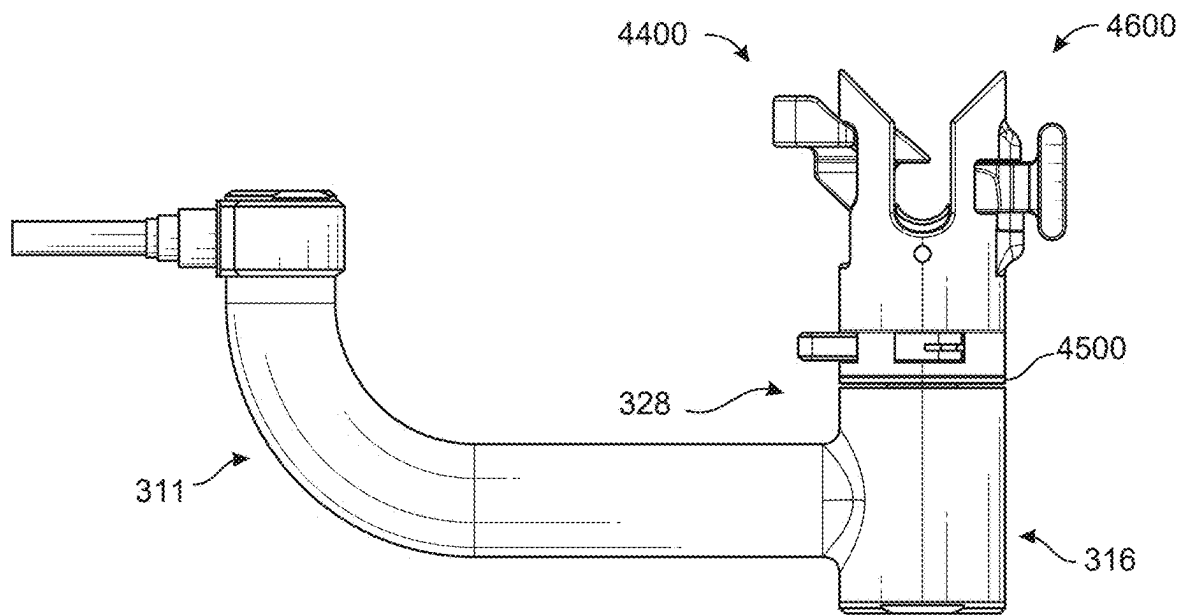


FIG. 44A

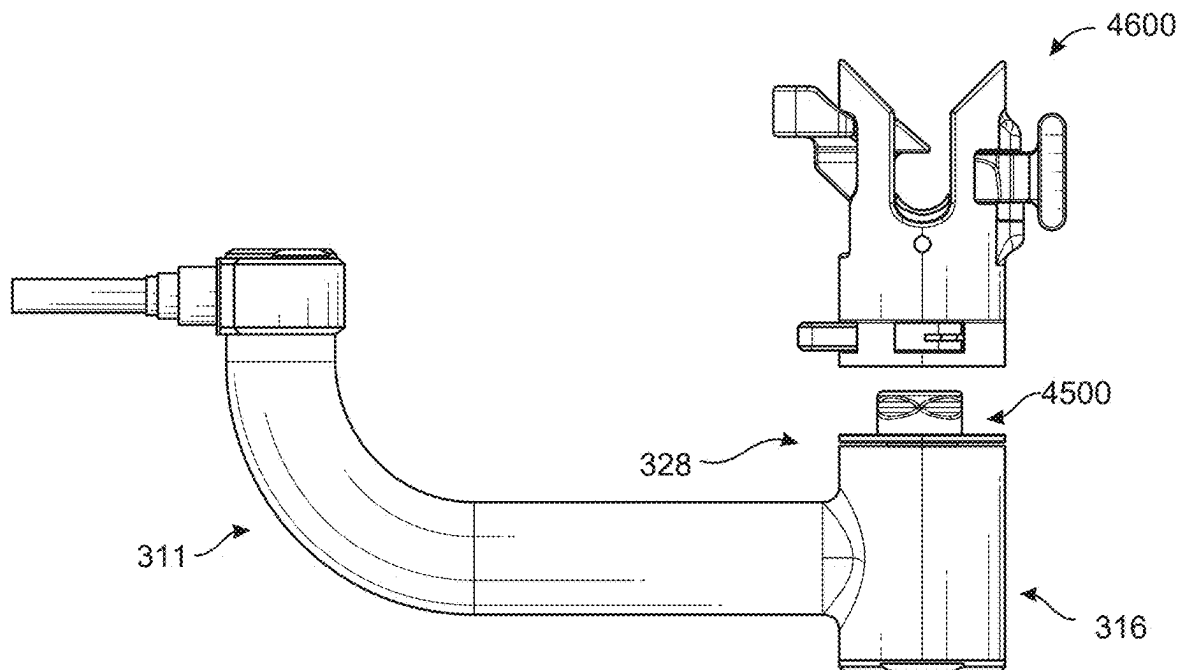


FIG. 44B

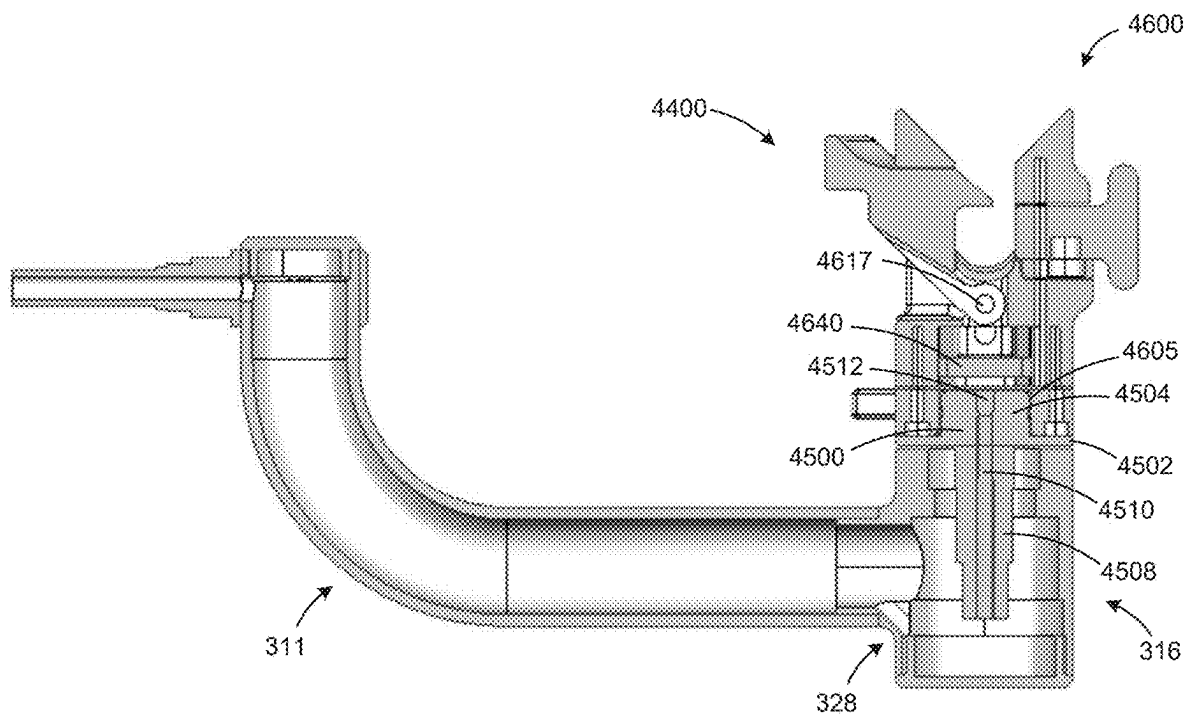


FIG. 44C

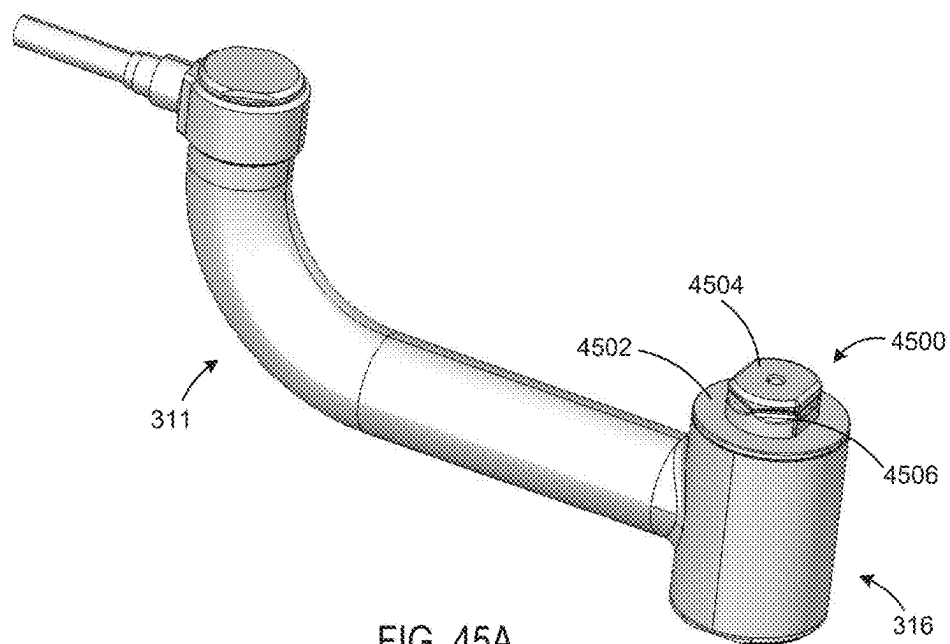


FIG. 45A

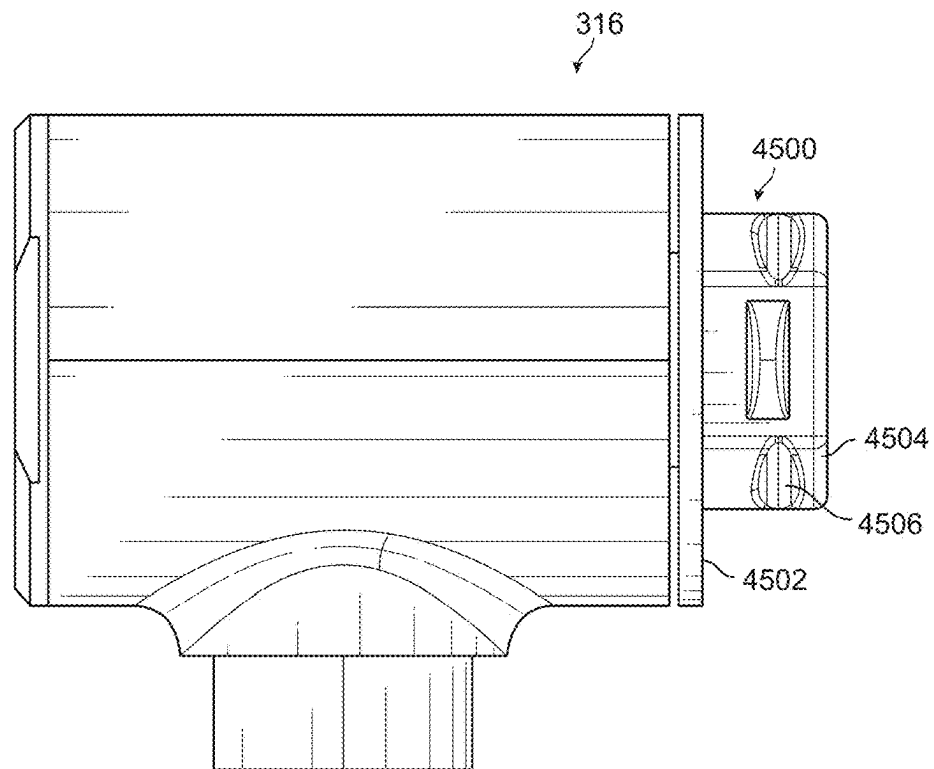


FIG. 45B

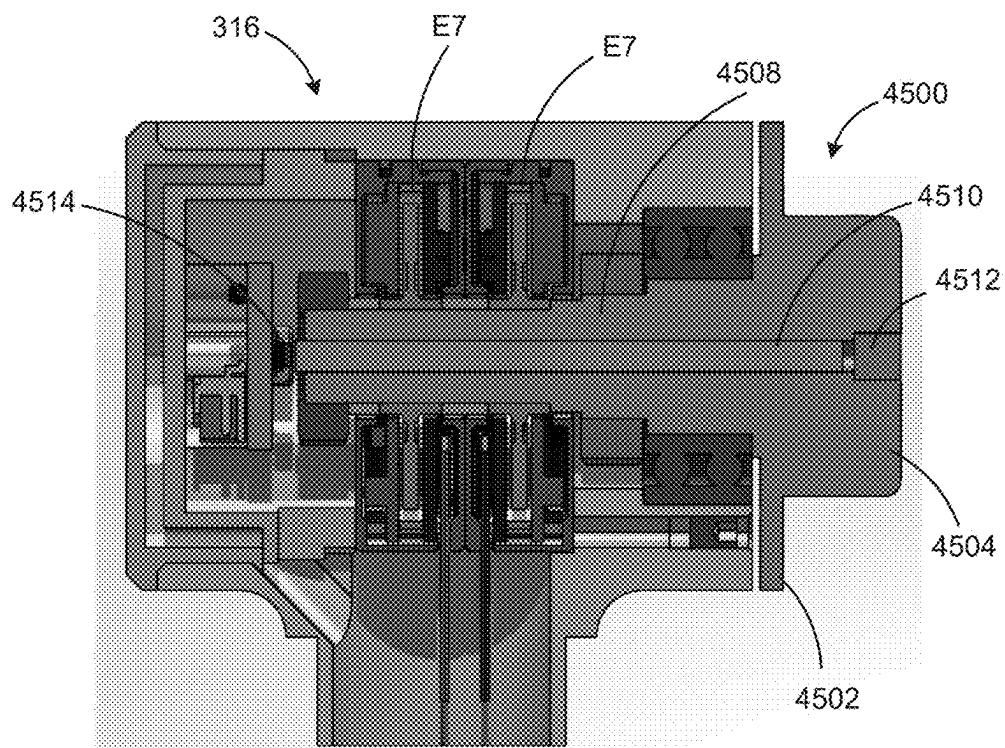


FIG. 45C

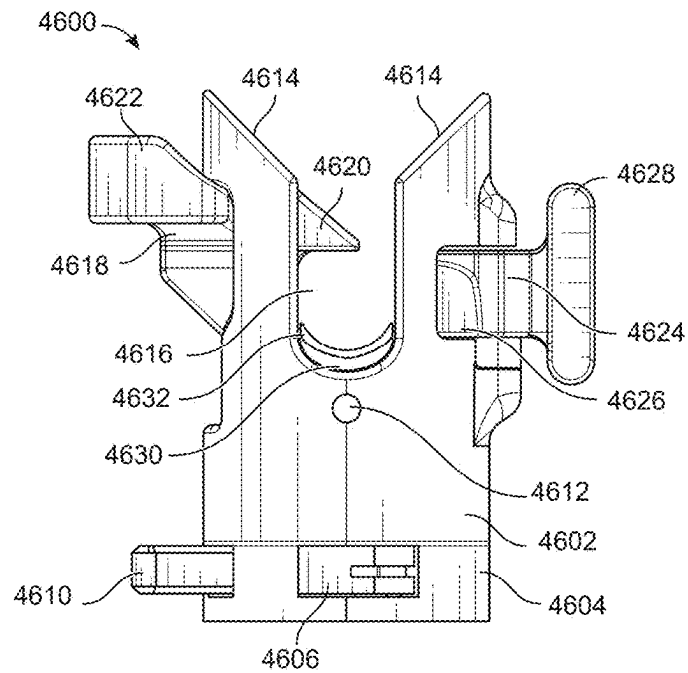


FIG. 46A

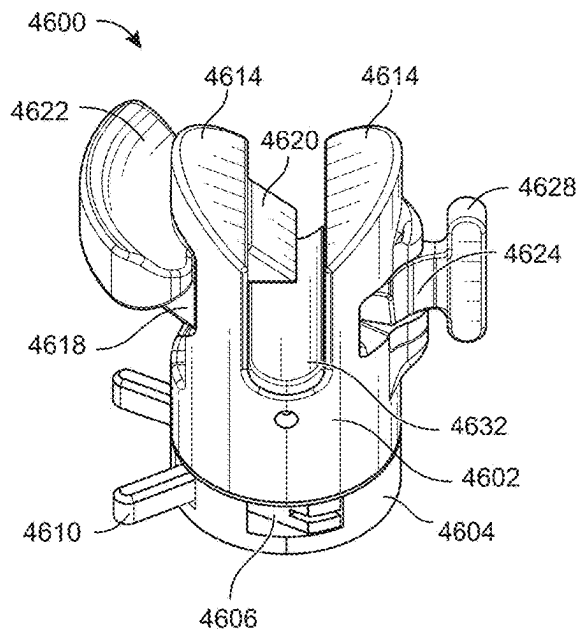


FIG. 46B

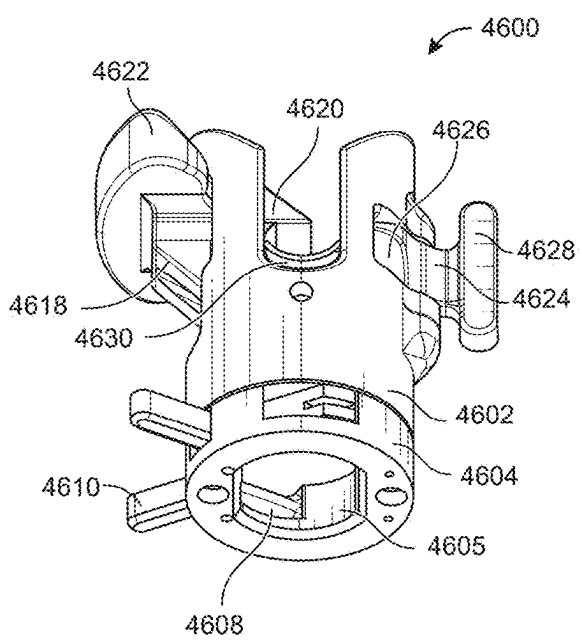


FIG. 46C

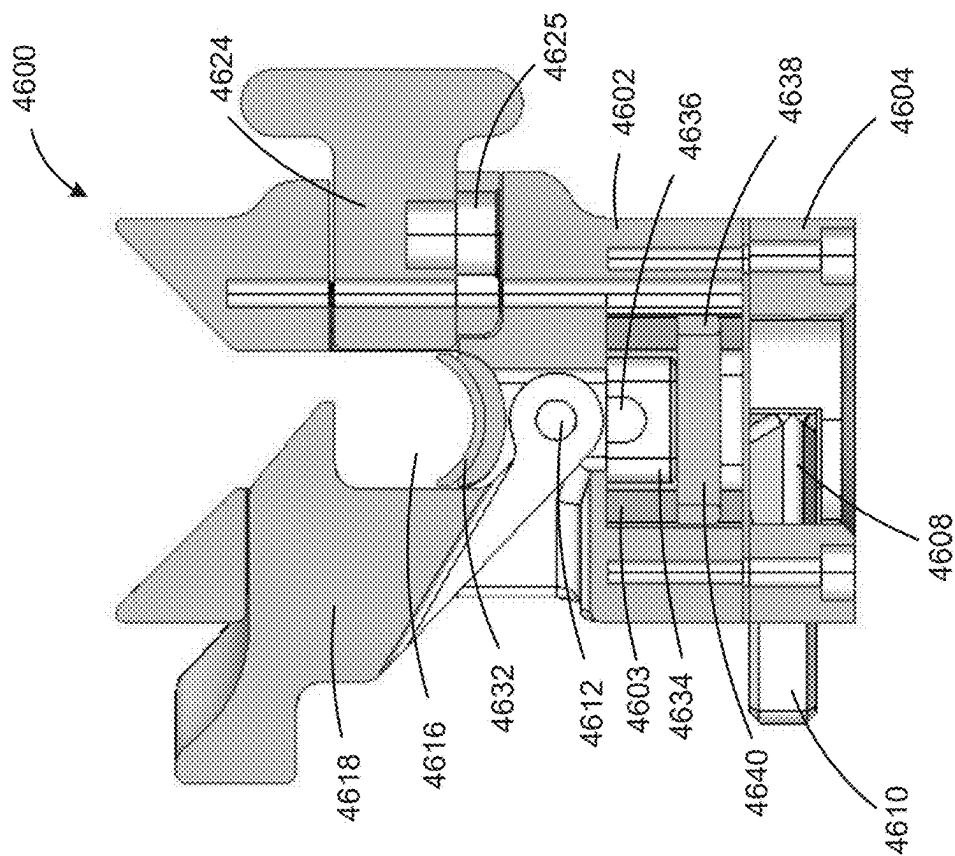


FIG. 46E

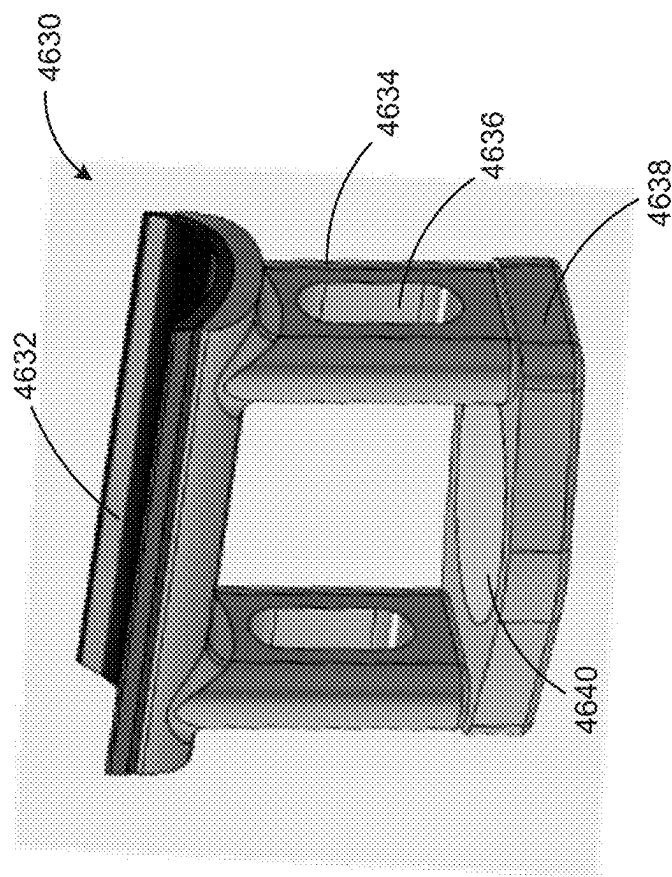


FIG. 46D

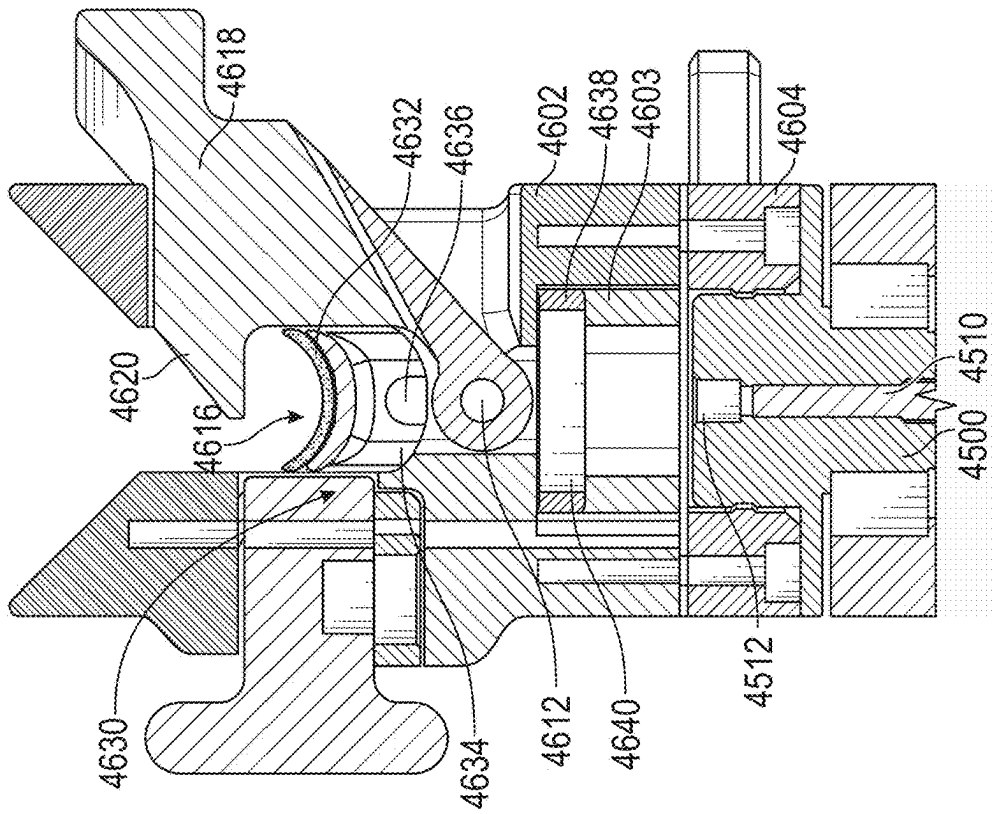


FIG. 47B

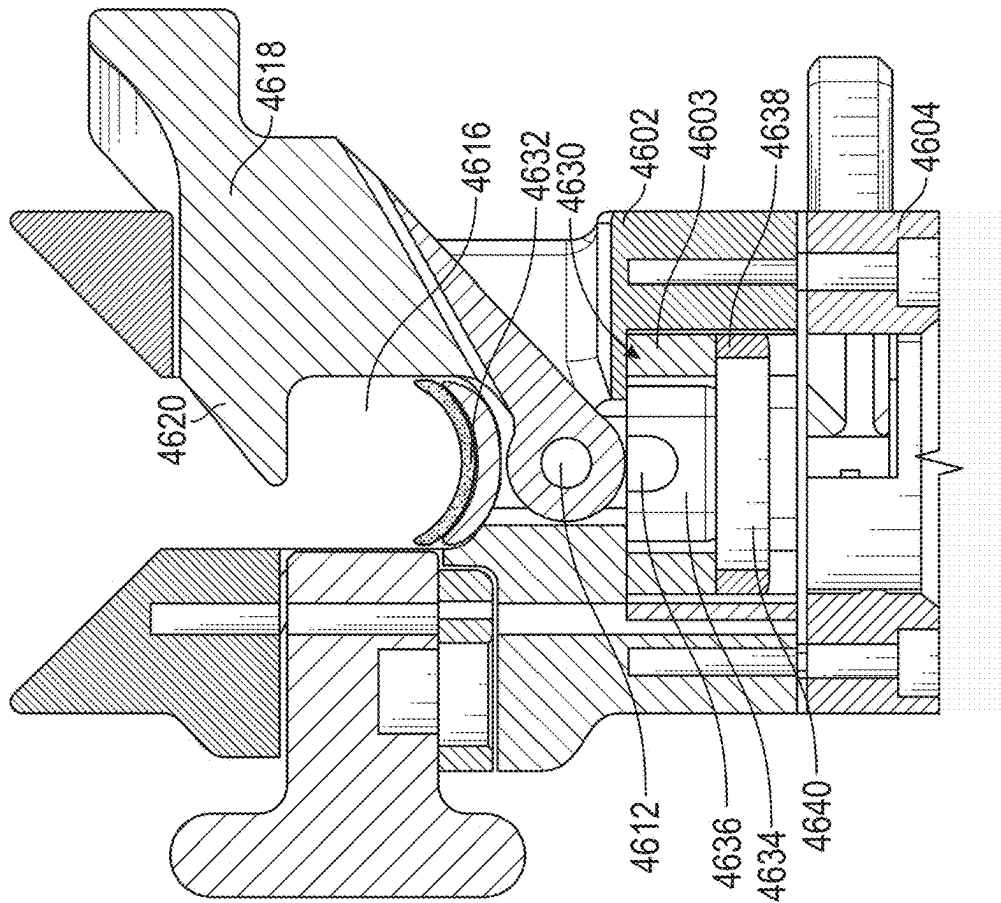


FIG. 47A

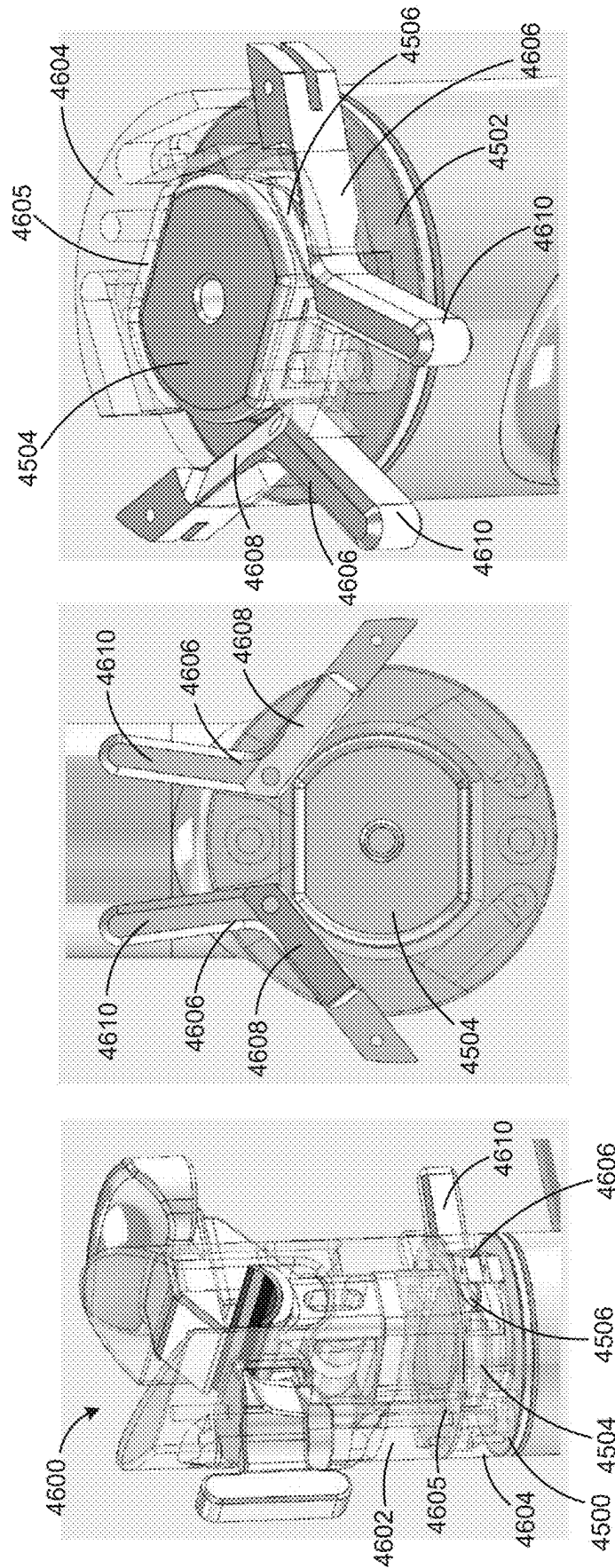


FIG. 48A

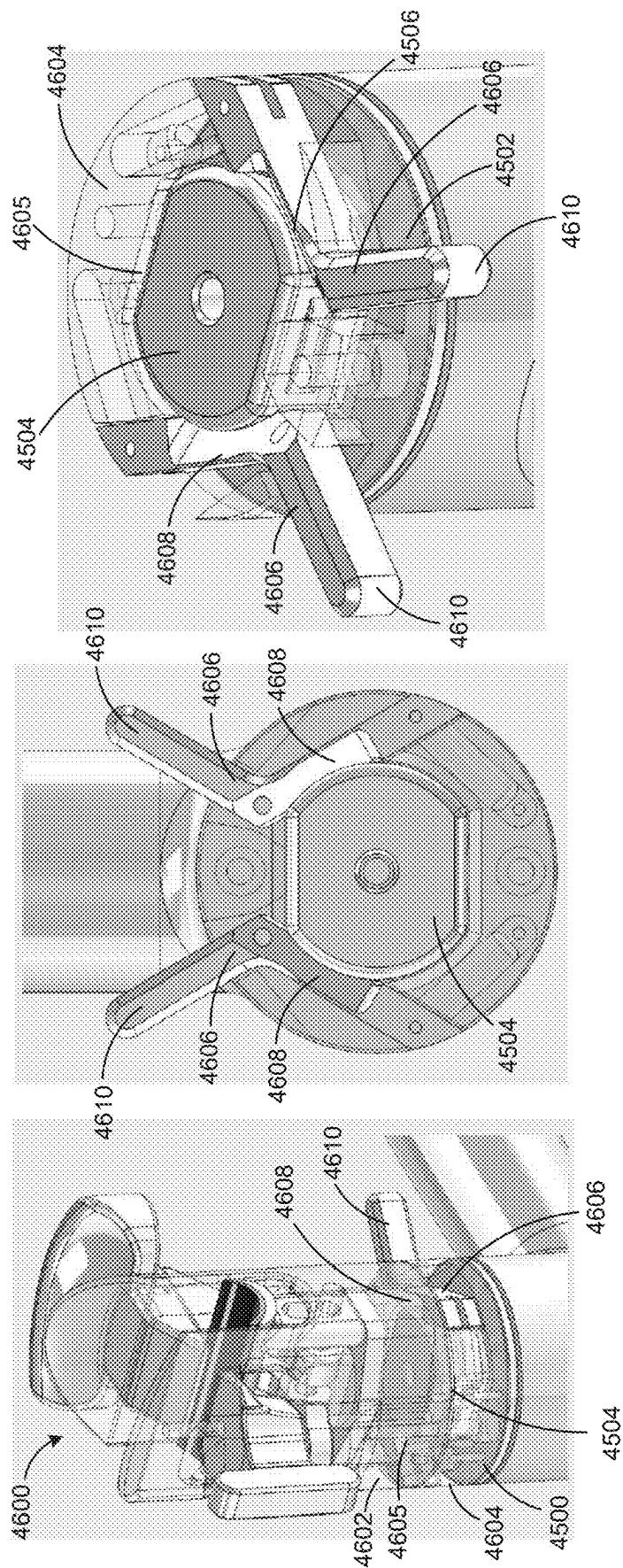


FIG. 48B

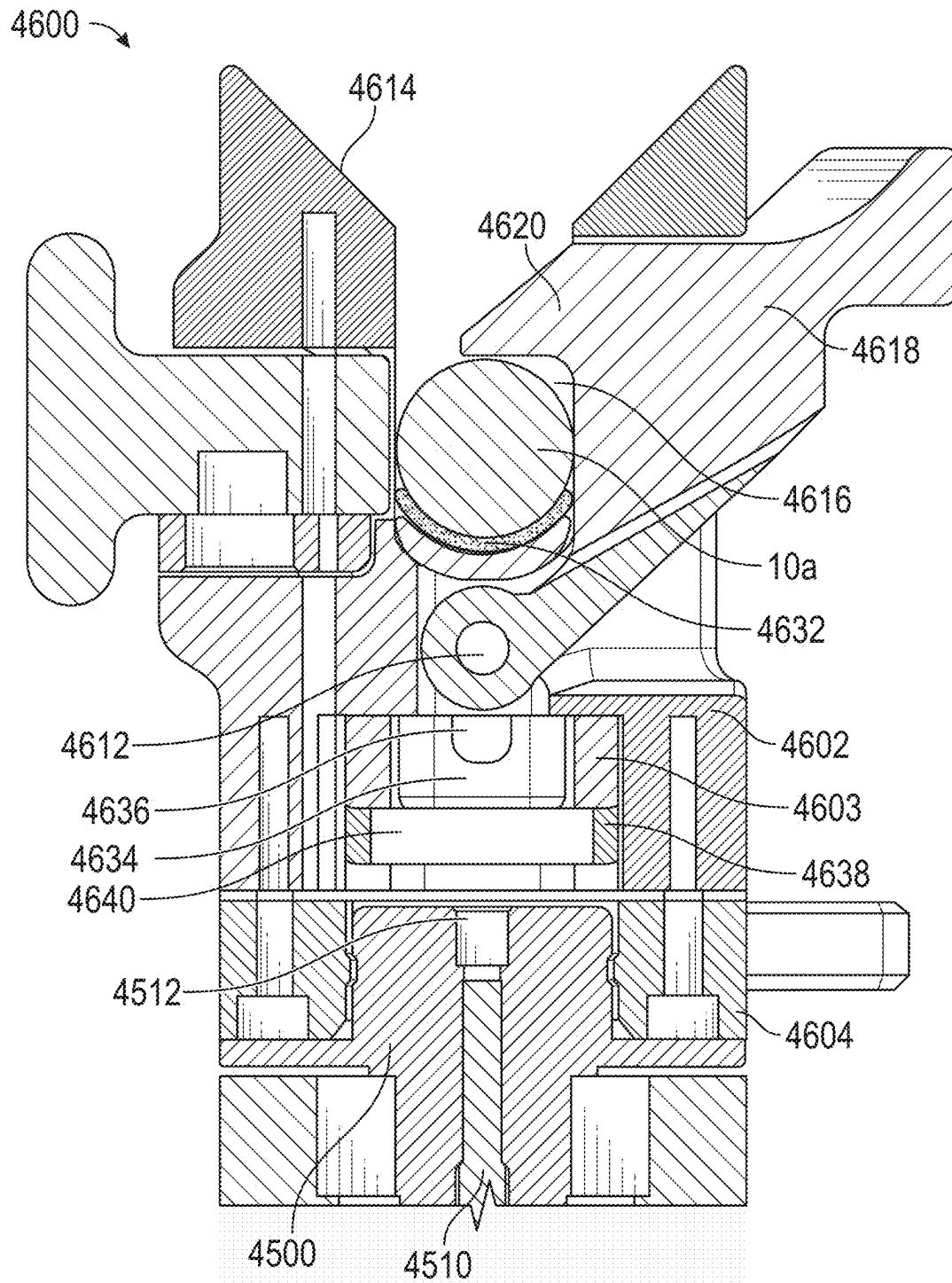


FIG. 49

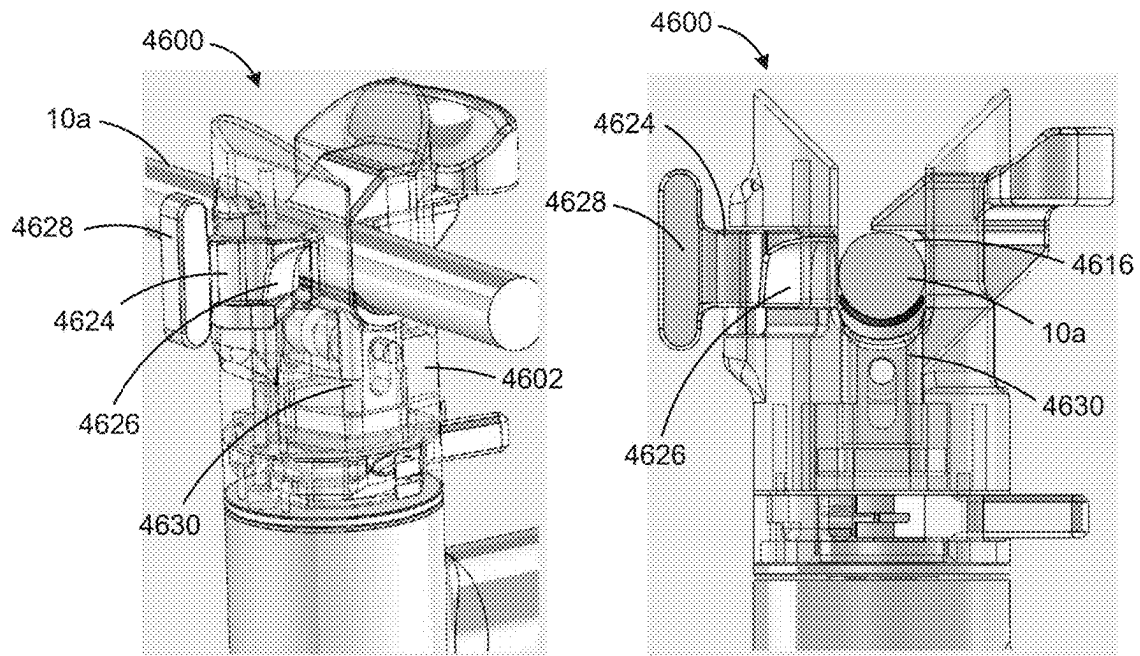


FIG. 50A

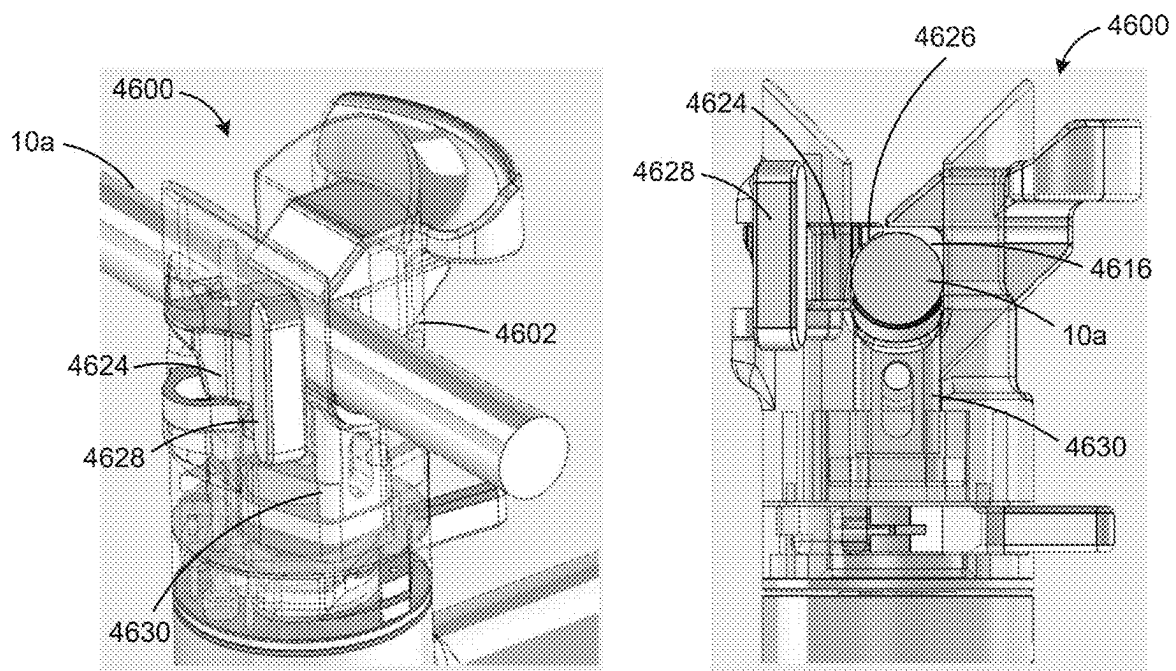


FIG. 50B

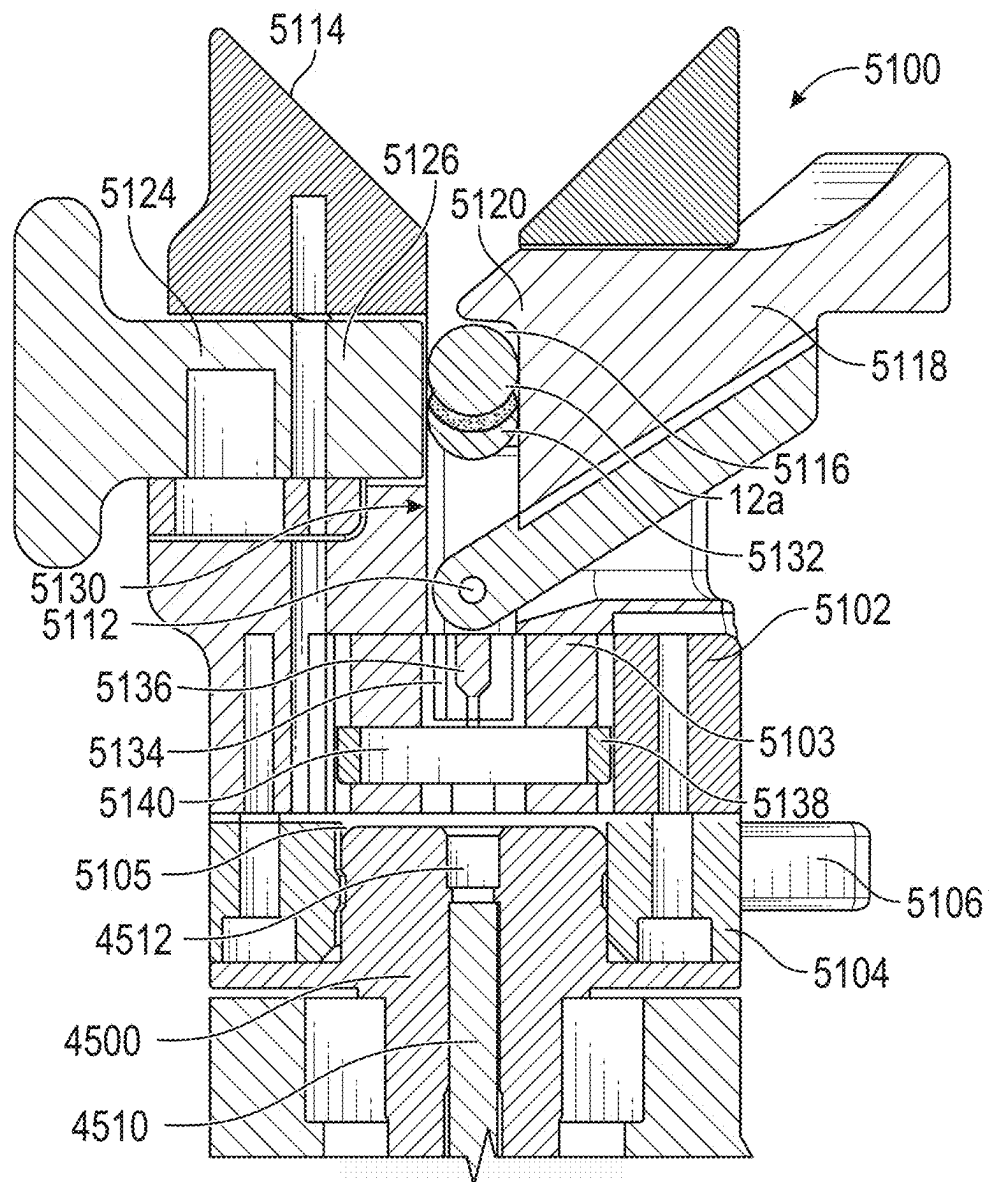


FIG. 51

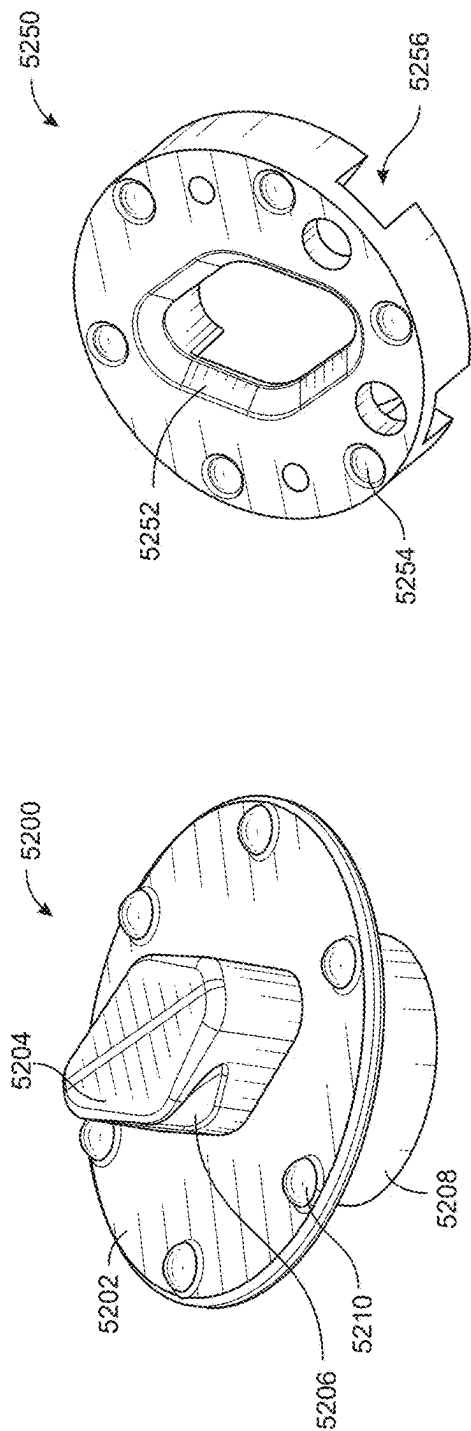


FIG. 52A

FIG. 52B

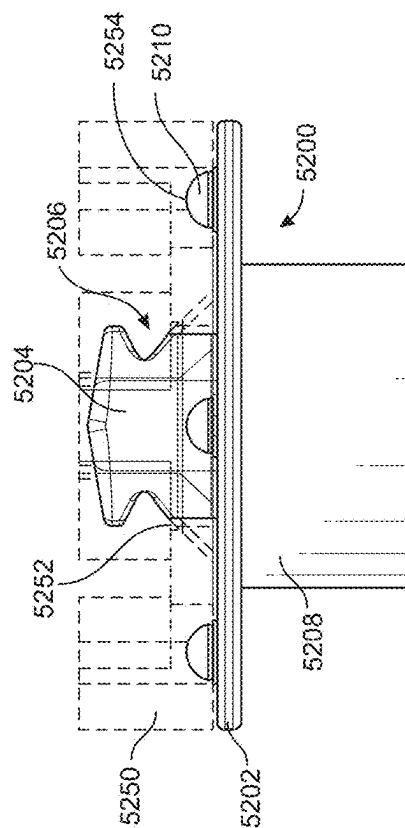


FIG. 52C

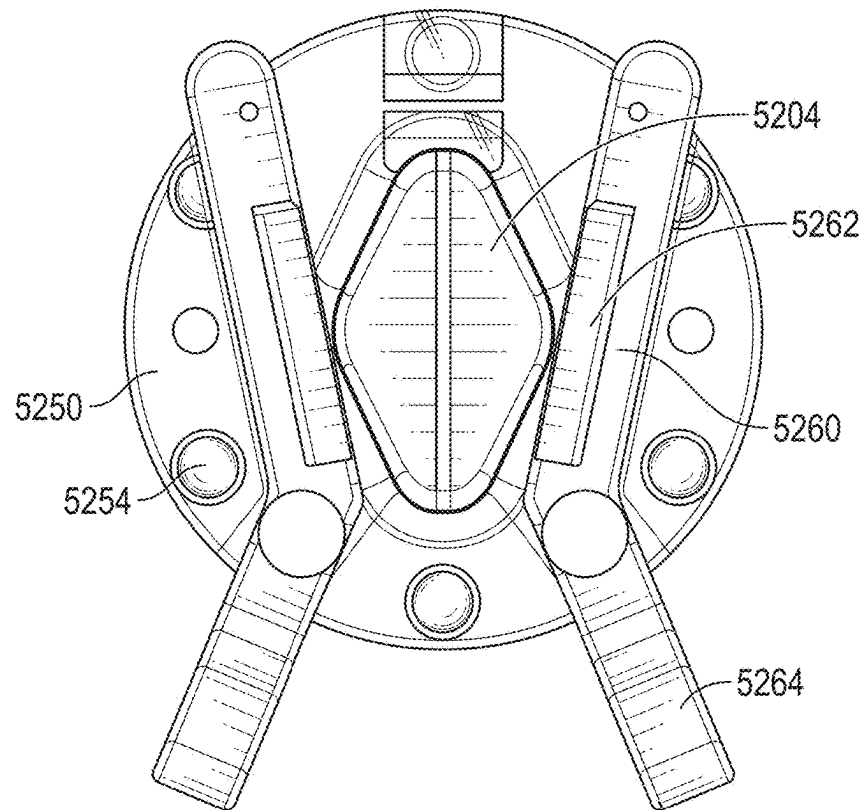


FIG. 52D

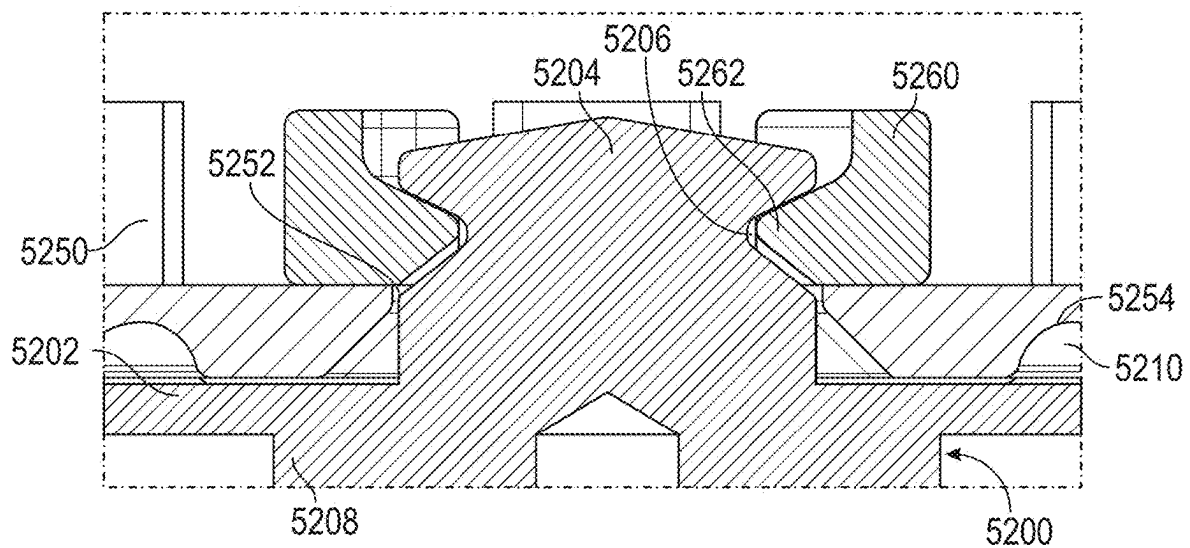


FIG. 52E

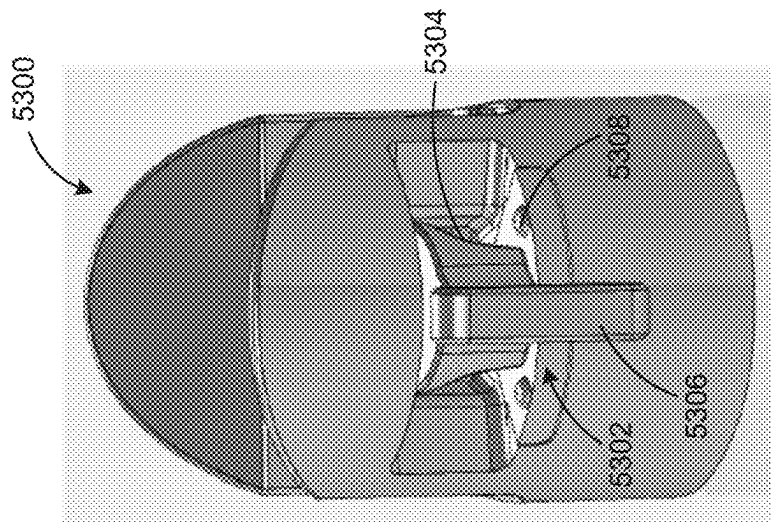


FIG. 53A

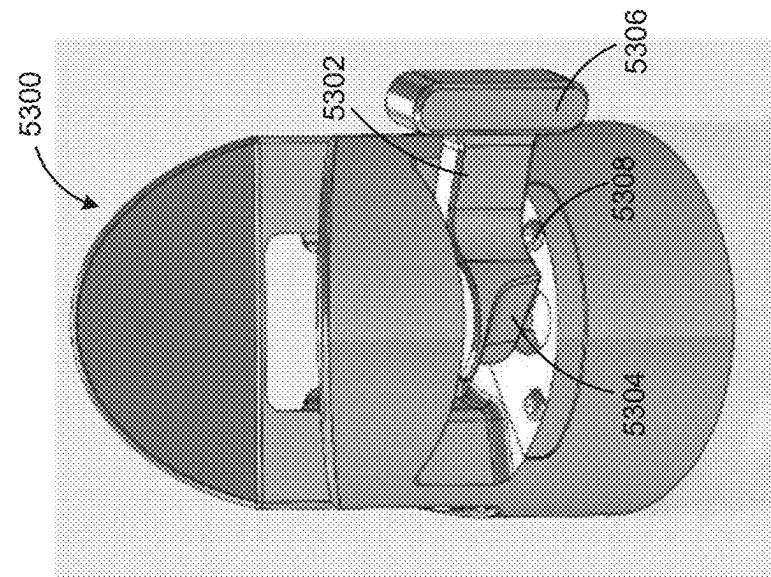


FIG. 53B

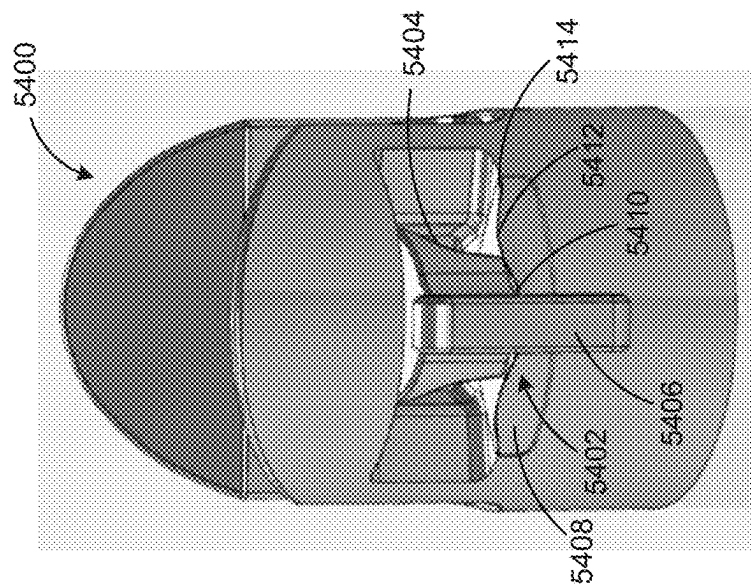


FIG. 54

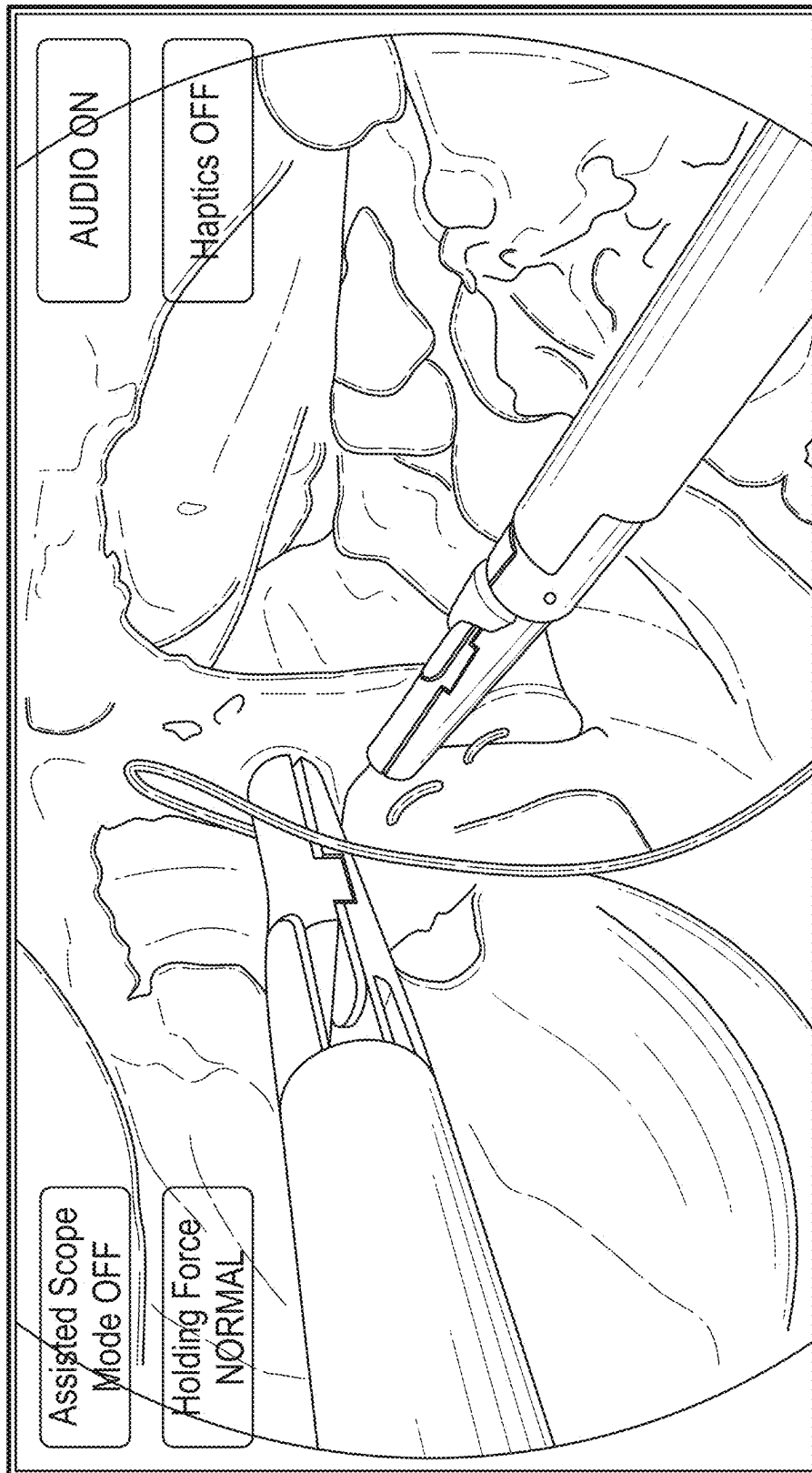


FIG. 55

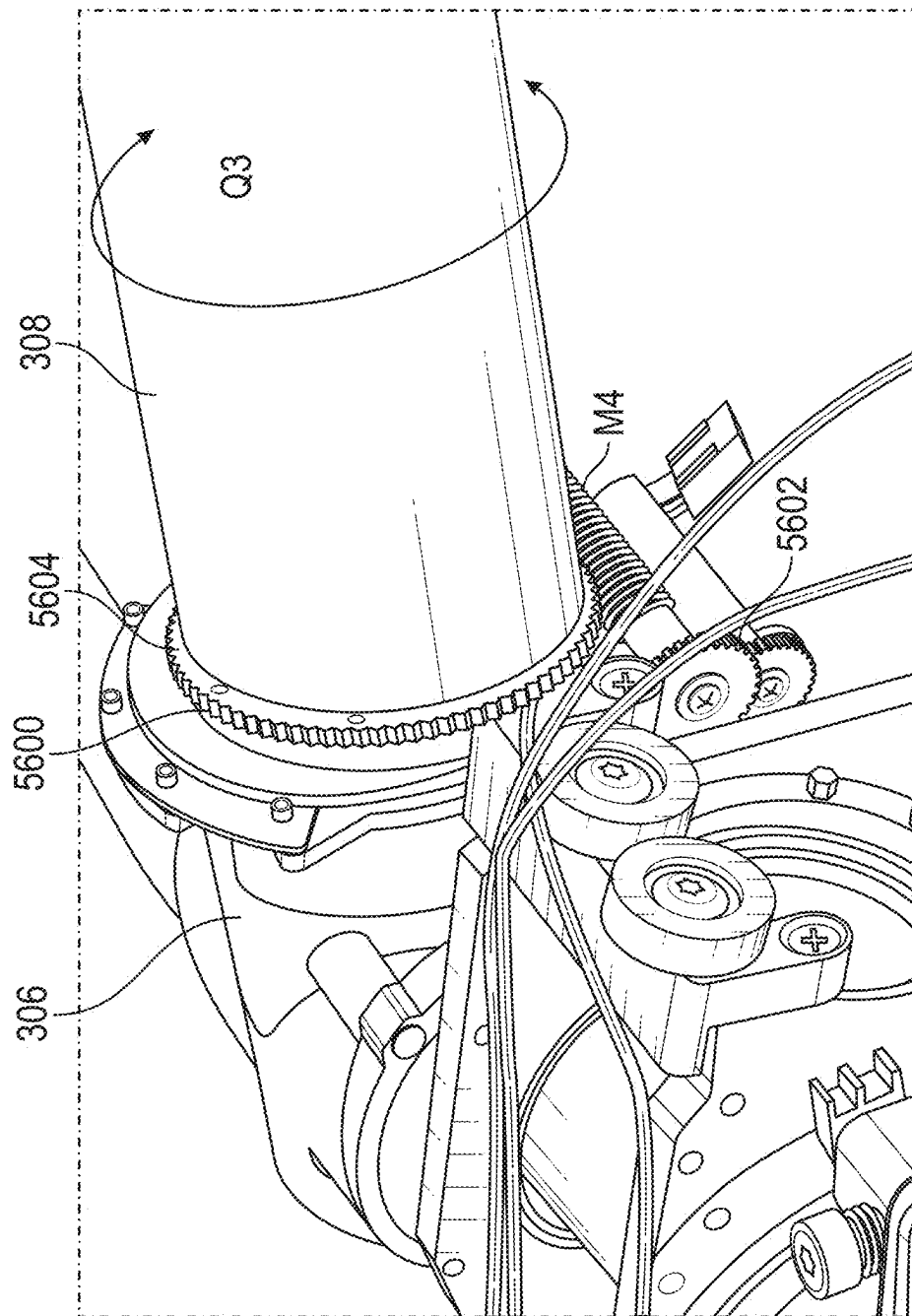


FIG. 56

**CO-MANIPULATION SURGICAL SYSTEM
HAVING MULTIPLE OPERATIONAL MODES
FOR USE WITH SURGICAL INSTRUMENTS
FOR PERFORMING SURGERY**

CROSS-REFERENCE TO RELATED
APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/743,763, filed Jun. 14, 2024, now U.S. Pat. No. 12,178,418, which is a continuation-in-part application of U.S. patent application Ser. No. 18/318,699, filed May 16, 2023, now U.S. Pat. No. 12,011,149, which is a continuation of U.S. patent application Ser. No. 18/057,191, filed Nov. 18, 2022, now U.S. Pat. No. 11,812,938, which is a continuation-in-part application of U.S. patent application Ser. No. 17/815,885, filed Jul. 28, 2022, now U.S. Pat. No. 11,504,197, which is a continuation application of PCT Patent Appl. No. PCT/IB2022/052989, filed Mar. 30, 2022, which claims priority to EP patent application Ser. No. 21/306,904.0, filed Dec. 22, 2021, EP patent application Ser. No. 21/306,905.7, filed Dec. 22, 2021, EP patent application Ser. No. 21/305,929.8, filed Jul. 5, 2021, and EP patent application Ser. No. 21/305,417.4, filed Mar. 31, 2021, the entire contents of each of which are incorporated herein by reference. U.S. patent application Ser. No. 18/743,763, filed Jun. 14, 2024, now U.S. Pat. No. 12,178,418, is also a continuation application of PCT Patent Appl. No. PCT/IB2023/059981, filed Oct. 4, 2023, which claims the benefit of priority of U.S. Provisional Patent Appl. No. 63/378,434, filed Oct. 5, 2022, and EP patent application Ser. No. 22/306,496.5, filed Oct. 5, 2022, the entire contents of each of which are incorporated herein by reference. U.S. patent application Ser. No. 18/743,763, filed Jun. 14, 2024, now U.S. Pat. No. 12,178,418, is related to U.S. patent application Ser. No. 17/816,958, filed Aug. 2, 2022, now U.S. Pat. No. 11,622,826, the entire contents of which are incorporated herein by reference.

FIELD OF USE

The present disclosure is directed to co-manipulation robotic systems, such as those having a coupling mechanism for removably attaching a surgical instrument.

BACKGROUND

Managing vision and access during a laparoscopic procedure is a challenge. The surgical assistant paradigm is inherently imperfect, as the assistant is being asked to anticipate and see with the surgeon's eyes, without standing where the surgeon stands, and similarly to anticipate and adjust how the surgeon wants the tissue of interest exposed, throughout the procedure. For example, during a laparoscopic procedure, one assistant may be required to hold a retractor device to expose tissue for the surgeon, while another assistant may be required to hold a laparoscope device to provide a field of view of the surgical space within the patient to the surgeon during the procedure, either one of which may be required to hold the respective tools in an impractical position, e.g., from between the arms of the surgeon while the surgeon is actively operating additional surgical instruments.

Various attempts have been made at solving this issue. For example, a rail-mounted orthopedic retractor, which is a purely mechanical device that is mounted to the patient bed/table, may be used to hold a laparoscope device in

position during a laparoscopic procedure, and another rail-mounted orthopedic retractor may be used to hold a retractor device in position during the laparoscopic procedure. However, the rail-mounted orthopedic retractor requires extensive manual interaction to unlock, reposition, and lock the tool in position.

Complex robot-assisted systems such as the Da Vinci Surgical System (made available by Intuitive Surgical, Sunnyvale, California) have been used by surgeons to enhance laparoscopic surgical procedures by permitting the surgeon to tele-operatively perform the procedure from a surgeon console remote from the patient console holding the surgical instruments. Such complex robot-assisted systems are very expensive, and have a very large footprint and take up a lot of space in the operating room. Moreover, such robot-assisted systems typically require unique system-specific surgical instruments that are compatible with the system, and thus surgeons may not use standard off-the-shelf surgical instruments that they are used to. As such, the surgeon is required to learn an entirely different way of performing the laparoscopic procedure.

In view of the foregoing drawbacks of previously known systems and methods, there exists a need for a system that provides the surgeon with the ability to seamlessly position and manipulate various surgical instruments as needed, thus avoiding the workflow limitations inherent to both human and mechanical solutions.

SUMMARY

The present disclosure overcomes the drawbacks of previously-known systems and methods by providing a co-manipulation surgical system to assist with laparoscopic surgery performed using a surgical instrument having a handle, an operating end, and an elongated shaft therebetween. The co-manipulation surgical system may include a robot arm comprising a proximal end, a distal end configured to be removably coupled to the surgical instrument, a plurality of links, and a plurality of joints between the proximal end and the distal end, the distal end of the robot arm comprising a coupler interface. The co-manipulation surgical system further may include a coupler body configured to be removably coupled to the coupler interface. The coupler body comprises a lumen sized and shaped to receive the elongated shaft of the surgical instrument therein, and may be configured to transition between an open state where the elongated shaft is slidably moveable within the lumen, and a closed state where longitudinal movement of the elongated shaft relative to the coupler body is inhibited while rotational movement of the elongated shaft relative to the coupler body is permitted responsive to movement at the handle of the surgical instrument. When the coupler body is coupled to the coupler interface, the coupler body may be configured to rotate relative to the distal end of the robot arm via the coupler interface to self-align the lumen with the elongated shaft as the elongated shaft is inserted into the lumen. Moreover, when the coupler body is coupled to the coupler interface in the closed state, the robot arm may be permitted to be freely moveable responsive to movement at the handle of the surgical instrument for performing laparoscopic surgery. The coupler body may be disposable after a single laparoscopic surgery.

The co-manipulation surgical system further may include a switch configured to transition between an unlocked position and a locked position. The switch may include an engagement portion configured to engage with the elongated shaft when the elongated shaft is disposed within the lumen

3

and the switch is in the locked position to thereby secure the elongated shaft within the lumen. Accordingly, when the coupler body is coupled to the coupler interface and the elongated shaft is disposed within the lumen, the robot arm may be configured to be freely moveable responsive to movement at the handle of the surgical instrument. Moreover, when the elongated shaft is disposed within the lumen and the switch is in the locked position, the engagement portion may apply a friction force against the elongated shaft, the friction force configured to permit rotational movement of the elongated shaft relative to the coupler body, while inhibiting longitudinal movement of the elongated shaft relative to the coupler body. The switch may include a handle portion configured to be actuated to transition the switch between the unlocked position and the locked position.

Moreover, the coupler body may comprise a ramped surface having a first valley configured to engage the switch in the unlocked position, a second valley configured to engage the switch in the locked position, and a crest between the first and second valleys. The crest may be configured to permit the switch to transition between the unlocked and locked positions when a force applied to the switch exceeds a predetermined force threshold. The coupler body further may include a holder slidably disposed within the coupler body, the holder comprising a contact surface configured to define at least a portion of the lumen. In addition, the holder may be biased in a direction toward the lumen such that, when the elongated shaft is disposed within the lumen, the contact surface engages with the elongated shaft. In some embodiment, the coupler interface may include a repulsion magnet, and the holder may include a magnet, such that the repulsion magnet applies a magnetic force to the magnet to thereby cause the holder to be biased in the direction toward the lumen. Further, the holder may include a harness configured to be coupled to the magnet, the harness sized and shaped to be slidably disposed within a channel of the coupler body. The co-manipulation surgical system further may include a clamp pivotally coupled to the coupler body via a rod. The clamp may be configured to transition between an unlocked state where the lumen is permitted to receive the elongated shaft and a locked state where the clamp secures the elongated shaft within the lumen. Moreover, the holder may include one or more cradles coupled to the contact surface. Each of the one or more cradles may have a channel sized and shaped to slidably receive the rod therethrough, such that the holder may be slidably disposed within the coupler body along the rod.

In addition, the coupler interface may include a protrusion, and the coupler body may include a groove configured to receive the protrusion of the coupler interface. Moreover, the protrusion may include one or more indentations, and the coupler body may include one or more locking arms configured to transition between a locked configuration where at least a portion of the one or more locking arms extend within the groove of the coupler body, and an unlocked configuration where the one or more locking arms do not extend within the groove of the coupler body. Accordingly, the protrusion of the coupler interface may be received by the groove of the coupler body when the one or more locking arms are in the unlocked configuration, and the at least a portion of the one or more locking arms may extend within the one or more indentations of the protrusion when the protrusion is disposed within the groove and the locking arms are in the locked configuration to thereby secure the coupler body to the coupler interface. The one or more locking arms may be biased toward the locked configura-

4

tion. In addition, each of the one or more locking arms may have a handle portion configured to be actuated to transition the one or more locking arms from the locked configuration to the unlocked configuration.

The protrusion of the coupler interface may have a first geometry and the groove of the coupler body may have a second geometry corresponding to the first geometry, such that, when the protrusion is received by the groove, rotational movement between the coupler body and the coupler interface is prohibited. Moreover, the coupler interface may comprise one or more additional protrusions having a first geometry, and the coupler body may comprise one or more additional grooves having a second geometry, such that, when the one or more additional protrusions are received by the one or more additional grooves, rotational movement between the coupler body and the coupler interface is prohibited. The coupler body and the coupler interface may be configured to receive a sterile drape therebetween, such that the sterile drape prevents contact between the surgical instrument and the robot arm during the laparoscopic surgery.

Additionally, the coupler body may comprise one or more tapered surfaces configured to guide the elongated shaft into the lumen and facilitate self-alignment of the lumen with the elongated shaft by causing the coupler body to rotate relative to the distal end of the robot arm via the coupler interface as the elongated shaft is inserted along the one or more tapered surfaces into the lumen. The coupler body further may include a clamp configured to transition between an unlocked state where the lumen is permitted to receive the elongated shaft and a locked state where the clamp secures the elongated shaft within the lumen. The clamp may be biased toward the locked state. Moreover, at least a portion of the clamp may comprise a tapered surface configured to guide the elongated shaft into the lumen and to facilitate transitioning of the clamp from the locked state to the unlocked state responsive to a force applied to the tapered surface by the elongated shaft as the elongated shaft is inserted into the lumen.

In accordance with another aspect of the present disclosure, a method using a robot arm configured to be removably coupled to a surgical instrument having a handle, an operating end, and an elongated shaft therebetween, e.g., for assisting with laparoscopic surgery, is provided. The method may include: removably coupling a coupler body to a coupler interface at a distal end of the robot arm; inserting the elongated shaft of the surgical instrument into a lumen of the coupler body; transitioning the coupler body from an open state where the elongated shaft is slidably moveable within the lumen to a closed state where longitudinal movement of the elongated shaft relative to the coupler body is inhibited while rotational movement of the elongated shaft relative to the coupler body is permitted responsive to movement at the handle of the surgical instrument; and freely moving the robot arm by moving the handle of the surgical instrument when the coupler body is coupled to the coupler interface in the closed state, e.g., to perform the laparoscopic surgery. The coupler body may rotate relative to the distal end of the robot arm via the coupler interface to self-align the lumen with the elongated shaft as the elongated shaft is inserted into the lumen.

For example, removably coupling the coupler body to the coupler interface may include: actuating one or more locking arms of the coupler body to transition the one or more locking arms from a locked configuration where at least a portion of the one or more locking arms extend within a groove of the coupler body, to an unlocked configuration

5

where the one or more locking arms do not extend within the groove; inserting a protrusion of the coupler interface within a groove of the coupler body; and releasing the one or more locking arms to transition the one or more locking arms from the unlocked configuration to the locked configuration, such that the at least a portion of the one or more locking arms extend within one or more indentations of the protrusion to thereby secure the coupler body to the coupler interface. Moreover, inserting the elongated shaft of the surgical instrument within the lumen of the coupler body may include guiding the elongated shaft along one or more tapered surfaces of the coupler body into the lumen. For example, guiding the elongated shaft along one or more tapered surfaces of the coupler body into the lumen may cause the coupler body to rotate relative to the distal end of the robot arm via the coupler interface to self-align the lumen with the elongated shaft as the elongated shaft is inserted along the one or more tapered surfaces into the lumen.

In addition, inserting the elongated shaft of the surgical instrument within the lumen of the coupler body may include: actuating a clamp of the coupler body to transition the clamp from a locked state to an unlocked state where the lumen is permitted to receive the elongated shaft; inserting the elongated shaft of the surgical instrument within the lumen; and releasing the clamp to transition the clamp from the unlocked state to the locked state, such that the clamp secures the elongated shaft within the lumen. Moreover, transitioning the coupler body from the open state to the closed state may include transitioning a switch of the coupler body from an unlocked position where the elongated shaft is slidably moveable within the lumen, to a locked position where an engagement portion of the switch engages with the elongated shaft disposed within the lumen to thereby inhibit longitudinal movement of the elongated shaft relative to the coupler body while permitting rotational movement of the elongated shaft relative to the coupler body. Accordingly, when the elongated shaft is disposed within the lumen and the switch is in the locked position, the engagement portion may apply a friction force against the elongated shaft, the friction force configured to permit rotational movement of the elongated shaft relative to the coupler body, while inhibiting longitudinal movement of the elongated shaft relative to the coupler body. The method further may include positioning a sterile drape between the coupler body and the coupler interface prior to removably coupling the coupler body to the coupler interface.

In accordance with another aspect of the present disclosure, a coupler device for removably coupling a surgical instrument having a handle and an elongated shaft to a distal end of a robot arm of a co-manipulation surgical system to assist with laparoscopic surgery performed using the surgical instrument, the distal end of the robot arm comprising a coupler interface configured to be removably coupled to the coupler device is provided. The coupler device may include a lumen sized and shaped to receive the elongated shaft of the surgical instrument therein, and a coupler body configured to transition between an open state where the elongated shaft is slidably moveable within the lumen, and a closed state where longitudinal movement of the elongated shaft relative to the coupler body is inhibited while rotational movement of the elongated shaft relative to the coupler body is permitted responsive to movement at the handle of the surgical instrument. Accordingly, when the coupler body is coupled to the coupler interface, the coupler body may be configured to rotate relative to the distal end of the robot arm

6

via the coupler interface to self-align the lumen with the elongated shaft as the elongated shaft is inserted into the lumen.

The coupler device further may include a switch configured to transition between an unlocked position and a locked position. The switch may comprise an engagement portion configured to apply a friction force against the elongated shaft when the elongated shaft is disposed within the lumen and the switch is in the locked position to thereby permit rotational movement of the elongated shaft within the lumen, while prohibiting translational movement of the elongated shaft relative to the coupler body. In addition, the coupler device may include a ramped surface having a first valley configured to engage the switch in the unlocked position, a second valley configured to engage the switch in the locked position, and a crest disposed between the first and second valleys. The crest may be configured to permit the switch to transition between the unlocked and locked positions when a force applied to the switch exceeds a predetermined force threshold. Moreover, the switch may have a handle configured to be actuated to transition the switch between the unlocked position and the locked position. When the coupler body is coupled to the coupler interface and the elongated shaft is disposed within the lumen, the robot arm may be configured to be freely moveable responsive to movement at the handle of the surgical instrument.

In addition, the coupler body may comprise one or more tapered surfaces configured to guide the elongated shaft into the lumen and facilitate self-alignment of the lumen with the elongated shaft by causing the coupler body to rotate relative to the distal end of the robot arm via the coupler interface as the elongated shaft is inserted along the one or more tapered surfaces into the lumen. The coupler device further may include a clamp configured to transition between an unlocked state where the lumen is permitted to receive the elongated shaft and a locked state where the clamp secures the elongated shaft within the lumen. The clamp may be biased toward the locked state. Moreover, at least a portion of the clamp may comprise a tapered surface configured to guide the elongated shaft into the lumen and to facilitate transitioning of the clamp from the locked state to the unlocked state responsive to a force applied to the tapered surface by the elongated shaft as the elongated shaft is inserted into the lumen. Additionally, the clamp may have a handle portion configured to be actuated to transition the clamp from the locked state to the unlocked state.

In addition, the coupler device may include a holder slidably disposed within the coupler body, the holder comprising a friction pad configured to define at least a portion of the lumen. The holder may be biased in a direction toward the lumen such that, when the elongated shaft is disposed within the lumen, the friction pad engages with the elongated shaft. In some embodiments, the coupler interface may include a repulsion magnet, and the holder may include a magnet, such that the repulsion magnet applies a magnetic force to the magnet to thereby cause the holder to be biased in the direction toward the lumen. The holder may comprise a harness configured to be coupled to the magnet, the harness sized and shaped to be slidably disposed within a channel of the coupler body. In addition, the coupler device may include a clamp pivotally coupled to the coupler body via a rod, the clamp configured to transition between an unlocked state where the lumen is permitted to receive the elongated shaft and a locked state where the clamp secures the elongated shaft within the lumen. Moreover, the holder may comprise one or more cradles coupled to the friction

pad, each of the one or more cradles comprising a channel sized and shaped to slidably receive the rod therethrough, such that the holder is configured to be slidably disposed within the coupler body along the rod.

Moreover, the coupler body may include a groove configured to receive a protrusion of the coupler interface. The coupler device further may include one or more locking arms configured to transition between a locked configuration where at least a portion of the one or more locking arms extend within the groove of the coupler body, and an unlocked configuration where the one or more locking arms do not extend within the groove, such that the protrusion of the coupler interface is permitted to be received by the groove when the one or more locking arms are in the unlocked configuration. Additionally, the at least a portion of the one or more locking arms may extend within one or more indentations of the protrusion of the coupler interface when the protrusion is disposed within the groove and the locking arms are in the locked configuration to thereby secure the coupler body to the coupler interface. The one or more locking arms may be biased toward the locked configuration. Each of the one or more locking arms may comprise a handle portion configured to be actuated to transition the one or more locking arms from the locked configuration to the unlocked configuration. In addition, the protrusion of the coupler interface may have a first geometry and the groove of the coupler body may have a second geometry corresponding to the first geometry, such that, when the protrusion is received by the groove, rotational movement between the coupler body and the coupler interface is prohibited. The coupler body may be configured to receive a sterile drape between the coupler body and the coupler interface when the coupler body is coupled to the coupler interface.

In accordance with another aspect of the present disclosure, another co-manipulation surgical system to assist with laparoscopic surgery performed using a surgical instrument having a handle, an operating end, and an elongated shaft therebetween is provided. The system may include a robot arm comprising a proximal end, a distal end configured to be removably coupled to the surgical instrument, a plurality of links, and a plurality of joints between the proximal end and the distal end, the distal end of the robot arm comprising a coupler interface, and a coupler body configured to be removably coupled to the coupler interface. The coupler body may include a lumen sized and shaped to receive the elongated shaft of the surgical instrument therein, and may be configured to transition between an open state where the elongated shaft is slidably moveable within the lumen, and a closed state where longitudinal movement of the elongated shaft relative to the coupler body is inhibited while rotational movement of the elongated shaft relative to the coupler body is permitted responsive to movement at the handle of the surgical instrument. Moreover, when the coupler body is coupled to the coupler interface in the closed state, the robot arm may be permitted to be freely moveable responsive to movement at the handle of the surgical instrument for performing laparoscopic surgery. In some embodiments, the coupler body may be disposable after a single laparoscopic surgery. Alternatively, the coupler body may be sterilizable, such that it may be reused for multiple surgical procedures.

The system further may include a switch configured to transition between an unlocked position and a locked position. The switch may include an engagement portion configured to engage with the elongated shaft when the elongated shaft is disposed within the lumen and the switch is in the locked position to thereby secure the elongated shaft

within the lumen. Accordingly, when the coupler body is coupled to the coupler interface, the elongated shaft is disposed within the lumen, and the switch is in the locked position, the robot arm may be configured to be freely moveable responsive to movement at the handle of the surgical instrument. Moreover, when the elongated shaft is disposed within the lumen and the switch is in the locked position, the engagement portion may be configured to apply a friction force against the elongated shaft, the friction force configured to permit rotational movement of the elongated shaft relative to the coupler body, while inhibiting longitudinal movement of the elongated shaft relative to the coupler body. The switch may include a handle portion configured to be actuated to transition the switch between the unlocked position and the locked position.

The coupler body further may include a holder slidably disposed within the coupler body. The holder may include a contact surface configured to define at least a portion of the lumen, and may be configured to be biased in a direction toward the lumen such that, when the elongated shaft is disposed within the lumen, the contact surface is configured to engage with the elongated shaft. Additionally, the coupler interface may include a repulsion magnet, and the holder may include a magnet, such that the repulsion magnet may be configured to apply a magnetic force to the magnet to thereby cause the holder to be biased in the direction toward the lumen. Moreover, the holder may include a harness configured to be coupled to the magnet, the harness sized and shaped to be slidably disposed within a channel of the coupler body. In addition, the contact surface may be configured to apply a friction force against the elongated shaft when the elongated shaft is disposed within the lumen and the switch is in the locked position, such that the friction force may be configured to facilitate permission of rotational movement of the elongated shaft relative to the coupler body, while inhibiting translational movement of the elongated shaft relative to the coupler body.

The coupler body further may include a clamp configured to transition between an unlocked state where the lumen is permitted to receive the elongated shaft and a locked state where the clamp secures the elongated shaft within the lumen. The clamp may be pivotally coupled to the coupler body via a rod, and may be configured to be biased toward the locked state. Accordingly, the clamp further may include a handle portion configured to be actuated to cause the clamp to transition from the locked state to the unlocked state, e.g., to release the surgical instrument from the coupler body. Moreover, at least a portion of the clamp may include a tapered surface configured to guide the elongated shaft into the lumen and to facilitate transitioning of the clamp from the locked state to the unlocked state responsive to a force applied to the tapered surface by the elongated shaft as the elongated shaft is inserted into the lumen. In addition, the holder may include one or more cradles coupled to the contact surface, each of the one or more cradles comprising a channel sized and shaped to slidably receive the rod therethrough, such that the holder may be configured to be slidably disposed within the coupler body along the rod.

The coupler interface further may include a protrusion, and the coupler body may include a groove configured to receive the protrusion of the coupler interface. For example, the protrusion may include one or more indentations, and the coupler body may include one or more locking arms configured to transition between a locked configuration where at least a portion of the one or more locking arms extend within the groove of the coupler body, and an unlocked configuration where the one or more locking arms do not extend

within the groove of the coupler body. The protrusion of the coupler interface may be configured to be received by the groove of the coupler body when the one or more locking arms are in the unlocked configuration, and at least a portion of the one or more locking arms may extend within the one or more indentations of the protrusion when the protrusion is disposed within the groove and the locking arms are in the locked configuration to thereby secure the coupler body to the coupler interface. The one or more locking arms may be biased toward the locked configuration. In addition, each of the one or more locking arms may include a handle portion configured to be actuated to transition the one or more locking arms from the locked configuration to the unlocked configuration.

Moreover, the protrusion of the coupler interface may have a first geometry and the groove of the coupler body may have a second geometry corresponding to the first geometry, such that, when the protrusion is received by the groove, rotational movement between the coupler body and the coupler interface is prohibited. In some embodiments, the coupler interface may include one or more additional protrusions having a first geometry, and the coupler body may include one or more additional grooves having a second geometry, such that, when the one or more additional protrusions are received by the one or more additional grooves, rotational movement between the coupler body and the coupler interface is prohibited. In addition, the coupler body and the coupler interface may be configured to receive a sterile drape therebetween, such that the sterile drape prevents contact between the surgical instrument and the robot arm during the laparoscopic surgery. The coupler body further may include one or more tapered surfaces configured to guide the elongated shaft into the lumen and facilitate self-alignment of the distal end of the robot arm relative to the surgical instrument by causing the coupler body and the coupler interface to rotate to align the lumen with the elongated shaft as the elongated shaft is inserted along the one or more tapered surfaces into the lumen.

In accordance with another aspect of the present disclosure, a method for using the robot arm configured to be removably coupled to a surgical instrument having a handle, an operating end, and an elongated shaft therebetween, e.g., for assisting with laparoscopic surgery, is provided. The method may include: removably coupling a coupler body to a coupler interface at a distal end of the robot arm; inserting the elongated shaft of the surgical instrument into a lumen of the coupler body; transitioning the coupler body from an open state where the elongated shaft is slidably moveable within the lumen to a closed state where longitudinal movement of the elongated shaft relative to the coupler body is inhibited while rotational movement of the elongated shaft relative to the coupler body is permitted responsive to movement at the handle of the surgical instrument; and freely moving the robot arm by moving the handle of the surgical instrument when the coupler body is coupled to the coupler interface in the closed state, e.g., to perform the laparoscopic surgery.

For example, removably coupling the coupler body to the coupler interface may include: actuating one or more locking arms of the coupler body to transition the one or more locking arms from a locked configuration where at least a portion of the one or more locking arms extend within a groove of the coupler body, to an unlocked configuration where the one or more locking arms do not extend within the groove; inserting a protrusion of the coupler interface within a groove of the coupler body; and releasing the one or more locking arms to transition the one or more locking arms from

the unlocked configuration to the locked configuration, such that the at least a portion of the one or more locking arms extend within one or more indentations of the protrusion to thereby secure the coupler body to the coupler interface.

Moreover, inserting the elongated shaft of the surgical instrument within the lumen of the coupler body may include guiding the elongated shaft along one or more tapered surfaces of the coupler body into the lumen, e.g., by rotating the coupler body and the coupler interface to facilitate self-alignment of the lumen with the elongated shaft as the elongated shaft is inserted along the one or more tapered surfaces into the lumen.

In addition, inserting the elongated shaft of the surgical instrument within the lumen of the coupler body may include: actuating a clamp of the coupler body to transition the clamp from a locked state to an unlocked state where the lumen is permitted to receive the elongated shaft; inserting the elongated shaft of the surgical instrument within the lumen; and releasing the clamp to transition the clamp from the unlocked state to the locked state, such that the clamp secures the elongated shaft within the lumen. Transitioning the coupler body from the open state to the closed state may include transitioning a switch of the coupler body from an unlocked position where the elongated shaft is slidably moveable within the lumen, to a locked position where an engagement portion of the switch engages with the elongated shaft disposed within the lumen to thereby inhibit longitudinal movement of the elongated shaft relative to the coupler body while permitting rotational movement of the elongated shaft relative to the coupler body. The method further may include positioning a sterile drape between the coupler body and the coupler interface prior to removably coupling the coupler body to the coupler interface.

In accordance with another aspect of the present disclosure, a method for using a robot arm configured to be removably coupled to a surgical instrument having a handle, an operating end, and an elongated shaft therebetween, e.g., for assisting with laparoscopic surgery, the method performed bedside adjacent to a bed holding a patient, is provided. The method may include: positioning the robot arm bedside, e.g., for performing the laparoscopic surgery; coupling the surgical instrument to the robot arm, the robot arm comprising a proximal end, a distal end, and a plurality of links and a plurality of joints between the proximal end and the distal end; and freely moving the robot arm by moving the handle of the surgical instrument coupled thereto while the robot arm remains positioned bedside, e.g., to perform the laparoscopic surgery using the surgical instrument. The robot arm may remain bedside while using the surgical instrument coupled to the robot arm. Accordingly, a practitioner performing the laparoscopic surgery may remain bedside while performing the laparoscopic surgery using the surgical instrument coupled to the robot arm. Coupling the surgical instrument to the robot arm may comprise coupling the surgical instrument to the robot arm using a pure mechanical coupling, coupling the robot arm to only the elongated shaft of the surgical instrument, coupling the surgical instrument to the robot arm while keeping a handle of the surgical instrument completely exposed for contact with a surgeon's hand, coupling the surgical instrument to the robot arm while the robot arm is positioned bedside, and/or removably coupling a coupler body to a coupler interface disposed at the distal end of the robot arm, and removably coupling the surgical instrument to the coupler body.

The method further may include transitioning the coupler body from an open state where the elongated shaft is slidably

11

moveable within a lumen of the coupler body, and a closed state where the robot arm is permitted to be freely moveable responsive to movement at the handle of the surgical instrument, e.g., for performing the laparoscopic surgery. Accordingly, when the coupler body is coupled to the coupler interface in the closed state, longitudinal movement of the elongated shaft relative to the coupler body may be inhibited while rotational movement of the elongated shaft relative to the coupler body may be permitted responsive to movement at the handle of the surgical instrument. Moreover, when the coupler body is coupled to the coupler interface in the closed state, the coupler body may apply a friction force against the elongated shaft sufficient to permit rotational movement of the elongated shaft relative to the coupler body, while inhibiting longitudinal movement of the elongated shaft relative to the coupler body. In addition, removably coupling the surgical instrument to the coupler body may comprise removeably coupling the coupler body to a fixed point along the elongated shaft to provide a consistent point of reference for force calculations of the surgical instrument. The method further may include positioning a sterile drape between the coupler body and the coupler interface prior to removably coupling the coupler body to the coupler interface. Further, the method may include disposing of the coupler body after a single laparoscopic surgery. The robot arm may not be teleoperated via user input received at a remote surgeon console.

Moreover, freely moving the robot arm by moving the handle of the surgical instrument coupled thereto may comprise applying a force at the robot arm via the surgical instrument that exceeds a predetermined threshold to thereby automatically switch the robot arm to a co-manipulation mode where an impedance is applied to the robot arm to account for weight of the surgical instrument and the robot arm. The method further may include adjusting, via a graphical user interface operatively coupled to the robot arm, the predetermined threshold of force applied at the robot arm to cause the robot arm to automatically switch to the co-manipulation mode. In addition, the plurality of joints of the robot arm may comprise one or more motorized joints operatively coupled to one or more motors disposed in a base coupled to the proximal end of the robot arm. Accordingly, the method may include measuring current of the one or more motors, the current indicative of the force applied at the robot arm via the surgical instrument. Additionally, the impedance applied to the robot arm to account for weight of the surgical instrument and the robot arm may be applied via the one or more motorized joints of the robot arm.

The method further may include maintaining movement of the robot arm within a predetermined amount for at least a predetermined dwell time period to thereby automatically switch the robot arm to a passive mode where the robot arm maintains a static position. Moreover, the method may include adjusting, via a graphical user interface operatively coupled to the robot arm, at least one of the predetermined amount of movement of the robot arm or the predetermined dwell time period to cause the robot arm to automatically switch to the passive mode. In addition, the method may include moving the robot arm by moving the handle of the surgical instrument outside a predefined haptic barrier to thereby automatically switch the robot arm to a haptic mode where an impedance is applied to the robot arm to make movement of the robot arm responsive to movement at the handle of the surgical instrument more viscous in the haptic mode. The method further may include adjusting, via a graphical user interface operatively coupled to the robot arm, a position of the predefined haptic barrier.

12

The method further may include selecting, via a graphical user interface operatively coupled to the robot arm, an identity of the surgical instrument coupled to the robot arm. Moreover, the proximal end of the robot arm may be coupled to a base, the base coupled to a platform via a stage assembly configured to move the base in at least two degrees of freedom relative to the platform. Accordingly, the method may include adjusting at least one of a vertical height or a horizontal position of the robot arm relative to the platform via the stage assembly. For example, adjusting at least one of the vertical height or the horizontal position of the robot arm via the stage assembly may comprise providing user input via at least one of a graphical user interface operatively coupled to the stage assembly or application of force by a user at a distal region of the robot arm in at least one of the at least two degrees of freedom.

The platform may comprise a plurality of wheels, such that positioning the robot arm bedside for performing the laparoscopic surgery may comprise moving the platform relative to the bed via the plurality of wheels. The method further may include disengaging a braking mechanism of the plurality of wheels to permit movement of the platform via the plurality of wheels. Moreover, the method may include displaying a virtual map comprising graphical representations of the platform relative to the bed within an area surrounding the platform to facilitate positioning of the robot arm bedside for performing the laparoscopic surgery. In addition, the method may include adjusting at least one of a height or orientation of an optical sensor having a field of view of a surgical scene comprising at least one of the robot arm, the surgical instrument coupled to the robot arm, or the bed, to thereby optimize the field of view of the optical sensor. The method further may include selecting a laparoscope, a retractor tool, a grasper tool, or a surgical cutting tool, such that coupling the surgical instrument to the robot arm may include coupling the laparoscope, the retractor tool, the grasper tool, or the surgical cutting tool to the robot arm.

In accordance with another aspect of the present disclosure, another co-manipulation surgical system is provided. The co-manipulation surgical system may include a robot arm comprising a proximal end, a distal end configured to be removably coupled to the surgical instrument, a plurality of links, and a plurality of joints, a coupler interface disposed at the distal end of the robot arm, and a coupler body configured to be removably coupled to the coupler interface and to the elongated shaft of the surgical instrument. The coupler interface may include a ferrous rod and one or more sensors, e.g., one or more Hall effect sensors, configured to measure a magnetic field of the ferrous rod. The coupler body may include a magnet slidably disposed within the coupler body. For example, the magnet may be configured to induce a magnetic field in the ferrous rod based on a position of the magnet relative to the ferrous rod within the coupler body. In addition, the co-manipulation surgical system may include a controller operatively coupled to the robot arm and the one or more sensors, such that the controller may be programmed to determine whether the coupler body is coupled to the coupler interface based on the magnetic field of the ferrous rod measured by the one or more sensors.

Moreover, the controller may be programmed to determine whether the surgical instrument is coupled to the coupler body when the coupler body is coupled to the coupler interface based on the magnetic field of the ferrous rod measured by the one or more sensors. In addition, the controller may be programmed to identify a size of the surgical instrument coupled to the coupler body based on the magnetic field of the ferrous rod measured by the one or

13

more sensors. The magnet may be biased in a direction away from coupler interface when the coupler body is coupled to the coupler interface.

When the coupler body is coupled to the coupler interface and no surgical instrument is coupled to the coupler body, the magnet may be disposed within a predefined position within the coupler body, such that the controller may be configured to determine that the coupler body is coupled to the coupler interface without the surgical instrument coupled to the coupler body based on the magnetic field induced in the ferrous rod by the magnet at the predefined position measured by the one or more sensors. Moreover, when the coupler body is coupled to the coupler interface and the surgical instrument is coupled to the coupler body, the magnet may be disposed within a predefined position within the coupler body, such that the controller may be configured to determine that the coupler body is coupled to the coupler interface and the surgical instrument is coupled to the coupler body based on the magnetic field induced in the ferrous rod by the magnet at the predefined position measured by the one or more sensors.

The coupler body may include a holder slidably disposed within the coupler body. The holder may be configured to be coupled to the magnet and may include a friction pad configured to engage with the elongated shaft when the elongated shaft is coupled to the coupler body and the coupler body is coupled to the coupler interface, such that, when the elongated rod is coupled to the coupler body, the elongated shaft applies a force to the friction pad to thereby move the magnet to the predefined position within the coupler body via the holder. The coupler interface may include a repulsion magnet configured to apply a magnetic force to the magnet to bias the magnet in the direction away from coupler interface when the coupler body is coupled to the coupler interface. Alternatively, the magnet may be coupled to a compression spring configured to bias the magnet in the direction away from coupler interface.

The controller may be configured to permit the robot arm to be freely moveable responsive to movement at the handle of the surgical instrument for performing laparoscopic surgery using the surgical instrument when the surgical instrument is coupled to the coupler interface via the coupler body. The co-manipulation surgical system further may include a sterile drape configured to cover the coupler interface and the robot arm during surgery, such that the coupler body may be configured to be coupled to the coupler interface with the sterile drape sandwiched therebetween. The coupler body may be configured to be disposed of after a single surgery while the robot arm and the coupler interface may be configured to be reusable for additional surgeries.

In accordance with another aspect of the present disclosure, a method for using a robot arm comprising a proximal end, a distal end having a coupler interface configured to be removably coupled to a surgical instrument via a coupler body, a plurality of links, and a plurality of joints, e.g., assisting with laparoscopic surgery, is provided. The method may include: measuring, via one or more sensors, a magnetic field of a ferrous rod extending within the distal end of the robot arm; and determining, via a controller operatively coupled to the one or more sensors, whether the coupler body is coupled to the coupler interface at the distal end of the robot arm based on the magnetic field of the ferrous rod measured by the one or more sensors, wherein the coupler body comprises a magnet slidably disposed therein, such that the magnetic field of the ferrous rod measured by the one or more sensors varies based on a position of the magnet relative to the ferrous rod, and wherein the magnet is biased

14

in a direction away from the coupler interface when the coupler body is coupled to the coupler interface.

In accordance with another aspect of the present disclosure, a device for removably coupling a surgical instrument having an elongated shaft to a distal end of a robot arm of a co-manipulation surgical system to assist with laparoscopic surgery performed using the surgical instrument, the distal end of the robot arm comprising a coupler interface having a protrusion, is provided. The device may include a coupler body configured to be removably coupled to the coupler interface and to the elongated shaft of the surgical instrument. The coupler body may include a groove configured to receive the protrusion of the coupler interface, an opening sized and shaped to receive the elongated shaft therein, and a switch configured to transition between an unlocked position and a locked position. The switch may comprise an engagement portion configured to engage with the elongated shaft when the elongated shaft is disposed within the opening and the switch is in the locked position to thereby secure the elongated shaft within the opening. Accordingly, when the coupler body is coupled to the coupler interface, the elongated shaft is disposed within the opening, and the switch is in the locked position, the robot arm may be configured to be freely moveable responsive to movement at the handle of the surgical instrument.

Moreover, when the elongated shaft is disposed within the opening and the switch is in the locked position, the engagement portion may apply a friction force against the elongated shaft, the friction force configured to permit rotational movement of the elongated shaft within the opening, while prohibiting translational movement of the elongated shaft relative to the coupler body. In addition, the coupler body may comprise one or more tapered surfaces configured to guide the elongated shaft into the opening. The one or more tapered surfaces may be configured to facilitate self-alignment of the distal end of the robot arm relative to the surgical instrument by causing the coupler body and the coupler interface to rotate to align the opening with the elongated shaft as the elongated shaft is inserted along the one or more tapered surfaces into the opening. The device further may include a clamp configured to transition between an unlocked state where the opening is configured to receive the elongated shaft and a locked state where the clamp secures the elongated shaft within the opening. The clamp may be biased toward the locked state. Additionally, at least a portion of the clamp may comprise a tapered surface configured to guide the elongated shaft into the opening and to facilitate transitioning of the clamp from the locked state to the unlocked state responsive to a force applied to the tapered surface by the elongated shaft as the elongated shaft is inserted into the opening.

The coupler body further may include a holder slidably disposed within the coupler body. The holder may include a friction pad configured to define at least a portion of the opening, and the holder may be biased in a direction toward the opening such that, when the elongated shaft is disposed within the opening, the friction pad engages with the elongated shaft. In some embodiments, the coupler interface may include a repulsion magnet, and the holder may include a magnet, such that the repulsion magnet applies a magnetic force to the magnet to thereby cause the holder to be biased in the direction toward the opening. Moreover, the holder may comprise a harness configured to be coupled to the magnet, the harness sized and shaped to be slidably disposed within a channel of the coupler body. Additionally, the friction pad may be configured to apply a friction force against the elongated shaft when the elongated shaft is

15

disposed within the opening and the switch is in the locked position, the friction force configured to permit rotational movement of the elongated shaft within the opening, while prohibiting translational movement of the elongated shaft relative to the coupler body.

The device further may include a clamp pivotally coupled to the coupler body via a rod, the clamp configured to transition between an unlocked state where the opening is configured to receive the elongated shaft and a locked state where the clamp secures the elongated shaft within the opening. Moreover, the holder may comprise one or more cradles coupled to the friction pad, each of the one or more cradles comprising a channel sized and shaped to slidably receive the rod therethrough, such that the holder may be slidably disposed within the coupler body along the rod. In some embodiments, the holder may be coupled to a compression spring configured to apply a spring force to the holder to thereby cause the holder to be biased in the direction toward the opening. The clamp may include a handle portion configured to be actuated to transition the clamp from the locked state to the unlocked state. Additionally, the switch may include a handle configured to be actuated to transition the switch between the unlocked position and the locked position.

The protrusion may include one or more indentations, and the coupler body may include one or more locking arms configured to transition between a locked configuration where at least a portion of the one or more locking arms extend within the groove of the coupler body, and an unlocked configuration where the one or more locking arms do not extend within the groove of the coupler body. Accordingly, the protrusion of the coupler interface may be received by the groove of the coupler body when the one or more locking arms are in the unlocked configuration, and the at least a portion of the one or more locking arms may extend within the one or more indentations of the protrusion when the protrusion is disposed within the groove and the locking arms are in the locked configuration to thereby secure the coupler body to the coupler interface. The one or more locking arms may be biased toward the locked configuration. In addition, each of the one or more locking arms may include a handle portion configured to be actuated to transition the one or more locking arms from the locked configuration to the unlocked configuration. The protrusion of the coupler interface may have a first geometry and the groove of the coupler body may have a second geometry corresponding to the first geometry, such that, when the protrusion is received by the groove, rotational movement between the coupler body and the coupler interface is prohibited. In addition, the coupler body may be configured such that a sterile drape may be disposed between the coupler body and the coupler interface when the coupler body is coupled to the coupler interface.

In accordance with another aspect of the present disclosure, another co-manipulation surgical system is provided. The co-manipulation surgical system may include a robot arm comprising a proximal end operatively coupled to a base, a distal end configured to be removably coupled to the surgical instrument, a plurality of links, and a plurality of joints, a platform coupled to the base, the platform configured to move the base in at least one degree of freedom, a plurality of motors operatively coupled to at least some joints of the plurality of joints, and one or more sensors configured to collect sensor data comprising at least one of 3D depth data or pixel image data. The co-manipulation surgical system further may include a controller operatively coupled to the robot arm and the one or more sensors, and

16

configured to permit the robot arm to be freely moveable responsive to movement at the handle of the surgical instrument for performing laparoscopic surgery using the surgical instrument, the controller programmed to: identify at least one of a position or orientation of one or more objects, e.g., a surgical bed, within an operating room based on the sensor data from the one or more sensors; estimate a relative distance between the one or more objects and at least one of the base or the robot arm as the at least one of the base or the robot arm moves within the operating room; and apply torque or impedance to the at least some joints of the plurality of joints of the robot arm via the plurality of motors to reposition or stop movement of the robot arm to avoid a collision between the one or more objects and the at least one of the base or the robot arm if the estimated relative distance approaches a predetermined threshold.

Moreover, the controller may be configured to: detect a movement at the distal end of the robot arm in a first direction responsive to a first force applied to the distal end of the robot arm by a user; cause the platform to move the base in the first direction responsive to the detection of the movement at the distal end of the robot arm in the first direction; and cause the platform to stop movement of the base in the first direction if the first force applied to the distal end of the robot arm by the user falls below a predetermined threshold. The controller may be configured to cause the platform to move the base in the first direction if the first force applied to the distal end of the robot arm exceeds a predetermined force threshold. The controller may be configured to identify a plane of the one or more objects within the operating room based on the sensor data from the one or more sensors, and estimate the relative distance between the one or more objects and the at least one of the base or the robot arm based on the plane of the one or more objects. In addition, the controller may be configured to: determine a type of laparoscopic surgery to be performed; identify at least one of a position or orientation of a trocar port based on the sensor data from the one or more sensors; and apply torque to the at least some joints of the plurality of joints of the robot arm via the plurality of motors to automatically position the robot arm in a predetermined configuration relative to the trocar port based on the type of laparoscopic surgery to be performed.

In accordance with another aspect of the present disclosure, another co-manipulation surgical system is provided. The co-manipulation surgical system may include a robot arm comprising a proximal end, a distal end configured to be removably coupled to the surgical instrument, a plurality of links, and a plurality of joints, one or more sensors configured to collect sensor data comprising at least one of 3D depth data or pixel image data, and a controller operatively coupled to the robot arm and the one or more sensors, and configured to permit the robot arm to be freely moveable responsive to movement at the handle of the surgical instrument for performing laparoscopic surgery using the surgical instrument. The controller may be programmed to: determine at least one of a position or orientation of a trocar port relative to the robot arm based on the sensor data from the one or more sensors; detect movement of the trocar port based on the sensor data from the one or more sensors when the operating end of the surgical instrument is inserted through the trocar port; and reposition the robot arm to maintain a position of the operating end of the surgical instrument relative to the trocar port during the movement of the trocar port. For example, the controller may be configured to detect movement of the trocar port responsive to movement of a surgical bed. Moreover, the controller may

be configured to detect movement of the trocar port responsive to movement of a patient's body responsive to breathing by the patient. In some embodiments, the controller may be configured to cause the distal end of the robot arm to retract the operating end of the surgical instrument within the trocar port prior to repositioning the robot arm to maintain the position of the operating end of the surgical instrument relative to the trocar port during the movement of the trocar port.

In accordance with another aspect of the present disclosure, a co-manipulation surgical system that may be configured to calibrate a new robot arm is provided. The co-manipulation surgical system may include a robot arm comprising a proximal end configured to be removably coupled to a cart, a distal end configured to be removably coupled to the surgical instrument, a plurality of links, and a plurality of joints, an optical scanner configured to measure depth data, and a controller operatively coupled to the robot arm and the optical scanner, and configured to permit the robot arm to be freely moveable responsive to movement at the handle of the surgical instrument for performing laparoscopic surgery using the surgical instrument. The controller may be programmed to: cause the robot arm to move in an intended predefined pattern of movement relative to the cart in accordance with a preprogrammed routine; compare depth data from the optical scanner indicative of an actual movement of the robot arm responsive to the preprogrammed routine with the intended predefined pattern of movement, and generate a degree of error indicative of a deviation between the actual movement of the robot arm and the intended predefined pattern of movement; and execute an optimization algorithm configured to reduce the degree of error such that the deviation between the actual movement of the robot arm and the intended predefined pattern of movement decreases. The controller may be configured to permit the robot arm to be freely moveable responsive to movement at the handle of the surgical instrument for performing laparoscopic surgery using the surgical instrument when the degree of error falls below a predetermined threshold.

In accordance with another aspect of the present disclosure, a co-manipulation surgical system that may be configured to track surgical instruments and overlay a virtual menu on a video feed is provided. The co-manipulation surgical system may include a robot arm comprising a proximal end, a distal end configured to be removably coupled to the surgical instrument, a plurality of links, and a plurality of joints, and a controller operatively coupled to the robot arm and a laparoscope configured to generate a video feed. The controller may be configured to permit the robot arm to be freely moveable responsive to movement at the handle of the surgical instrument for performing laparoscopic surgery using the surgical instrument. Moreover, the controller may be programmed to: cause a virtual menu to overlay on the video feed displayed on a display; track movement of the operating end of the surgical instrument responsive to movement at the handle of the surgical instrument within the video feed to detect one or more predetermined gestural patterns of movement of the operating end; and actuate a function of the co-manipulation surgical system associated with the virtual menu based on the detection of the one or more predetermined gestural patterns of movement of the operating end relative to the virtual menu. The virtual menu may include one or more menu options overlaid in at least one corner of the video feed.

The function of the co-manipulation surgical system associated with the virtual menu may include, for example,

adjusting a holding force threshold required to be exceeded to cause the robot arm to switch from a passive mode where the controller causes the robot arm to maintain a static position to a co-manipulation mode where the controller permits the robot arm to be freely moveable responsive to movement at the handle of the surgical instrument for performing laparoscopic surgery using the surgical instrument. In addition, the function of the co-manipulation surgical system associated with the virtual menu may include actuation of an assisted scope mode where the controller causes the laparoscope to automatically adjust at least one of field of view or position to assist with the laparoscopic surgery. The controller may be configured to cause the virtual menu to overlay on the video feed displayed on the display responsive to user input received via a graphical user interface operatively coupled to the controller, responsive to a voice command by a user, and/or responsive to actuation of an actuator disposed on the robot arm. In addition, the controller may be configured to track movement of the operating end of the surgical instrument responsive to user input received via a graphical user interface operatively coupled to the controller, responsive to a voice command by a user, and/or responsive to actuation of an actuator disposed on the robot arm.

In accordance with another aspect of the present disclosure, a co-manipulation surgical system that may be configured to provide indications via haptic feedback is provided. The co-manipulation surgical system may include a robot arm comprising a proximal end, a distal end configured to be removably coupled to the surgical instrument, a plurality of links, and a plurality of joints, and a controller operatively coupled to the robot arm. The controller may be programmed to: cause the robot arm to automatically switch between a co-manipulation mode wherein the controller permits the robot arm to be freely moveable responsive to movement at the handle of the surgical instrument for performing laparoscopic surgery using the surgical instrument, and a passive mode where the controller causes the robot arm to maintain a static position; and cause a vibration at the distal end of the robot arm, the vibration indicative of the robot arm switching from the co-manipulation mode to the passive mode. For example, the vibration may be configured to be perceivable by a user holding the handle of the surgical instrument, while causing negligible movement at the operating end of the surgical instrument. In addition, the controller may be configured to cause a second vibration at the distal end of the robot arm when the surgical instrument is coupled to distal end of the robot arm, wherein the second vibration may be indicative that the surgical instrument is coupled to the distal end of the robot arm.

The controller may be configured to cause the robot arm to switch to the passive mode responsive to determining that movement of the robot arm due to movement at the handle of the surgical instrument is less than a predetermined amount for at least a predetermined dwell time period. Moreover, the controller may be configured to cause the robot arm to switch to the co-manipulation mode responsive to determining that force applied at the robot arm due to force applied at the handle of the surgical instrument exceeds a predetermined threshold. The controller may be configured to apply a first impedance to the robot arm in the co-manipulation mode to account for weight of the surgical instrument and the robot arm. In addition, the controller may be configured to generate an audible alert indicative of the robot arm switching from the co-manipulation mode to the passive mode. The robot arm further may include a base operatively coupled to the proximal end of the robot arm,

and the system further may include a plurality of motors disposed within the base, wherein the plurality of motors are operatively coupled to at least some joints of the plurality of joints. Accordingly, the controller may be programmed to actuate at least one motor of the plurality of motors to cause the vibration at the distal end of the robot arm.

In accordance with another aspect of the present disclosure, a co-manipulation surgical system that may be configured for automated scope detection is provided. The co-manipulation surgical system may include a robot arm comprising a proximal end, a distal end configured to be removably coupled to the laparoscope, a plurality of links, and a plurality of joints, an optical scanner configured to measure depth data, and a controller operatively coupled to the robot arm and the optical scanner. The controller may be programmed to: compare movement of the laparoscope based on depth data from the optical scanner with movement of the field of view of the laparoscope during the movement of the laparoscope based on the video feed collected from the operating end of the laparoscope; and identify a type of the laparoscope based on the movement of the field of view of the laparoscope during the movement of the laparoscope. The controller may be configured to execute a preprogrammed routine in a calibration mode to cause the movement of the laparoscope in a predefined pattern of movement in accordance with the preprogrammed routine. For example, the predefined pattern of movement may include a circular motion. Alternatively, the movement of the laparoscope may be responsive to movement at the handle of the laparoscope by a user.

The type of laparoscope may be an angular degree of the operating end of the laparoscope. For example, the controller may be configured to identify the type of the laparoscope as a flat tipped laparoscope when the movement of the laparoscope comprises a circular motion and the movement of the field of view of the laparoscope during the circular movement of the laparoscope comprises a corresponding circular motion. Moreover, the controller may be configured to identify the type of the laparoscope as a flat tipped laparoscope when the movement of the laparoscope comprises a circular motion and the movement of the field of view of the laparoscope during the circular movement of the laparoscope comprises no change in depth of the field of view. In addition, the controller may be configured to identify the type of the laparoscope as an angled-tipped laparoscope when the movement of the laparoscope comprises a circular motion and the movement of the field of view of the laparoscope during the circular movement of the laparoscope comprises a change in depth of the field of view. The controller may be configured to permit the robot arm to be freely moveable responsive to movement at the handle of the laparoscope for performing laparoscopic surgery using the laparoscope.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A and 1B illustrate a traditional laparoscopic procedure performed by a surgeon and one or more assistants.

FIG. 2 illustrates an exemplary co-manipulation surgical system constructed in accordance with the principles of the present disclosure.

FIG. 3 illustrates an exemplary robot arm of the system of FIG. 2 constructed in accordance with the principles of the present disclosure.

FIGS. 4A and 4B illustrate an exemplary wrist portion of the robot arm constructed in accordance with the principles of the present disclosure.

FIG. 4C is a close-up view of an exemplary surgical instrument coupling mechanism of the wrist portion of FIGS. 4A and 4B.

FIG. 4D is a close-up view of an exemplary robot arm coupler interface of the surgical instrument coupling mechanism of FIG. 4C constructed in accordance with the principles of the present disclosure.

FIGS. 5A and 5B illustrate an exemplary surgical instrument coupler body of the surgical instrument coupling mechanism of FIG. 4C constructed in accordance with the principles of the present disclosure.

FIG. 6A illustrates an alternative exemplary surgical instrument coupler body constructed in accordance with the principles of the present disclosure.

FIGS. 6B-6D illustrate attachment of the coupler body of FIG. 6A to a surgical retractor device in accordance with the principles of the present disclosure.

FIG. 7A illustrates another alternative exemplary surgical instrument coupler body constructed in accordance with the principles of the present disclosure.

FIGS. 7B-7D illustrate attachment of the coupler body of FIG. 7A to a surgical laparoscope device in accordance with the principles of the present disclosure.

FIGS. 8A and 8B illustrate the robot arms in a sterile-drape ready configuration.

FIGS. 9A and 9B illustrate the robot arms covered in a sterile drape.

FIGS. 10A-10D illustrate rotation of the shoulder link of the robot arm in accordance with the principles of the present disclosure.

FIG. 11A illustrates an exemplary co-manipulation surgical system having an optical scanner in accordance with the principles of the present disclosure, and FIG. 11B illustrates the optical scanner of FIG. 11A.

FIG. 11C illustrates an exemplary co-manipulation surgical system having a plurality of optical scanners in accordance with the principles of the present disclosure.

FIG. 12 illustrates a user operating the co-manipulation surgical system of FIG. 11A in accordance with the principles of the present disclosure.

FIG. 13A illustrates a field of view of the optical scanner during a laparoscopic surgical procedure, and FIG. 13B illustrates a depth map of the field of view the optical scanner of FIG. 13A.

FIG. 14 shows some example components that may be included in a co-manipulation robot platform in accordance with the principles of the present disclosure.

FIG. 15 is a flow chart illustrating operation of the co-manipulation surgical system in accordance with the principles of the present disclosure.

FIG. 16 is a flow chart illustrating surgical instrument calibration of the co-manipulation surgical system in accordance with the principles of the present disclosure.

FIG. 17 is a flow chart illustrating operation of the robot arm in accordance with the principles of the present disclosure.

FIGS. 18A and 18B are free-body diagrams illustrating forces applied to the surgical instrument coupled to the robot arm during a laparoscopic surgical procedure.

FIG. 19 is a table of example values related to some arrangements of a passive mode of the robot arm in accordance with the principles of the present disclosure.

FIG. 20 illustrates an example overview of some features and capabilities of the co-manipulation surgical system in accordance with the principles of the present disclosure.

FIG. 21 is a schematic overview of some electrical components and connectivity of the co-manipulation surgical system in accordance with the principles of the present disclosure.

FIG. 22 is a flow chart illustrating an example process of acquisition and processing of data from an optical scanner and an example application of the data in accordance with the principles of the present disclosure.

FIG. 23 is a schematic overview of data flow of the co-manipulation surgical system in accordance with the principles of the present disclosure.

FIG. 24 is another schematic overview of data flow the co-manipulation surgical system in accordance with the principles of the present disclosure.

FIG. 25 is a schematic overview of data flow and output control of the co-manipulation surgical system in accordance with the principles of the present disclosure.

FIG. 26 is a schematic overview of data flow in a network of co-manipulation surgical systems in accordance with the principles of the present disclosure.

FIGS. 27A-27D illustrate vertical and horizontal movement of the robot arms in accordance with the principles of the present disclosure.

FIGS. 28A-28D illustrate an exemplary graphical user interface of the co-manipulation surgical system.

FIG. 29 is a schematic of an alternative co-manipulation surgical system constructed in accordance with the principles of the present disclosure.

FIGS. 30A-43 illustrate various alternative surgical instrument coupling mechanisms constructed in accordance with the principles of the present disclosure.

FIGS. 44A and 44B illustrates another exemplary surgical instrument coupling mechanism constructed in accordance with the principles of the present disclosure.

FIG. 44C is a cross-sectional view of the surgical instrument coupling mechanism of FIG. 44A.

FIGS. 45A and 45B illustrate a coupler interface of the surgical instrument coupling mechanism of FIGS. 44A-44C.

FIG. 45C is a cross-sectional view of the coupler interface of FIG. 45B.

FIGS. 46A-46C illustrate a coupler body of the surgical instrument coupling mechanism of FIGS. 44A-44C.

FIG. 46D illustrates a magnet holder of the coupler body of FIGS. 46A-46C.

FIG. 46E is a cross-sectional view of the coupler body of FIG. 46A.

FIG. 47A is a cross-sectional view of the coupler body of FIG. 46A when the coupler body is detached from the coupler interface.

FIG. 47B is a cross-sectional view of the surgical instrument coupling mechanism of FIG. 44A when the coupler body is coupled to the coupler interface.

FIGS. 48A and 48B illustrate various views of the surgical instrument coupling mechanism of FIG. 44A when coupling the coupler body to the coupler interface in accordance with the principles of the present disclosure.

FIG. 49 is a cross-sectional view of the surgical instrument coupling mechanism of FIG. 44A when a surgical instrument is coupled to the coupler body.

FIGS. 50A and 50B illustrate various views of the surgical instrument coupling mechanism of FIG. 44A when coupling the surgical instrument to the coupler body in accordance with the principles of the present disclosure.

FIG. 51 is a cross-sectional view of an alternative exemplary surgical instrument coupling mechanism when a surgical instrument is coupled to the coupler body, constructed in accordance with the principles of the present disclosure.

FIGS. 52A to 52E illustrate an alternative exemplary surgical instrument coupling mechanism constructed in accordance with the principles of the present disclosure.

FIGS. 53A and 53B illustrate an alternative exemplary coupler body constructed in accordance with the principles of the present disclosure.

FIG. 54 illustrates another alternative exemplary coupler body constructed in accordance with the principles of the present disclosure.

FIG. 55 illustrates an exemplary virtual overlay of a graphical user interface of the co-manipulation surgical system.

FIG. 56 illustrates a motorized joint of an alternative exemplary robot arm constructed in accordance with the principles of the present disclosure.

DETAILED DESCRIPTION

Disclosed herein are co-manipulation surgical robot systems for assisting an operator, e.g., a surgeon, in performing a surgical procedure, e.g., a laparoscopic procedure, and methods of use thereof. Currently, laparoscopic procedures typically require a surgeon and one or more assistants. For example, as shown in FIG. 1A, during a laparoscopic procedure assistant A1 may be required to hold retractor device 12 to expose tissue for surgeon S, while another assistant A2 may be required to hold laparoscope device 10 to provide a field of view of the surgical space within the patient to surgeon S via a display (not shown) during the procedure. As shown in FIG. 1A, assistant A2 may be required to hold laparoscope device 10 in an impractical position, e.g., from between the arms of surgeon S while the surgeon actively operates additional surgical instruments, e.g., surgical instruments 14 and 16. As further shown in FIG. 1A, surgeon S may need to let go of surgical instrument 16 in order to guide/reposition laparoscope device 10 held by assistant A2 in order to achieve the field of view desired by the surgeon.

As shown in FIG. 1B, rail-mounted orthopedic retractors 18 may be used to hold one or more surgical instruments in position during the laparoscopic procedure, in attempt to free hands of the surgeon and/or assistant for other tasks, as well as for stability. As shown in FIG. 1B, first rail-mounted orthopedic retractor 18a may include retractor end 20a for engaging with and holding laparoscope device 10 in position upon actuation of lock 22a. For example, lock 22a may be disengaged such that retractor 18a may be manually positioned at a desired location relative to the patient, and re-engaged to lock retractor 18a, and accordingly laparoscopic device 10 coupled thereto, in the desired position. As shown in FIG. 1B, second rail-mounted orthopedic retractor 18b having retractor end 20b may be used during the procedure to engage with and hold another surgical instrument in position upon actuation of lock 22b. Thus, retractors 18a and 18b require extensive manual interaction with locks 22a and 22b, and with retractors 18a and 18b themselves, to reposition and lock the respective tools in position.

The co-manipulation surgical robot systems described herein provide superior control and stability such that the surgeon and/or assistant may seamlessly position various off-the-shelf surgical instruments as needed, thus avoiding the workflow limitations inherent to both human and mechanical solutions. For example, the robot arms of the

co-manipulation surgical robot system may provide surgical assistance by holding a first surgical instrument, e.g., a laparoscope, via a first robot arm, and a second surgical instrument, e.g., a retractor, via a second robot arm, stable throughout the procedure to provide an optimum view of the surgical site and reduce the variability of force applied by the surgical instruments to the body wall at the trocar point. As will be understood by a person having ordinary skill in the art, the robots arms of the co-manipulation surgical robot systems described herein may hold any surgical instrument, preferably having a long and thin instrument shaft, used for surgical procedures such as laparoscopic procedures including, e.g., endoscopes/laparoscopes, retractors, graspers, surgical scissors, needle holders, needle drivers, clamps, suturing instruments, cautery tools, staplers, clip applicators, etc.

The co-manipulation surgical robot system further allows the surgeon to easily maneuver both tools when necessary, providing superior control and stability over the procedure and overall safety. Any implementations of the systems described herein enable a surgeon to directly co-manipulate instruments while remaining sterile at the patient bedside. For example, the system may include two robot arms that may be used by the surgeon to hold both a laparoscope and a retractor. During a surgical procedure, the system may seamlessly reposition either instrument to provide optimal visualization and exposure of the surgical field. Both instruments may be directly coupled to the robot arms of the system and the system may constantly monitor and record the position of the two instruments and/or the two robot arms throughout the procedure. Moreover, the system may record information such as the position and orientation of surgical instruments attached to the robot arm, sensor readings related to force(s) applied at proximal and distal ends of the surgical instruments attached to robot arms, force required to hold each instrument in position, endoscopic video streams, algorithm parameters, operating room 3D stream captured with an optical scanning device, including, e.g., position(s) of surgical entry port(s), position and movements of the surgeon's hands, surgical instrument(s) position and orientation, whether or not attached to robot arms, patient position, and patient table orientation and height.

Such data may be used to develop a database of historical data that may be used to develop the algorithms used in some implementations to control one or more aspects of an operation of the system. In addition, such data may be used during a procedure to control of one or more aspects of an operation of the system per one or more algorithms of the system. For example, the data may be used to assess a level of fatigue of a user of the system.

As the operator manipulates a robot arm of the co-manipulation surgical robot system by applying movement to the surgical instrument coupled to the robot arm, the system may automatically transition the robot arm between various operational modes upon determination of predefined conditions. For example, the system may transition the robot arm to a passive mode responsive to determining that movement of the robot arm due to movement at the handle of the surgical instrument is less than a predetermined amount for at least a predetermined dwell time period, such that in the passive mode, the robot arm maintains a static position, e.g., to prevent damage to the equipment and/or injury to the patient. Additionally, the system may transition the robot arm to a co-manipulation mode responsive to determining that force applied at the robot arm due to force applied at the handle of the surgical instrument exceeds a predetermined threshold, such that in the co-manipulation mode, the robot arm is permitted to be freely moveable

responsive to movement at the handle of the surgical instrument for performing laparoscopic surgery using the surgical instrument, while a first impedance is applied to the robot arm in the co-manipulation mode to account for weight of the surgical instrument and the robot arm. Moreover, the system may transition the robot arm to a haptic mode responsive to determining that at least a portion of the robot arm is outside a predefined haptic barrier, such that in the haptic mode, a second impedance greater than the first impedance is applied to the robot arm, thereby making movement of the robot arm responsive to movement at the handle of the surgical instrument more viscous in the haptic mode than in the co-manipulation mode. The system further may transition the robot arm to a robotic assist mode responsive to detecting various conditions that warrant automated movement of the robot arm to guide the surgical instrument attached thereto, e.g., along a planned trajectory or to avoid a collision with another object or person in the surgical space.

Referring now to FIG. 2, co-manipulation surgical robot system **200** is provided. As shown in FIG. 2, system **200** may include platform **100**, e.g., a surgical cart, sized and shaped to support or more robot arms **300**, e.g., robot arm **300a** and robot arm **300b**, each of robot arms **300** having surgical instrument coupler interface **400** for removably coupling to a surgical instrument, and a computing system operatively coupled to platform **100** and robot arms **300**. As shown in FIG. 2, system **200** further may include graphical user interface display **110** for displaying operational information as well as receiving user input.

In addition, each of robot arms **300** further may include indicators **334** for visually indicating the operational mode associated with the respective robot arm in real-time. For example, indicators **334** may be positioned on at least the elbow joint of the robot arm. Additionally or alternatively, indicators **334** may be placed elsewhere on system **200**, e.g., on platform **100**, on display **110**, on one or more links and/or joints, etc. Moreover, indicators **334** may include lights, e.g., LED lights, that may illuminate in a variety of distinct colors and in distinct patterns, e.g., solid on or blinking. For example, each operational mode of system **200** may be associated with a uniquely colored light, such as red, yellow, blue, green, purple, white, orange, etc. Accordingly, indicators **334** may indicate a transition from one operational mode to another operational mode. Additionally or alternatively, transitions from one operational mode to another operational mode may be indicated to a user via haptic feedback, e.g., a vibration delivered to the distal end of robot arm **300**, and accordingly to the surgical instrument coupled thereto. For example, the distal end of robot arm **300** may vibrate as robot arm **300** transitions from co-manipulation mode to static mode to assure the user that robot arm **300** is in static mode and will remain in position upon release by the user. Additionally or alternatively, an audible alert may be emitted to indicate to the user when robot arm **300** transitions from one operational mode to another operational mode.

As shown in FIG. 2, platform **100** may include one or more stages coupled to the base portion of one or more robot arms, e.g., base portion **302a** of robot arm **300a** and base portion **302b** of robot arm **300b**, for providing movement to the respective robot arm, e.g., in at least the horizontal and vertical directions relative to platform **100**. Each stage may include vertical extenders, e.g., vertical extender **106a** or vertical extender **106b**, for independently moving robot arm **300a** or robot arm **300b**, respectively, vertically relative to platform **100**, and horizontal extenders, e.g., horizontal

25

extender **108a** or horizontal extender **108b**, for independently moving robot arm **300a** or robot arm **300b**, respectively, horizontally relative to platform **100**, to thereby permit the operator flexibility in positioning robot arms **300** relative to the patient.

Moreover, platform **100** may include a plurality of wheels **104**, e.g., castor wheels, to provide mobility of platform **100**, and accordingly, robot arms **300**, within the operating room. Wheels **104** may each include a braking mechanism which may be actuated to prevent movement of platform **100** via wheels **104**. Accordingly, platform **100** may independently move each of robot arm **300a** and robot arm **300b** in any direction, including a first or vertical direction toward and away from the floor, a second or horizontal direction toward and away from the patient, and/or a third direction or horizontal direction along a length of the patient. In some embodiments, platform **100** may move robot arm **300a** and robot arm **300b** in the same direction simultaneously, and further may cause rotation movement of robot arm **300a** and robot arm **300b**. When ready for operation, platform **100** may be moved to a desired position at the side of the patient bed and locked in place via wheels **104**, and the vertical and horizontal positions of robot arms **300a** and **300b** may be adjusted to an optimum position relative to the patient for the procedure via vertical extenders **106a**, **106b** and horizontal extenders **108a**, **108b**, responsive to user input received by graphical user interface display **110**. As described in further detail below, platform **100** may automatically move robot arm **300a** and robot arm **300b** responsive to detection of, e.g., potential collisions with other objects and/or persons within the operating room and/or user input applied via the robot arms, during a laparoscopic procedure and/or during setup of the robot arms.

Surgical robot system **200** is configured for co-manipulation, such that system **200** may assist the user or operator, e.g., a surgeon and/or surgical assistant, by permitting the user to freely move robot arm **300a** and/or robot arm **300b** due to manipulation of one or more surgical instruments coupled with the robot arms in response to force inputs provided by the user to the surgical instruments. Accordingly, system **200** may be configured so that it is not controlled remotely, such that robot arms **300** move directly responsive to movement of the surgical instrument coupled thereto by the operator, while compensating for the mass of the surgical instrument and of the respective robot arm and providing localized impedance along the robot arm, thereby increasing the accuracy of the movements or actions of the operator as the operator manipulates the surgical instrument.

System **200** may be particularly useful in laparoscopic surgical procedures and/or other surgical procedures that utilize long and thin instruments that may be inserted, e.g., via cannulas, into the body of a patient to allow surgical intervention. As will be understood by a person having ordinary skill in the art, system **200** may be used for any desired or suitable surgical operation. Moreover, system **200** may be used in conjunction or cooperation with video monitoring provided by one or more cameras and/or one or more endoscopes so that an operator of system **200** may view and monitor the use of the instrument coupled with robot arms **300a**, **300b** via respective coupler interfaces, as described in further detail below. For example, robot arm **300a** may be removeably coupled with and manipulate an endoscope, while robot arm **300b** may be removeably coupled with and manipulate a surgical instrument.

Referring now to FIG. 3, a surgical support arm, e.g., robot arm **300**, is provided. As described above, system **200** may include a plurality of robot arms, e.g., robot arm **300a**

26

and robot arm **300b**. However, as each robot arm may be constructed identically, only a single robot arm is described with regard to FIG. 3 for brevity, collectively as robot arm **300**. Aspects of the robot arms described herein may utilize structures from U.S. Pat. No. 10,118,289 to Louveau, the entire contents of which are incorporated herein by reference. Robot arm **300** may include a plurality of arm segments/links and a plurality of articulation joints extending from a base portion. For example, robot arm **300** may include a base portion, a shoulder portion, an elbow portion, and a wrist portion, thereby mimicking the kinematics of a human arm. As shown in FIG. 3, robot arm **300** may include a base, which includes base portion **302** rotatably coupled to shoulder portion **304** at base joint **303**. For example, shoulder portion **304** may sit on top of base portion **302**, and may be rotated relative to base portion **302** about axis **Q1** at base joint **303**. In some embodiments, robot arms **300** may be interchanged, swapped, or coupled with the base in any desired arrangement.

Robot arm **300** further may include shoulder link **305**, which includes proximal shoulder link **306** rotatably coupled to distal shoulder link **308**. A proximal end of proximal shoulder link **306** may be rotatably coupled to shoulder portion **304** of the base at shoulder joint **318**, such that proximal shoulder link **306** may be rotated relative to shoulder portion **304** about axis **Q2** at shoulder joint **318**. As shown in FIG. 3, axis **Q2** may be perpendicular to axis **Q1**. The distal end of proximal shoulder link **306** may be rotatably coupled to the proximal end of distal shoulder link **308** at joint **320**, such that distal shoulder link **308** may be rotated relative to proximal shoulder link **306** about axis **Q3** at joint **320**. As shown in FIG. 3, axis **Q3** may be parallel to the longitudinal axis of shoulder link **305**. In addition, robot arm **300** may include actuator **330**, e.g., a lever, button, collar, or switch, operatively coupled to distal shoulder link **308** and/or proximal shoulder link **306**, such that distal shoulder link **308** may only be rotated relative to proximal shoulder link **306** upon actuation of actuator **330**. Accordingly, axis **Q3** may be a "setup" axis, such that distal shoulder link **308** may be rotated and fixed relative to proximal shoulder link **306** during a setup stage prior to operating stage where robot arm **300** is used in a surgical procedure, as described in further detail with regard to FIGS. 10A to 10D. As shown in FIG. 3, actuator **330** may be disposed on elbow link **310**.

In some embodiments, upon actuation of actuator **330**, distal shoulder link **308** may be manually rotated in predefined increments relative to proximal shoulder link **306**. Alternatively, upon actuation of actuator **330**, distal shoulder link **308** may be automatically rotated relative to proximal shoulder link **306** until actuator **330** is released, as described in further detail below with regard to FIG. 56, and as described in U.S. patent application Ser. No. 18/331,060 to Noonan, the entire contents of which are incorporated herein by reference. For example, actuator **330** may be a button, collar, or switch operatively coupled to a motor operatively coupled to distal shoulder link **308** and/or proximal shoulder link **306**, such that upon actuation of actuator **330**, the associated motor causes distal shoulder link **308** to rotate relative to proximal shoulder link **306**. The motor is disposed within the base of robot arm **300**, or alternatively, the motor may be disposed adjacent to joint **320**, e.g., on shoulder link **305**. Accordingly, actuator **330** may be a button, collar, or switch that permits dual actuation, e.g., a first actuation to cause distal shoulder link **308** to rotate in a first direction relative to shoulder link **306**, and a second actuation to cause distal shoulder link **308** to rotate in a second direction opposite to the first direction. In some

embodiments, the button or switch may be located on a graphical user interface such as display 110. In addition, in some embodiments, distal shoulder link 308 may be automatically rotated relative to proximal shoulder link 306 by a processor of the co-manipulation robot platform, e.g., during setup, to avoid collisions, as described in further detail below.

Robot arm 300 further may include elbow link 310. A proximal end of elbow link 310 may be rotatably coupled to a distal end of distal shoulder link 308 at elbow joint 322, such that elbow link 310 may be rotated relative to distal shoulder link 308 about axis Q4 at elbow joint 322. Robot arm 300 further may include wrist portion 311, which may include proximal wrist link 312 rotatably coupled to the distal end of elbow link 310 at wrist joint 324, middle wrist link 314 rotatably coupled to proximal wrist link 312 at joint 326, and distal wrist link 316 rotatably coupled to middle wrist link 314 at joint 328, as further shown in FIGS. 4A and 4B. Accordingly, wrist portion 311 may be rotated relative to elbow link 310 about axis Q5 at wrist joint 324, middle wrist portion 314 may be rotated relative to proximal wrist link 312 about axis Q6 at joint 326, and distal wrist link 316 may be rotated relative to middle wrist link 314 about axis Q7 at joint 328.

Robot arm 300 further may include wrist portion 311, which may include proximal wrist link 312 rotatably coupled to the distal end of elbow link 310 at wrist joint 324, middle wrist link 314 rotatably coupled to proximal wrist link 312 at joint 326, and distal wrist link 316 coupled to/extending from middle wrist link 314, which may be rotatably coupled to a surgical instrument coupler interface (not shown) at joint 328, e.g., coupler interface 400 as further shown in FIGS. 4A and 4B, coupler interface 4500 as further shown in FIGS. 44A and 44B and/or coupler interface 5200 as further shown in FIGS. 52A to 52E. Accordingly, wrist portion 311 may be rotated relative to elbow link 310 about axis Q5 at wrist joint 324, middle wrist portion 314 may be rotated relative to proximal wrist link 312 about axis Q6 at joint 326, and the surgical instrument coupler interface may be rotated relative to distal wrist link 316, and accordingly middle wrist link 314, about axis Q7 at joint 328.

In addition, as shown in FIGS. 3 and 4B, robot arm 300 may include actuator 332, e.g., a lever, button, or switch, operatively coupled to elbow link 310 and/or proximal wrist link 312 at joint 324, such that proximal wrist link 312 may only be rotated relative to elbow link 310 upon actuation of actuator 332. Accordingly, axis Q5 may be a “setup” axis, such proximal wrist link 312 may be rotated and fixed relative to elbow link 310 during a setup stage, upon actuation of actuator 332, e.g., prior to operating stage where robot arm 300 is used in a surgical procedure. When actuator 332 is in an unactuated state, setup joint 324 prevents relative movement between proximal wrist link 312 and elbow link 310, such that proximal wrist link 312 is fixed relative to elbow link 310. In some preferred embodiments, upon actuation of actuator 332, proximal wrist link 312 may be manually rotated in predefined increments relative to elbow link 310, thereby removing the necessity of having additional motors and/or electronics at the distal region of robot arm 300. Alternatively, upon actuation of actuator 332, proximal wrist link 312 may be automatically rotated relative to elbow link 310 until actuator 332 is released, e.g., via a motor operatively coupled to proximal wrist link 312 and/or elbow link 310 at joint 324.

As shown in FIG. 3, robot arm 300 may include a plurality of motors, e.g., motors M1, M2, M3, which may all be

disposed within the base of robot arm 300, and M4, which preferably may be disposed adjacent to joint 320. Alternatively, motor M4 also may be disposed within the base of robot arm 300. Each of motors M1, M2, M3, may be operatively coupled to a respective motorized joint of robot arm 300, e.g., base joint 303, shoulder joint 318, and elbow joint 322, to thereby apply a localized impedance at the respective joint. For example, motors M1, M2, M3 may produce an impedance/torque at any of base joint 303, shoulder joint 318, and elbow joint 322, respectively, to thereby effectively apply an impedance at the distal end of robot arm, e.g., at the attachment point with the surgical instrument, to improve the sensations experienced by the operator during manipulation of the surgical instrument as well as the actions of the operator during surgical procedures. For example, impedance may be applied to the distal end of robot arm 300, and accordingly the surgical instrument coupled thereto, to provide a sensation of a viscosity, a stiffness, and/or an inertia to the operator manipulating the surgical instrument. Moreover, applied impedances may simulate a tissue density or stiffness, communicate surgical boundaries to the operator, and may be used to direct a surgical instrument along a desired path, or otherwise. In some embodiments, the motors may actuate the respective joints to thereby cause movement of robot arm 300 about the respective joints. Accordingly, axis Q1, axis Q2, and axis Q4 may each be a “motorized” axis, such that motors M1, M2, M3 may apply an impedance/torque to base joint 303, shoulder joint 318, and elbow joint 322, respectively, to inhibit or actuate rotation about the respective axis. As described in further detail below, motors M1, M2, M3 may be controlled by a processor of the co-manipulation robot platform. With three motorized axes, some implementations of robot arm 300 may apply force/torque at the distal end of robot arm 300 in three directions to thereby move the surgical instrument coupled to the distal end of robot arm 300 in three degrees of freedom.

Motor M4 may be operatively coupled to setup joint 320 to thereby apply a torque to joint 320 to actuate rotation of distal shoulder link 308 relative to proximal shoulder link 306 about axis Q3, as described above. Unlike the other motorized joints described herein, e.g., base joint 303, shoulder joint 318, and elbow joint 322, motorized joint 320 is preferably not “back-drivable,” in that the user cannot actuate motorized joint 320, e.g., via movement of the surgical instrument coupled to the robot arm when the system is in co-manipulation mode. Instead, as described above, actuation of motorized joint 320 may be conducted via one or more actuators, e.g., actuator 330, that may be actuated to automatically cause rotation of distal shoulder link 308 relative to proximal shoulder link 306.

Axis Q6 and axis Q7 may each be a “passive” axis, such that middle wrist link 314 may be rotated relative to proximal wrist link 312 at passive joint 326 without any applied impedance from system 200, and the surgical instrument coupler interface may be rotated relative to distal wrist link 316 at passive joint 328 without any applied impedance from system 200. The distal end of distal wrist link 316 may be rotatably coupled to the surgical instrument coupler interface for removably coupling with a surgical instrument, e.g., via coupler body 500 as further shown in FIGS. 4A and 4B, coupler body 4600 as further shown in FIGS. 44A and 44B, coupler body 5100 as further shown in FIG. 51, coupler body 5300 as further shown in FIGS. 53A and 53B, and/or coupler body 5400 as further shown in FIG. 54, which may be removeably coupled to the surgical instrument and to the coupler interface, as described in further detail below. Alter-

29

natively, wrist portion **311** may include a passive ball joint at the attachment point with the surgical instrument, as described in U.S. Pat. No. 10,582,977, the entire disclosure of which is incorporated herein by reference.

Referring again to FIG. 3, robot arm **300** further may include a plurality of encoders, e.g., encoders **E1-E7**, disposed on at least some of the plurality of joints of robot arm **300**. For example, encoder **E1** for measuring angulation between base portion **302** and shoulder portion **304** may be disposed on or adjacent to base joint **303** within the base, encoder **E2** for measuring angulation between shoulder portion **304** and proximal shoulder link **306** may be disposed on or adjacent to shoulder joint **318** within the base, encoder **E3** for measuring angular rotation between proximal shoulder link **306** and distal shoulder link **308** may be disposed on or adjacent to joint **320**, encoder **E4** for measuring angulation between distal shoulder link **308** and elbow link **310** may be disposed adjacent to motor **M3** operatively coupled to elbow joint **322** within the base as transmission of rotational motion at elbow joint **322** is achieved via a connection rod extending from the base to elbow joint **322**, encoder **E5** for measuring angular rotation between elbow link **310** and proximal wrist link **312** may be disposed on or adjacent to wrist joint **324**, encoder **E6** for measuring angulation between proximal wrist link **312** and middle wrist link **314** may be disposed on or adjacent to joint **326**, and encoder **E7** for measuring angulation of between distal wrist link **316** and the surgical instrument coupler interface may be disposed on or adjacent to joint **328**. Alternatively, encoder **E4** may be disposed on or adjacent to elbow joint **322**. The encoders may be absolute encoders or other position/angulation sensors configured to generate data for accurately determining the position and/or angulation of corresponding links at the respective joint and/or the exact position of the surgical instrument coupled to the distal end of robot arm **300**. Accordingly, the exact position of each link, joint, and the distal end of robot **300** may be determined based on measurements obtained from the plurality of encoders. Preferably, a redundant encoder is disposed at each location along robot arm **300** where an encoder is placed, to provide more accurate position data, as well as, to detect a fault condition, as described in further detail below.

Prior to attachment with a surgical instrument, robot arm **300** may be manually manipulated by a user, e.g., to position robot arm **300** in a desired position for coupling with the surgical instrument. For example, the user may manually manipulate robot arm **300** via wrist portion **311**, actuator **330**, and/or actuator **332**. Upon actuation of actuator **330**, the user may manually or automatically rotate distal shoulder link **308**, and upon actuation of actuator **332**, the user may manually manipulate proximal wrist portion **312**. Moreover, robot arm **300** may further be manually moved by application of a force directly on the other links and/or joints of robot arm **300**.

In some embodiments, responsive to force applied to robot arm **300**, e.g., at wrist portion **311**, wrist joint **324**, elbow link **310**, etc., by the user, e.g., in a predetermined amount or pattern, in a given direction, the processor of the co-manipulation robot platform may cause the stage coupled to base portion **302** of robot arm **300** to move robot arm **300** in the same direction, e.g., via the vertical and horizontal extenders of the stage, until the force applied to robot arm **300** by the user is detected by the system to drop below a predetermined threshold, e.g., when the user releases robot arm **300**. In some embodiments, the system may stop movement of robot arm **300** in the same direction as the force applied by the user when the user applies a counter

30

force to robot arm **300**, e.g., in a direction opposite to the direction of movement of robot arm **300**, to facilitate setup of robot arm **300** relative to the patient.

For example, the user may apply a force that exceeds a predetermined force threshold on wrist portion **311** in a first direction, which causes the stages of platform **100** to move robot arm **300** in that same direction until the user stops movement of wrist portion **311**, e.g., by letting go of robot arm **300** or by applying a counter force to robot arm **300**, such that the system stops movement of the stages of platform **100**. Accordingly, movement of the distal end of the robot arms, e.g., wrist portion **311**, wrist joint **324**, elbow link **310**, etc., may serve as an input for motion generated in particular directions of the robot arms via the stages coupled thereto. Such automated movement of the stages of platform **100** responsive to force applied to the distal end of robot arm **300** by the user may be limited to when the system is in a predefined operating mode, which may be entered in during setup and/or during a surgical procedure, e.g., upon actuation at GUI **110** and/or via voice control.

Upon attachment to the surgical instrument, robot arm **300** may still be manipulated manually by the user exerting force, e.g., one or more linear forces and/or one or more torques, directly to robot arm **300**; however, during the laparoscopic procedure, the operator preferably manipulates robot arm **300** only via the handle of the surgical instrument, which applies force/torque to the distal end of the robot arm **300**, and accordingly the links and joints of robot arm **300**. As the operator applies a force to the surgical instrument attached to robot arm **300**, thereby causing movement of the surgical instrument, robot arm **300** will move responsive to the movement of the surgical instrument to provide the operator the ability to freely move surgical instrument relative to the patient. As described in further detail below, robot arm **300** may apply an impedance to account for weight of the surgical instrument and of robot arm **300** itself, e.g., gravity compensation, as the operator moves the surgical instrument, thereby making it easier for the operator to move the instrument despite gravitational forces and/or inertial forces being exerted on the robot arm and/or the surgical instrument. As will be understood by a person having ordinary skill in the art, robot arm **300** may include less or more articulation joints than is shown in FIG. 3, as well as a corresponding number of motors and encoders/sensors.

Referring now to FIG. 4C, a close-up view of the coupling mechanism of coupler interface **400** and coupler body **500** is provided. Coupler interface **400** may be coupled to the distal end of distal wrist link **316** using any suitable fasteners or connectors, e.g., magnets, screws, pins, clamps, welds, adhesive, rivets, and/or any other suitable faster or any combination of the foregoing. As shown in FIG. 4C, coupler interface **400** may be rotatably coupled with the distal end of distal wrist portion **316** using fastener **410** which may be threaded or have other features that enable fastener **410**, and accordingly coupler interface **400** to be selectively attached to distal wrist portion **316**. Fastener **410** may be coupled with insert element **408** having an opening therein to receive fastener **410**, positioned at or in the distal end of distal wrist portion **316**. In some embodiments, fastener **410** may be a pin or may have other features such as a ball, a latch, or otherwise to permit fastener **410** to selectively couple with distal wrist portion **316**.

Coupler body **500**, which may have opening **514** sized and shaped to slidably and releasably receive the elongated shaft of a surgical instrument therethrough, may be removably coupled with coupler interface **400**. For example,

31

coupler body 500 may be removeably coupled to coupler body 500 via a magnetic connection, to thereby facilitate efficient attachment and detachment between coupler body 500 and coupler interface 400, e.g., by overcoming the magnetic coupling force between coupler body 500 and coupler interface 400. Accordingly, as shown in FIG. 4C, coupler body 500 may have one or more magnets 506 extending away from a surface of coupler body 500 that, in an assembled state, contacts a surface of coupler interface 400. Alternatively, in embodiments that do not have a coupler interface, magnets 506 may directly contact the distal end of distal wrist portion 316. Accordingly, coupler interface 400 or the distal end of distal wrist portion 316 may have a ferrous base component configured to receive and magnetically couple with magnets 506 of coupler body 500 so that coupler body 500 may be removeably coupled with coupler interface 500 and/or the distal end of distal wrist portion 316.

FIG. 4D illustrates surgical instrument coupler interface 400. As shown in FIG. 4D, coupler interface 400 may have recessed portion 404 sized and shaped to receive the complementary geometry of coupler body 500, defined by ridges 402. Accordingly, when the complementary geometry of coupler body 500 is received in recessed portion 404 in an assembled state, rotational movement of coupler body 500 relative to coupler interface 400 may be limited or otherwise prevented. In addition, coupler interface 400 may have one or more recesses or depressions 406 sized and shaped to receive one or more magnets 506 therein. Coupler interface 400 may have a ferrous base component or magnets within recesses 406 to magnetically couple with magnets 506. For example, the magnets within recesses 406 may have a south magnetic pole and magnets 506 may have a north magnetic pole, or vice versa. Moreover, the polarity of the magnets can ensure appropriate coupling orientation. Recesses 406 may be sized and shaped to limit or otherwise prevent movement between coupler body 500 and coupler interface 400 in any direction that is radial or normal to an axial (e.g., longitudinal) centerline of magnets 506 when coupler body 500 is in an assembled state with coupler interface 400. As will be understood by a person having ordinary skill in the art, coupler interface 400 may have less or more than two recesses 406, such that coupler body 500 will have a corresponding amount of magnets.

Referring now to FIGS. 5A and 5B, coupler body 500 is provided. As shown in FIG. 5A, coupler body 500 may have one or more magnets 506 disposed on portion 502 having a geometry complementary to recessed portion 404 of coupler interface 400, as described above, to facilitate alignment between coupler body 500 and coupler interface 400. In addition, coupler body 500 may have one or more grooves 504 sized and shaped to engage with complementary ridges 402 of coupler interface 400. Grooves 504 and ridges 402 may interact to assist with the alignment of coupler body 500 with coupler interface 400 by limiting or otherwise preventing movement between coupler body 500 and coupler interface 400 in at least two directions D1 and D2, as shown in FIG. 4C. Accordingly, in an assembled state, coupler body 500 may be prevented from moving in any axial direction relative to coupler interface 400.

As shown in FIGS. 5A and 5B, coupler body 500 may have first portion 508 and second portion 510. First portion 508 may be coupled with, or integrally formed with, second portion 510, e.g., via hinge 512, which may be a living hinge formed from the same material as first and second portions 508, 510 and/or integrally formed with first and second portions 508, 510 so that second portion 510 may be moved

32

or rotated relative to first portion 508 to cause opening 514 defined by first portion 508 and second portion 510 to expand (increase in size) or contract (decrease in size). First portion 508 and second portion 510 may form a clamp that may constrict about the elongated shaft of a surgical instrument that is positioned in opening 514 as screw 516, e.g., a thumb screw, is tightened, to couple the instrument 112 with the coupler body 141. Accordingly, coupler body 500 may transition between a first, unsecured/open state or position and a second, secured/closed state or position.

The diameter of opening 514 may be selected based on the surgical instrument to be coupled to coupler body 500. For example, a coupler body may be selected from a plurality of coupler bodies, each coupler body having an opening sized and shaped to receive the elongate shaft of a specific surgical instrument having a predefined elongated shaft diameter such as a laparoscopic or other surgical instrument including surgical instruments used for orthopedic and trauma surgery (OTS), a needle holder, clamp, scissors, etc. Coupler body 500 may be coupled with the surgical instrument at any desired axial position on the surgical instrument.

As shown in FIG. 4C, coupler body 500 may include recess 520 extending through second portion 510 and recess 522 extending through at least a portion of first portion 508. Recess 520 is aligned with recess 522 for receiving locking portion 518 of screw 516. For example, locking portion 518 may have a male threaded surface, and recesses 520, 522 may have a female threaded surface to engage with locking portion 518. Screw 516 may be loosened by hand to open or expand opening 514 so that the surgical instrument may be removed, repositioned, rotated, and/or slid, etc. Once coupler body 500 is coupled with the surgical instrument, e.g., via screw 516, coupler body 500 and the surgical instrument that is coupled with the coupler body 500 may be removeably coupled with coupler interface 400, via magnets 506.

Opening 514 may be defined by a first semi-circular cutout in first portion 508 and a second semi-circular cutout in the second portion 510 of coupler body 500, to thereby engage with the circular outer surface of the elongate shaft of a surgical instrument. Opening 514 may include, e.g., rubber pads, sheets, bumps, O-rings, projections, or other components or features configured to contact and grip the outer surface of the elongated shaft of the surgical instrument. For example, the rubber material may be a silicone rubber or any other suitable type of rubber. Accordingly, once coupler body 500 is coupled with the surgical instrument, e.g., by securing screw 516, the surgical instrument may be at least inhibited or otherwise prevented from moving axially, e.g., the direction along the longitudinal axis of the surgical instrument, or, in some embodiments, moving axially and rotationally, relative to coupler body 500 in the secured state. Preferably, the surgical instrument coupled with coupler body 500 may be freely rotated by an operator relative to coupler body 500, while axial movement of the surgical instrument relative to coupler body 500 is inhibited or otherwise prevented in the secured state. For example, the frictional force between the outer surface of the elongated shaft of the surgical instrument and the inner surface of coupler body 500 defining opening 514 may be selected such that rotation of the surgical instrument relative to coupler body 500 requires less force than axial movement of the surgical instrument relative to coupler body 500 in the secured state. Accordingly, coupler 500 may be configured to account for diametric variations and surface variations (including variations in a coefficient of friction of the surface) of the surgical instruments.

In some embodiments, the surgical instrument may be moved in an axial direction relative to coupler body 500 upon the application of at least a threshold force on the surgical instrument relative to coupler body 500, or upon actuation of a release or a state change of coupler body 500. For example, such actuation may be achieved by, e.g., pressing a button, loosening a locking screw such as locking screw 516 or other connector, moving a dial, or otherwise changing coupler body 500 and/or coupler interface 400 from a second, secured state to a first, unsecured state. Accordingly, the surgical instrument may be axially repositioned relative to coupler body 500 by loosening screw 516 or other hand-operated fastener or fastening mechanism such as a clamp in coupler body 500, repositioning the surgical instrument in the desired axial position, and re-tightening screw 516 or other hand-operated fastener or fastening mechanism. Coupler body 500 may be disposable, or alternatively, may be sterilizable such that it may be sterilized between surgical procedures.

As described above, the diameter of the opening of the coupler body may be selected based on the surgical instrument to be coupled to the coupler body. Most commonly used laparoscopic surgical instruments have a predefined, known elongated shaft diameter, and thus the numerous coupler bodies may be provided, each having an opening sized and shaped to receive and engage with a specific surgical instrument. For example, FIG. 6A illustrates coupler body 600 having opening 614 sized and shaped to receive a 5 mm diameter surgical instrument, e.g., retractor device 12. Coupler body 600 may be constructed similar to coupler body 500. For example, coupler body 600 may include first portion 608 coupled to second portion 610 via hinge portion 612, and recesses 620, 622 for securely receiving locking portion 618 of screw 616. As shown in FIG. 6B, coupler body 600 may receive elongated shaft 12a of retractor 12 through opening 614, e.g., from the operating end of retractor 12, such that coupler body 600 may be slid over elongated shaft 12a until coupler body 600 engages with proximal portion 12b of retractor 12, as shown in FIG. 6C. Preferably, coupler body 600 is coupled to retractor 12 when coupler body 600 contacts proximal portion 12b as this point along retractor 12 is fixed, thereby providing a consistent point of reference for calculating force measurements, as described in further detail below. Accordingly, when coupler body 600 is in the desired location along the elongated shaft of retractor 12, e.g., adjacent to proximal portion 12b, screw 616 may be coupled to coupler body 600 to secure coupler body 600 to retractor 12. As described above, coupler body 600 is secured to retractor 12 such that rotational movement of retractor 12 relative to coupler body 600 is permitted, while axial movement of retractor 12 relative to coupler body 600 is constrained, e.g., the force required to move retractor 12 relative to coupler body 600 is much higher than the force required to rotate retractor 12 relative to coupler body 600.

FIG. 7A illustrates coupler body 700 having opening 714 sized and shaped to receive a 10 mm diameter surgical instrument, e.g., laparoscope device 10. Coupler body 700 may be constructed similar to coupler body 600. For example, coupler body 700 may include first portion 708 coupled to second portion 710 via hinge portion 712, and recesses 720, 722 for securely receiving locking portion 718 of screw 716. As shown in FIG. 7B, coupler body 700 may receive elongated shaft 10a of laparoscope device 10 through opening 714, e.g., from the operating end of laparoscope 10, such that coupler body 700 may be slid over elongated shaft 10a until coupler body 700 engages with

proximal portion 10b of laparoscope 10, as shown in FIG. 7C. Preferably, coupler body 700 is coupled to laparoscope 10 when coupler body 700 contacts proximal portion 10b as this point along laparoscope 10 is fixed, thereby providing a consistent point of reference for calculating force measurements, as described in further detail below. Accordingly, when coupler body 700 is in the desired location along the elongated shaft of laparoscope 10, e.g., adjacent to proximal portion 10b, screw 716 may be coupled to coupler body 700 to secure coupler body 700 to laparoscope 10. As described above, coupler body 700 is secured to laparoscope 10 such that rotational movement of laparoscope 10 relative to coupler body 700 is permitted, while axial movement of laparoscope 10 relative to coupler body 700 is constrained, e.g., the force required to move laparoscope 10 relative to coupler body 700 is much higher than the force required to rotate laparoscope 10 relative to coupler body 700.

With the appropriate sized coupler body coupled to the selected surgical instrument, the coupler body may be removeably coupled to coupler interface 400 of robot arm 300. Coupler body 500 and coupler interface 400 may be configured for single-handed coupling, such that an operator may couple coupler body 500, and accordingly the surgical instrument coupled thereto, to coupler interface 400 of robot arm 300 using a single hand. Preferably, a surgical drape may be pinched or clamped between the coupler body and coupler interface 400, and draped over robot arm 300 to maintain sterility of the surgical space and prevent contact with non-sterile components of robot arm 300. Accordingly, the sterile drape may pass continuously (e.g., without a hole, a slit, or any other type of opening) between the coupler body and the coupler interface such that the coupler body is on a first side of the sterile drape and the coupler interface, robot arm 300, and/or other components of system 200 are on the other side of the sterile drape. In some embodiments, the coupler body may be integrated with the surgical drape. Additionally or alternatively, the surgical drape may include an adapter integrated therewith, such that coupler body 500 may be coupled to coupler interface 400 via the adapter, e.g., the adapter may be positioned between coupler body 500 and coupler interface 400.

Referring now to FIGS. 8A and 8B, robot arm 300 may be positioned in a surgical drape-ready configuration. As shown in FIG. 8A, robot arm 300 may be extended such that wrist portion 311, elbow link 310, and shoulder link 305 extend away from shoulder portion 304 of the base to permit a surgical/sterile drape to be draped over each component of robot arm 300. Moreover, as shown in FIG. 8B, when there are two robot arms, e.g., robot arm 300a and robot arm 300b, robot arm 300a and robot arm 300b may be angled away from each other, e.g., by rotating shoulder portion 304a relative to base portion 302a of robot arm 300a and by rotating shoulder portion 304b relative to base portion 302b of robot arm 300b, such that wrist portion 311a, elbow link 310a, and shoulder link 305a extend away from wrist portion 311b, elbow link 310b, and shoulder link 305b. This configuration permits efficient and accessible draping of the respective robot arms with a surgical/sterile drape. Moreover, in the extended position, the robot arms may be outside the virtual haptic boundary, such that the robot arms are in the haptic mode and a high level of impedance is applied to the robot arms thereby making movement of the robot arms more viscous, which makes it easier for the operator to drape the robot arms, yet provide movement thereto if necessary. For example, FIG. 9A illustrates a single robot arm 300

35

draped with sterile drape **800**, and FIG. 9B illustrates robot arms **300a**, **300b** draped with sterile drapes **800a**, **800b**, respectively.

Sterile drape **800** may be completely closed at an end portion thereof. In some embodiment, sterile drape **800** may have an opening (that can optionally have a sterile seal or interface) in a distal portion thereof that a portion of robot arm **300**, coupler interface **400**, coupler body **500**, and/or the surgical instrument may pass through. Drapes having a sealed end portion without any openings, and being sealed along a length thereof may provide a better sterile barrier for system **200**. Accordingly, all of robot arm **300** may be located inside sterile drape **800** and/or be fully enclosed within sterile drape **800**, except at an opening at a proximal end of sterile drape **800**, e.g., near the base of robot arm **300**. In some embodiments, coupler body **500** and coupler interface **400** may have electrical connectors to produce an electronic connection between robot arm **300** and the surgical instrument. Accordingly, the electrical signals may be transmitted through sterile drape **800**. Alternatively, sterile drape **800** may include an opening such that electrical wires or other components may pass through the opening to provide a wired communication channel to electrical components that may include, e.g., memory chips for calibration, radiofrequency probes for ablation, cameras, and other electronic components. The surgical instrument and the coupler body may instead be passive or non-electronic such that no electrical wires need pass through sterile drape **800**.

Referring now to FIGS. 10A to 10D, rotation of distal shoulder link **308** relative to proximal shoulder link **306** of shoulder link **305** is provided. As described above, axis Q3 may be a "setup" axis, such that distal shoulder link **308** may be rotated relative to proximal shoulder link **306** upon actuation of actuator **330** during a setup stage of robot arm **300**, e.g., prior to operation of robot arm **300** in a surgical procedure. As shown in FIG. 10A, shoulder portion **304** optionally may be initially rotated relative to base portion **302** to a desired position, thereby causing rotation of all the link distal to proximal shoulder link **306**, which is coupled to shoulder portion **304**, to rotate relative to base portion **302** and provide ample space for rotation of robot arm **300** about joint **320**. Moreover, as shown in FIG. 10A, wrist portion **311** may be at least partially extended away from base portion **302** so as to not collide with any components of robot arm **300** upon rotation of robot arm **300** about joint **320**. As shown in FIG. 10B, actuator **330** must be actuated to permit rotation of distal shoulder link **308** relative to proximal shoulder link **306** at joint **320**. As described above, actuator **330** may be constructed as described in U.S. patent application Ser. No. 18/331,060. FIG. 10C illustrates robot arm **300** in a desirable location for a specific laparoscopic procedure upon rotation of distal shoulder link **308** relative to proximal shoulder link **306**. FIG. 10D illustrates robot arm **300a** in the desirable location upon rotation of distal shoulder link **308a** relative to proximal shoulder link **306a**, relative to robot arm **300b**. In some embodiments, joint **320** may be operatively coupled to a motor, such that distal shoulder link **308** may be automatically rotated relative to proximal shoulder link **306**, as described in further detail with regard to FIG. 56.

Referring now to FIGS. 11A and 11B, an exemplary co-manipulation robot surgical system having an optical scanner is provided. As shown in FIG. 11A, the system may be constructed similar to system **200** of FIG. 2, having a plurality of robot arms, e.g., robot arm **300a** and robot arm **300b**. As described above, although only two robot arms are shown in FIG. 11A, less or more robot arms may be used in

36

conjunction with optical scanner **1100**. In addition, the system may include one or more optical scanners **1100**, e.g., a LiDAR scanner or other suitable optical scanning device such as an RGBD camera or sensor, RGB camera with machine learning, a time-of-flight depth camera, structured light, multiple projection cameras, a stereo camera, ultrasound sensors, laser scanner, other type of coordinate measuring area scanner, or any combination of the foregoing. For example, the LiDAR camera/scanner may be capable of recording both color (RGB) and the Depth (D) of the surgical field, and may include, for example, an Intel RealSense LiDAR Camera L515 or an Intel RealSense Depth Camera D435i (made available by Intel, Santa Clara, California) or other LiDAR or depth cameras having similar or suitable specifications including, without limitation, any of the following specifications: (i) range: 25 cm to 500 cm; depth accuracy: 5 mm or approximately 5 mm; depth field of view: 70×55 or approximately 70×55 (degrees); depth output resolution: 1024×768 pixels or approximately 1024×768 pixels; depth/RGB frame rate: 30 frames per second; RGB frame resolution: 1920×1080; and/or RGB field of view: 70×43 degrees or approximately 70×43 degrees. The LiDAR scanner or optical scanner further may include both a ¼-20 UNC thread or 2× M3 thread mounting points. As will be understood by a person having ordinary skill in the art, optical scanner **1100** may be used in other co-manipulation robot surgical systems described herein, e.g., system **200**, or any variations thereof.

As shown in FIG. 11A, the platform supporting robot arms **300a**, **300b** may support optical scanner **1100**, and any other electronics, wiring, or other components of the system, such that optical scanner **1100** is mounted in a fixed location relative to the other objects in the surgical space, and the position and orientation of optical scanner **1100** is known or may be determined with respect to the global coordinate system of the system, and accordingly, the robot arms. This allows all data streams to be transformed into a single coordinate system for development purposes. For example, optical scanner **1100** may be supported on a rod or shaft, e.g., rod **1102**, which may have an adjustable height or otherwise be adjustable in any direction, e.g., up/down, left/right, toward/away from the patient, to allow optical scanner **1100** to gain an optimum field-of-view or position relative to the other components of the system, for example, robot arms **300a**, **300b**, the surgical instruments attached thereto, the surgeon, and/or surgical assistant. Moreover, telemetry data captured by optical scanner **1100**, e.g., indicative of the movements of the surgeon's hands, other body parts, the patient bed, the trocar, the surgical instruments, and other components of the system, may be recorded to provide a rich and detailed dataset describing the precise movements and forces applied by the surgeon throughout the procedure.

For example, the data obtained may be used to optimize the procedures performed by the system including, e.g., automatic servoing (i.e., moving) of one or more portions of robot arm **300**. By tracking the tendency of the surgeon to keep the tools in a particular region of interest and/or the tendency of the surgeon to avoid moving the tools into a particular region of interest, the system may optimize the automatic servoing algorithm to provide more stability in the particular region of interest. In addition, the data obtained may be used to optimize the procedures performed by the system including, e.g., automatic re-centering of the field of view of the optical scanning devices of the system. For example, if the system detects that the surgeon has moved or predicts that the surgeon might move out of the field of view,

the system may cause the robot arm supporting the optical scanning device, e.g., a laparoscope, to automatically adjust the laparoscope to track the desired location of the image as the surgeon performs the desired procedure. This behavior may be surgeon-specific and may require an understanding of a particular surgeon's preference for an operating region of interest. Thus, the system may control the robot arms pursuant to specific operating requirements and/or preferences of a particular surgeon. Moreover, if the system detects that the robot arms are in an extended position for a period of time exceeding a predetermined threshold, the system may cause the stages coupled to the base portions of the robot arms to move the robot arms in a manner to ease extension of the robot arms, and thereby provide additional range for extension of the robot arms by the user.

Referring now to FIG. 11C, another exemplary co-manipulation robot surgical system having a plurality of optical sensors is provided. As shown in FIG. 11C, system 200 has a plurality of robot arms, e.g., robot arm 300a and robot arm 300b, supported by platform 100 having a plurality of wheels for providing mobility to platform 100. As described above, the plurality of wheels may each include a braking mechanism which may be actuated to be engaged and prevent movement of platform 100. For example, the braking mechanism may be operatively coupled to a controller of system 200. Moreover, system 200 may include a plurality of optical sensors, e.g., optical scanners 1100a, 1100b, and 1100c, disposed on platform 100. For example, optical scanner 1100a may be disposed on top of platform 100, as described above with regard to optical scanner 1100 of FIG. 11A, and optical scanners 1100b and 1100c may be disposed on the sides of platform 100. Additionally or alternatively, one or more optical scanners may be disposed underneath platform 100. Optical scanners 1100a, 1100b, and 1100c are configured to capture depth data. For example, optical scanners 1100a, 1100b, and 1100c may be, e.g., a depth camera, a stereo RGB camera, a LIDAR device, and/or an electromagnetic, capacitive, or infrared proximity sensor, etc.

The depth data generated by the plurality of optical sensors may be used by the controller of system 200 to generate a virtual map, e.g., a "bird's eye view", of the area surrounding platform 100, e.g., within the operating room, in real-time. For example, the virtual map may illustrate the operating room from a top perspective. Moreover, as shown in FIG. 11C, the virtual map may include graphical representations of platform 100 (including robot arms 300a, 300b), as well as one or more objects, e.g., patient table PT, and/or one or more persons, e.g., operator O, person P1, and person P2, within the area surrounding platform 100. Specifically, the virtual map may graphically illustrate the proximity between platform 100 and the one or more objects/persons, e.g., as platform 100 is being moved through the operating room by operator O. The controller may cause display 110 to display the virtual map, such that operator O can view the virtual map on display 110 in real-time as operator O moves platform 100 through the operating room. Accordingly, operator O may see objects and/or persons in the area surrounding platform 100 that operator O could not otherwise see with their own eyes, e.g., due to platform 100 and/or robot arms 300a, 300b obstructing the view of operator O, and avoid collisions between platform 100 and/or robot arms 300a, 300b with the objects/persons in the operating room. Moreover, the controller may cause display 110 to display an alert, e.g., a visual or audible alert, when the virtual map indicates that platform 100 and/or robot arms 300a, 300b are approaching or within a

predetermined distance from the one or more objects/persons within the operating room.

In some embodiments, the controller may only cause display 110 to display the virtual map while platform 100 is being moved within the operating room. For example, platform 100 may include one or more actuators, e.g., a button, lever, or handlebar, that may be operatively coupled to the braking mechanism of the wheels of platform 100, such that upon actuation of the actuator, the braking mechanism is disengaged such that mobility of platform 100 is permitted. Accordingly, when the actuator is not actuated, the braking mechanism is engaged such that mobility of platform 100 is prevented. Thus, upon actuation of the actuator, the controller may automatically cause display 110 to display the virtual map, such that operator O can view the area surrounding platform 100 before, during, or after movement of platform 100 while the braking mechanism is disengaged. Once the actuator is released, such that the braking mechanism is reengaged, display 110 may stop displaying the virtual map. In some embodiments, when the virtual map indicates that platform 100 and/or robot arms 300a, 300b are approaching or within the predetermined distance from the one or more objects/persons within the operating room, the controller may override actuation of the actuator by the operator and reengage the braking mechanism to thereby prevent further movement of platform 100. Accordingly, the actuator may need to be released and re-actuated by the operator to disengage the braking mechanism and permit further movement of platform 100.

Moreover, the system may process color and/or depth data obtained from optical scanners 1100a, 1100b, and/or 1100c to identify objects within the operating room, e.g., the patient bed or the trocar, as well as the planes associated with the identified objects. With knowledge of the location platform 100 and robot arms 300a, 300b relative to the identified objects, the system may cause the stages coupled to the base portions of robot arms 300a, 300b to automatically move (or stop movement of) robot arms 300a, 300b to avoid collision with the identified objects during setup, e.g., when robot arms 300a, 300b approaches a predetermined distance threshold relative to the identified objects. In addition, the system may generate and emit, e.g., an audible alert indicative of the proximity of the stages of platform 100 and/or robot arms 300a, 300b relative to the identified objects. For example, the audible alert may change in amplitude and/or frequency as the distance between the stages of platform 100 and/or robot arms 300a, 300b and the identified objects decreases, as perceived by the system based on the depth data.

Additionally, with knowledge of the location platform 100 and robot arms 300a, 300b relative to the patient and the trocar, in combination with knowledge of where robot arms 300a, 300b are positioned relative to the patient and the trocar for a given surgical procedure, the system may automatically position robot arms 300a, 300b in a setup configuration relative to the patient and the trocar for the given surgical procedure. For example, the system may automatically position the distal end of the robot arm adjacent to the trocar, and further may arrange the robot arm in a predetermined configuration, e.g., via the motorized joints of the robot arm, that is preferred for the given surgical procedure.

In addition, with knowledge of the location platform 100 and robot arms 300a, 300b relative to the trocar, if the system detects that the position of the patient bed, and accordingly the trocar, is changing, e.g., via adjustment by a user, the system may automatically adjust the arrangement

of the robot arm to accommodate the movement of the patient bed and maintain relative position between the distal end of the robot arm and the trocar. In some embodiments, upon detection of movement of the patient bed, the system may automatically move the robot arm to retract the surgical instrument coupled thereto within the trocar, prior to automatically adjusting the arrangement of the robot arm to maintain relative position between the distal end of the robot arm and the trocar, such that the distal end of the surgical instrument is positioned within the trocar and away from anatomical structures within the patient.

FIG. 12 shows the system having optical scanner 1100 in operation during a laparoscopic procedure. As shown in FIG. 12, an optional additional optical scanner, e.g., camera 1200, may be utilized to provide an additional point of view, e.g., redundant measurement of the movements of the instruments held by the robot arms, and/or provide a video stream of the surgical scene, e.g., via streaming, for monitoring and analysis. As shown in FIG. 12, the system may include two robot arms, e.g., robot arms 300a, 300b, such that robot arm 300a holds laparoscope 10 in a fixed position relative to the patient, while the surgeon operates and manipulates retractor 12, which is coupled to the distal end of robot arm 300b. Moreover, during the surgical procedure, robot arms 300a, 300b may be draped with sterile drapes 800a, 800b, respectively. As described above, the surgeon may freely manipulate retractor 12 while retractor 12 is coupled to robot arm 300b, thereby causing movement of robot arm 300b due to movement of retractor 12 by the surgeon, and while robot arm 300b accounts for weight of retractor 12 and robot arm 300b. During the surgical procedure, optical scanner 1100 may be used to monitor an identity, position, orientation, and/or movement of the surgical instrument coupled to robot arm 300a, e.g., laparoscope 10, and an identity, position, orientation, and/or movement of the surgical instrument coupled to robot arm 300b, e.g., retractor 12, as well as if either surgical instrument is detached from the respective robot arm, either intentionally or unintentionally. Moreover, optical scanner 1100 may be used to monitor an identity, position, orientation, and/or movement/displacement of any of trocars Tr to ensure proper alignment of the robot arms and/or surgical instruments relative to the respective trocars. The system may be used in a surgical procedure having one, two, three, four, or more trocars, depending on the surgical procedure intended to be performed by the system.

FIGS. 13A and 13B illustrate exemplary data produced by optical scanner 1100. For example, FIG. 13A illustrates image data captured by optical scanner 1100, and FIG. 13B illustrates a depth map of at least some objects within the surgical space generated from the data captured by optical scanner 1100. Specifically, optical scanner 1100 may create a depth map, e.g., point clouds, where each pixel's value is related to the distance from optical scanner 1100. For example, the difference between pixels for a first object (such as a first surgical instrument) and a second object (for example, a trocar) will enable the system to calculate the distance between the surgical instrument and the trocar. Moreover, the difference between pixels for a first object (such as a first surgical instrument) at a first point in time and the first object at a second point in time will enable the system to calculate whether the first object has moved, the trajectory of movement, the speed of movement, and/or other parameters associated with the changing position of the first object.

As shown in FIGS. 13A and 13B, surgeon S is manipulating surgical tools and/or the draped robot arm (DA) and the undraped robot arm (UA) that are positioned relative to

insufflated abdomen (A). As described above, the data streams from the robot arms, the camera feed from the laparoscope, the data acquired from optical scanner 1100, as well as data optionally captured from one or more imaging devices disposed on a structure adjacent to the robot arms, the walls, ceiling, or other structures within the operating room, may be recorded, stored, and used individually or in combination to understand and control the surgical system and procedures of the surgical system. The foregoing components, devices, and combinations thereof are collectively referred to herein as optical scanners or optical scanning devices.

For example, the system may measure and record any of the following within the coordinate space of the system: motion of the handheld surgical instruments manipulated by the surgeon (attached to or apart from a robot arm); the presence/absence of other surgical staff (e.g., scrub nurse, circulating nurse, anesthesiologist, etc.); the height and angular orientation of the surgical table; patient position and volume on the surgical table; presence/absence of the drape on the patient; presence/absence of trocar ports, and if present, their position and orientation; gestures made by the surgical staff; tasks being performed by the surgical staff; interaction of the surgical staff with the system; surgical instrument identification; attachment or detachment "action" of surgical instruments to the system; position and orientation tracking of specific features of the surgical instruments relative to the system (e.g., camera head, coupler, fiducial marker(s), etc.); measurement of motion profiles or specific features in the scene that allow for the phase of the surgery to be identified; position, orientation, identity, and/or movement of any other instruments, features, and/or components of the system or being used by the surgical team.

The system may combine measurements and/or other data described above with any other telemetry data from the system and/or video data from the laparoscope to provide a comprehensive dataset with which to improve the overall usability, functionality, and safety of the co-manipulation robot-assisted surgical systems described herein. For example, as the system is being setup to start a procedure, optical scanner 1100 may detect the height and orientation of the surgical table. This information may allow the system to automatically configure the degrees of freedom of platform 100 supporting robot arms 300 to the desired or correct positions relative to the surgical table. Specifically, optical scanner 1100 may be used to ensure that the height of platform 100 is optimally positioned to ensure that robot arms 300 overlap with the intended surgical workspace. In addition, as described above, the system may automatically reconfigure the degrees of freedom of platform 100 as well as the arrangement of robot arms 300 responsive to movement of the surgical table, and accordingly the trocar(s), to maintain relative position between the distal end of the robot arms and the trocar(s).

Moreover, based on the data obtained by optical scanner 1100, the system may alert the surgical staff of a potential collision (either during setup or intra-operatively) between the system and other pieces of capital equipment in the operating room, e.g., the surgical table, a laparoscopic tower, camera booms, etc., as well as with a member of the surgical staff, e.g., an inadvertent bump by the staff member. The system may use this information to recommend a repositioning of platform 100 and/or other components of the system, the surgical table, and/or patient, and/or prevent the robot arm from switching to the co-manipulation mode as a result of the force applied to the robot arm by the collision

with the staff member, even if the force exceeds the predetermined force threshold of the robot arm.

In addition, the data obtained from optical scanner **1100** may be used to monitor the progress of setup for a surgical procedure and may be combined with the known state of the system to inform remote hospital staff (e.g., the surgeon) of the overall readiness to start the procedure. Such progress steps may include: (i) patient on table; (ii) patient draped; (iii) sterile instruments available; (iv) robot arm draped; (v) trocar ports inserted; and (vi) confirmation that instruments (e.g., a laparoscope and retractor) are attached to the robotic arms of system. For example, the data obtained from optical scanner **1100** may include detected gestures indicative of the system state (e.g., system is draped), readiness to start the procedure, etc., and further may be used to prepare the system for the attachment or detachment of a surgical instrument.

In addition, optical scanner **1100** may identify the specific surgeon carrying out the procedure, such that the system may use the surgeon's identity to load a system profile associated with the particular surgeon into the system. The system profile may include information related to a surgeon's operating parameter and/or preferences, a surgeon's patient list having parameters for each patient, the desired or required algorithm sensitivity for the surgeon, the degree of freedom positioning of the support platform, etc. Examples of algorithm sensitivities that may be surgeon-specific include: adapting/adjusting the force required to transition from passive mode to co-manipulation mode (e.g., from low force to high force), adapting/adjusting the viscosity felt by the surgeon when co-manipulating the robot arm (e.g., from low viscosity to high viscosity), etc. Moreover, the surgeon's preferences may include preferred arrangements of robot arm **300**, e.g., the positioning of the links and joints of robot arm **300** relative to the patient, with regard to specific surgical instruments, e.g., the preferred arrangement may be different between a laparoscope and a retractor.

In some embodiments, the surgeon's preferences may be learned based on data from past procedures and/or sensors collecting information about current procedure including a surgeon's current pose, a surgeon's height, a surgeon's hand preference, and other similar factors. For example, the system may record when a user interacts with the system and also record what the user does with the system, such that the dataset may allow for surgeon preferences to be "learned" and updated over time. This learning may be done either via traditional algorithmic methods (i.e., trends over time, averaging, optical flow, etc.) or via machine learning approaches (classification, discrimination, neural networks, reinforcement learning, etc.). FIG. **24** illustrates data flow **2400** for updating the system configurations based on learned behaviors of the user. As shown in FIG. **24**, the system may be connected to an online database that may store a surgeon profile and each of a plurality of possible data sources, which may include optical sensors, encoders, and/or other sensors, and/or a database of manually entered user input. The data sources may be associated with a given surgeon, their preferred robot arm arrangement and operating parameters, and each procedure performed with the system, which may allow the recording and analysis of the system configuration and how it changes from procedure to procedure, and within the procedure. In the case of machine learning, the co-manipulation capability of the system may be leveraged such that the user's actions may be used to annotate the data to create a training dataset.

Regarding the degree of freedom positioning, a height of a surgical table is typically adjusted to accommodate the

height of the surgeon in some operating rooms. Thus, by detecting the surgeon and loading the surgeon's specific profile, the system may position the platform at a height that is suitable for the respective surgeon to accommodate the preferred height of the surgical table. In addition, the horizontal translation of a robot arm may depend on the size of the patient. Thus, by accessing the patient list, the system may adjust the position of the arm based on the patient's body mass index ("BMI"). For example, for a patient with a high BMI, the system may move the robot arm away from the operating table and, for a patient with a low BMI, the system may move the robot arm closer to the operating table. Accordingly, the system permits the surgical team to fine-tune the position of the robot arm relative to the patient as necessary. The system further may be configured to access a hospital medical record database to access the procedure type and any other medical data available (e.g., CT scan images, x-ray images, MRI images, and/or other patient specific information), which may be used to inform positioning of the trocar ports, and the position and orientation of platform **100** relative to the patient.

Based on the data captured by optical scanner **1100**, the system may generate a virtual model of the pieces of capital equipment and/or other objects in an operating room that are within a range of movement of the robot arms in the same co-ordinate space as the robot arms and surgical instruments coupled thereto, such that the virtual model may be stored and monitor, e.g., to detect potential collisions. Additionally, the system may track the position and orientation of each virtual model, and the objects within the virtual models as the objects move relative to each other, such that the system may alert the user if the proximity of (i.e., spacing between) any of the virtual models or objects falls below a predefined threshold, e.g., within 50 mm, 75 mm, from 30 mm or less to 100 mm, or more. In some embodiments, the distance threshold may be based off the Euclidean distance between the closest points on two virtual models, the normal distance between two surfaces of the virtual models, etc. Moreover, the system may stop or inhibit (e.g., prevent) further movement of a robot arm, e.g., freeze the robot arm, if the proximity of any of the virtual models or objects, e.g., a robot arm reaches or falls below the predefined threshold relative to a laparoscopic tower, or the surface of the surgical table, or other objects within the surgical space. In addition, the system may freeze the robot arm if the system detects that the proximity between an object, e.g., capital equipment or a member of the surgical staff other than the surgeon, moving toward a respective robot arm reaches or falls below the predefined threshold, to thereby prevent the inadvertent movement of the robot arm that may otherwise result from such a collision or inadvertent force, e.g., an inadvertent bump from a member of the staff or another piece of capital equipment, etc.

In addition, based on the data captured by optical scanners **1100a**, **1100b**, **1100c**, the system may generate a virtual map with graphical representations of objects and/or persons that are within a predefined area surrounding the platform and robot arms in an operating room in the same co-ordinate space as the platform and robot arms, such that the virtual map may be stored and displayed to a user, e.g., to detect potential collisions while the user moves the platform throughout the operating room. Additionally, the system may track the position and orientation of the graphical representations within the virtual map, such that the system may alert the user if the proximity between any of the objects and/or persons from the platform and/or robot arms

falls within a predetermined threshold, e.g., within 50 mm, 75 mm, from 30 mm or less to 100 mm, or more.

Moreover, based on the data captured by optical scanner **1100**, the system may track the motion of the handheld surgical instruments that are directly and independently controlled by the surgeon, that are not coupled with the robot arm. For example, the optical scanner **1100** may track a clearly defined feature of the instrument, a fiducial marker attached to the instrument or to the gloves (e.g., the sterile gloves) of the surgeon, the coupler between the robot arm and the instrument, a distal tip of the instrument, and/or any other defined location on the instrument. For example, fiducial markers may include Manus virtual reality gloves (made available by Manus, The Netherlands) or other wearables, and/or the OptiTrack systems (made available by NaturalPoint, Corvallis, Oregon). The following are examples of uses and purposes of the motion data: (i) closing a control loop between a handheld instrument and the robot arm holding the camera, thus allowing the surgeon to servo (i.e., move) the camera by “pointing” with a handheld instrument; (ii) tracking information that may be used independently or in combination with other data streams to identify the phase of the surgical procedure; (iii) to identify the dominant hand of the surgeon; (iv) to monitor metrics associated with the experience of the surgeon; (v) to identify which tools the surgeon is using and when to change them for other tools; and/or (vi) tracking of the skin surface of the patient, as well as the number, position and orientation of the trocar ports. This data and information also may be used and computed by the system as part of the co-manipulation control paradigm. By measuring the true position and orientation of the trocar ports, the system may be provided an additional safety check to ensure that the system level computations are correct, e.g., to ensure that the actual motion of the robot arms or instrument matches a commanded motion of the robot arms or instrument in robotic assist mode.

Based on the data captured by optical scanner **1100**, the system further may track the which instrument is being used in a respective port, how often instruments are swapped between ports, which ports have manually held instruments versus instruments coupled to the robot arm, to monitor and determine if additional trocar ports are added, if the system is holding the instruments in place while the patient or surgical table is moving (in which case, the system may change the operational mode of the robot arms to a passive mode and accommodate the movement by repositioning robot arm **300** and/or platform **100**), and/or other conditions or parameters of the operating room or the system. The knowledge of the position and orientation of the skin surface and trocar ports relative to the robot arms may facilitate the implementation of “virtual boundaries” as described in further detail below.

Moreover, based on the data obtained by optical scanner **1100**, e.g., tracked movements of the distal end of a laparoscope coupled to robot arm **300**, in addition to image data captured by the laparoscope, the system may identify the type of laparoscope coupled to robot arm **300**. For example, laparoscopes commonly used during laparoscopic procedures include flat-tipped laparoscopes and angled-tipped laparoscopes, e.g., a laparoscope having a 30 degree angled tip. The system may determine which laparoscope type is currently coupled to robot arm **300** by comparing the image data obtained by optical scanner **1100** of a predefined pattern of movement of the laparoscope, e.g., moving the distal end of the laparoscope in a circular pattern in a plane perpendicular to the longitudinal axis of the laparoscope, with the

image data obtained by the laparoscope as the laparoscope is being moved in the predefined pattern of movement. For example, for a flat-tipped laparoscope, the image data captured by the laparoscope as the distal end of the laparoscope is moved in a circular pattern in the plane perpendicular to the longitudinal axis of the laparoscope should move along a circular planar path, e.g., there will be no change in depth of the field of view of the laparoscope; whereas, for an angled-tipped laparoscope, the image data captured by the laparoscope as the distal end of the laparoscope is moved in a circular pattern in the plane perpendicular to the longitudinal axis of the laparoscope will observe a change of depth of the field of view of the laparoscope.

In addition, the system may calibrate a new robot arm when a current robot arm is replaced, e.g., during a surgical procedure, based on the data obtained by optical scanner **1100**, with or without utilizing a tracker at the distal end of the new robot arm, to ensure the system is accurately aware of the kinematics of the new robot arm. Specifically, the system may calibrate optical scanner **1100** to platform **100**, calibrate the new robot arm with respect to the base portion of the new robot arm, and calibrate the new robot arm with respect to platform **100** when the new robot arm is coupled to platform **100**. For example, based on the telemetry data obtained by optical scanner **1100**, the system may compare the actual real-time movements of the new robot arm as captured by optical scanner **1100** to the movements expected based on commands sent to the new robot arm by the system, e.g., to execute a preprogrammed routine intended to move the new robot arm in specific positions, and generate a degree of error indicative of a deviation between the actual real-time movements of the new robot arm and the expected movements of the robot arm based on the preprogrammed routine. The system further may execute an optimization algorithm to reduce or eliminate the degree of error between the actual real-time movements and the expected movements, e.g., until the degree of error falls below a predetermined threshold. This calibration process may occur when the system is in a predefined calibration mode, or alternatively, in real-time during a surgical procedure after the new robot arm is coupled to platform **100**.

Based on the data obtained by optical scanner **1100**, e.g., knowledge of the position and orientation of the surgical bed relative to platform **100** and/or the trocar port(s), the system may automatically position robot arms **300** in a preferred configuration relative to the patient during setup via the motorized joints of robot arm **300** and/or the stages of platform **100** based on the surgical procedure to be performed, e.g., responsive to actuation by a user via GUI **110**, while avoiding collisions between the stages of platform **300**, robot arm **300**, and objects in the operating room such as the surgical bed. Accordingly, the system may store preset robot arm configurations, e.g., based on stored surgeon preferences, for various surgical procedures.

Referring now to FIG. **14**, components that may be included in co-manipulation robot platform **1400** are described. Platform **1400** may include one or more processors **1402**, communication circuitry **1404**, power supply **1406**, user interface **1408**, and/or memory **1410**. One or more electrical components and/or circuits may perform some of or all the roles of the various components described herein. Although described separately, it is to be appreciated that electrical components need not be separate structural elements. For example, platform **1400** and communication circuitry **1404** may be embodied in a single chip. In addition, while platform **1400** is described as having memory **1410**, a memory chip(s) may be separately provided.

Platform **1400** may contain memory and/or be coupled, via one or more buses, to read information from, or write information to, memory. Memory **1410** may include processor cache, including a multi-level hierarchical cache in which different levels have different capacities and access speeds. The memory also may include random access memory (RAM), other volatile storage devices, or non-volatile storage devices. Memory **1410** may be RAM, ROM, Flash, other volatile storage devices or non-volatile storage devices, or other known memory, or some combination thereof, and preferably includes storage in which data may be selectively saved. For example, the storage devices can include, for example, hard drives, optical discs, flash memory, and Zip drives. Programmable instructions may be stored on memory **1410** to execute algorithms for, e.g., calculating desired forces to be applied along robot arm **300** and/or the surgical instrument coupled thereto and applying impedances at respective joints of robot arm **300** to effect the desired forces.

Platform **1400** may incorporate processor **1402**, which may consist of one or more processors and may be a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any suitable combination thereof designed to perform the functions described herein. Platform **1400** also may be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

Platform **1400**, in conjunction with firmware/software stored in the memory may execute an operating system (e.g., operating system **1446**), such as, for example, Windows, Mac OS, QNX, Unix or Solaris 5.10. Platform **1400** also executes software applications stored in the memory. For example, the software may be programs in any suitable programming language known to those skilled in the art, including, for example, C++, PHP, or Java.

Communication circuitry **1404** may include circuitry that allows platform **1400** to communicate with an image capture devices such as optical scanner and/or endoscope. Communication circuitry **1404** may be configured for wired and/or wireless communication over a network such as the Internet, a telephone network, a Bluetooth network, and/or a WiFi network using techniques known in the art. Communication circuitry **1404** may be a communication chip known in the art such as a Bluetooth chip and/or a WiFi chip. Communication circuitry **1404** permits platform **1400** to transfer information, such as force measurements on the body wall at the trocar insertion point locally and/or to a remote location such as a server.

Power supply **1406** may supply alternating current or direct current. In direct current embodiments, power supply may include a suitable battery such as a replaceable battery or rechargeable battery and apparatus may include circuitry for charging the rechargeable battery, and a detachable power cord. Power supply **1406** may be a port to allow platform **1400** to be plugged into a conventional wall socket, e.g., via a cord with an AC to DC power converter and/or a USB port, for powering components within platform **1400**. Power supply **1406** may be operatively coupled to an emergency switch, such that upon actuation of the emergency switch, power stops being supplied to the components within platform **1400** including, for example, the braking mechanism disposed on at least some joints of the plurality

of joints of robot arm **300**. For example, the braking mechanisms may require power to disengage, such that without power supplied to the braking mechanisms, the braking mechanisms engage to prevent movement of robot arm **300** without power.

User interface **1408** may be used to receive inputs from, and/or provide outputs to, a user. For example, user interface **1408** may include a touchscreen, display, switches, dials, lights, etc. Accordingly, user interface **1408** may display information such as selected surgical instrument identity and force measurements observed during operation of robot arm **300**. Moreover, user interface **1408** may receive user input including adjustments to the predetermined amount of movement at the handle of the surgical instrument or the predetermined dwell time period to cause the robot arm to automatically switch to the passive mode, the predetermined threshold of force applied at the handle of the surgical instrument to cause the robot arm to automatically switch to the co-manipulation mode, a position of the predefined haptic barrier, an identity of the surgical instrument coupled to the distal end of the robot arm, a vertical height of the robot arm, a horizontal position of the robot arm, etc., such that platform **1400** may adjust the information/parameters accordingly. In some embodiments, user interface **1408** is not present on platform **1400**, but is instead provided on a remote, external computing device communicatively connected to platform **1400** via communication circuitry **1404**.

Memory **1410**, which is one example of a non-transitory computer-readable medium, may be used to store operating system (OS) **1446**, surgical instrument identification module **1412**, surgical instrument calibration module **1414**, encoder interface module **1416**, robot arm position determination module **1418**, trocar position detection module **1420**, force detection module **1422**, impedance calculation module **1424**, motor interface module **1426**, optical scanner interface module **1428**, gesture detection module **1430**, passive mode determination module **1432**, co-manipulation mode determination module **1434**, haptic mode determination module **1436**, robotic assist mode determination module **1438**, fault detection module **1440**, indicator interface module **1442**, and fatigue detection module **1444**. The modules are provided in the form of computer-executable instructions/algorithms that may be executed by processor **1402** for performing various operations in accordance with the disclosure.

For example, during a procedure, the system may continuously run the algorithms described herein based on the data collected by the system. That data may be collected and/or recorded using any of the components and methods disclosed herein, including, e.g., from sensors/encoders within the robots, from optical scanning devices in communication with the other components of the robotic system, and/or from manual inputs by an operator of the system. Accordingly, the algorithms, the data, and the configuration of the system may enable the user to co-manipulate the robot arms with minimal impact and influence from the weight of the robot arms and/or surgical instruments coupled thereto, force of gravity, and other forces that traditional robot arms fail to compensate for. Some of the parameters of the algorithms described herein may control an aspect of the behavior of the system including, e.g., robustness of detected features, sensitivity to false positives, robot control gains, number of features to track, dead zone radius, etc.

Surgical instrument identification module **1412** may be executed by processor **1402** for identifying the surgical instrument coupled to each of the robot arms, and loading the appropriate calibration file into the controller system.

For example, the calibration file for each surgical instrument may be stored in a database accessible by surgical instrument identification module **1412**, and may include information associated with the surgical instrument such as, e.g., instrument type, weight, center of mass, length, instrument shaft diameter, etc. Accordingly, when the appropriate calibration file is loaded, and the associated surgical instrument is coupled to robot arm **300**, the system will automatically account for the mass of the surgical instrument, e.g., compensate for gravity on the surgical instrument, when the surgical instrument is attached to robot arm **300** based on the data in the calibration file, such that robot arm **300** may hold the surgical instrument in position after the surgical instrument is coupled to the robot arm and the operator lets go of the surgical instrument. For example, surgical instrument identification module **1412** may identify the surgical instrument based on user input via user interface **1408**, e.g., the operator may select the surgical instrument from a database of surgical instruments stored in memory **1410**.

In some embodiments, surgical instrument identification module **1412** may automatically identify the surgical instrument coupled with the robotic arm via the coupler body and the coupler interface using, e.g., an RFID transmitter chip and reader or receiver (e.g., placing an RFID sticker or transmitter on the surgical instrument that may transmit information about the surgical instrument to a receiver of the system), an near field communication ("NFC") device such as a near field magnetic induction communication device, a barcode and scanner or other optical device, a magnet based communication system, reed switches, a Bluetooth transmitter, the weight of the instrument and/or data gathered from the optical scanner and a lookup table, and/or any other features or mechanisms described herein or suitable for identification of the surgical instrument. As described above, the coupler body may be selected based on the size and shape of the lumen extending therethrough to accommodate and engage with a surgical instrument having a known elongated shaft diameter. Accordingly, surgical instrument identification module **1412** may automatically identify the surgical instrument based on the coupler body that is coupled to the surgical instrument via the magnetic connection between the coupler body and the coupler interface.

In some embodiments, e.g., when using coupling mechanism **4400** described below with regard to FIGS. **44A-50B**, surgical instrument identification module **1412** may identify the surgical instrument, e.g., the type of surgical instrument, based on data obtained from a sensor, e.g., a Hall effect sensor within distal wrist link **316**, indicative of a magnetic field strength induced by a magnet displaced responsive to coupling of the surgical instrument to the coupler body coupled to the coupler interface. Based on the strength of the detected magnetic field, the system may determine whether the coupler body is coupled to the coupler interface, whether a surgical instrument is coupled to the coupler body, and the diameter of the shaft of the surgical instrument, e.g., a 5 mm or 10 mm surgical instrument, as described in further detail below.

Surgical instrument identification module **1412** further may cause the distal end of robot arm **300**, when coupled to a laparoscope, to move in a predefined pattern, e.g., in a circular motion in a plane perpendicular to the longitudinal axis of the laparoscope, and compare the image data captured by the laparoscope as the distal end of the laparoscope is moved in the predefined circular pattern with image data obtained by optical scanner **1100** of the laparoscope as the laparoscope is being moved in the predefined pattern to

identify whether the laparoscope is a flat-tipped laparoscope or an angled-tip laparoscope, as described above.

In some embodiments, surgical instrument identification module **1412** may identify the surgical instrument, e.g., the type of surgical instrument, based on data obtained by optical scanner **1100** via optical scanner interface module **1428** described in further detail below. For example, the data may include measurement data associated with the specific instrument, such that surgical instrument identification module **1412** may compare such data with information contained within the database to identify the instrument and load the appropriate calibration file into the controller system. Similarly, surgical instrument identification module **1412** may detect if the instrument is removed and return the calibration parameters to a default configuration.

Surgical instrument calibration module **1414** may be executed by processor **1402** for calibration a surgical instrument, e.g., a surgical instrument that does not currently have an associated calibration file in the database stored in memory **1410**. Accordingly, surgical instrument calibration module **1414** may calculate measurements and specifications of a surgical instrument when it is coupled to robot arm **300** and the system is in calibration mode, as described in further detail below with regard to FIG. **16**, based on force measurements of robot arm **300** applied by the surgical instrument via force detection module **1422**. For example, surgical instrument calibration module **1414** may generate a calibration file for the surgical instrument including information such as instrument type, weight, center of mass, length, instrument shaft diameter, a viscosity parameter of the surgical instrument, etc. At least some of the surgical instrument information in the calibration file may be provided by user input via user interface **1408**, e.g., the instrument type, or may be detected by optical scanner interface module **1428**, e.g., the instrument type, the center of mass of the instrument, the instrument length, and the instrument diameter.

If surgical instrument calibration module **1414** determines that re-calibration results are consistently different from the configurations already loaded into the system, surgical instrument calibration module **1414** may replace existing information or add to its list of known tools without any user inputs and load them automatically. Surgical instrument calibration module **1414** may determine that the calibration factors are not adequate to compensate for the force of gravity if, e.g., when a surgical instrument is coupled with the robot arm, the robot arm moves due only to forces of gravity acting on the robot arm and/or the surgical instrument, which may be done when the surgical instrument is positioned completely outside of the patient's body. Moreover, surgical instrument calibration module **1414** may automatically update or adjust the calibration factors (e.g., the forces applied to the joints of the robot arm) if it determines that the calibration factors are not adequate to compensate for the force of gravity. Thus, surgical instrument calibration module **1414** may update the calibration factors for the particular surgical instrument and store the updated calibration factors for the particular surgical instrument in the associated calibration file for future use.

Encoder interface module **1416** may be executed by processor **1402** for receiving and processing angulation measurement data from the plurality of encoders of robot arm **300**, e.g., encoders **E1-E7**, in real time. For example, encoder interface module **1416** may calculate the change in angulation over time of the links of robot arm **300** rotatably coupled to a given joint associated with the encoder. As described above, the system may include redundant encod-

ers at each joint of robot arm 300, to thereby ensure safe operation of robot arm 300. Moreover, additional encoders may be disposed on platform 100 to measure angulation/position of each robot arm relative to platform 100, e.g., the vertical and horizontal position of the robot arms relative to platform 100. Accordingly, an encoder may be disposed on platform 100 to measure movement of the robot arms along the vertical axis of platform 100 and another encoder may be disposed on platform 100 to measure movement of the robot arms along the horizontal axis of platform 100.

Robot arm position determination module 1418 may be executed by processor 1402 for determining the position of robot arm 300 and the surgical instrument attached thereto, if any, in 3D space in real time based on the angulation measurement data generated by encoder interface module 1416. For example, robot arm position determination module 1418 may determine the position of various links and joints of robot arm 300 as well as positions along the surgical instrument coupled to robot arm 300. Based on the position data of robot arm 300 and/or the surgical instrument, robot arm position determination module 1418 may calculate the velocity and/or acceleration of movement of robot arm 300 and the surgical instrument attached thereto in real time. For example, by determining the individual velocities of various joints of robot arm 300, e.g., via the encoder associated with each joint of the various joints, robot arm position determination module 1418 may determine the resultant velocity of the distal end of robot arm 300, which may be used by passive mode determination module 1432 to determine whether movement of the distal end of robot arm 300 is within a predetermined threshold for purposes of transitioning system 200 to passive mode, as described in further detail below.

Trocar position detection module 1420 may be executed by processor 1402 for determining the position and/or orientation of one or more trocar port inserted within the patient. The position and/or orientation of a trocar port may be derived based on data obtained from, e.g., inertial measurement units and/or accelerometers, optical scanners, electromechanical tracking instruments, linear encoders, the sensors and data as described above. For example, the position of the trocar ports on the patient may be determined using a laser pointing system that may be mounted on one or more of the components of the system, e.g., wrist portion 311 of the robot arm, and may be controlled by the system to point to the optimal or determined position on the patient's body to insert the trocar. Moreover, upon insertion of the surgical instrument that is attached to robot arm 300 through a trocar, virtual lines may continuously be established along the longitudinal axis of the surgical instrument, the alignment/orientation of which may be automatically determined upon attachment of the surgical instrument to coupler interface 400 via the coupler body via the magnetic connection as described above, in real time as the surgical instrument moves about the trocar point. Moreover, when the surgical instrument is inserted within the trocar port, it will be pointing toward the trocar point, and accordingly, distal wrist link 316 will also point toward the trocar point, the angle of which may be measured by an encoder associated therewith. Accordingly, the trocar point may be calculated as the intersection of the plurality of virtual lines continuously established along the longitudinal axis of the surgical instrument. In this manner, the calculated trocar point will remain fixed relative to the patient as the surgical instrument is maneuvered about the trocar port, e.g., rotated or moved in or out of the patient. In addition, the orientation of the trocar port and its position relative to robot

arm 300 may be determined based on image data received from one or more optical scanners, e.g., a LiDAR camera and/or an RGBD camera.

Based on the known position and/or orientation of a trocar port in addition to the known position of the distal end of robot arm 300 from robot arm position determination module 1418, the system may maintain the position of the distal end of robot arm 300 relative to the trocar point as robot arm 300 moves, e.g., via vertical or horizontal adjustment thereof by platform 100, or as the patient table height is adjusted, thereby causing the height of the patient's abdomen to move, thereby keeping the surgical instrument within the patient's body and coupled to robot arm 300 steady during these external movements. To achieve this, the known position of the distal end of robot arm 300 from robot arm position determination module 1418 is calculated in the global frame of the system by adding position of platform 100 to the kinematics calculations (e.g., the "forward kinematics" of robot arm 300 in the context of serial chain robotic manipulators).

With the position of the distal end of robot arm 300 known globally, the system may hold that position steady by applying appropriate forces to robot arm 300 during the external movements that minimize the error between its current and desired positions. Accordingly, for example, when a surgical instrument coupled to the distal end of robot arm 300 is inserted through a trocar port such that the tip of the instrument is inside of the patient, and a user adjusts the height of the patient table, the system may apply forces/torques to robot arm 300 to reconfigure robot arm 300 and/or cause movement of the stages of platform 100 to maintain the relative position between the distal end of robot arm 300, and accordingly the surgical instrument, and the trocar port. In some embodiments, the system may cause the distal end of robot arm 300 to retract slightly such that the tip of the surgical instrument is positioned within the trocar port and out of contact with anatomical structures within the patient's body prior to reconfiguring robot arm 300 to maintain the relative position between the surgical instrument and the trocar port.

Force detection module 1422 may be executed by processor 1402 for detecting forces applied on robot arm 300, e.g., at the joints or links of robot arm 300 or along the surgical instrument, as well as applied on the trocar, e.g., body wall forces. For example, force detection module 1422 may receive motor current measurements in real time at each motor, e.g., M1, M2, M3, disposed within the base of robot arm 300, which are each operatively coupled to a joint of robot arm 300, e.g., base joint 303, shoulder joint 318, elbow joint 322, wrist joint 332. The motor current measurements are indicative of the amount of force applied to the associated joint. Accordingly, the force applied to each joint of robot arm 300 as well as to the surgical instrument attached thereto may be calculated based on the motor current measurements and the position data generated by robot arm position determination module 1418 and/or trocar position detection module 1420.

Due to the passive axes at the distal end of robot arm 300, the force applied by the instrument coupled with the robot arm on the trocar may remain generally consistent throughout the workspace of the robot arm. The force on the trocar may be affected by the interaction of the distal tip of the instrument with tissue within the body. For example, if a tissue retractor advanced through the trocar is engaged with (e.g., grasping) bodily tissue or another object inside the body, the force exerted on the end of the instrument from the bodily tissue or other object may cause a change in the force

51

applied to the trocar. In some aspects, the force on the trocar may be a function of how much weight is being lifted by the instrument being used.

Impedance calculation module **1424** may be executed by processor **1402** for determining the amount of impedance/torque needed to be applied to respective joints of robot arm **300** to achieve the desired effect, e.g., holding robot arm **300** in a static position in the passive mode, permitting robot arm **300** to move freely while compensating for gravity of robot arm and the surgical instrument attached thereto in the co-manipulation mode, applying increased impedance to robot arm **300** when robot arm **300** and/or the surgical instrument attached thereto is within a predefined virtual haptic barrier in the haptic mode, etc.

For example, impedance calculation module **1424** may determine the amount of force required by robot arm **300** to achieve the desired effect based on position data of robot arm **300** generated by robot arm position determination module **1418** and the position data of the trocar generated by trocar position detection module **1420**. For example, by determining the position of the distal end of robot arm **300**, as well as the point of entry of the surgical instrument into the patient, e.g., the trocar position, and with knowledge of one or more instrument parameters, e.g., mass and center of mass of the surgical instrument stored by surgical instrument calibration module **1414**, impedance calculation module **1424** may calculate the amount of force required to compensate for gravity of the surgical instrument (compensation force), as described in further detail below with regard to FIG. **18A**. Accordingly, the amount of compensation force required to compensate for the gravity of the surgical instrument may be converted to torque to be applied at the joints of robot arm **300**, e.g., by the motors operatively coupled to the joints of robot arm **300**, as indicated by the motor current measurements.

Moreover, by determining the position of the distal end of robot arm **300**, and accordingly, a change in position of the distal end of robot arm **300** over time, for example, due to an external force applied to the distal end of robot arm **300**, e.g., by tissue held by the operating end of the surgical instrument, and with knowledge of one or more instrument parameters, e.g., mass, center of mass, and length of the surgical instrument stored by surgical instrument calibration module **1414**, impedance calculation module **1424** may calculate the amount of force required to maintain the surgical instrument in a static position (hold force), as described in further detail below with regard to FIG. **18B**. Accordingly, the amount of hold force required to resist the change in position of the distal end of robot arm **300**, in addition to the amount of compensation force required to compensate for the gravity of the surgical instrument, may be converted to torque to be applied at the joints of robot arm **300** to maintain robot arm **300** in a static position, e.g., by the motors operatively coupled to the joints of robot arm **300**, as indicated by the motor current measurements. In addition, impedance calculation module **1424** and/or force detection module **1422** may calculate the amount of force applied by the surgical instrument to the patient at the point of entry, e.g., at the trocar, as well as the amount of force applied to the operating end of the surgical instrument, e.g., the grasper end of a surgical instrument, based on the compensation force, the hold force, one or more parameters of the surgical instrument such as the mass, center of mass, and length of the surgical instrument, and the distance from the center of mass to the point of entry.

Additionally or alternatively, by determining the forces applied on robot arm **300** via force detection module **1422**,

52

as well as the position/velocity/acceleration of the distal end of robot arm **300** in 3D space via robot arm position determination module **1418**, the desired force/impedance to be applied to robot arm **300** to compensate for the applied forces may be calculated, e.g., for gravity compensation or to hold robot arm **300** in a static position in the passive mode. Accordingly, the desired force may be converted to torque to be applied at the joints of robot arm **300**, e.g., by the motors operatively coupled to the joints of robot arm **300**. For example, the robot Jacobian may be used for this purpose. Jacobian is a matrix that is computer at each given post of the robot arm, and relates the velocities at the joints to the velocity at the distal end of robot arm **300**:

$$V = J * q_{dot}$$

Here, V is the velocity vector at the distal end of robot arm **300**, J is its Jacobian matrix, and q_{dot} is its joint velocities expressed in vector form. Using the energy principle, and assuming negligible masses for the links of robot arm **300** and negligible friction/dampening, the power of the system may be determined by multiplying its force and velocity:

$$F \cdot V = \tau \cdot q_{dot}$$

$$\Rightarrow$$

$$F \cdot (J * q_{dot}) = \tau \cdot q_{dot}$$

Here, F is the generalized force vector at the distal end of robot **300**. Further, vector manipulation results in:

$$(J^T * F) \cdot q_{dot} = \tau \cdot q_{dot}$$

$$\Rightarrow$$

$$\tau = J^T * F$$

Here, t denotes the transpose of the matrix, such that the forces at the distal end of robot arm **300** may be converted to torques to be applied at the joints using the Jacobian matrix.

Motor interface module **1426** may be executed by processor **1402** for receiving motor current readings at each motor, e.g., M1, M2, M3, disposed within the base of robot arm **300**, and for actuating the respective motors, e.g., by applying a predetermined impedance to achieved the desired outcome as described herein and/or to cause the joints operatively coupled to the respective motors to move, such as in the robotic assist mode. In some embodiments, when joint **320** is operatively coupled to a motor, e.g., M4, such that distal shoulder link **308** may be automatically rotated relative to proximal shoulder link **306**, as described in further detail with regard to FIG. **56**, motor interface module **1426** may actuate M4 to cause rotation of distal shoulder link **308** relative to proximal shoulder link **306**.

Optical scanner interface module **1428** may be executed by processor **1402** for receiving depth data obtained by optical scanner **1100** and processing the depth data to detect, e.g., predefined conditions therein. Moreover, optical scanner interface module **1428** may generate depth maps indicative of the received depth data, which may be displayed to the operator, e.g., via a monitor. For example, optical

scanner interface module **1428** may map the location of the trocar ports in 3D space, such that the mapping of trocar ports may be communicated to the operator, e.g., via display or user interface **1408**. Based on depth data and/or color data received from optical scanner **1100**, optical scanner interface

module **1428** may determine the relative distances between, e.g., the stages of platform **100**, robot arm **300**, any surgical instruments attached thereto, and objects/persons in the operating room such as the surgical table, drapes, etc. Optical scanner interface module **1428** further may receive image data from additional optical scanning devices as defined herein, including for example, an endoscope operatively coupled to the system. Moreover, optical scanner interface module **1428** may receive depth data obtained by optical scanners **1100a**, **1100b**, **1100c** coupled to platform **100** and process the depth data to generate a virtual map of the area surrounding platform **100**, as described above with regarding to FIG. **11C**, which may be displayed to the operator via a monitor, e.g., display **110**. For example, optical scanner interface module **1428** may generate graphical representations of system **200** including platform **100** and robot arms **300a**, **300b**, and any objects and/or persons within the area surrounding platform **100** for display in the virtual map.

Gesture detection module **1430** may be executed by processor **1402** for detecting predefined gestural patterns as user input, and executing an action associated with the user input. The predefined gestural patterns may include, for example, movement of a surgical instrument (whether or not attached to robot arm **300**), movement of robot arm **300** or other components of the system, e.g., foot pedal, buttons, etc., and/or movement of the operator in a predefined pattern. For example, movement of the surgical instrument back and forth in a first direction (e.g., left/right, up/down, forward/backward, in a circle) may be associated with a first user input requiring a first action by the system and/or back and forth in a second direction (e.g., left/right, up/down, forward/backward, in a circle) that is different than the first direction may be associated with a second user input requiring a second action by the system. Similarly, pressing the foot pedal or a button operatively coupled with the system in a predefined manner may be associated with a third user input requiring a third action by the system, and movement of the operator's head back and forth or up and down repeatedly may be associated with a fourth user input requiring a fourth action by the system. Various predefined gestural patterns associated with different components or operators of the system may be redundant such that the associated user input may be the same for different gestural patterns. The predefined gestural patterns may be detected by, e.g., an optical scanning device such as a laparoscope or optical scanner **1100** via optical scanner interface module **1428** or directly by force applied to robot arm **300** via force detection module **1422** or other components of the system.

Actions responsive to user input associated with predefined gestural patterns may include, for example, enabling tool tracking to servo (i.e., move) the laparoscope based on the motion of a handheld tool; engaging the brakes on (e.g., preventing further movement of) the robot arm; engaging a software lock on the robot arm; dynamically changing the length of time that the robot arm takes to transition between states from a default setting; loading a virtual menu overlay on the video feed whereby a surgical instrument in the field of view of the laparoscope functions as a pointer to trigger further actions available from the virtual menu; and/or identifying which member of the surgical staff is touching the robot arm, if any. This information may be used to ensure

that the system does not move if the surgeon is not touching the robot arm, e.g., to avoid the scenario where an external force is acting on the robot arm (e.g., a light cable or other wire being pulled across the robot arm) and the system perceives the force to be intentional from the surgeon. The same information may be used to detect the gaze direction of the surgeon, e.g., whether the surgeon is looking at the video feed or somewhere else in the room, such that the system may freeze the robot arm if the surgeon's gaze is not in the direction it should be. Additionally, the system may reposition a field of view of a camera based on, for example, the direction a surgeon is facing or based on the objects that the surgeon appears to be looking at, based on the data from the optical scanner **1100**.

As described above, responsive to detection of a predefined gestural pattern by the user, e.g., a predefined pattern of movement of the distal tip of the surgical instrument within the field of view of the laparoscope, gesture detection module **1430** may cause a virtual menu to overlay on the video feed, such that the surgical instrument within the field of view of the laparoscope functions as a pointer, as shown in FIG. **52**. Moreover, gesture detection module **1430** may detect further predefined patterns of movement of the distal end of the surgical instrument, e.g., two quick movements in the same direction or a circular movement over a select area of the virtual menu, which may be interpreted as a selection actuation, e.g., a click on the virtual menu. For example, as shown in FIG. **52**, the virtual menu overlay on the video feed may include menu options in the corners of the video feed, e.g., "hot corners", such as: turning on/off assisted scope mode where the system automatically moves the robot arm coupled to a laparoscope to follow the surgical instrument and/or zoom in or out to change the field of view of the laparoscope; adjusting the holding force of robot arm coupled to a retractor, e.g., the amount of force that may be applied to the distal tip of the surgical instrument before the system transitions from passive mode to co-manipulation mode; turning on/off audio; and turning on/off haptic feedback. As will be understood by a person having ordinary skill in the art, more or less menu options may be provided via the virtual menu. In some embodiments, initiation of the display of the virtual menu overlay on the video feed may be triggered by, e.g., actuation of an external actuator such as a foot pedal, a predefined pattern of force applied to the robot arm such double tapping wrist portion **311** and/or the surgical instrument coupled to the robot arm as detected by encoders at the distal end of the robot arm, voice activation, wireless buttons, hot buttons, etc.

In some embodiments, the operator may actively switch the system to a command mode, e.g., via user interface **1408**, where particular movements or gestures of the robot arm, surgical instrument, operator, or otherwise as described herein are monitored by gesture detection module **1430** to determine if they are consistent with a predefined gestural pattern associated with a predefined user input.

Passive mode determination module **1432** may be executed by processor **1402** for analyzing the operating characteristics of robot arm **300** to determine whether to switch the operational mode of robot arm **300** to the passive mode where the system applies impedance to the joints of robot arm **300** via motor interface module **1426** in an amount sufficient to maintain robot arm **300**, and accordingly a surgical instrument attached thereto, if any, in a static position, thereby compensating for mass of robot arm **300** and the surgical instrument, and any other external forces acting of robot arm **300** and/or the surgical instrument. If robot arm **300** is moved slightly while in the passive mode,

55

but not with enough force to switch out of the passive mode, the system may adjust the amount of impedance applied the robot arm 300 to maintain the static position, and continue this process until robot arm 300 is held in a static position. For example, passive mode determination module 1432 may determine to switch the operational mode of robot arm 300 to the passive mode if movement of the robot arm due to movement at the handle of the surgical instrument as determined by force detection module 1422 is less than a predetermined amount, e.g., no more than 1 to 5 mm, for at least a predetermined dwell time period associated with robot arm 300. The predetermined dwell time period refers to the length of time that robot arm 300 and/or the surgical instrument attached thereto, if any, are held in a static position. For example, the predetermined dwell time may range between, e.g., 0.1 to 3 seconds or more, and may be adjusted by the operator. FIG. 19 illustrates a table or exemplary values of the threshold dwell times for a range of sample instrument types.

In some embodiments, passive mode determination module 1432 may determine to switch the operational mode of robot arm 300 to the passive mode if movement of the distal end of the robot arm due to movement at the handle of the surgical instrument as determined by force detection module 1422 has a velocity that is less than a predetermined dwell velocity/speed. For example, if passive mode determination module 1432 determines that the distal end of the robot arm 300 and/or the surgical instrument attached thereto, if any, moves at a speed that is lower than the predetermined dwell speed during an entire predetermined dwell period, then passive mode determination module 1432 may switch the operational mode of robot arm 300 to the passive mode. FIG. 19 illustrates a table or exemplary values of the threshold dwell speeds for a range of sample instrument types. For example, for surgical instruments such as scopes and tissue manipulation devices, the threshold dwell speeds may be, e.g., 3-5 mm/second, and for surgical instruments such as suturing instruments, needle drivers, high force instruments, staplers, and clip applicators, the threshold dwell speeds may be, e.g., 1-2 mm/second. In some embodiments, passive mode determination module 1432 may determine to switch the operational mode of robot arm 300 to the passive mode based on the identity of the surgical instrument upon attachment of the surgical instrument to robot arm 300 and/or responsive detachment of the surgical instrument from robot arm 300.

Co-manipulation mode determination module 1434 may be executed by processor 1402 for analyzing the operating characteristics of robot arm 300 to determine whether to switch the operational mode of robot arm 300 to the co-manipulation mode where robot arm 300 is permitted to be freely moveable responsive to movement at the handle of the surgical instrument for performing laparoscopic surgery using the surgical instrument, while the system applies an impedance to robot arm 300 via motor interface module 1426 in an amount sufficient to account for mass of the surgical instrument and robot arm 300. Moreover, the impedance applied to robot arm 300 may provide a predetermined level of viscosity perceivable by the operator. FIG. 19 illustrates a table or exemplary values of viscosity levels for a range of sample instrument types. In some embodiments, the viscosity level may be a function of the speed that the surgical instrument is being moved and the distance of the tip of the instrument from the trocar point. For example, co-manipulation mode determination module 1434 may determine to switch the operational mode of robot arm 300 to the co-manipulation mode if force applied at robot arm

56

300 due to force applied at the handle of the surgical instrument exceeds a predetermined threshold associated with robot arm 300 (e.g., a "breakaway force"). The predefined force threshold may be, e.g., at least 7 Newtons, approximately 7 Newtons, at least 7 Newtons, 4-15 Newtons, 4-10 Newtons. The predefined force threshold may be dependent on the type of surgical instrument that is being used and/or whether there is an external force being applied to the surgical instrument.

FIG. 19 illustrates a table or exemplary values of the predefined force thresholds for a range of sample instrument types. As shown in FIG. 19, the predefined force thresholds may reflect the typical external tissue forces that may be exerted on the surgical instrument. In some embodiments, predefined force threshold may be increased if a force is exerted on the surgical instrument by tissue or an organ or otherwise, depending on the direction of the breakaway force. For example, if the breakaway force is in the same direction as the force exerted on the surgical instrument from the tissue or organ, the predefined force threshold may be increased by an amount equal to or commensurate with the force exerted on the surgical instrument from the tissue or organ. In some embodiments, the predefined force threshold for a respective robot arm be adjusted based on a patient's body mass index ("BMI"). For example, a patient with a higher BMI may have a heavier liver that would likely exert a greater force on the instrument. Accordingly, the predefined force threshold may be selected to be higher for the patients with a higher BMI. Accordingly, the operation may actuate a "high force mode," e.g., via user interface 1408, where predefined force threshold is increased to accommodate for engaging with heavier tissue or organs. For example, the predefined force threshold may be selectively increased by 20-100% or more.

Moreover, the force exerted by the user on the surgical instrument and any external tissue forces applied to the surgical instrument may be directionally dependent. For example, if the force exerted by the user on the surgical instrument is in the same direction as an external tissue force applied to the surgical instrument, the two forces may be additive such that the amount of force exerted by the user on the surgical instrument needed to overcome the predefined force threshold may be reduced by the magnitude of the external tissue force such that a lower force than the predefined force threshold would be required to exit the passive mode and enter the co-manipulation mode. On the other hand, if the force exerted by the user on the surgical instrument is in a direction opposite to an external tissue force applied to the surgical instrument, then the necessary amount of force exerted by the user on the surgical instrument needed to overcome the predefined force threshold may be increased by the magnitude of the external tissue force such that a higher force than the predefined force threshold would be required to exit the passive mode and enter the co-manipulation mode.

In addition, if the force exerted by the user on the surgical instrument is in a direction that is perpendicular to an external tissue force applied to the surgical instrument, then the necessary amount of force exerted by the user on the surgical instrument needed to overcome the predefined force threshold may not be affected by the magnitude of the external tissue force such that the necessary force exerted by the user on the surgical instrument needed to exit the passive mode and enter the co-manipulation mode will equal the predefined force threshold. For other directions, the force vectors of the applied forces may be added to or offset by the force vectors of the external tissue forces to overcome

predefined force threshold values for the system or the particular surgical instrument that is coupled with the robot arm, depending on the direction of the external tissue force, if any, and the force applied by the user. In some embodiments, co-manipulation mode determination module **1434** may determine to switch the operational mode of robot arm **300** to the co-manipulation mode based on the identity of the surgical instrument.

Haptic mode determination module **1436** may be executed by processor **1402** for analyzing the operating characteristics of robot arm **300** to determine whether to switch the operational mode of robot arm **300** to the haptic mode where the system applies an impedance to robot arm **300** via motor interface module **1426** in an amount higher than applied in the co-manipulation mode, thereby making movement of robot arm **300** responsive to movement at the handle of the surgical instrument more viscous in the co-manipulation mode. For example, haptic mode determination module **1436** may determine to switch the operational mode of robot arm **300** to the haptic mode if at least a portion of robot arm **300** and/or the surgical instrument attached thereto is within a predefined virtual haptic boundary. Specifically, a virtual haptic boundary may be established by the system, such that the robot arm or the surgical instrument coupled thereto should not breach the boundary. For example, a virtual boundary may be established at the surface of the patient to prevent any portion of the robot arms or the instruments supported by the robot arms from contacting the patient, except through the one or more trocars. Similarly, the virtual haptic boundary may include a haptic funnel to help guide the instrument into the patient as the operator inserts the instrument into a trocar port. Accordingly, based on position data of robot arm **300** and/or the surgical instrument coupled thereto, e.g., received by robot arm position determination module **1418** and/or trocar position detection module **1420**, haptic mode determination module **1436** may determine if robot arm **300** and/or the surgical instrument is within the predefined virtual haptic boundary, and accordingly transition robot arm **300** to the haptic mode where processor **1402** may instruct associated motors to apply an effective amount of impedance to the joints of robot arm **300** perceivable by the operator to communicate to the operator the virtual haptic boundary. Accordingly, the viscosity of robot arm **300** observed by the operator will be much higher than in co-manipulation mode. In some embodiments, haptic mode determination module **1436** may determine to switch the operational mode of robot arm **300** to the haptic mode based on the identity of the surgical instrument.

Robotic assist mode determination module **1438** may be executed by processor **1402** for analyzing the operating characteristics of robot arm **300** to determine whether to switch the operational mode of robot arm **300** to the robotic assist mode where processor **1402** may instruct associated motors via motor interface module **1426** to cause movement of corresponding link and joints of robot arm **300** to achieve a desired outcome. For example, robotic assist mode determination module **1438** may determine to switch the operational mode of robot arm **300** to the robotic assist mode if a predefined condition exists based on data obtained from, e.g., optical scanner interface module **1428**.

For example, robotic assist mode determination module **1438** may determine that a condition exists, e.g., the field of view of a laparoscope coupled to robot arm **300** or optical scanner **1100** is not optimal for a given surgical procedure, e.g., due to blocking by the surgeon or assistant or another component of the system, based on image data obtained

from the laparoscope or optical scanner **1100** via optical scanner interface module **1428**, such that the robot arm coupled to the laparoscope or optical scanner **1100** should be repositioned or zoom in/out to optimize the field of view of the surgical site for the operator. Thus, in robotic assist mode, processor **1402** may instruct robot arm **300**, either automatically/quasi-automatically or responsive to user input by the operator, to move to reposition the laparoscope and/or cause the laparoscope to zoom in or zoom out, or to increase a resolution of an image, or otherwise. For example, the user input by the operator may be determined by gesture detection module **1430**, as described above, such that movement of the robot arm or a surgical instrument in a predefined gestural pattern in a first direction causes the endoscope to increase resolution or magnification and in a second direction causes the endoscope to decrease resolution or magnification, and movement in another predefined gestural pattern causes the robot arm holding the laparoscope to retract away from the patient's body.

In addition, robotic assist mode determination module **1438** may determine that a condition exists, e.g., that one or more trocars are not in an optimal position, for example, due to movement of the patient, such that robot arm **300** should be repositioned to maintain the trocar in the optimal position, e.g., in an approximate center of the movement range of robot arm **300**, thereby minimizing the risk of reaching a joint limit of the robot arm during a procedure. Thus, in robotic assist mode, processor **1402** may instruct system to reposition robot arm **300**, e.g., via vertical/horizontal adjustment by platform **100** or via the joints and links of robot arm **300**, to better align the surgical instrument workspace.

Robotic assist mode determination module **1438** may determine that a condition exists, e.g., the distance between an object and robot arm **300** is within a predetermined threshold, based on image data obtained from the laparoscope or optical scanner **1100** via optical scanner interface module **1428**, such that the robot arm should be frozen to avoid collision with the object. Thus, in robotic assist mode, processor **1402** may instruct robot arm **300** apply the brakes to slow down the robot arm or inhibit or prevent movement within a predetermined distance from the other object.

Robotic assist mode determination module **1438** further may determine that a condition exists, e.g., robot arm **300** is in an extended position for a period of time exceeding a predetermined threshold during a surgical procedure, such that the robot arm should be repositioned to provide the user more available workspace in the vicinity of the surgical instrument coupled to the extended robot arm. Thus, in robotic assist mode, processor **1402** may instruct the system to reposition robot arm **300**, e.g., via vertical/horizontal adjustment by platform **100** and/or via the joints and links of robot arm **300**, to move robot arm **300** closer to the surgical instrument.

Fault detection module **1440** may be executed by processor **1402** for analyzing the data indicative of the operating characteristics of the system, e.g. position data generated by robot arm position determination module **1418** and/or trocar position detection module **1420** and/or force measurement calculated by force detection module **1422**, to detect whether a fault condition is present. For example, fault detection module **1440** may a fault condition of the system and determine whether the fault condition is a "minor fault," a "major fault," or a "critical fault," wherein each category of fault condition may be cleared in a different predefined manner.

For example, fault detection module **1440** may detect a minor fault condition such as robot arm **300** being moved

with a velocity exceeding a predetermined velocity threshold, which may be cleared, e.g., by slowing down the movement of robot arm 300. In some embodiments, the system may automatically apply additional impedance to robot arm 300 when robot arm 300 is moving too fast to thereby force the operator to slow down movement of robot arm 300. Moreover, fault detection module 1440 may detect a major fault condition such as an inadvertent bump of robot arm 300 as indicated by a large force applied to robot arm 300 by a person other than the operator. In response to detection of a major fault condition, fault detection module 1440 may actuate the braking mechanism associated with each motorized joint of robot arm 300 (or at least the joints associated with the major fault condition), to thereby freeze robot arm 300 and inhibit further movement of robot arm 300. Such a major fault condition may be cleared by the operator actuating a “clear” option displayed on user interface 1408. Fault detection module 1440 may detect a critical fault condition such as redundant encoders associated with a given joint of robot arm 300 generating different angulation measurements with a delta exceeding a predetermined threshold. In response to detection of a critical fault condition, fault detection module 1440 may actuate the braking mechanism associated with each motorized joint of robot arm 300 to thereby freeze robot arm 300 and inhibit further movement of robot arm 300. Such a critical fault condition may be cleared by the operator restarting the system. Upon restart of the system, if the critical fault condition is still detected by fault detection module 1440, robot arm 300 will remain frozen until the critical fault condition is cleared.

Indicator interface module 1442 may be executed by processor 1402 for causing indicators 334 to communicate the state of the system, e.g., the operational mode of robot arm 300, to the operator or other users, based on, for example, determinations made by passive mode determination module 1432, co-manipulation mode determination module 1434, haptic mode determination module 1436, and/or robotic assist mode determination module 1438. For example, indicator interface module 1442 may cause indicators 334 to illuminate in specific color light associated with a specific state of the system. For example, indicator interface module 1442 may cause indicators 334 to illuminate in a first color (e.g., yellow) to indicate that no surgical instrument is attached to the robot arm, and that the robot arm may be moved freely such that the system compensates for the mass of the robot arm; in a second color (e.g., purple) to indicate that a surgical tool is attached to the robot arm, and that the robot arm may be moved freely such that the system compensates for the mass of the robot arm and the mass of the surgical instrument coupled to the robot arm; in a third color (e.g., blue) to indicate that a surgical instrument is attached to the robot arm, and that the robot arm is in the passive mode as determined by passive mode determination module 1432; in a fourth color (e.g., pulsing orange) to indicate that at least a portion of the robot arm and/or the surgical instrument attached thereto is within the virtual haptic boundary, e.g., 1.4 m or more above the ground; in a fifth color (e.g., pulsing red) to indicate that a fault has been detected by the system by fault detection module 1440. As will be understood by a person having ordinary skill in the art, different colors and patterns may be communicated by indicators 334 to indicate the states of the system described above.

Additionally, indicators 334 may be illuminated in other distinct colors and/or patterns to communicate additional maneuvers by robot arm 300, e.g., when robot arm 300 retracts the surgical arm in the robotic assist mode, or

performs another robotically-assisted maneuver in the robotic assist mode. As described above, indicators 334 further may include devices for emitting other alerts such as an audible alert or text alert. Accordingly, indicator interface module 1442 may cause indicators 334 to communicate the state of the system to the operator using audio or text, as well as or instead of light. For example, indicator interface module 1442 may cause one or more speakers to emit an audible alert that changes in, e.g., amplitude and/or frequency, as robot arm 300 approaches a potential collision with one or more objects/persons within the operating room.

Additionally or alternatively, indicator interface module 1442 may communicate the state of the system, e.g., transition from co-manipulation mode to passive mode, via haptic feedback at the distal end of robot arm 300, and accordingly on the surgical instrument coupled thereto. For example, when the surgical instrument is held in a position for the predetermined dwell time such that the system switches to passive mode, the user may feel a vibration at the surgical instrument indicating that the system has transitioned to passive mode and that the user may let go of the surgical instrument. As another example, the user may feel a vibration after the surgical instrument is coupled to the coupler body to indicate that the surgical instrument is successfully coupled to the robot arm. The vibration may be strong enough to be felt by the user, but weak enough such that any movement at the distal tip of the surgical instrument resulting therefrom is negligible.

Fatigue detection module 1444 may be executed by processor 1402 for detecting user fatigue that may occur during operation of robot arm 300 in a surgical procedure, as described in further detail below with regard to FIG. 25. For example, based on data from, e.g., robot arm position determination module 1418, force detection module 1422, impedance calculation module 1424, fatigue detection module 1444 may determine the level of fatigue of the operator using the surgical instrument coupled to robot arm 300, and compare the level of fatigue with a predetermined fatigue threshold. For example, fatigue detection module 1444 may assess an overall score for a given procedure to determine the level of fatigue based on, e.g., operator hand tremor, distance/minimum path travelled by the instrument tip, time to achieve procedure steps, and/or time to complete the procedure. Based on the data generated by fatigue detection module 1444, impedance calculation module 1422 may determine an amount of impedance necessary to apply to robot arm 300 to, e.g., reduce tremor of the operator, such that motor interface module 1426 may cause the associated motors to apply the requisite impedance to robot arm 300. Moreover, based on the data generated by fatigue detection module 1444, motor interface module 1426 may cause the associated motors to move the links of robot arm 300 to guide the operator’s manipulation of the surgical instrument attached thereto.

The co-manipulation surgical robot systems described herein may include additional modules within memory 1410 of platform 200 for executing additional tasks based on the data obtained. For example, the system may determine that a surgical instrument has been attached to robot arm 300 by detecting a rapid or sudden change in force (a “snapping motion”) applied to robot, e.g., due to the attraction force of the magnetic connection between the coupler body and coupler interface 400, via force detection module 1422. For example, the attractive forces of the magnets on the coupler body and coupler interface 400 may cause a sudden movement on at least an end portion of the robot arm, and/or a sudden rotation of the last joint of the robot arm when the

61

magnets are aligning. Accordingly, this sudden movement may be detected and may trigger surgical instrument identification module **1412** to determine that an instrument has been attached or detached from the robot arm. Similarly, surgical instrument identification module **1412** may determine that the surgical instrument has been detached from robot arm **300**, e.g., when subsequent motions of the distal end of robot arm **300** are accompanied by little to no rotation in the distal-most joint of robot arm **300**.

Additionally, the system may determine if the surgical instrument has been detached from robot arm **300** based on data indicative of the position of the distal end of robot arm **300** relative to the trocar point generated by trocar position detection module **1420**, as well as the direction of an instrument shaft and/or an orientation of the distal-most link of robot arm **300**, e.g., distal wrist link **316**. For example, if the instrument is pointing directly at the trocar, then there is a higher probability that a tool is attached to the robot arm. Moreover, axis **Q7** of robot arm **300** may indicate the pointing direction of the instrument and, if the instrument is passing through the trocar port, the distal wrist link **316** will point in a direction of the trocar port. Therefore, if distal wrist link **316** is not pointing toward the trocar port, then the system may determine that the robot arm is not supporting an instrument or the instrument is not advanced through the trocar port. For example, when an instrument is detached from robot arm **300** and robot arm **300** is moved, the computed direction of the instrument shaft (e.g., the direction that the instrument would point if attached to robot arm **300**) may no longer point to the trocar entry point and likely will not point to the trocar entry point. Accordingly, the system may alert a user if the system determines that no tool is coupled with robot arm **300**, e.g., via indicators **334**.

In addition, the system may identify when a user may be attempting to remove or decouple a surgical instrument from robot arm **300** and adjust the removal force required to decouple the surgical instrument, and accordingly the coupler body, from coupler interface **400**. For example, where one or more magnets are used to provide a biasing force to bias the surgical coupler body to the coupler interface, a force greater than the attraction force provided by the one or more magnets in a direction opposing the force provided by the one or more magnets must be exerted on the surgical instrument and/or the coupler body that is coupled with the surgical instrument to overcome the attracting force and decouple the coupler body and surgical instrument from the coupler interface. For example, the removal force may be 30-60 Newtons.

Moreover, the system may gather and analyze telemetry data regarding forces being applied to the robot arm to assess or estimate whether a user is attempting to remove a tool from the robot arm and, if so, reduce the coupling force between the coupler body and the coupler interface to make it easier for the user to disengage the surgical instrument from the robot arm. For example, the coupling/removal force may be reduced by 50-80%. Based on historical data and user feedback, as well as on data such as whether a user replaces the instrument without adjusting a location of the instrument, which could indicate inadvertent removal of the instrument, the system may estimate the optimal times to reduce a coupling force between the coupler body and the coupler interface. Moreover, the coupling force may be increased during operation to prevent inadvertent removal of surgical instrument from the robot arm.

Additionally, the system may determine an optimum positioning of robot arms **300** and its joints, the surgical instruments coupled with the robot arms, or other compo-

62

nents of the robot arms and/or the system based on data obtained from the optical scanning devices used with the system, and provide guidance to the operator of the system to achieve the optimum positioning. Data indicative of the optimum positioning further may be used by processor **1402** to instruct the motors to cause corresponding links and joints of robot arm **300** to move, e.g., in robotic assist mode, to automatically reposition robot arm **300** and/or the optical scanning devices in the optimum position, e.g., during the setup stage or thereafter.

In addition, the system may collect data from sensors, e.g., position data of robot arm **300** or the surgical instrument attached thereto via the encoders or optical scanning devices and/or position data of the operator via body sensors or optical scanning devices, during a procedure, e.g., during setup or operation of robot arm **300**, such that processor **1402** may detect deviations of movements or processes of the current user as compared to a model or optimal movement pattern, and communicate the deviations to the current user in real-time. For example, processor **1402** may cause a monitor to display the deviations to the current user in real-time, as well as the optimal and/or actual movement pattern. Additionally, or alternatively, indicator interface module **1440** may cause indicators **334** to indicate deviations from the model or optimal movement pattern, e.g., by illuminating a specific color and/or in a specific pattern. Additionally, or alternatively, motor interface module **1426** may apply impedance to robot arm **30** perceivable by the operator as haptic feedback including vibrations, restrictions on movement, or sensations to indicate deviations from the model or optimal movement pattern. Accordingly, the system may be used as a training tool for new users as such data may be used to optimize the position of a surgical device in real-time.

The system further may analyze the depth map generated by the optical scanning devices and cluster different groups of (depth) pixels into unique objects, a process which is referred to as object segmentation. Examples of such algorithms for segmentation may include: matching acquired depth map data to a known template of an object to segment; using a combination of depth and RGB color image to identify and isolate relevant pixels for the object; and/or machine learning algorithms trained on a real or synthetic dataset to objects to identify and segment. Examples of such segmentation on a depth map may include: locating the robot arms or determining the position of the robot arms; identifying patient ports (e.g., trocar ports) and determining a distance from the instruments to the trocar ports; identifying the surgeon and distinguishing the surgeon from other operators in the room; and/or identifying the surgeon in the sensor's field of view. Moreover, the system may use object segmentation algorithms to uniquely identify the surgeon and track the surgeon with respect to, for example, a surgical table, a patient, one or more robot arms, etc. In addition, the system may use object segmentation algorithms to determine if a surgeon is touching or handling either of the robot arms and, if so, identify which robot arm is being touched or handled by the surgeon. The system further may use object segmentation to locate the surgical instrument and the distal end of the robot arm in 3D space, such that the system may determine whether the surgical instrument is attached to the distal end of the robot arm, e.g., based on proximity between the surgical instrument and the distal end of the robot arm.

Referring now to FIG. **15**, operation **1500** of the co-manipulation surgical robot systems described herein is provided. As shown in FIG. **15**, at step **1502**, the operator may couple a selected surgical instrument to coupler inter-

63

face **400** of robot arm **300** via a coupler body, e.g., coupler body **500**, **600**, **700**. As described above, the operator may select a coupler body sized and shaped to couple with the selected surgical instrument, e.g., based on the elongated shaft diameter of the surgical instrument. When the surgical instrument and coupler body are ready to be coupled to robot arm **300**, the operator may load the calibration file of the selected surgical instrument, e.g., via user interface **1408**, such that information associated with the selected surgical instrument, e.g., a laparoscope or retractor, is loaded into the system. For example, the operator may select the calibration file from a database of calibration files for a variety of surgical instruments. The calibration files may be stored from previous procedures, and may be pre-loaded to include calibration files of commonly used laparoscopic instruments.

If the calibration file for the selected surgical instrument is not available in the database, the operator may self-calibrate the surgical instrument using the system. For example, FIG. **16** illustrates surgical instrument calibration process **1600** for calibrating a surgical instrument, e.g., to determine the center of mass of the surgical instrument, which may be used in calculating accurate force measurements on the surgical instrument and robot arm **300** during operation. At step **1601**, the operator may actuate the “startup” option on user interface **1408**. At step, **1602**, the operator may select the “load tool calibration” to begin the calibration process. At step **1603**, the system does not apply any impedance to robot arm **300** for gravity compensation of a surgical instrument. The system may apply impedance to robot arm **300** to account for the weight of robot arm **300**, e.g., to prevent robot arm **300** from dropping to the ground. At step **1604**, the surgical instrument is coupled to coupler interface **400** of robot arm **300** via the appropriate sized coupler body, which may cause wrist portion **411** of robot arm **300** to rotate about axis **Q7** to engage with the coupler body.

At step **1605**, the system compensates for the gravity of the surgical instrument and the force applied by the hand of the operator, e.g., by measuring the force applied to the distal end of robot arm **300** due to the mass of the surgical instrument. As described above, the force applied to the distal end of robot arm **300** may be measured by measuring the motor current across the motors disposed in the base of robot arm **300**. If the system overcompensates for the gravity of the surgical instrument, at step **1606**, robot arm **300** may “runaway”, e.g., drift upward. The runaway effect may be detected at step **1607**, and at step **1608**, indicators **334** may blink to indicate to the operator of the runaway. At step **1609**, the system may identify the runaway as a minor fault, and accordingly apply additional impedance to robot arm **300** and freeze robot arm **300** when robot arm **300** slows down before removing the additional impedance. Once the minor fault is addressed, calibration process **1600** may return to step **1603**.

After step **1605**, when the system compensates for the gravity of the surgical instrument, if the surgical instrument is detached, either accidentally or manually by the operator at step **1611**, at step **1610**, the system detected the detachment of the surgical instrument from robot arm **300**. As a result, the system will stop compensating for the gravity of the surgical instrument, and calibration process **1600** may return to step **1603**. After step **1605**, when the system compensates for the gravity of the surgical instrument, calibration process **1600** is ready to enter calibration mode at step **1612**. For example, the operator may initiate calibration mode via user interface **1408** at step **1613**. At step

64

1614, the system may indicate to the operator, e.g., via user interface **1408** and/or blinking of indicators **334**, that it is safe to let go of surgical instrument, such that the operator may let go of the surgical instrument at step **1616**. At step **1615**, the system calibrates the surgical instrument.

Referring again to FIG. **15**, when the surgical instrument and coupler body are ready to be coupled to robot arm **300**, and the appropriate calibration file is loaded, the operator may easily place the coupler body near coupler interface **400**, such that the magnetic connection between the coupler body and coupler interface **400** automatically aligns and coupled the surgical instrument to robot arm **300**. The system will now accurately compensate for the gravity of the selected surgical instrument. At step **1504**, the user may use the co-manipulation surgical system by freely manipulating the surgical instrument coupled to robot arm **300** in the ordinary manner that the operator would without robot arm **300** coupled thereto. As shown in FIG. **15**, as the operator manipulates the surgical instrument, and accordingly robot arm **300** coupled thereto, the system may automatically switch between, e.g., co-manipulation mode **1506**, passive mode **1508**, haptic mode **1510**, and robotic assist mode **1512** (collectively referred to as “operational modes”), upon detection of predefined conditions, as described below with regard to FIG. **17**. In some embodiments, the system may automatically switch between only co-manipulation mode **1506**, passive mode **1508**, and haptic mode **1510**. In some embodiments, the operator may select which operational mode to set the system in prior to using the co-manipulation surgical system at step **1504**.

For example, an operator may exert a particular force on the distal end of robot arm **300**, e.g. by manipulating the surgical instrument coupled to robot arm **300**, to indicate that the operator wishes to change the operational mode of the particular robot arm. Sensors and/or motor current readings may be used to detect the force applied to the distal end of robot arm **300** and to determine if the force matches a predefined force signature associated with an operational change, e.g., by comparing the force with one or more predefined force signatures stored in the system. If there is a match, then the system may change the operational mode of the robot arm to the particular operational mode that matches the force signature.

As described above, during operation of the co-manipulation surgical system, the system may continuously monitor the robot arm and forces applied thereto to detect predefined conditions that require switching the operational modes of the system, as described in method **1700** of FIG. **17**. As shown in FIG. **17**, at step **1702**, the system continuously collects data related to a first operating characteristic of the robot arm and/or of the surgical instrument coupled with the robot arm. For example, as described above, the system may measure motor current of the motors operatively coupled to the joints of the robot arm as well as angulations of the links of the robot arm based on measurements by the encoders of the robot arm to calculate the position of the robot arm and the surgical instrument as well as the forces acting on any portion of the robot arm as well as on the surgical instrument, if any, in real time. At step **1704**, the system may analyze the data related to the first operating characteristic to determine if a first condition is present. For example, based on the position and force data of the robot arm and/or surgical instrument, the system may determine if the movement of the robot arm due to movement of the surgical instrument coupled thereto is within a predetermined movement threshold of the robot arm for a period of time longer than the predetermined dwell time of the robot arm. Upon

detection of this first condition, at step 1706, the system may modify a first operating parameter of the robot arm. For example, the system may switch the operational mode of the robot arm to the passive mode, where the robot arm maintains the surgical instrument in a static position.

For example, a first robot arm may be coupled to a laparoscope, and the operator may manipulate the laparoscope within the patient until a desirable field of view is provided by the laparoscope, e.g., via a monitor displaying the image feed from the laparoscope. In order to freely move the laparoscope coupled to the first robot arm in the co-manipulation mode, the operator must apply a sufficient force to the laparoscope that exceeds a predetermined force threshold. The predetermined force threshold should be low enough such that it does not require much force by the operator to freely move the laparoscope. Moreover, the predetermined force threshold may be selected so as to resist inadvertent movement away from the passive mode. As the operator freely moves the laparoscope in the co-manipulation mode, as described above, the system will apply enough impedance to the first robot arm to compensate for the effects of mass (i.e., inertia) and/or gravity of the first robot arm and the laparoscope during the movement, such that a mass or weight of the first robot arm is not detectable by the operator or is otherwise significantly attenuated. In some embodiments, if when the operator couples the laparoscope to the first robot arm, the laparoscope is not already positioned within the body of the patient, the system may determine that there are no external forces acting on the surgical instrument and may automatically switch the first robot arm to the haptic mode in order to guide the operator to move the laparoscope to the appropriate location through the trocar port, e.g., via a virtual haptic funnel established about the trocar port.

When the laparoscope is in the desired position relative to the patient and the surgical site within the patient, the system will automatically switch from co-manipulation mode to passive mode upon detection that movement of the first robot arm due to movement of the surgical instrument is within a predetermined movement threshold for a period of time exceeding a predetermined dwell time. For example, upon reaching the desired position, the operator will hold the laparoscope in the desired position, e.g., for at least a quarter of the second. Thus, if the predetermined dwell time is a quarter of a second, holding the laparoscope in the desired position for any longer than the predetermined dwell period will cause the system to automatically switch to passive mode. Moreover, as the operator may not be able to hold the laparoscope perfectly still, at least some movement of the laparoscope is permitted for the duration of the predetermined dwell time to enter into the passive mode. As described above, in passive mode, the first robot arm will hold the laparoscope in a static position, e.g., by the system applying enough impedance to the first robot arm to compensate for all external forces acting on the laparoscope.

Similarly, a second robot arm may be coupled to a retractor, and the operator may freely manipulate the retractor within the patient in the co-manipulation mode, e.g., to grasp tissue within the patient and retract the tissue to provide a clear field of view of the surgical site by the laparoscope coupled to the first robot arm, by applying a sufficient force to the second robot arm due to force applied at the retractor exceeding the predetermined force threshold of the second robot arm. As the operator grasps/lifts/retracts the tissue with retractor, the system may only compensate for the gravity of the second robot arm and/or the instrument and not of the tissue being grasped, such that the operator

may feel any other forces acting on the retractor, including without limitation the forces acting on the instrument from the tissue. In this optional configuration. Accordingly, the haptics associated with the tissue being grasped may be preserved.

When the retractor sufficiently grasps and retracts the tissue, the system may automatically transition to the passive mode upon the operator holding the retractor in position, e.g., with movement not exceeding a predetermined movement threshold of the second robot arm, for a period of time exceeding the predetermined dwell period of the second robot arm. Accordingly, when the retractor is retracting the tissue within the patient in the passive mode, the second robot arm will account for the mass of the tissue in addition to the mass of the retractor and the second robot arm. Thus, the predetermined force threshold to cause the second robot arm to switch out of the passive mode must be greater than the force applied to second robot arm due to force applied to the tip of the retractor by the tissue, such that if the force applied by the tissue to the surgical instrument exceeds the predetermined first threshold of the second robot arm, the system will automatically cause the second robot arm to switch out of the passive mode and into, e.g., the co-manipulation mode. However, the predetermined force threshold should not be so high that it is very difficult for the operator to move the retractor. As described above, the operator may adjust the predetermined force threshold via, e.g., user interface 1408.

Upon retraction of the tissue via the retractor coupled to the second robot arm, the operator may need to readjust the field of view of the laparoscope coupled to the first robot arm. Accordingly, the operator may apply a force to the laparoscope that exceeds the predetermined force threshold of the first robot arm, such that the system automatically switches the first robot arm from the passive mode to the co-manipulation mode. When the new desired position of the laparoscope is achieved, the first robot arm may automatically switch back to the passive mode if the predefined conditions described above are met. Alternatively, to readjust the laparoscope or to reposition the links of the first robot arm to avoid potential collisions during the laparoscopic procedure or to switch the laparoscope to a different robot arm altogether, the operator may elect to decouple the laparoscope, readjust the robot arm and/or laparoscope, and reattach the laparoscope to the first robot arm (or to the other robot arm). Upon reattachment of the laparoscope to the first robot arm, the first robot arm may automatically switch to the passive mode if the predefined conditions described above are met.

Moreover, as the operator freely moves the retractor in the co-manipulation mode, e.g., prior to inserting the tip of the retractor through the trocar within the patient, if the operator moves the tip of the retractor too close to the patient's skin away from the trocar port, and a virtual haptic boundary has been established by the system on the skin of the patient outside the trocar ports, the system may automatically switch to the haptic mode. Accordingly, the system may apply an impedance to the second robot arm that is much higher than the impedance applied to the second robot arm in co-manipulation mode to indicate to the operator that they are approaching or within the virtual haptic boundary. For example, movement of the retractor by the operator may feel much more viscous in the haptic mode. The system may remain in the haptic mode until the operator moves the retractor out of the virtual haptic boundary. In some embodiments, in the haptic mode, the second robot arm may reduce the effects of gravity, eliminate tremor of the instrument tip,

67

and apply force feedback to avoid critical structures as defined by the virtual haptic boundary. Accordingly, the system does not replace the operator, but rather augments the operator's capabilities through features such as gravity compensation, tremor removal, haptic barriers, force feedback, etc.

In some embodiments, the system may switch the second robot arm to the robotic assist mode. For example, as the operator attempts to retract the tissue, if more force is required to retract the tissue than the operator is able or willing to apply to the retractor, the operator may provide user input to the system indicating that the operator wants the second robot arm to assist in the retraction of the tissue. For example, as described above, the operator may perform a predefined gestural pattern that may be detected by, e.g., optical scanner 1100, such that the system switches the second robot arm to the robotic assist mode and causes the motors of the second robot arm to move the second robot arm, and accordingly the retractor, to provide the additional force required to retract the tissue.

In addition, instead of manually manipulating the laparoscope coupled to the first robot arm as described, the operator may provide another user input to the system indicating that the operator wants the system to reposition the laparoscope. For example, if the operator is actively manipulating a surgical scissor, which may or may not be coupled to a robot arm of the system, such that the tip of the surgical scissor is within the field of view of the laparoscope coupled to the first robot arm, the operator may perform a predefined gestural pattern with the tip of the surgical scissor, e.g., moving the surgical scissor quickly back in forth in a particular direction. The predefined gestural pattern of the surgical scissor may be captured as image data by the laparoscope, and based on the data, the system may detect and associated the predefined gestural pattern with a predefined user input requiring that the system switch the first robot arm from the passive mode to the robotic assist mode, and cause the first robot arm to reposition itself, and accordingly the laparoscope, to adjust the field of view in the direction of the pattern motion of the surgical scissor. As described above, additional gestural patterns may be performed via the surgical scissor within the field of view of the laparoscope to cause the first robot arm to retract the laparoscope and/or to cause the laparoscope itself to zoom in or zoom out or improve resolution. In some embodiments, based on the image data captured by the laparoscope, using object tracking of the additional tools in the field of view of the laparoscope, e.g., the surgical scissors actively operated by the operator, the system may cause the first robot arm coupled to the laparoscope to automatically switch to the robotic assist mode and cause the first robot arm to reposition itself to adjust the field of view to ensure that the tip of the surgical scissors remain within an optimum position within the field of view of the laparoscope during the procedure.

The operational mode of any one of the robot arms may be changed independent of the operational mode of the other robot arms of the system. In addition, the operational parameters of each robot arm may be tailored to the specific surgical instrument coupled thereto. For example, the predetermined force threshold for the robot arm coupled to the retractor device may be higher than the predetermined force threshold for the robot arm coupled to the laparoscope, as the retractor will endure higher forces during the procedure. The sensors, motors, etc. of the system may be active in all modes, but may act very differently in each mode, e.g., including acting as if inactive. As will be understood by a

68

person having ordinary skill in the art, the system may include more than two robot arms, such that the operator may couple a third surgical instrument, e.g., a grasper device, to a third robot arm and a fourth surgical instrument, e.g., a surgical scissor device, to a fourth robot arm for operation during the laparoscopic procedure.

In some embodiments, the operational mode of a robot arm may be changed responsive to user input provided by the operator. For example, the operator may selectively change the operational mode of the robot arm by actuating a button, dial, or switch located on the robot arm, a foot pedal or foot switch, voice command, an input on a touch-screen, or using gestures or force signatures as described above. In some embodiments, the operational mode of a robot arm may be changed based only on the coupling of the surgical instrument to the coupler interface via the coupler body. As described above, the system may automatically identify the surgical instrument based on the coupling of the coupler body to the coupler interface. Accordingly, based on the identity of the surgical instrument coupled to the robot arm, the system may automatically switch the operational mode of the robot arm to a predetermined operational mode, e.g., passive mode if the surgical instrument is an endoscope, or if the robot arm is already in the passive mode, the system will remain in the passive mode upon coupling of the endoscope with the robot arm.

Similarly, based on the identity of the surgical instrument upon attachment of the surgical instrument to the robot arm, the system may automatically switch the operational mode of the robot arm to the co-manipulation mode, e.g., is the surgical instrument identity indicates that it is a tool that will be actively operated by the operator during the laparoscopic procedure. Additionally, based on the identity of the surgical instrument upon attachment of the surgical instrument to the robot arm, the system may automatically switch the operational mode of the robot arm to the robotic assist mode, e.g., if the surgical instrument identity indicates that it is a tool that the operator desires to be completely robotically controlled such as an irrigation device. Accordingly, upon attachment of the irrigation device to the robot arm, the system will switch to the robotic assist mode and cause the robot arm to position the irrigation device in the desired position within the body.

Moreover, the system may be instructed by the operator, e.g., via user interface 1408, to operate the robot arm in less than the four operational modes discussed above. For example, the operator may deactivate any one of the operational modes for a given procedure. In some embodiments, the system may cause the robot arm to operate in an additional operational mode, such as a locking mode, which may be similar to the passive mode, except that the predetermined force threshold of the robot arm to switch out of passive/locking mode may be so high that the robot arm is effectively frozen so as to protect the robot arm from inadvertently switching out of the passive/locking mode, e.g., to avoid movement due to inadvertent bumps of the robot arm. In this locking mode, if the force from the inadvertent bump is sufficiently high to cause even a slight movement of the robot arm, the system may cause the robot arm to reposition itself to the position it was in prior to the inadvertent bump.

In addition, when no surgical instrument is coupled to the distal end of a robot arm of the system, the system is still capable of automatically switching the operational modes of the robot arm responsive to movement of the robot arm by an operator upon detection of the predefined conditions described above. Accordingly, the system will apply an

69

impedance to the joints of the robot arm to compensate for the mass of the robot arm such that the robot arm may remain in a static position when in the passive mode, and will permit the robot arm to be freely moveably by the operator in the co-manipulation mode if the system detects that the force applied to the robot arm by the operator exceeds the predetermined force threshold of the robot arm. Additionally, the system will switch the robot arm to the haptic mode if the operator attempts to move any portion of the robot arm within a predefined virtual haptic barrier. At step 1514, when the laparoscopic procedure is complete, the operator may remove the surgical instruments from the respective robot arms.

Referring now to FIGS. 18A to 18C, force measurements during operation of robot arm 300 are provided. As described above, upon attachment of the surgical instrument to coupler interface 400 via the coupler body coupled to the surgical instrument, the orientation of the surgical instrument may be automatically determined based on the magnetic connection between the coupler interface and the coupler body. Moreover, as described above, the calibration file of the surgical instrument coupled to robot arm 300 loaded on the system may include information of the surgical instrument including, e.g., the mass of the surgical instrument, the center of mass of the surgical instrument, and the length of the surgical instrument, such that distance D3 between the center of mass and the instrument tip may be derived. In addition, as described above, the position of the surgical instrument at the trocar, e.g., where the surgical instrument enters the patient's body, may be calculated in real-time, such that distance D2 between the center of mass of the surgical instrument and the trocar may be derived in real time. Additionally, as described above, the coupler body is preferably coupled to the surgical instrument at a fixed, known position along the elongated shaft of the surgical instrument (which may be included in the calibration file), e.g., adjacent to the proximal portion of the surgical instrument, and thus distance D1 between the center of mass of the surgical instrument and the coupler body, e.g., the point of attachment to the distal end of robot arm 300, may be derived. Alternatively or additionally, as described above, optical scanning devices may be used determine any one of D1, D2, or D3.

As shown in FIG. 18A, when the surgical instrument is positioned through trocar Tr, without any additional external forces acting on the surgical instrument other than at trocar Tr, e.g., the surgical instrument is not lifting or retracting tissue within the patient, the force applied to the surgical instrument at trocar Tr by the body wall (e.g., the "body wall force" or the "trocar force") may be calculated with the following equation:

$$F_{eff} + W + F_{tr} = 0 = > F_{tr} = -W - F_{eff}$$

Where F_{eff} is the force at the distal end of robot arm 300 (e.g., the "end-effector force" of robot arm 300), W is the weight vector of the surgical instrument ($=mgz$), and F_{tr} is the trocar force. Accordingly, F_{eff} is the desired force sent to the system, which is the sum of all the forces generated in the algorithm pipeline including, e.g., gravity compensation, hold, etc.

As shown in FIG. 18B, when the surgical instrument is positioned through trocar Tr and holding/retracting tissue, such that an external force is applied to the tip of the surgical instrument, there are two forces to resolve: F_{tr} and $F_{tr'}$.

70

Accordingly, two equations are needed to solve for the two unknown vectors, which may be the balances of forces and also the balance of moments around the center of mass of the surgical instrument, e.g., L_{cg} .

$$W + F_{eff} + F_{tr} + F_{tr'} = 0$$

$$F_{eff} \times D1 + F_{tr} \times D2 + F_{tr'} \times D3 = 0$$

Here, distances D1 and D3 are known as described above, and D2 may be derived based on the known position of the distal end of robot arm 300 and the calculated position of trocar Tr. As shown in FIG. 18B, the center of mass Leg of the surgical instrument is behind the point of attachment of the coupler body to the distal end of robot arm 300.

As described above, the system may alert the operator if the forces, e.g., force F_{tr} applied to the tip of the instrument and/or force $F_{tr'}$ applied by the instrument at the trocar using, are greater than the respective threshold forces, and accordingly freeze the system if the calculated force is greater than the threshold force, and/or reduce the force exerted at the trocar point at the body wall or at the tip of the instrument by automatically applying brakes or stopping forces to robot arm 300, by slowing or impeding further movement of the instrument in the direction that would increase forces applied at the tip of the instrument or the trocar, and/or automatically moving the robotic arm in a direction that reduces the force being exerted at the instrument tip and/or at the trocar point at the body wall.

Referring now to FIG. 20, a high level example 2000 of the different combinations of data inputs for the various sensors and devices of the systems disclosed herein, e.g., system 200, and the multiple features and capabilities that any implementations of the systems disclosed herein may have and can produce based at least in part on the multiple possible data inputs is provided. As shown in FIG. 20, some implementations of the system may be configured to gather data from at least three monitoring sources 2002, including telemetry from the system (which may include force data from the robot arms, position data from the robot arms, etc.), video from the laparoscopic tower, and/or data from optical scanner 1100. The data gathered from the monitoring sources 2002 may undergo data processing steps 2004 using one or more processors in the system. The data processing steps may include, e.g., data fusion (e.g., fusion of the data gathered from the monitoring sources 2002) and data analysis, which may include algorithm computations. In addition, the data from the monitoring sources 2002 may undergo processing 2004 for the development of system usability features 2006, system safety features 2008, and system performance features 2010. The system may provide the features in real-time. For example, the system usability features may include identifying the surgeon and adjusting the platform height based on the surgeon's profile, detecting the skin surface of the patient and creating a virtual boundary around the skin surface to prevent inadvertent contact with the skin surface of the patient, detecting an instrument type and automatically loading the calibration file appropriate for the particular instrument, etc. In addition, the system safety features may include displaying a virtual map of the area surrounding platform 100, e.g., as an operator moves platform 100 throughout the operating room, to provide the operator with a view of the area surrounding platform 100,

71

such that the operator may avoid collisions between platform **100** and any objects and/or persons within the area surrounding platform **100**.

Referring to FIG. **21**, a schematic overview of the electrical components of the electrical system and connectivity **2100** of the system is provided. This includes the flow of energy throughout the illustrated portion of the system, the ports that may be used for connectivity, and other details related to the various electronic components. For example the system may include non-real time computer **2102** that may be used to acquire data from the optical scanning devices and perform other functions. Non-real time computer **2102** also may control the graphical user interface of the system for the surgeon to interact with. As described above, the graphical user interface may include a touch screen. Non-real time computer **2102** may include, e.g., a 10th Gen Intel® Core™ i7-10700 processor, 32 GB of RAM (which can optionally be 2x16 GB, DDR4, 2933 Mhz), a standard keyboard and a 512 GB PCIe M.2 SSD+1 TB SATA 7200 RPM hard drive, a wireless and Bluetooth card such as the Killer™ Wi-Fi 6 AX1650i (2x2) 802.11ax Wireless and Bluetooth 5.1, and/or a NVIDIA® GeForce RTX™ 2060 6 GB GDDR6 graphics card. The system further may include real-time computer **2104** that may be used to operate and control the robot arms and the related robot controllers and/or other functions, such as acquiring data and information from the optical scanning devices. Real-time computer **2104** may include, e.g., an Intel Core i7 (8th Gen) processor, 32 GB of RAM for memory, a 500 GB SSD hard drive, and/or two or more RJ45 connectors for Ethernet connectivity.

Referring now to FIG. **22**, a flow chart of process **2200** for the acquisition and processing of data from an optical scanning device is provided. As shown in FIG. **22**, at step **2202**, depth data may be acquired from one or more optical scanning devices, e.g., optical scanner **1100**. At step **2204**, filtering/other signal processing algorithms may be performed, e.g., median filter, Gaussian noise removal, anti-aliasing algorithms, morphological operations, ambient light adjustments, etc. At step **2206**, 3D object segmentation may be performed using, e.g., template matching, machine learning, Brute force matching, color plus depth segmentation, 2D-3D registration, pixel value thresholding, etc. At step **2208**, object coordinates may be transformed to task space. For example, transforming object coordinates to task space may include converting a position and an orientation of an object from the optical scanning device's coordinate frame to the coordinate frame of the task needed (e.g., a robot frame for robot control, a cart frame for system setup, etc.). Additionally or alternatively, transforming object coordinates to task space may include using known optical scanning device to the support platform (e.g., a cart) transformations, the surgical robot transformations, and/or the user interface screen transformations, and generating new transformations for specific tasks such as tracking the surgeon's body (e.g., face, hands, etc.) with respect to different elements of the system (e.g., support platform, robot arms, screen, etc.), tracking the surgical table with respect to the cart platform, tracking patient orientation for system setup, tracking trocar port location and orientation for setup, and tracking the position of operating room staff for safety. At step **2210**, the desired task may be performed, e.g., moving the robot arms into the vicinity of the patient/trocar port for easy setup, tracking operating room staff to ensure the system only responds to surgeon commands, recording the surgeon's hand movements during different phases of surgery, etc.

72

In addition, FIG. **22** illustrates a flow chart of process **2212** for the acquisition and processing of data from an optical scanning device. At step **2214**, depth data may be acquired from one or more optical scanning devices, e.g., optical scanner **1100**. At step **2216**, specular noise filtering may be performed. At step **2218**, patient/trocar port segmentation and identification may be performed. At step **2218**, tracked port coordinates may be transformed to robot coordinate space. At step **2222**, the robot arms may be moved to a desired vicinity of the patient/trocar port.

Referring now to FIG. **23**, an example data flow **2300** of the system is provided. As shown in FIG. **23**, non-real-time computer **2302** may gather data from an optical scanning device, e.g., optical scanner **1100** and/or from a camera feed from a laparoscope. Non-real-time computer **2302** also may receive data from real-time computer **2308** having a robot controller, including telemetry information such as positions of the robot arms, forces applied to the various motors/sensors of the robot arms, operational mode information, etc. Non-real-time computer **2302** also may receive data from patient database **2310** having information specific to the patient in the procedure including, e.g., CT scan data, relevant health conditions, and other information that may be desired by the surgeon.

Non-real-time computer **2302** further may provide user feedback **2312** to the user via user interface **2314**. User feedback may include, e.g., collision notifications, positioning information and/or recommendations regarding the various components of the system, the operational mode that has been detected by the system, etc. Non-real-time computer **2302** further may provide commands **2318**, e.g., high level commands, to real-time computer **2308**. High-level commands may include, e.g., mode changes, trajectories, haptic barriers, user configurations, etc. Real-time computer **2308** may include robot controller **2320** programmed to provide robot commands **2322**, e.g., motion or force commands, to the one or more robot arms **2324**, e.g., robot arms **300**. Robot controller **2320** may receive robot feedback data **2326**, e.g., motion, force, and/or touchpoint data, etc., from the one or more robotic arms **2324**.

Referring now to FIG. **25**, method **2500** for estimating user fatigue during a surgical procedure using robot arm **300** is provided. As described above, the algorithms for gravity compensation, viscosity, and/or effects of mass may be used to account for user fatigue. Specifically, during a laparoscopic procedure, a surgeon may be subject to fatigue and may experience hand tremor or erroneous tool motion for surgical tools such as, e.g., scissors, needle drivers, cautery tools, graspers, as the procedure progresses. As shown in FIG. **25**, at step **2502**, the system may receive and monitor data indicative of the operator's performance, e.g. from optical scanner **1100** such as a LiDAR camera, robot telemetry, and/or an endoscope, during the surgical procedure while the operator maneuvers the surgical instruments coupled to robot arm **300**. Learning from a large dataset of clinical procedures and/or gathering and analyzing data during a procedure or a portion of a procedure may allow the system to infer a level of competency of the surgeon as the procedure progresses, at step **2504**, and further may allow the system to adapt algorithm parameters in order to help the surgeon to move more effectively while co-manipulating the surgical instruments attached to the robot arm. For example, at step **2506**, the system may adjust one or more operating parameters of robot arm **300** to change its behavior. If the fatigue level goes above a specific threshold, at step **2608**, the system may warn the surgeon. In addition, ranking procedures may be used to allow the system to provide the

surgeon a summary of their performance for a given procedure and show their overall progress, procedure after procedure.

In some embodiments, the system may collect data during a procedure indicative of at least one of operator hand tremor, distance/minimum path travelled by the instrument tip, time to achieve procedure steps, and/or time to complete the procedure, and compare such data with threshold or predefined values for each of the factors to determine whether a magnitude of any one of the factors has reached a level sufficient to cause the system to warn the operator and/or sufficient to cause the system to adjust one or more operating parameters to mitigate the user's fatigue. For example, the system may eliminate or reduce tremor of the instrument tip by exerting forces on the instrument to increase the impedance or viscosity of the instrument, to avoid critical structures, and/or to apply force feedback. User fatigue may be identified when, for example, a procedure time increases beyond a threshold value for a particular procedure, the number of movements of the surgical instrument increases beyond a threshold value for a particular procedure or otherwise indicates errant or uncontrolled movements, if an operator moves an instrument into a haptic barrier a predefined number of times, if an operator exerts an excessive force on the trocar one or a predetermined number of times, etc. As described above, such data may be collected using the sensors on the robot arms and/or one or more optical scanning devices. When a particular level of user fatigue is identified by the system, the system may increase a viscosity or impedance of the instrument and/or the robot arm associated with the instrument to reduce a magnitude of movements and/or a number of movements of the surgical instrument and/or the robot arm.

Additionally, the system may collect data regarding the speed and frequency with which the operator moves the various instruments/laparoscopes along with estimates of how much tremor is involved in the movements, estimate the required added viscosity to reduce tremors while not hindering their motions or adding unnecessary fatigue to the operator. In some embodiments, a controller of robot arm 300 may iteratively adjust a viscosity value for a particular instrument, collect data related to the movement of the instrument, and to assess whether an additional adjustment is needed to the viscosity applied to the instrument. Moreover, the system may use additional algorithms to adopt an iterative approach to optimizing a particular operational characteristic or parameter of robot arm 300, including collecting data related to a particular operational characteristic or parameter, changing operational characteristic or parameter, collecting additional data related to the operational characteristic or parameter, and analyzing the data to determine if additional changes to the operational characteristic or parameter should be made, which may be based on, e.g., deviations between the actual data values and preferred or optimal values of an operational characteristic or parameter.

Referring now to FIG. 26, dataflow 2600 of a distributed network of co-manipulation surgical robot systems is provided. For example, a distributed network of co-manipulation robotic ("cobot") surgical systems may be used in multiple hospitals, each of which may be connected to an online database. This arrangement may provide considerably more data and user information that may be used by any of the cobot systems in operation. The systems may aggregate the data from the distributed network of systems to identify the optimum configuration based on factors such as procedure type, surgeon experience, patient attributes etc.

Through analytics or clinician input, the cobot systems may identify a routine procedure versus a procedure that may be more complicated. This information may be used to provide advice or guidance to novice surgeons.

Moreover, centralizing procedure data may enable the running of large data analytics on a wide range of clinical procedures coming from different users. Analysis of data may result in optimized settings for a specific procedure, including, e.g., optimized system positioning, optimal ports placement, optimal algorithms settings for each robot arm and/or detection of procedure abnormalities (e.g., excessive force, time, bleeding, etc.). These optimal settings or parameters may depend on patient and tool characteristics. As described above, a surgeon may load and use optimal settings from another surgeon or group of surgeons. This way, an optimal setup may be achieved depending on, e.g., the surgeon's level of expertise. To keep track of the various users in the distributed network of cobot systems, it may be beneficial to identify each user. As such, the user may log into the cobot system and access their profile online as necessary. This way the user may have access to their profile anywhere and will be able to perform a clinical procedure with their settings at a different hospital location.

An example user profile may contain the user's specific settings and information, including, e.g., username; level of expertise; different procedures performed, and/or region of clinical practice. In addition, the clinical procedure may require a user to store specific settings such as clinical procedure (e.g., cholecystectomy, hernia, etc.), table orientation and height, preferred port placement, settings per assistant arm for each algorithm, patient characteristics (e.g., BMI, age, sex), and/or surgical tools characteristics and specifications (e.g., weights, length, center of gravity, etc.). The user may be able to enable his own profile, and optionally may enable another user's profile, such as the profile of a peer, the most representative profile of a surgeon of the user's area of practice, the most representative profile of a surgeon with a specific level of expertise, and/or the recommended profile according to patient characteristics.

The identification of a user may be performed via password, RFID key, facial recognition, etc. Learning from a large number of procedures may result in a greater level of optimization of the cobot system setup for a given procedure. This may include, e.g., cart position, individual robot arm position, surgical table height and orientation, port placement, and/or setup joints position. These settings may be based on patient height, weight, and sex, and further may be interdependent. For example, the optimal port placement may depend on patient table orientation.

Additionally, a clinical procedure may be described as a sequence of clinical procedures steps. Learning these different steps may allow the cobot system to infer in real time the actual step for a given procedure. For example learning clinical steps from procedures may allow or enable: adjustment of algorithm settings, the system to give the practical custom reminders, the system to notify staff of an estimate procedure end time, the system to alert staff if necessary equipment is not available in the room, and/or the system to alert staff of the occurrence of an emergency situation.

During a clinical procedure, the surgeon will often realize simple and routine surgical tasks such as grasping, retracting, cutting etc. Learning these different tasks may allow the cobot system to infer in real time preferences and habits of the surgeon regarding a sequence of a procedure in real time. Some algorithms of the cobot system may be tuned (i.e., adjusted and optimized) during the procedure based on this sequence recognition and help the user to be better at this

simple surgical task. An example of such a task is the automated retraction of a liver during a gall bladder procedure. By aggregating the information over many cases, the optimized force vectors may be developed.

Further, some complications may occur during a clinical procedure that may result in unexpected steps or surgical acts. Learning how to discriminate these unexpected events would help the cobot system to enable some specific safety features. In case of emergency, the robot arms may be stopped or motion restricted depending on the level of emergency detected by the system.

Referring now to FIGS. 27A to 27D, setup of the co-manipulation surgical system is provided. Platform 2700 may be constructed similar to platform 100, such that platform 2700 supports one or more robot arms, e.g., robot arm 300a' and robot arm 300b', and may cause the robot arms to move relative to platform 2700. As shown in FIG. 27A, platform 2700 may be moved to a desirable position relative to patient table PT by a user, e.g., via wheels 104', while robot arms 300a', 300b' are in their respective stowed configurations.

As platform 2700 is being moved toward the patient, the scene may be directly observed by a depth mapping sensor, e.g., optical scanner 1100', which may be mounted on platform 2700. From the depth maps observed and generated by optical scanner 1100', key features may be identified such as, for example, the height and/or location of patient table PT, the surface of the patient's abdomen, position and other characteristics of the surgeon, including the surgeon's height, and the trocar port(s), the base of robot arms 300a', 300b', e.g., base portions 302a', 302b' and shoulder portions 304a', 304b', robot arms 300a', 300b', and/or one or more surgical instruments coupled with the robot arms. Identification of such key features may be carried out using standard computer vision techniques such as template matching, feature tracking, edge detection, etc. As each feature is registered, its position and orientation may be assigned a local co-ordinate system and transformed into the global co-ordinate system the system using standard transformation matrices. Once all features are transformed into a single global co-ordinate system, an optimization algorithm, e.g., least squares and gradient descent, may be used to identify the most appropriate vertical and horizontal positions of robot arms 300a', 300b', which may be adjusted via platform 2700, to maximize the workspace of the robot arms with respect to the insertion point on the patient. The optimal workspace may be dependent on the surgical operation to be performed and/or the surgeon's preferred position.

As shown in FIG. 27B, when platform 2700 is in its desired position relative to patient table PT, such that wheels 104' are locked, robot arms 300a', 300b' may be extended away from their respective stowed configurations. As shown in FIG. 27C, the vertical position of the robot arms relative to platform 2700 may be adjusted to the desired position, and as shown in FIG. 27D, the horizontal position of the robot arms relative to platform 2700 may be adjusted to the desired position.

Referring now to FIGS. 28A to 28D, screenshots of exemplary graphical user interface 2800 are provided. Exemplary graphical user interface 2800 may be configurable by a user and may be integrated with display 110. FIG. 28A illustrates an exemplary start menu. The operator may initiate operation of the co-manipulation system by actuating the "start" option. FIG. 28B illustrates an exemplary system setup screen. As shown in FIG. 28B, when the system includes two robot arms, graphical user interface 2800 may identify which robot arm is to be used with which instru-

ment, e.g., retractor arm 2806 and endoscope arm 2808, as well as the procedure to be completed. Graphical user interface 2800 may permit the user to pre-load specific calibration files or setup joint positions based on the procedure being performed and/or the surgeon performing the procedure. For example, if the user inputs that a procedure is a laparoscopic cholecystectomy, the system may pre-load tool types known to be associated with that procedure. Populating these pre-loaded settings may be achieved by monitoring which tools a user manually selects for a given procedure. If a given tool is consistently selected for a predetermined number of procedures, the system may automatically pre-populate that tool the next time the procedure is selected by the user.

In addition, the operator may adjust the vertical and horizontal position of each robot arm, as shown in FIGS. 27C and 27D above. As shown in FIG. 28B, to adjust the vertical and/or horizontal position of the robot arm that will be or is currently coupled to the retractor device, the operator may toggle adjustment actuator 2802, and to adjust the vertical and/or horizontal position of the robot arm that will be or is currently coupled to the endoscope device, the operator may toggle adjustment actuator 2804. In some embodiments, the user may adjust the horizontal and vertical position of the robot arms by using the robot arm as a force sensitive input device. For example, the robot arm may be configured to sense the user's intention by measuring the force applied by the user onto the robot arm. If the user applies a force in the positive horizontal direction, platform may move the robot arm in that direction until the user no longer applies a force. A similar approach be taken for the other directions, e.g., negative horizontal, positive vertical, and negative vertical. As shown in FIG. 28B, graphical user interface 2800 may indicate whether an error, e.g., fault condition, is detected by the system during setup or operation of the system, via error notification 2810.

As shown in FIG. 28C, graphical user interface 2800 may display information associated with the selected surgical instruments, as described above. For example, graphical user interface 2800 may display, for each instrument to be coupled to each robot arm, the instrument type, overall length, distance between the coupler body and the instrument tip, distance between the center of mass to the instrument tip, mass, and the preset unlocking force required to unlock the instrument. As shown in FIG. 28C, graphical user interface 2800 may permit the operator to select between a high or low unlocking force of the surgical instrument. In addition, graphical user interface 2800 may permit the operator to initiate a surgical instrument calibration, e.g., for a new surgical instrument that does not already have an associated calibration file stored in the system. FIG. 28D illustrates an exemplary screen during operation of the system, e.g. during a surgical procedure. As shown in FIG. 28D, graphical user interface 2800 may display the trocar force and the force being applied to the tip of the surgical instrument, e.g., by tissue within the patient's body.

Referring now to FIG. 29, an alternative co-manipulation surgical robot system is provided. System 2900 may be constructed similar to system 200 of FIG. 2. For example, platform 1400', base portion 302', shoulder portion 304', encoders E1', E2', E3', E5', E6', E7', motor M1', shoulder joint 318', shoulder link 305', elbow joint 322', elbow link 310', wrist portion 311', and coupler interface 400' for coupling surgical instrument SI to the robot arm, may be constructed similar to platform 1400, base portion 302, shoulder portion 304, encoders E1, E2, E3, E5, E6, E7, motor M1, shoulder joint 318, shoulder link 305, elbow joint

322, elbow link 310, wrist portion 311, and coupler interface 400, respectively. System 2900 differs from system 200 in that system 2900 includes motors disposed at the joints of the robot arm. For example, system 2900 may include motor M2' disposed at elbow joint 318' and motor M3' disposed at elbow joint 322', configured to rotate the associated links to manipulate the robot arm. In addition, encoder E4' may be positioned on or adjacent to elbow joint 322'.

Some implementations of the systems described herein may be configured to be controlled or manipulated remotely, e.g., via joystick or other suitable remote control device, computer vision algorithm, force measuring algorithm, and/or by other means. However, in a preferred embodiment, the systems described herein operate without any telemetry, e.g., the robot arm is not teleoperated via a remote surgeon console separate from the robot arm, but instead the robot arm moves in response to movement applied to the surgical instrument coupled thereto. Any robot-assisted movements applied to the surgical instrument by the system, e.g., in the robotic assist mode, are not responsive to user input received at a remote surgeon console.

FIG. 30A illustrates a top view of coupler 3000 for coupling surgical instrument SI to the robot arm, showing coupler body 3002 (also referred to herein as a body) coupled with coupler interface 3001 (also referred to herein as an interface). FIG. 2B illustrates a top view of coupler 3000 of FIG. 30A, showing coupler body 3002 decoupled from coupler interface 3001. As shown in FIGS. 30A and 30B, coupler 3000 may have coupler body 3002 and coupler interface 3001. Coupler interface 3001 may be coupled with robotic arm 300 and may be configured such that coupler body 3002 may be removably coupled with coupler interface 3001. Coupler body 150 may be coupled with surgical instrument SI at any desired axial position on surgical instrument SI. Once coupler body 3002 is coupled with surgical instrument SI, coupler body 3002 and surgical instrument SI that is coupled with coupler body 3002 may be coupled with coupler interface 3001. Coupler body 3002 may be configured such that, once coupler body 3002 is coupled with surgical instrument SI, surgical instrument SI may be at least inhibited (e.g., prevented) from moving axially or, in some embodiments, moving axially and rotationally relative to coupler body 3002. Coupler 3000 may be configured such that coupler body 3002 may be at least inhibited (e.g., prevented) from moving in any axial direction relative to coupler interface 3001. In some embodiments, coupler 3000 may be configured such that coupler body 3002 is free to rotate relative to coupler interface 3001. In this configuration, surgical instrument SI coupled with coupler body 3002 may be free to rotate relative to coupler interface 3001 that coupler body 3002 is coupled with, and may be at least inhibited from (e.g., prevented from) any axial movement relative to coupler interface 3001 that coupler body 3002 is coupled with.

In other embodiments, coupler 300 may be configured such that surgical instrument SI may be moved in an axial direction relative to coupler body 3002 upon the application of at least a threshold force on surgical instrument SI relative to coupler body 3002 or upon actuation of a release or a state change of coupler body 3002. Such actuation may be achieved in some embodiments by, e.g., pressing a button, loosening a locking screw or other connector, moving a dial, or otherwise changing coupler 3000, coupler body 3002, and/or coupler interface 3001 from a second, secured state to a first, unsecured state. For example, in some embodiments, surgical instrument SI may be axially repositioned relative to coupler 3000 by loosening one or more thumb-

screws 3010 or other hand-operated fastener or fastening mechanism such as a clamp in coupler body 3002, repositioning surgical instrument SI in the desired axial position, and re-tightening thumbscrew 3010 or other hand-operated fastener or fastening mechanism.

As shown in FIG. 30B, coupler interface 3001 may have recess 3003 sized and shaped to receive coupler body 3002. Recess 3003 may inhibit (e.g., prevent) an axial movement or, in some embodiments, an axial and a rotational movement of coupler body 3002 relative to coupler interface 3001 while permitting free rotational movement of coupler body 3002 relative to coupler interface 3001. Coupler 3000 may be configured such that surgical instrument SI may be at least inhibited (e.g., prevented) from rotational movement relative to coupler 3000. This may be achieved by at least inhibiting (e.g., preventing) the rotational movement between surgical instrument SI and coupler 3000, or between coupler body 3002 and coupler interface 3001. In some embodiments, a surgical drape may be pinched or clamped between coupler body 3002 and coupler interface 3001.

FIG. 30C illustrates an end view of coupler body 3002 and surgical instrument SI, showing coupler body 3002 in the first, unsecured or open state in which surgical instrument SI may be removed and replaced or repositioned relative to coupler body 3002. FIG. 30D illustrates an end view of coupler body 3002 of FIG. 30C, showing coupler body 3002 in the second, secured or closed state in which surgical instrument SI may be at least inhibited (e.g., prevented) from axial movement or, in some embodiments, axial and rotational movement relative to coupler body 3002. In some embodiments, coupler body 3002 may have first portion 3004 and second portion 3006. In some embodiments, first portion 3004 may be rigidly coupled with second portion 3006 via hinge 3005 or shaft or otherwise. In some embodiments, first and second portions 3004, 3006 may have a semicircular cut out or recess 3008 therein sized and shaped to receive surgical instrument SI therein. Fastener 3010 may be used to couple first portion 3004 with second portion 3006, such as when surgical instrument SI is positioned in recesses 3008, as shown in FIG. 30D. As described above, coupler body 3002 may be configured to at least substantially inhibit (e.g., prevent) an axial movement or, in some embodiments, an axial and a rotational movement of surgical instrument SI relative to coupler body 3002. Rubber pads, sheets, bumps, O-rings, projections, or other components or features configured to grip an outside of surgical instrument SI may be used with any of the coupler embodiments disclosed herein. For example, the rubber interface may be positioned within the recess or recesses of the coupler body, such as recesses 3008 of first portion 3004 and/or second portion 3006 of coupler body 3002 and may be coupled to coupler body 3002. The rubber may be a silicone rubber or any other suitable type of rubber.

FIGS. 31A to 31D illustrate another embodiment of coupler 3100 that may be used with any robotic system embodiments disclosed herein to couple an instrument to an end portion of a robot arm. Coupler 3100 may include coupler body 3101 and coupler interface 3120 that may have a recess or depression 3190 configured to receive coupler body 3101 therein. Coupler interface 3120 may be coupled with an end portion of robot arm 300. Coupler 3100 may have coupler body 3101 that removably or nonremovably couples directly with an end portion of robot arm 300.

As shown in FIG. 31A, coupler body 3101 may have cylindrical body portion 3102 having annular flange 3104 projecting away from the surface of cylindrical body portion

3102. Body portion **3102** may have opening **3106** extending axially through body portion **3102**. Opening **3106** may be sized and shaped to receive surgical instrument SI therein. Opening **3106** may be slightly larger than a diameter or outside size of surgical instrument SI. Coupler body **3101** may have one or more deflectable tabs **3108** (two being shown), or four or more deflectable tabs **3108** that may be configured to deflect radially inwardly so that, when tabs **3108** are deflected radially inwardly, tabs **3108** exert a force on an outside surface of surgical instrument SI. Coupler **3100** may be configured such that, when coupler body **3101** is positioned within recess **3109** of coupler interface **3120** and coupler interface **3120** is in a second, closed or secured state, coupler interface **3120** may exert a force or otherwise deflect tabs **3108** radially inward so as to grip surgical instrument SI and at least inhibit (e.g., prevent) an axial movement or axial and rotational movement of surgical instrument SI relative to coupler body **3101**. For example, tabs **3108** may have a greater thickness near distal end **3110** of tabs **3108** such that, in a relaxed state or in the first, open state, distal end **3110** of tabs **3108** may project or protrude away from an outside surface of body portion **3102** of coupler body **3101**. In this configuration, when coupler body **3101** is positioned within recess **3109** of coupler interface **3120**, moving coupler interface **3120** to the second, closed state may cause a force to be applied to distal end portions **3110** of the tabs **3108** to thereby deflect tabs **3108** inwardly against an outside surface of surgical instrument SI.

In some embodiments, recess **3109** may have enlarged portion **3111** sized and shaped to receive annular flange **3104** therein and to permit a rotational movement of flange **3104**, while also restricting or at least inhibiting (e.g., preventing) an axial movement of coupler body **3101** by providing an axial limit to the movement of annular flange **3104**. In this arrangement, surgical instrument SI may be axially advanced through opening **3106** of coupler body **3101** to any desired location. Thereafter, surgical instrument SI with coupler body **3101** coupled thereto may be positioned within recess **3109** of coupler interface **3120**. Coupler interface **3120** may be removably or non-removably coupled with an end portion of robot arm **300** of any of the co-manipulation surgical systems disclosed herein.

As shown in FIG. **31C**, rubber pads, sheets, bumps, O-rings, projections, or other gripping features **3112** (O-rings being shown) configured to grip an outside of surgical instrument SI may be positioned within coupler body **3101** to increase a frictional force between surgical instrument SI and coupler body **3101**. In some embodiments, one or more tabs **3108** may be configured to exert a force on gripping features **3112** when one or more tabs **3108** are deflected inwardly.

As shown in FIG. **31D**, coupler interface **3120** may have first portion **3105** that may be coupled with second portion **3103**. In some embodiments, first and second portions **3105**, **3103** may be rigid and may be coupled to one another via mechanical hinge **3107**. Alternatively, a living hinge, a shaft, one or more fasteners, or other components or features may be used to couple first and second portions **3105**, **3103** together. In some embodiments, second portion **3103** may be flexible and may be configured to extend over surgical instrument SI and/or a coupler body **3101** supported within recess **3109**, such as an elastically elongatable or an elastically rigid strap. Additional fasteners, clamps, clasps, or other components or features may be used in conjunction with or in place of hinge **3107** to securely couple first and second portions **3105**, **3103** together once coupler body

3101 is received within recess **3109** of coupler interface **3120** to securely couple surgical instrument SI with coupler **3100**.

In some embodiments, the coupler may include a coupler body and a coupler interface having a recess configured to receive the coupler body. The coupler body may have an opening extending axially therethrough configured to receive an instrument and an annular flange extending around an outside surface thereof. The recess in the coupler interface may have an enlarged portion configured to receive the annular flange and to permit a rotational movement of the flange while at least inhibiting (e.g., preventing) an axial movement of the coupler body by providing an axial limit to the movement of the annular flange. The coupler interface may be configured to couple with an end portion of a robotic arm.

FIGS. **32A** and **32B** illustrate coupler body **3200** that may be used with any robotic system embodiments disclosed herein to couple an instrument to an end portion of a robot arm. Coupler body **3200** may have any of the components, features, and/or other details of any of the other embodiments of the coupler body disclosed herein, in any combination with any of the components, features, and/or other details of the embodiment of coupler body **3200** shown in FIGS. **32A** and **32B**. Any of the other embodiments of the coupler body disclosed herein may have any of the components, features, and/or other details of coupler body **3200**, in any combination with any of the components, features, and/or other details of the other coupler body embodiments disclosed herein.

Coupler body **3200** may have opening **3202** axially therethrough sized and shaped to receive a surgical instrument therein and clamping mechanism **3204** configured to reduce an inside diameter of opening **3202** as clamping mechanism **3204** is actuated so as to cause coupler body **3200** to move from the first, unsecured or open state as shown in FIG. **32A** to the second, secured or closed state as shown in FIG. **32B**. In this arrangement, coupler body **3200** may be positioned around an outside surface of the surgical instrument while coupler body **3200** is in the first, open or unsecured state. Thereafter, clamping mechanism **3204** may be actuated so as to cause coupler body **3200** to secure itself to an outside surface of a surgical instrument. Then, coupler body **3200** may be coupled with a coupler interface sized and configured to receive and support coupler body **3200**.

FIGS. **33A** and **33B** illustrate coupler body **3300** that may be used with any robotic system embodiments disclosed herein to couple an instrument to an end portion of a robot arm. Coupler body **3300** may have any of the components, features, and/or other details of any of the other embodiments of the coupler body disclosed herein, in any combination with any of the components, features, and/or other details of the embodiment of coupler body **3300**. Any of the other embodiments of the coupler body disclosed herein may have any of the components, features, and/or other details of coupler body **3300**, in any combination with any of the components, features, and/or other details of the other coupler body embodiments disclosed herein.

Coupler body **3300** may have an opening **3302** axially therethrough sized and shaped to receive a surgical instrument therethrough and clamping mechanism **3304** having a first and second handle member or tab configured to reduce an inside diameter of opening **3302** as clamping mechanism **3304** is actuated so as to cause coupler body **3300** to move from the first, unsecured or open state as shown in FIG. **33A** to the second, secured or closed state as shown in FIG. **33B**. In this arrangement, coupler body **3300** may be positioned

81

around an outside surface of the surgical instrument while coupler body 3300 is in the first, open or unsecured state. Coupler body 3300 may be moved to the first, open or unsecured state by squeezing or moving the handles of clamping mechanism 3204 together, as shown in FIG. 33A. Thereafter, clamping mechanism 3204 may be released so as to cause coupler body 3300 to secure itself to an outside surface of a surgical instrument. Coupler body 3300 may then be coupled with a coupler interface sized and configured to receive and support coupler body 3300.

FIGS. 34A to 34C illustrate coupler 3400 that may be used with any robotic system embodiments disclosed herein to couple an instrument to an end portion of a robot arm. Coupler 3400 may have any of the components, features, and/or other details of any of the other coupler embodiments disclosed herein, in any combination with any of the components, features, and/or other details of the embodiment of coupler 3400. Any of the other coupler embodiments disclosed herein may have any of the components, features, and/or other details of coupler 3400, in any combination with any of the components, features, and/or other details of the other coupler embodiments disclosed herein.

Coupler 3400 may have one or more coupler bodies 3402 (two being shown) coupled with coupler interface 3404. Coupler bodies 3402 may be slidably received within openings 3406 in coupler interface 3404. Coupler interface 3404 may have recess 3408 which may have a semicircular cross-sectional shape or other cross-sectional shape that matches a shape of an outside surface of the surgical instrument extending along a length thereof that may be configured to receive an outside surface of surgical instrument SI therein. Coupler bodies 3402 may have a curved end portion 3410 sized and shaped to route or curve at least partially around an outside surface of surgical instrument SI. In this configuration, coupler bodies 3402 when in a second, secured or closed position as shown in FIG. 34A, may be used to selectively secure surgical instrument SI in recess 3408 or otherwise secure surgical instrument SI to coupler interface 3404. Springs or other biasing mechanisms 3412 may be used to bias coupler bodies 3402 in the second, closed or secured position, as shown in FIG. 34A. The user may push coupler bodies 3402 in the axial direction indicated by arrow A1 so as to move coupler bodies 3402 from the second, closed or secured position to the first, open or unsecured position. The force exerted on coupler bodies 3402 should be greater than the spring or biasing force from the spring or biasing mechanisms 3412 coupled with each of coupler bodies 3402.

As shown in FIG. 34B, coupler bodies 3402 may have sloped end surface 3414. Sloped end surface 3414 may be configured such that a space between coupler end surface 3414 and an adjacent surface of coupler interface 3404 is greater at a position of coupler end surface 3414 that is further away from the recess such that, as surgical instrument SI is advanced laterally toward recess 3408 in coupler interface 3404, an outside surface of surgical instrument SI may contact end surface 3414 of the coupler body and the slope of end surface 3414 of coupler body 3400 will cause coupler body 3400 to move from the second, closed or secured state toward a first, open or unsecured state to permit surgical instrument SI to be received within recess 3408. Coupler body 3400 may have a spring or other biasing mechanism 3416 configured to bias coupler body 3400 to the second, closed or secured state or position.

As shown in FIG. 34C, sloped end surface 3414 of any embodiments of coupler bodies 3402 may be sloped such that, as surgical instrument SI is advanced in a downward

82

direction relative to end surface 3418 of coupler body 3400, such interaction between an outside surface of surgical instrument SI and sloping surface 3418 of coupler body 3402 may cause coupler body 3400 to rotate about pivot point 3420 away from recess 3408 and permit surgical instrument SI to be received within recess 3408.

FIGS. 35A to 35D illustrate coupler 3500 that may be used with any robotic system embodiments disclosed herein to couple an instrument to an end portion of a robot arm. Coupler 3500 may have any of the components, features, and/or other details of any of the other coupler embodiments disclosed herein, in any combination with any of the components, features, and/or other details of the embodiment of coupler 3500. Any of the other coupler embodiments disclosed herein may have any of the components, features, and/or other details of the coupler 3500, in any combination with any of the components, features, and/or other details of the other coupler embodiments disclosed herein.

Coupler 3500 may have coupler body 3502 that may be coupled with or engaged with coupler interface 3504. For example, coupler body 3502 may be slidably received within recess 3506 formed in coupler interface 3504. Coupler body 3502 also may have recess 3505 that may have a semicircular cross-sectional shape or other cross-sectional shape that matches a shape of an outside surface of the surgical instrument extending along a length of coupler body 3502 that may be configured to receive and at least partially surround, or in some embodiments fully surround, an outside surface of surgical instrument SI at least when coupler 3500 is in the second state, as shown in FIG. 35B.

Coupler body 3502 may be made from a flexible material, such as rubber including neoprene. Coupler body 3502 may have a width that is greater than a width of the recess and may be biased toward a planar or generally planar shape, as shown in FIG. 35A. Coupler body 3502 may be flexible enough such that, when coupler body 3502 is forced toward a distal surface 3506a of recess 3506, coupler body 3502 will bend or fold about a middle portion or other portion adjacent to recess 3505. Once coupler body 3502 is fully advanced into recess 3506 of coupler interface 3504, coupler 3500 may be configured to bias coupler body 3502 to remain within the second, secured position within recess 3506. In this configuration, to secure surgical instrument SI in coupler 3500, an operator can advance surgical instrument SI into recess 3505 of coupler body 3502, and continue to advance surgical instrument SI and/or coupler body 3502 toward distal surface 3506a. Some embodiments of coupler 3500 may be configured such that, once coupler body 3502 and surgical instrument SI have been advanced into recess 3506 of coupler interface 3504, surgical instrument SI will be axially and/or rotationally secured to coupler 3500. Thereafter, coupler 3500 may be coupled with an end portion of robot arm 300 such that robot arm 300 may be coupled with surgical instrument SI. In any embodiments, the recess may have sloped, curved, or otherwise tapered leading edge surfaces 3507 leading into the recess to facilitate the advancement of coupler body 3502 into recess 3506 of coupler interface 3504.

As shown in FIG. 35E, surgical drape 800 may be positioned between surgical instrument SI and coupler body 3302. In other embodiments, surgical drape 800 may be integrated into coupler body 3052 so that coupler body 3502 may form a portion of the surgical drape, as shown in FIG. 35F. Coupler body 3502 may be flexible enough to return to the original shape of coupler body 3502 once coupler body 3502 is removed from recess 3506. In any embodiments

disclosed herein, the coupler body or other components or features of the coupler can be configured to radially restrain the instrument.

As shown in FIG. 35C, coupler 3500 may be configured such that coupler body 3502 has a projection 3503 configured to extend into recess 3506 of coupler interface 3504 even when coupler body 3502 is in the first, open or unsecured state as shown in FIG. 35C. Projection 3503 may help bias coupler body 3502 to remain engaged with recess 3506 of coupler interface 3504 even when coupler body 3502 is in the first, open or unsecured state. As shown in FIG. 35D, coupler body 3502 also may have protrusions, flanges, handles, tabs, or other projections 3509 at a proximal end portion thereof configured to facilitate gripping and removal of coupler body 3502 from recess 3506.

In some embodiments, the coupler may include a coupler body made from a flexible material and a coupler interface having a recess configured to receive the coupler body. The coupler body may have a recess having a curved profile along a length of a first main surface thereof that is configured to receive an instrument therein. The coupler body may be flexible enough such that, when the coupler body is forced toward a distal surface of the recess, the coupler body will fold about a portion thereof adjacent to the recess, thereby at least axially and radially restraining the instrument. The coupler body may be flexible enough to return to the original shape of coupler body 3502 once the coupler body is removed from the recess.

FIG. 36 illustrates coupler 3600 that may be used with any robotic system embodiments disclosed herein to couple an instrument to an end portion of a robot arm. Coupler 3600 may have any of the components, features, and/or other details of any of the other coupler embodiments disclosed herein, in any combination with any of the components, features, and/or other details of the embodiment of coupler 3600. Any of the other coupler embodiments disclosed herein may have any of the components, features, and/or other details of coupler 3600, in any combination with any of the components, features, and/or other details of the other coupler embodiments disclosed herein.

Coupler 3600 may have a coupler body 3602 that may be coupled with or engaged with coupler interface 3604. For example, coupler body 3602 may be received within recess 3606 formed in coupler interface 3604. Coupler body 3602 also may have recess 3615 that may have a semicircular cross-sectional shape or other cross-sectional shape that matches a shape of an outside surface of the surgical instrument extending along a length of coupler body 3602 that may be configured to receive and at least partially surround, or in some embodiments fully surround, an outside surface of surgical instrument SI at least when coupler 3600 is in the second state.

Coupler body 3202 may be made from a flexible material, such as rubber including neoprene. Other embodiments of coupler body 3202 may be made from multiple materials, including first layer 3608 made from a flexible material that may have increased gripping such as a rubber and second layer 3610 that may be a backing layer or support layer for first layer 3608 may be made from a more rigid material, such as plastic, metal, or otherwise. Recess 3615 may be formed in first layer 3608. Recess 3615 may be formed in a middle portion of first layer 3608. Some embodiments of second layer 3610 may have hinge 3612 in or attached to a middle portion thereof. In some embodiments, hinge 3612 may run generally parallel to recess 3615 formed in first layer 3608 and recess 3606 formed in coupler interface 3604. In some embodiments, coupler body 3602 may fold or

hinge between the first, open state and the second, closed or secured state about surgical instrument SI by folding or hinging about hinge 3612.

Coupler body 3600 may have a width that is greater than a width of recess 3606. Coupler body 3602 may be configured such that, when coupler body 3602 is forced toward distal surface 3606a of recess 3606, coupler body 3602 will bend or fold about hinge 3612 so as to collapse or close about surgical instrument SI positioned within recess 3615 of coupler body 3602 so as to secure surgical instrument SI within coupler body 3602 and coupler interface 3604.

Some embodiments of coupler interface 3604 may have one or more rollers 3614 (two being shown) at proximal end 3606b of the recess 3606 formed in coupler interface 3604. The one or more rollers 3614 may facilitate the movement of coupler body 3602 into recess 3606 by permitting coupler body 3602 to roll on the rollers as coupler body 3602 is advanced into recess 3606. Some embodiments of coupler interface 3604 may have additional rollers 3616 along the side wall surfaces 3606c of recess 3606 to continue to facilitate the advancement of coupler body 3602 into recess 3606. In some embodiments, recess 3606 may have a generally rectangular shape. In other embodiments, recess 3606 may have a tapered or narrowing profile.

Once coupler body 3602 is fully advanced into recess 3606 of coupler interface 3604, some embodiments of coupler 3600 may be configured to bias coupler body 3602 to remain within the second, secured position within recess 3606. In this configuration, to secure surgical instrument SI in coupler 3600, an operator may advance surgical instrument SI into recess 3615 of coupler body 3602, and continue to advance surgical instrument SI and/or coupler body 3602 toward distal surface 3606a of recess 3606. Some embodiments of coupler 3600 may be configured such that, once coupler body 3602 and surgical instrument SI have been advanced into recess 3606 of coupler interface 3604, surgical instrument SI will be axially and/or rotationally secured to coupler 3600. Thereafter, coupler 3600 may be coupled with an end portion of robot arm 300 such that robot arm 300 may be coupled with surgical instrument SI.

FIG. 37 illustrates coupler 3700 that may be used with any robotic system embodiments disclosed herein to couple an instrument to an end portion of a robot arm. Coupler 3700 may have any of the components, features, and/or other details of any of the other coupler embodiments disclosed herein, in any combination with any of the components, features, and/or other details of the embodiment of coupler 3700. Any of the other coupler embodiments disclosed herein may have any of the components, features, and/or other details of coupler 3700, in any combination with any of the components, features, and/or other details of the other coupler embodiments disclosed herein.

Coupler 3700 may have coupler body 3702 that may be coupled with or engaged with coupler interface 3704. Coupler body 3702 may be received within recess 3796 formed in coupler interface 3704. Coupler body 3702 also may have recess 3705 that may have a semicircular cross-sectional shape or other cross-sectional shape that matches a shape of an outside surface of the surgical instrument extending along a length of coupler body 3702 that may be configured to receive and at least partially surround, or in some embodiments fully surround, an outside surface of surgical instrument SI at least when coupler 3704 is in the second state, as shown in FIG. 37.

Coupler body 3702 may be made from multiple materials, including first layer 3710 made from a flexible material that may have increased gripping such as a rubber and second

85

layer 3712 that may be a backing layer or support layer for first layer 3710 may be made from a more rigid material, such as plastic, metal, or otherwise. Recess 3705 may be formed in first layer 3710. In some embodiments, recess 3705 may be formed in a middle portion of first layer 3710. Some embodiments of second layer 3712 may have hinge 3714 in or attached to a middle portion thereof. In some embodiments, hinge 3714 may run generally parallel to recess 3705 formed in first layer 3710 and recess 3706 formed in coupler interface 3704. In some embodiments, coupler body 3702 may fold or hinge between the first, open state and the second, closed or secured state about surgical instrument SI by folding or hinging about hinge 3714.

Coupler body 3702 may have a width that is greater than a width of recess 3706. Coupler body 3702 may be configured such that, when coupler body 3702 is forced toward a distal surface 3706a of the recess 3706, coupler body 3702 will bend or fold about hinge 3714 so as to collapse or close about surgical instrument SI positioned within recess 3705 of coupler body 3702 so as to secure surgical instrument SI within coupler body 3702 and coupler interface 3704. In some embodiments, second layer 3712 may have wings or tabs 3716 that may be used to facilitate removal of coupler body 3702 from recess 3706. Tabs 3716 may be formed such that, when coupler body 3702 is in the second position, as shown in FIG. 37, tabs 3716 may be spaced apart from first surface 3704a (which can be an upper surface when coupler interface 3704 is positioned as shown in FIG. 37) such that a gap or space 3720 exists between tabs 3716 and upper surface 3704a of coupler interface 3704. Space 3720 may be large enough to permit tabs 3716 to move toward first surface 3704a when a force is applied to tabs 3716 in the direction of first surface 3704a. As tabs 3716 are deflected toward first surface 3704a, such movement of tabs 3716 may force a remainder of coupler body 3702 to move away from a distal surface 3706a of recess 3706, thereby allowing coupler body 3704 to be removed from recess 3706.

Some embodiments of coupler interface 3704 may have one or more rollers 3717 (two being shown) at proximal end 3706b of recess 3706 formed in coupler interface 3704. The one or more rollers 3717 may facilitate the movement of coupler body 3702 into recess 3706 by permitting coupler body 3702 to roll on the rollers as coupler body 3702 is advanced into recess 3706. Some embodiments of coupler interface 3704 may have additional rollers 3718 along the side wall surfaces 3706c of recess 3706 to continue to facilitate the advancement of coupler body 3702 into recess 3706.

Once coupler body 3702 is fully advanced into recess 3706 of coupler interface 3704, some embodiments of coupler 3700 may be configured to bias coupler body 3702 to remain within the second, secured position within recess 3706. In this configuration, to secure surgical instrument SI in coupler 3700, an operator may advance surgical instrument SI into recess 3705 of coupler body 3703, and continue to advance surgical instrument SI and/or coupler body 3702 toward distal surface 3706a of recess 3706. Some embodiments of coupler 3700 may be configured such that, once coupler body 3702 and surgical instrument SI have been advanced into recess 3706 of coupler interface 3704, surgical instrument SI will be axially and/or rotationally secured to coupler 3700. Thereafter, coupler 3700 may be coupled with an end portion of robot arm 300 such that robot arm 300 may be coupled with surgical instrument SI.

FIGS. 38A and 38B illustrate coupler 3800 that may be used with any robotic system embodiments disclosed herein to couple an instrument to an end portion of a robot arm.

86

Coupler 3800 may have any of the components, features, and/or other details of any of the other coupler embodiments disclosed herein, in any combination with any of the components, features, and/or other details of the embodiment of coupler 3800. Any of the other coupler embodiments disclosed herein may have any of the components, features, and/or other details of coupler 3800, in any combination with any of the components, features, and/or other details of the other coupler embodiments disclosed herein.

Coupler 3800 may have coupler body 3802 that may be coupled with or engaged with a coupler interface (not shown) or may be coupled with or engaged with a robot arm without the presence of a coupler interface (e.g., the coupler body of any embodiments disclosed herein can directly engage or interface with an end portion of robot arm 300). Coupler body 3802 may have first portion 3804 and second portion 3806 coupled with first portion 3804. In some embodiments, first portion 3804 may be hingedly or rotatably coupled with second portion 3806. For example, coupler body 3802 may have a hinge or joint 3810 that may couple first and second portions 3804, 3806 together.

In some embodiments, first portion 3804 of coupler body 3802 may have proximal portion 3804a and distal portion 3804b that is integrally formed with or coupled with proximal portion 3804a. First portion 3804 of coupler body 3802 may have recess 3812 and second portion 3806 of coupler body 3802 may have recess 3814, each of which can have a semicircular cross-sectional shape or other cross-sectional shape that matches a shape of an outside surface of the surgical instrument extending along a length of coupler body 3802 that may be configured to receive and at least partially surround, or in some embodiments fully surround, an outside surface of surgical instrument SI at least when coupler 3800 is in the second state. The second state of coupler body 3802 is shown FIG. 38B. In some embodiments, second portion 3806 may be similarly situated and may be a mirror copy of first portion 3804, with proximal portion 3806a and distal portion 3806b that is integrally formed with or coupled with the proximal portion 3806a.

Some embodiments of coupler 3800 may be configured to be bistable in that the coupler 3800 will be biased toward either the first, open or unsecured state or the second, closed or secured state and is unstable in any position or state except the first and second states. In the first state, distal portion 3804b of first portion 3804 of coupler 3800 is in contact with the distal portion 3806b of second portion 3806 of coupler 3800 and proximal portion 3804a of first portion 3804 of coupler 3800 is rotated away and spaced apart from proximal portion 3806a of second portion 3806 of coupler 3800. In the first, open or unsecured state, surgical instrument SI may be loaded into or removed from coupler 3800. In the second state, proximal portion 3804a of first portion 3804 of coupler 3800 is in contact with proximal portion 3806a of second portion 3806 of coupler 3800 and distal portion 3804b of first portion 3804 of coupler 3800 is rotated away and spaced apart from distal portion 3806b of second portion 3806 of coupler 3800. In the second, closed or secured state, surgical instrument SI loaded into coupler 3800 may be secured or supported by coupler 3800 such that surgical instrument SI may be at least inhibited (e.g., prevented) from an axial movement or, in some embodiments, an axial and a rotational movement relative to the coupler 3800.

In this configuration, when coupler 3800 is in the first, open state as shown in 38A, after positioning surgical instrument SI in either recess 3812 with recess 3814, the operator may change coupler 3800 to the second, closed

state by pinching or moving the proximal portion **3804a** of first portion **3804** toward proximal portion **3806a** of second portion **3806**, such as by exerting a force on proximal portions **3804a**, **3806a** of first and second portions **3804**, **3806** along the directions **A3** and **A4**, as shown in FIG. **38A** (e.g., by squeezing the proximal portions **3804a**, **3806a** of first and second portions **3804**, **3806** together). When coupler **3800** is in the second, closed state as shown in FIG. **38B**, the operator may change coupler **3800** to the first, open state by pinching or moving distal portion **3804b** of first portion **3804** toward distal portion **3806b** of second portion **3806**, such as by exerting a force on distal portions **3804b**, **3806b** of first and second portions **3804**, **3806** along the directions **A5** and **A6**, as shown in FIG. **38B** (e.g., by squeezing distal portions **3804b**, **3806b** of first and second portions **3804**, **3806** together).

FIGS. **39A** and **39B** illustrate coupler **3900** that may be used with any robotic system embodiments disclosed herein to couple an instrument to an end portion of a robot arm. Coupler **3900** may have any of the components, features, and/or other details of any of the other coupler embodiments disclosed herein, in any combination with any of the components, features, and/or other details of the embodiment of coupler **3900**. Any of the other coupler embodiments disclosed herein may have any of the components, features, and/or other details of the coupler **3900**, in any combination with any of the components, features, and/or other details of the other coupler embodiments disclosed herein.

Coupler **3900** may have a coupler body **3902** that may be coupled with or engaged with a coupler interface (not shown) or may be coupled with or engaged with a robotic arm without the presence of a coupler interface. Coupler body **3902** may have one or more projections **3903** (two being shown) that may be used to center or position coupler body **3902** relative to the coupler interface. For example, projections **3903** may be conical projections configured to engage with depressions or openings in the coupler interface to align coupler body **3902** with the coupler interface. In some embodiments, the coupler interface may have an equal number or a different number of depressions or openings as compared to the number of projections **3903**. In other embodiments, projections **3903** may be cylindrically shaped. In some embodiments, coupler body **3902** may have three or more projections **3903**.

Coupler body **3902** may have first tab **3904** hingedly or rotatably coupled with coupler body **3902** and second tab **3906** hingedly or rotatably coupled with coupler body **3902**. For example, coupler body **3902** may have a first hinge or joint **3910** that may couple first tab **3904** with coupler body **3902** and a second hinge or joint **3911** that may couple second tab **3906** with coupler body **3902**. First tab **3904** may have proximal end portion **3904a** and distal end portion **3904b**, as shown in FIG. **39B**. Second tab **3906** may have proximal end portion **3906a** and distal end portion **3906b**.

Coupler body **3902** may have recess **3914** formed therein, first tab **3904** may have recess **3916** formed in a distal end portion thereof and second tab **3906** may have recess **3918** formed in a distal end portion thereof, each of which may have a semicircular cross-sectional shape or other cross-sectional shape that, all together, may match a shape of an outside surface of surgical instrument **SI** extending along a length of coupler body **3902**, first tab **3904**, and second tab **3906** and that may be configured to receive and at least partially surround, or in some embodiments fully surround, an outside surface of surgical instrument **SI** at least when coupler **3900** is in the second state. The second state of coupler body **3902** is shown in FIG. **39B**. In some embodi-

ments, second tab **3906** may be similarly situated and may be a mirror copy of first tab **3904**.

Some embodiments of coupler **3900** may be biased toward the second state, using springs or other torsional biasing elements. An operator may overcome the bias or otherwise move coupler body **3902** from the second state as shown in FIG. **39B** to the first state as shown in FIG. **39A** by squeezing together or toward one another proximal end portions **3904a**, **3906a** of first and second tabs **3904**, **3906**. In the first state, the operator may remove surgical instrument **SI** from coupler **3900**. To support a surgical instrument **SI** in coupler **3900**, while coupler **3900** is in the first, open state, the operator may position surgical instrument **SI** in contact with or near recess **3914** and release the force that was applied to first and second tab **3904**, **3906** or otherwise relax first and second tab **3904**, **3906** and allow first and second tabs **3904**, **3906** to return to the relaxed position of first and second tabs **3904**, **3906**.

FIGS. **40** to **43** illustrate additional couplers **4000**, **4100**, **4200**, **4300**. Couplers **4000**, **4100**, **4200**, **4300** may have any of the components, features, and/or other details of any of the other coupler embodiments disclosed herein, in any combination with any of the components, features, and/or other details of the embodiment of couplers **4000**, **4100**, **4200**, **4300**. Any of the other coupler embodiments disclosed herein may have any of the components, features, and/or other details of couplers **4000**, **4100**, **4200**, **4300** in any combination with any of the components, features, and/or other details of the other coupler embodiments disclosed herein.

As shown in FIG. **40**, coupler **4000** may have first body portion **4002** and second body portion **4004** that may be slidably coupled with or engaged with first body portion **4002**. Coupler **4000** may have a recess or opening **4006** that may be enlarged and may be configured to receive surgical instrument **SI** therein when second body portion **4004** is moved toward first body portion **4002**. A spring or other biasing mechanism **4008** may be used to bias coupler **4000** toward the second, closed or secured state so that, when an operator releases first and second body portions **4002**, **4004**, coupler **4000** may exert a force on a surgical instrument to secure the surgical instrument therein. Some embodiments of coupler **400** may be figured to axially restrain a surgical instrument therein, but to permit a rotation of the surgical instrument. Coupler **4000** may be coupled with a coupler interface or directly to an end portion of robot arm **300**.

As shown in FIG. **41**, coupler **4100** may have first body portion **4102** and second body portion **4104** that may be slidably coupled with or engaged with first body portion **4102**. Coupler **4100** may have a recess or opening **4106** that may be enlarged and may be configured to receive surgical instrument **SI** therein when second body portion **4104** is moved toward first body portion **4102**. Spring **4108** or other biasing mechanism may be used to bias coupler **4100** toward the second, closed or secured state so that, when an operator releases first and second body portions **4102**, **4104**, coupler **4100** may exert a force on a surgical instrument to secure the surgical instrument therein. Some embodiments of coupler **4100** may be figured to axially restrain a surgical instrument therein, but to permit a rotation of the surgical instrument. Coupler **4100** may be coupled with a coupler interface or directly to an end portion of robot arm **300**.

As shown in FIG. **42**, coupler **4200** may have first body portion **4202** having proximal end portion **4202a** and distal end portion **4202b** and second body portion **4204** having proximal end portion **4204a** and distal end portion **4204b** that may be rotatably coupled with or engaged with first

body portion **4202** about an axis or shaft **4207**. Coupler **4200** may have a recess or opening **4206** formed in distal end portions **4202b**, **4204b** that may be enlarged and may be configured to receive surgical instrument SI therein when distal end portion **4204b** of second body portion **4204** is rotated away from distal end portion **4202b** of first body portion **4202**. Spring **4208** or other biasing mechanism may be used to bias coupler **4200** toward the second, closed or secured state so that, when an operator releases first and second body portions **4202**, **4204**, coupler **4200** may exert a force on a surgical instrument to secure the surgical instrument therein. Some embodiments of coupler **4200** may be figured to axially restrain a surgical instrument therein, but to permit a rotation of the surgical instrument. Coupler **4200** may be coupled with a coupler interface or directly to an end portion of robot arm **300**.

As shown in FIG. 43, coupler **4300** may be configured to engage with a coupler interface or the distal end portion **4301** of a robot arm. Coupler **4300** may be constructed similar to coupler **4200**, with similar components having like-prime reference numerals. Coupler **4300** differs from coupler **4200** in that coupler **4300** may have projections **4302** extending inwardly from an inner surface of proximal end portion **4202a'** of first body portion **4202'** and an inner surface of proximal end portion **4204a'** of second body portion **4204'** that may be received within recesses **4304** formed in distal end portion **4301** of the robot arm when coupler **4300** is in the second, closed state.

Referring now to FIGS. 44A to 50B, another exemplary surgical instrument coupling mechanism is provided. FIGS. 44A to 44C illustrate coupling mechanism **4400** positioned at the distal end of distal wrist link **316** of wrist portion **311** of robot arm **300**. Coupling mechanism **4400** may include coupler interface **4500** at the distal end of the distal-most link of the robot arm (illustratively, link **316**), and coupler body **4600**, which may be configured to be removably coupled to a surgical instrument and to coupler interface **4500**, such that a sterile drape may be placed between coupler interface **4500** and coupler body **4600**. Accordingly, coupler body **4600** may be disposable, or alternatively, sterilizable between surgical procedures. Moreover, coupling mechanism **4400** may be operatively coupled to one or more sensors for detecting when coupler body **4600** is coupled to coupler interface **4500**, and when a surgical instrument is coupled to coupler body **4600** when coupler body **4600** is coupled to coupler interface **4500**, as well as the type/size of the surgical instrument coupled to coupler body **4600**, as described in further detail below.

FIGS. 45A to 45C illustrate coupler interface **4500** at the distal end of link **316** of the robot arm. As shown in FIGS. 45A and 45B, coupler interface **4500** may include protrusion **4504** extending from flat portion **4502**. Flat portion **4502** may have an outer diameter that coincides with the outer diameter of link **316**. Protrusion **4504** may have a non-circular profile, which corresponds to the geometry of groove **4605** of coupler body **4600**, as described in further detail below. Moreover, protrusion **4504** may include one or more locking portions **4506** disposed on the outer surface of the sidewall of protrusion **4504**. For example, locking portions **4506** may be indentations/grooves extending along the outer surface of protrusion **4504**, and sized and shaped to engage with locking arms **4606** of coupler body **4600**, as described in further detail below, for securing coupler body **4600** to coupler interface **4500**, and for securing the sterile drape between coupler body **4600** and coupler interface **4500**.

Preferably, protrusion **4504** includes at least a pair of locking portions **4506**. For example, as shown in FIG. 45A, protrusion **4504** may have a profile defined by two straight sides connected by two curved sides, such that a first pair of locking portion **4506** may be disposed on a portion of each of the curved sides adjacent to a first straight side of protrusion **4504**, and a second pair of locking portion **4506** may be disposed on a portion of each of the curved sides adjacent to the other straight side of protrusion **4504**. By having two pairs of locking portions **4506**, coupler body **4600** may be securely coupled to coupler interface **4500** in two orientations.

As shown in FIG. 45C, coupler interface **4500** may include extended portion **4508** configured to be inserted within link **316**. Moreover, coupler interface **4500** may include a metal rod, e.g., ferrous rod **4510**, extending through coupler interface **4500** from protrusion **4504** through extended portion **4508**. Link **316** may include one or more sensors, e.g., Hall effect sensors **4514**, configured to detect a magnetic field induced in ferrous rod **4510**. Preferably, link **316** includes at least two Hall effect sensors to provide redundancy for more accurate magnetic field measurements. As shown in FIG. 45C, Hall effect sensors **4514** may be positioned adjacent to a proximal end of ferrous rod **4510**. In addition, coupler interface **4500** may include repulsion magnet **4512** disposed within protrusion **4504** adjacent to the distal end of ferrous rod **4512**. Repulsion magnet **4512** is configured to apply a magnetic force to a magnet slidably disposed within coupler body **4600** to facilitate determination of when coupler body **4600** is coupled to coupler interface **4500** and no surgical instrument is coupled to coupler body **4600**, e.g., by causing the magnet to move to a position within coupler body **4600** with a maximum distance from ferrous rod **4510**, and/or to facilitate coupling of the surgical instrument to coupler body **4600**, as described in further detail below. Moreover, as described above, robot arm **300** may include one or more encoders E7 for measuring angulation of between middle wrist link **314** and distal wrist link **316** may be disposed on or adjacent to joint **328**, e.g., within link **316**. For example, encoders E7 may include two or more encoders positioned circumferentially around extended portion **4508** of coupler interface **4500**.

Referring now to 46A to 46E, coupler body **4600** is provided. Coupler body **4600** may be configured to be removably coupled to a surgical instrument having a pre-defined shaft diameter, e.g., a 10 mm surgical instrument. Coupler body **4600** is preferably designed to be locked to the distal end of the robot arm with a sterile drape therebetween such that the robot arm remains covered and sterile throughout a procedure. Further, coupler body **4600** also has a separate portion for locking to a surgical instrument (e.g., a commercially available laparoscopic instrument) to permit the clinician to perform the surgeries with the robot arm(s) as described herein. As shown in FIGS. 46A to 46C, coupler body **4600** may include coupler interface connection portion **4604** and surgical instrument connection portion **4602**. Coupler interface connection portion **4604** and surgical instrument connection portion **4602** may generally have an outer diameter that coincides with the outer diameters of flat portion **4502** of coupler interface **4504** and link **316**. As shown in FIG. 46C, coupler interface connection portion **4604** may include groove **4605** extending inward from a bottom surface of coupler body **4600**. Groove **4605** may have a geometry that corresponds with the profile shape of protrusion **4504** of coupler interface **4500**, such that protrusion **4504** may be received by groove **4605**. Accordingly, the geometry of groove **4605** may include two straight sides

connected by two curved sides. The sterile drape may be positioned between protrusion **4504** and groove **4605** when protrusion **4504** is disposed within groove **4605**. Preferably, the profile of protrusion **4504** and the corresponding geometry of groove **4605** are symmetrical such that protrusion **4504** may be received by groove **4605** in at least two orientations. Moreover, the profile of protrusion **4504** and the corresponding geometry of groove **4605** may guide the coupling of coupler body **4600** to coupler interface **4500** by the user.

Additionally, coupler interface connection portion **4604** may include a pair of locking arms **4606** configured to facilitate securing of coupler body **4600** to coupler interface **4500** when protrusion **4504** is disposed within groove **4605**. Each of locking arms **4606** may include handle portion **4610** sized and shaped to be actuated by the user's fingers, and connection portion **4608** sized and shaped to engage with locking portions **4506** of protrusion **4504**. For example, connection portion **4608** may have a tapered profile for securely engaging with locking portion **4506**. Locking arms **4606** may be pivotally coupled to coupler interface connection portion **4604**, such that locking arms **4606** may be transitionable between an unlocked state and a locked state. Moreover, locking arms **4606** may be pivotally coupled to coupler interface connection portion **4604** via a spring, e.g., a torsion spring, an extension spring, a compression spring, etc., such that locking arms **4606** are biased toward the locked state. Accordingly, handle **4610** may be actuated to transition locking arms **4606** from the locked state to the unlocked state.

As shown in FIGS. **46A** to **46C**, surgical instrument connection portion **4602** may include opening **4616** extending therethrough, sized and shaped to receive the shaft of a surgical instrument. For example, opening **4616** may be sized and shaped to receive a 10 mm surgical instrument shaft. Opening **4616** may be defined by a channel extending downward from an upper surface of surgical instrument connection portion **4602** such that a surgical instrument may be inserted into opening **4616** via the channel. As shown in FIGS. **46A** and **46B**, the upper surface of surgical instrument connection portion **4602** may include tapered portions **4614** that angle downward towards opening **4616**, thereby defining the channel into opening **4616**. Accordingly, tapered portions **4614** ensure that the shaft of the surgical instrument is properly inserted into opening **4616** in one of two orientations by rotating coupler body **4600** and accordingly distal wrist link **316** to align with the longitudinal axis of the surgical instrument in one of two orientations. For example, tapered portions **4614** may facilitate in "self-alignment" of the distal end of robot arm **300**, e.g., by causing distal wrist link **316**, which is coupled to coupler body **4600** via coupler interface **4500**, to automatically rotate relative to middle wrist link **314** about axis **Q7** at passive joint **328** as the instrument shaft is guided down tapered portions **4614**, such that the longitudinal axis of opening **4616** aligns with the longitudinal axis of the surgical instrument. Accordingly, the user does not need to align the instrument shaft to opening **4616**, but rather, opening **4616** rotates via rotation of coupler body **4600** and distal wrist link **316** relative to middle wrist link **314** to align with the longitudinal axis of the instrument shaft.

In addition, surgical instrument connection portion **4602** may include clamp **4618** pivotally coupled to surgical instrument connection portion **4602** about axis **4612**, such that clamp **4618** may be transitionable between an unlocked state and a locked state. Moreover, clamp **4618** may be pivotally coupled to surgical instrument connection portion **4602** via

a torsion spring, such that clamp **4618** is biased toward the locked state. Clamp **4618** may include locking portion **4620** configured to secure the surgical instrument within opening **4616** when clamp **4618** is in its locked state. For example, a lower surface of locking portion **4620** may define the upper surface of opening **4616** when clamp **4618** is in its locked state, such that locking portion **4620** prevents upward movement of the surgical instrument when the surgical instrument is positioned within opening **4616** and clamp **4618** is in its locked state.

The upper surface of locking portion **4620** may be tapered to facilitate guidance of the surgical instrument into opening **4616** along with tapered portions **4614**. For example, the upper surface of locking portion **4620** may have a tapered angle that more or less coincides with the tapered angles of tapered portions **4614**. Accordingly, in some embodiments, the tapered angle of locking portion **4620** may be alone sufficient to permit a surgical instrument to be inserted into opening **4616**, such that insertion of the surgical instrument towards opening **4616** applies a force against the tapered upper surface of locking portion **4620**, thereby causing clamp **4618** to rotate about axis **4612** from the locked state to the unlocked state to permit the surgical instrument to be received by opening **4616**. Clamp **4618** further may include handle **4622** sized and shaped to be actuated by the user's fingers to transition clamp **4618** from the locked state to the unlocked state. For example, handle **4622** may be actuated to transition clamp **4618** to the unlocked state for insertion of the surgical instrument into opening **4616**, and/or for removal of the surgical instrument from opening **4616**.

Moreover, coupler body **4600** further may include switch **4624** pivotally coupled to surgical instrument connection portion **4602**, and configured to facilitate securement of the surgical instrument within opening **4616**. For example, switch **4624** may include one or more surgical instrument engagement portions **4626**, each having a geometry that corresponds with the outer diameter of the shaft of the surgical instrument to be inserted within opening **4616**. In addition, switch **4624** may include handle portion **4628** sized and shaped to be actuated by the user's fingers to transition switch **4624** between an unlocked state and a locked state where surgical instrument engagement portion **4626** engages with the surgical instrument shaft within opening **4616** and applies a friction force to the surgical instrument shaft.

In some embodiments, switch **4624** may include two surgical instrument engagement portions **4626**, one on each side of switch **4624**, such that switch **4624** may be rotated from its unlocked state in either direction to transition to its locked state where one of the surgical instrument engagement portions will engage with the surgical instrument within opening **4616**. Accordingly, in its locked state, surgical instrument engagement portion **4626** further defines opening **4616**. Surgical instrument engagement portion **4626** may have a coefficient of friction, such that when the surgical instrument is disposed within opening **4616** and switch **4624** is in its locked state, surgical instrument engagement portion **4626** applies a friction force against the surgical instrument that prevents longitudinal movement of the surgical instrument relative to coupler body **4600**, while permitting rotational movement of the surgical instrument within opening **4616**. Accordingly, when the surgical instrument is disposed within opening **4616**, switch **4624** may be actuated to its unlocked state to permit the user to readjust/move the surgical instrument longitudinally relative to coupler body **4600** within opening **4616**, and back to its locked state to prevent longitudinal movement of the surgical

instrument relative to coupler body **4600**. Preferably, both switch **4624** and clamp **4618** must be in their unlocked states to permit removal of the surgical instrument from coupler body **4600**.

Coupler body **4600** further may include holder **4630** disposed within surgical instrument connection portion **4602**, as shown in FIG. **46E**. Referring now to FIG. **46D**, holder **4630** is provided. Holder **4630** is configured to be slidably disposed within channel **4603** within surgical instrument connection portion **4602**, e.g., toward or away from coupler interface connection portion **4604**. Moreover, holder **4630** is configured to hold magnet **4640**, and provide friction to the surgical instrument disposed within opening **4616** to prevent longitudinal movement of the surgical instrument within opening **4616** when switch **4624** is in its locked state. As shown in FIG. **46D**, holder **4630** may include a contact surface, e.g., friction pad **4632**, disposed on an upper portion of cradle **4634**, such that the upper surface of friction pad **4632** defines the lower surface of opening **4616**. The upper surface of friction pad **4632** may have a curved profile, which may coincide with the curvature of the surgical instrument. Friction pad **4632** may have a coefficient of friction, such that when the surgical instrument is disposed within opening **4616** and switch **4624** is in its locked state, friction pad **4632** applies a friction force against the surgical instrument that facilitates prevention of longitudinal movement of the surgical instrument relative to coupler body **4600** by switch **4624**, while permitting rotational movement of the surgical instrument within opening **4616**. Although FIG. **46D** shows friction pad **4632** as a single piece, friction pad **4632** may be formed of multiple pieces configured to contact the surgical instrument within opening **4616**, or alternatively, may be wrapped around the upper surface of holder **4630** or otherwise integrated with holder **4630**. When switch **4624** is moved to its unlocked state, the friction force of friction pad **4632** may not be sufficient to prevent longitudinal movement of the surgical instrument relative to coupler body **4600**.

As shown in FIG. **46D**, holder **4634** may include two cradles **4634** extending between friction pad **4632** and magnet harness **4638**. Accordingly, friction pad **4632** may extend longitudinally between cradles **4632**. Friction pad may be supported via a support extending between cradles **4634**. Moreover, each cradle **4634** of holder **4630** may include channel **4636** extending within cradle **4634** in a direction from magnet harness **4638** towards friction pad **4632**. Channels **4636** may be sized and shaped to slidably receive a longitudinally extending rod therethrough, such that the longitudinally extending rod extends along axis **4612** between channels **4636**. Clamp **4618** may be pivotally coupled to the longitudinally extending rod, such that clamp **4618** may rotate about axis **4612**, as described above. Axis **4612** may be fixed relative to surgical instrument connection portion **4602**, such that holder **4630** may move toward/away from coupler interface connection portion **4604** via movement of channel **4636** along the longitudinally extending rod.

As shown in FIG. **46D**, magnet harness **4638** may extend between cradles **4634**, and may be configured to securely hold magnet **4640**. Magnet harness **4638** may have a geometry that corresponds with the geometry of channel **4603**, such that magnet harness may slidably move within channel **4603**, but cannot move beyond the lower or upper surface of channel **4603**. Magnet **4640** may have a magnetic force such that when coupler body **4600** is coupled to coupler interface **4500**, magnet **4640** induces a magnetic field in ferrous rod **4510**, which may be detected by Hall effector sensors **4514**.

Accordingly, the strength of the magnetic field induced in ferrous rod **4510** will be proportional to the distance between magnet **4640** and ferrous rod **4510**, such that the magnetic field detected by Hall effector sensors **4514** may be indicative of the position of magnet **4640**, and accordingly holder **4630**, within coupler body **4600**. Similarly, when no magnetic field is induced in ferrous rod **4510** via magnet **4640**, Hall effector sensors **4514** may detect that coupler body **4600** is not coupled to coupler interface **4500**. Moreover, repulsion magnet **4512** of coupler interface **4500** may have a magnetic force such that when coupler body **4600** is coupled to coupler interface **4500**, repulsion magnet **4512** applies a magnetic force to magnet **4640** to thereby cause magnet **4640**, and accordingly holder **4630**, to move away from coupler interface connection portion **4604**. The position of holder **4600** relative to coupler body **4600** may be indicative of whether a surgical instrument is or is not coupled to coupler body **4600** when coupler body **4600** is coupled to coupler interface **4500**.

For example, FIG. **47A** illustrates coupler body **4600** when coupler body **4600** is not coupled to coupler interface **4500**. As shown in FIG. **47A**, without repulsion magnet **4512** of coupler interface **4500** within the vicinity of magnet **4640**, no magnetic force will be applied to magnet **4640** to cause displacement of holder **4630**, e.g., toward opening **4616**. Accordingly, holder **4630** may be in a neutral position, e.g., towards coupler interface connection portion **4604** due to gravity. As shown in FIG. **47B**, when coupler body **4600** is coupled to coupler interface **4500** and no surgical instrument is coupled to coupler body **4600**, repulsion magnet **4512** may apply a magnetic force to magnet **4640**, thereby causing magnet **4640**, and accordingly holder **4630**, to move towards opening **4616** and away from coupler interface **4500** within channel **4603**, e.g., to a position within coupler body **4600** with a maximum distance from ferrous rod **4510**. For example, when coupler body **4600** is coupled to coupler interface **4500**, the magnetic force of repulsion magnet **4512** may cause magnet harness **4638** of holder **4630** to be positioned adjacent to the upper surface of channel **4603**, as shown in FIG. **47B**. Thus, when coupler body **4600** is coupled to coupler interface **4500**, friction pad **4632** may be closer to locking portion **4620** of clamp **4618**, thereby reducing the size of opening **4616**.

Moreover, the magnetic field induced in ferrous rod **4510** by magnet **4640** when magnet **4640** is in the position within channel **4603** farthest away from ferrous rod **4510** responsive to the magnetic force of repulsion magnet **4512** when coupler body **4600** is coupled to coupler interface **4500** and no instrument is coupled to coupler body **4600**, may provide a clean signal that may be detected by Hall effect sensors **4514**, indicative of coupler body **4600** being coupled to coupler interface **4500** without a surgical instrument attached thereto. Accordingly, the system may determine that coupler body **4600** is coupled to coupler interface **4500** with no surgical instrument coupled to coupler body **4600**, based on the strength of the magnetic field induced in ferrous rod **4510** by magnet **4640**, e.g., when magnet **4640** is a maximum distance from ferrous rod **4510** within coupler body **4600**. In some embodiments, coupler interface **4500** may not have repulsion magnet **4512**, such that holder **4630** may be biased toward opening **4616** via a compression spring that applies a spring force to holder **4630**, e.g., via magnet harness **4638**.

In some embodiments, the magnetic force and/or the polarity of magnet **4640** may be selected based on the coupler body size, e.g., the size of the surgical instrument that may be coupled to the coupler body. Accordingly,

different sized coupler bodies may have different predefined magnetic forces based on the magnet used therein. Thus, upon coupling of the coupler body to the coupler interface, the magnetic field induced in the ferrous rod responsive to the predefined magnetic force of the magnet within the coupler body may be indicative of the surgical instrument size even before the surgical instrument is coupled to the coupler body.

FIGS. 48A and 48B illustrate the coupling of coupler interface connection portion 4604 of coupler body 4600 to coupler interface 4500. As described above, locking arms 4606 may be biased toward its locked state where connection portions 4608 are moved radially inward, such that at least a portion of connection portions 4608 are positioned within groove 4605 of coupler interface connection portion 4604. Prior to coupling coupler body 4600 to coupler interface 4500, a sterile drape may be positioned between coupler body 4600 and coupler interface 4500, such that the sterile drape may be draped over robot arm 300, as described above. Next, the user may apply a force to handle portions 4610 of locking arms 4606, e.g., pinch handle portions 4610 toward each other, to thereby cause connection portions 4608 to move away from each other towards the unlocked state and out of groove 4605, and provide clearance for protrusion 4504 to be received within groove 4605, as shown in FIG. 48A.

Accordingly, when locking arms 4606 are in their unlocked state, coupler body 4600 may be coupled to coupler interface 4500 such that protrusion 4504 is disposed within groove 4605. Once protrusion 4504 is disposed within groove 4605, the user may release handle portions 4610, such that locking arms 4606 move back towards their locked state and connection portion 4608 engages with locking portion 4506 of protrusion 4506, as shown in FIG. 48B. Accordingly, the engagement of connection portion 4608 and locking portion 4506 due to the corresponding geometries of connection portion 4608 and locking portion 4506 may prevent movement between coupler body 4600 and coupler interface 4500, to thereby securely couple coupler body 4600 to coupler interface 4500.

FIG. 49 illustrates when a surgical instrument is coupled to coupler body 4600 when coupler body 4600 is coupled to coupler interface 4500. As described above, coupler body 4600 may be configured to be coupled to, e.g., a 10 mm surgical instrument such as surgical instrument 10 described above with regard to FIGS. 7A to 7D. Accordingly, opening 4616, which is at least partially defined by the lower surface of locking portion 4620 of clamp 4618 and the moveable upper surface of friction pad 4632 of holder 4630, may be sized and shaped to accommodate shaft 10a of surgical instrument 10. As shown in FIG. 49, shaft 10a may be inserted through the channel defined by tapered portions 4614 of surgical instrument connection portion 4602, such that clamp 4618 transitions from its locked state to its unlocked state to permit shaft 10a to pass into opening 4616, and back to its locked state when shaft 10a is completely disposed within opening 4616.

As the user inserts shaft 10a within opening 4616, shaft 10a applies a downward force against friction pad 4632, thereby causing holder 4630 to move downward within channel 4603 and increasing the size of opening 4616 until shaft 10a is completely disposed within opening 4616 and clamp 4618 is permitted to transition back to its locked state, such that the shaft 10a is positioned between the lower surface of locking portion 4620 and friction pad 4632. Upon release of surgical instrument 10 by the user, friction pad 4632 applies an upward force against shaft 10a due to the

magnetic force of repulsion magnet 4512 applied against magnet 4640, such that shaft 10a is pinned between the lower surface of locking portion 4620 and friction pad 4632.

Accordingly, the magnetic field induced in ferrous rod 4510 by magnet 4640 when magnet 4640 is in the position within channel 4603 responsive to the magnetic force of repulsion magnet 4512 when coupler body 4600 is coupled to coupler interface 4500 as well as the force applied to holder 4630, and accordingly magnet 4640, by shaft 10a via friction pad 4632, may be detected by Hall effect sensors 4514, and which may be indicative of coupler body 4600 being coupled to coupler interface 4500, and surgical instrument 10 being coupled to coupler body 4600. Accordingly, the system may determine that coupler body 4600 is coupled to coupler interface 4500 and that surgical instrument 10 is coupled to coupler body 4600, based on the strength of the magnetic field induced in ferrous rod 4510 by magnet 4640.

Moreover, the position of magnet 4640 within channel 4603 will depend on the diameter size of the surgical instrument disposed within opening 4616 when coupler body 4600 is coupled to coupler interface 4500, such that the magnetic field induced in ferrous rod 4510 will vary based on the surgical instrument shaft size disposed within opening 4616. Accordingly, the system may identify the precise size of the surgical instrument shaft based on the strength of the magnetic field induced in ferrous rod 4610 by magnet 4640, as detected by Hall effect sensors 4514. Based on the identified type of surgical instrument coupled to coupler body 4600, the system may load the calibration file associated with the identified surgical instrument as described above.

FIGS. 50A and 50B illustrate further securement of the surgical instrument to surgical instrument connection portion 4602 of coupler body 4600. FIG. 50A illustrates switch 4624 in its unlocked state when shaft 10a of surgical instrument 10 is positioned within opening 4616. As shown in FIG. 50A, in its unlocked state, surgical instrument engagement portion 4626 of switch 4624 is not engaged with the outer surface of shaft 10a. When switch 4624 is in its unlocked state, surgical instrument 10 may be moved longitudinally relative to opening 4616 along shaft 10a, e.g., to readjust the position of surgical 10 relative to coupler body 4600. To secure shaft 10a within opening 4616, such that surgical instrument 10 cannot move longitudinally relative to coupler body 4600, the user may actuate switch 4624, e.g., via handle portion 4628, to rotate switch 4624 relative to coupler body 4600, and transition switch 4624 to its locked state where surgical instrument engagement portion 4626 engages with the outer surface of shaft 10a, as shown in FIG. 50B.

As described above, the friction force applied to shaft 10a by surgical instrument engagement portion 4626 facilitates securement of shaft 10a within coupler body 4600, such that longitudinal movement of surgical instrument 10 is prevented unless the longitudinal force applied to surgical instrument 10 exceeds at least the friction force applied to shaft 10a by surgical instrument engagement portion 4626. Moreover, the friction force applied to shaft 10a by surgical instrument engagement portion 4626 is such that the rotational force required to overcome the friction force and cause rotational of shaft 10a within opening 4616 is minimized.

Referring now to FIG. 51, another exemplary surgical instrument coupling mechanism is provided. Specifically, the surgical instrument coupling mechanism includes coupler interface 4500, as described above, and coupler body 5100 configured to be removably coupled to a surgical

instrument and to coupler interface 4500. Coupler body 5100 may be constructed similar to coupler body 4600. For example, surgical instrument connection portion 5102, channel 5103, coupler interface connection portion 5104, groove 5105, locking arms 5106, axis 5112, tapered portions 5114, opening 5116, clamp 5118, switch 5124, and holder 5130 of coupler body 5100 correspond with surgical instrument connection portion 4602, channel 4603, coupler interface connection portion 4604, groove 4605, locking arms 4606, axis 4612, tapered portions 4614, opening 4616, clamp 4618, switch 4624, and holder 4630 of coupler body 4600, respectively. Coupler body 5100 differs from coupler body 4600 in that coupler body 5100 may be configured to be removably coupled to a smaller diameter surgical instrument, e.g., a 5 mm surgical instrument such as surgical instrument 12 described above with regard to FIGS. 6A to 6D.

Accordingly, as shown in FIG. 51, opening 5116, defined at least partially by the lower surface of locking portion 5120 of clamp 5118 when clamp 5118 is in its locked state and the upper surface of friction pad 5132 of holder 5130, may be sized and shaped to receive shaft 12a of surgical instrument 12 therein. Upon actuation of switch 5124 to its locked, such that surgical instrument engagement portion 5126 of switch 5124 engages shaft 12a within opening 5116, longitudinal movement of surgical instrument 12 is prevented unless the longitudinal force applied to surgical instrument 12 exceeds at least the friction forces applied to shaft 12a by surgical instrument engagement portion 5126 and friction pad 5132. Moreover, the friction forces applied to shaft 12a by surgical instrument engagement portion 5126 and friction pad 5132 are such that the rotational force required to overcome the friction forces and cause rotational of shaft 12a within opening 5116 is minimized. When switch 5124 is moved to its unlocked state, the friction force of friction pad 5132 may not be sufficient to prevent longitudinal movement of the surgical instrument relative to coupler body 5100. Friction pad 5132 and surgical instrument engagement portion 5126 may have curved profiles that correspond with the outer surface of shaft 12a.

As shown in FIG. 51, when shaft 12a is positioned within opening 5116 and clamp 5118 is in its locked state, shaft 12a applies a downward force against friction pad 5132, thereby causing displacement of magnet 5140 coupled to magnet harness 5138 of holder 5130, which is coupled to friction pad 5132 via cradles 5136 of holder 5130, within channel 5103 of coupler body 5100. Moreover, repulsion magnet 4512 applies a magnetic force against magnet 5140, which causes friction pad 5132 to apply an upward force against shaft 12a, thereby maintaining the position of magnet 5140 within channel 5103. Accordingly, the magnetic field induced in ferrous rod 4510 by magnet 5140 when magnet 5140 is in the position within channel 5103 responsive to the magnetic force of repulsion magnet 4512 when coupler body 5100 is coupled to coupler interface 4500 as well as the force applied to holder 5130, and accordingly magnet 5140, by shaft 12a via friction pad 5132, may be detected by Hall effect sensors 4514, and which may be indicative of coupler body 5100 being coupled to coupler interface 4500, and surgical instrument 12 being coupled to coupler body 5100. Accordingly, the system may determine that coupler body 5100 is coupled to coupler interface 4500 and that surgical instrument 12 is coupled to coupler body 5100, based on the strength of the magnetic field induced in ferrous rod 4510 by magnet 5140.

Referring now to FIGS. 52A to 52E, an alternative exemplary surgical instrument coupling mechanism is pro-

vided. Coupler interface 5200 may be rotatably coupled to or otherwise integrated with link 316, and connection portion 5250 may be coupled to a coupler body, e.g., coupler body 4600, 5100, 5300, 5400, for removably coupling the coupler body to coupler interface 5200. As shown in FIG. 52A, coupler interface 5200 may include protrusion 5204 extending from flat portion 5202. Flat portion 5202 may have an outer diameter that coincides with the outer diameter of link 316. In addition, coupler interface 5200 may include extended portion 5208 extending from flat portion 5202 and configured to be inserted within link 316. Like protrusion 4504, protrusion 5204 may have a non-circular profile, which corresponds to the geometry of groove 5252 of connection portion 5250 of the coupler body, as described in further detail below. For example, as shown in FIG. 52A, protrusion 4504 may have a diamond-shaped profile. Accordingly, when protrusion 5202 is disposed within groove 5252 of connection portion 5250, rotational movement between coupler interface 5200 and connection portion 5250 is prevented.

Moreover, protrusion 5204 may include one or more locking portions 5206 disposed on the outer surface of the sidewall of protrusion 5204. For example, locking portions 5206 may be indentations/grooves extending along the outer surface of protrusion 5204, and sized and shaped to engage with locking arms 5260 of connection portion 5250, as described in further detail below, for securing the coupler body to coupler interface 5200, and for securing the sterile drape between connection portion 5250 and coupler interface 5200. Preferably, protrusion 5204 includes a pair of locking portions 5206. For example, as shown in FIGS. 52A and 52C, the pair of locking portions 5206 may be disposed on opposing apexes of the diamond-shaped profile of protrusion 5204. Accordingly, connection portion 5250 may be securely coupled to coupler interface 5200 in two orientations.

As shown in FIG. 52A, coupler interface 5200 may include one or more additional protrusions 5210, e.g., "mating dots," disposed on flat portion 5202. For example, coupler interface 5200 may include a plurality of protrusions 5210, preferably evenly spaced apart along flat portion 5202, e.g., adjacent to the outer edge of flat portion 5202. Protrusions 5210 may have a geometry that corresponds with the geometry of one or more additional grooves 5254 of connection portion 5250, as shown in FIG. 52B. For example, protrusions 5210 may have a semi-spherical shape, and grooves 5254 may have a corresponding semi-spherical shape. As shown in FIG. 52B, grooves 5254 may be disposed along connection portion 5250, such that grooves 5254 are aligned with protrusions 5210 so that protrusions 5210 may be disposed within grooves 5254 when connection portion 5250 is coupled to coupler interface 5200, as shown in FIG. 52C. Accordingly, when protrusion 5202 is disposed within groove 5252 of connection portion 5250, and protrusions 5210 are disposed within grooves 5254, rotational movement between coupler interface 5200 and connection portion 5250 is prevented. As will be understood by a person having ordinary skill in the art, coupler interface 5200 and connection portion 5250 may include more or less protrusions 5210 and grooves 5254, respectively, that are shown in FIGS. 52A and 52B. In addition, other coupler interfaces and coupler bodies described herein, e.g., coupler interface 4500, 5200 and coupler body 4600, 5100, 5300, 5400 may include similar additional protrusions and grooves for providing additional stabilization when the coupler interface is coupled to the coupler body.

As shown in FIG. 52D, connection portion 5250 may include a pair of locking arms 5260, which may be constructed similar to locking arms 4606 of connection portion 4604, for releasably securing connection portion 5250 to coupler interface 5200. For example, locking arms 5260 may include handle portion 5264 sized and shaped to be actuated by the user's fingers, and connection portion 5262 sized and shaped to engage with locking portions 5206 of protrusion 5204. Accordingly, locking arms 4606 may transition between an unlocked state where locking arms 5260 are disengaged from protrusion 5204, as shown in FIG. 52D, and a locked state where connection portion 5262 of locking arms 5260 are engaged with locking portions 5206 of protrusion 5204, as shown in FIG. 52E, such that locking arms 4606 are biased toward the locked state.

Referring now to FIGS. 53A and 53B, an alternative exemplary coupler body is provided. Coupler body 5300 may be constructed similar to coupler body 4600 and/or coupler body 5100. For example, coupler body 5300 may include switch 5302, which may be constructed similar to switch 4624 of coupler body 4600 and/or switch 5124 of coupler body 5100. As shown in FIGS. 53A and 53B, switch 5302 may include one or more surgical instrument engagement portions 5304, each having a geometry that corresponds with the outer diameter of the shaft of the surgical instrument to be inserted within the opening of coupler body 5300. In addition, switch 5302 may include handle portion 5306 sized and shaped to be actuated by the user's fingers to transition switch 5302 between an unlocked state, as shown in FIG. 53A, and a locked state, as shown in FIG. 53B, where surgical instrument engagement portion 5304 engages with the surgical instrument shaft within the opening of coupler body 5300 and applies a friction force to the surgical instrument shaft. As shown in FIGS. 53A and 53B, coupler body 5300 further may include one or more prot

Referring now to FIGS. 53A and 53B, an alternative exemplary coupler body is provided. Coupler body 5300 may be constructed similar to coupler body 4600 and/or coupler body 5100. For example, coupler body 5300 may include switch 5302, which may be constructed similar to switch 4624 of coupler body 4600 and/or switch 5124 of coupler body 5100. As shown in FIGS. 53A and 53B, switch 5302 may include one or more surgical instrument engagement portions 5304, each having a geometry that corresponds with the outer diameter of the shaft of the surgical instrument to be inserted within the opening of coupler body 5300. In addition, switch 5302 may include handle portion 5306 sized and shaped to be actuated by the user's fingers to transition switch 5302 between an unlocked state, as shown in FIG. 53A, and a locked state, as shown in FIG. 53B, where surgical instrument engagement portion 5304 engages with the surgical instrument shaft within the opening of coupler body 5300 and applies a friction force to the surgical instrument shaft.

As shown in FIGS. 53A and 53B, coupler body 5300 further may include one or more locking nubs 5308 for securing switch 5302 in the locked state. Locking nubs 5308 may have a height sufficient to permit switch 5302 to pass over locking nub 5308 when a force exceeding a predetermined threshold is applied to switch 5302, and to prevent switch 5302 from passing over locking nub 5308 when an insufficient amount of force is applied to switch 5302. In some embodiments, coupler body 5300 may include an internal spring coupled to switch 5302 to bias switch 5302 in a downward direction, e.g., toward the surface on which locking nubs 5308 are disposed. Accordingly, when a sufficient force is applied to switch 5302 against locking nub

5308, locking nub 5308 applies an upward force against switch 5302 to thereby cause the spring to compress and permit switch 5302 to pass over locking nub 5308. Moreover, the spring force applied to switch 5302 when switch 5302 is in its locked state prevents switch 5302 from passing over locking nub 5308 unless a sufficient amount of force is applied to switch 5302 to compress the spring beyond the height of locking nub 5308.

Referring now to FIG. 54, another alternative exemplary coupler body is provided. Coupler body 5400 may be constructed similar to coupler body 5300 and may include switch 5402 having engagement portion 5404 and handle portion 5406. Coupler body 5400 differs from coupler body 5300 in that instead of locking nubs 5308, coupler body 5400 may include ramped surface 5408. As shown in FIG. 54, ramped surface 5408 may have a wave-shaped profile, e.g., two crests disposed between three valleys. For example, when switch 5402 is in its unlocked state, switch 5402 may be disposed within valley 5410, and when switch 5402 is in its locked state, switch 5402 may pass over crest 5412 and disposed within valley 5414. Accordingly, crest 5412 may have a height sufficient to permit switch 5402 to pass over crest 5412 when a force exceeding a predetermined threshold is applied to switch 5402, and to prevent switch 5402 from passing over crest 5412 when an insufficient amount of force is applied to switch 5402. In some embodiments, coupler body 5400 may include an internal spring coupled to switch 5402 to bias switch 5402 in a downward direction, e.g., toward ramped surface 5408. Accordingly, when a sufficient force is applied to switch 5402 against crest 5412, crest 5412 applies an upward force against switch 5402 to thereby cause the spring to compress and permit switch 5402 to pass over crest 5412. Moreover, the spring force applied to switch 5402 when switch 5402 is in its locked state within valley 5414 prevents switch 5402 from passing over crest 5412 unless a sufficient amount of force is applied to switch 5402 to compress the spring beyond the height of crest 5412.

Referring now to FIG. 56, an alternative robot arm having a motorized joint about axis Q3 is provided. Motorized joint 5600 may be constructed similar to joint 320, such that distal shoulder link 308 may be rotated relative to proximal shoulder link 306 about axis Q3 at joint 5600, wherein axis Q3 may be parallel to the longitudinal axis of shoulder link 305. Motorized joint 5600 differs from joint 320 in that at least one motor, e.g., motor M4, which may be controlled by a processor of the co-manipulation robot platform, may be operatively coupled to joint 5600, to thereby apply a torque to joint 5600 to actuate rotation of distal shoulder link 308 relative to proximal shoulder link 306 about axis Q3. For example, as shown in FIG. 56, motor M4 may be operatively coupled to a motion transmission mechanism coupled to distal shoulder link 308, e.g., worm gear 5604, via gear 5602, such that actuation of motor M4 causes rotation of distal shoulder link 308 relative to proximal shoulder link 306 via engagement between gear 5602 and worm gear 5604.

As shown in FIG. 56, motor M4 may be positioned adjacent to joint 5600. Unlike the other motorized joints described herein, e.g., base joint 303, shoulder joint 318, and elbow joint 322, motorized joint 5600 is preferably not be "back-drivable," in that the user cannot actuate motorized joint 5600, e.g., via movement of the surgical instrument coupled to the robot arm when the system is in co-manipulation mode. Instead, actuation of motorized joint 5600 may be conducted via one or more actuators, e.g., an actuator disposed adjacent to motorized joint 5600 and/or an actuator

101

displayed on GUI 110, that may be actuated to automatically cause rotation of distal shoulder link 308 relative to proximal shoulder link 306. As described above, when forces applied at the distal end of robot arm 300 by the user serves as an input to the system, e.g., when the user applies a force exceeding a predetermined threshold in a predefined direction, the system may automatically actuate motorized joint 5600 to cause rotation of distal shoulder link 308 relative to proximal shoulder link 306 to facilitate movement of robot arm 300 in the predefined direction. For example, similar to how the system may cause the stages of platform 100 to move robot arm 300 responsive to movement of the distal end of robot arm 300 by the user, e.g., back/forth along the x-axis or up/down along the z-axis, as described above, the system may cause motorized joint 5600 to rotate distal shoulder link 308 relative to proximal shoulder link 306 to move robot arm 300 along the y-axis responsive to movement of the distal end of robot arm 300 by the user along the y-axis. Accordingly, the system may stop actuation of motorized joint 5600 when the force applied by the user to the distal end of robot arm 300 drops below a predetermined threshold.

Moreover, as described above, as the robot arm is moved, either manually or automatically during setup, based on depth data obtained from the one or more optical scanners, the system may detect when either the stages of platform 100 or the robot arm approaches a predetermined distance threshold relative to an object in the operating room, e.g., the surgical bed. Accordingly, the system may automatically reconfigure the robot arm to avoid a collision with the object, e.g., by automatically actuating motorized joint 5600 to rotate distal shoulder link 308 relative to proximal shoulder link 306. Similarly, system may automatically reconfigure the robot arm to avoid a collision with an object in the operating room by automatically actuating motorized joint 5600 during a surgical procedure.

While various illustrative embodiments of the invention are described above, it will be apparent to one skilled in the art that various changes and modifications may be made therein without departing from the invention. The appended claims are intended to cover all such changes and modifications that fall within the true scope of the invention.

What is claimed:

1. A co-manipulation surgical system to assist with surgery performed using a surgical instrument having a handle, an operating end, and an elongated shaft therebetween, the co-manipulation surgical system comprising:

a robot arm comprising a proximal end, a distal end configured to be removably coupled to the surgical instrument, a plurality of links, and a plurality of joints; and

a controller operatively coupled to the robot arm, the controller programmed to:

cause the robot arm to automatically switch between a co-manipulation mode wherein the controller permits the robot arm to be freely moveable responsive to movement at the handle of the surgical instrument for performing surgery using the surgical instrument, and a passive mode where the controller causes the robot arm to maintain a static position; and cause a vibration at the distal end of the robot arm, the vibration indicative of the robot arm switching from the co-manipulation mode to the passive mode.

2. The co-manipulation surgical system of claim 1, wherein the vibration is configured to be perceivable by a

102

user holding the handle of the surgical instrument, while causing negligible movement at the operating end of the surgical instrument.

3. The co-manipulation surgical system of claim 1, wherein the controller is configured to cause the robot arm to switch to the passive mode responsive to determining that movement of the robot arm due to movement at the handle of the surgical instrument is less than a predetermined amount for at least a predetermined dwell time period.

4. The co-manipulation surgical system of claim 1, wherein the controller is configured to cause the robot arm to switch to the co-manipulation mode responsive to determining that force applied at the robot arm due to force applied at the handle of the surgical instrument exceeds a predetermined threshold.

5. The co-manipulation surgical system of claim 1, wherein the controller is configured to apply a first impedance to the robot arm in the co-manipulation mode to account for weight of the surgical instrument and the robot arm.

6. The co-manipulation surgical system of claim 1, wherein the controller is configured to generate an audible alert, the audible alert indicative of the robot arm switching from the co-manipulation mode to the passive mode.

7. The co-manipulation surgical system of claim 1, wherein the controller is configured to generate a text alert, the text alert indicative of the robot arm switching from the co-manipulation mode to the passive mode, the co-manipulation surgical system further comprising:

a graphical user interface operatively coupled to the controller, the graphical user interface configured to display the text alert.

8. The co-manipulation surgical system of claim 1, wherein the controller is configured to generate a visual alert, the visual alert indicative of the robot arm switching from the co-manipulation mode to the passive mode.

9. The co-manipulation surgical system of claim 8, wherein the visual alert comprises a plurality of colors, the co-manipulation surgical system further comprising:

one or more indicators operatively coupled to the controller, each of the one or more indicators configured to illuminate in the plurality of colors,

wherein the controller is configured to cause the one or more indicators to illuminate in a first color of the plurality of colors when the robot arm is in the co-manipulation mode and in a second color of the plurality of colors when the robot arm is in the passive mode.

10. The co-manipulation surgical system of claim 1, further comprising:

a plurality of motors operatively coupled to at least some joints of the plurality of joints,

wherein the controller is configured to actuate at least one motor of the plurality of motors to cause the vibration at the distal end of the robot arm.

11. The co-manipulation surgical system of claim 10, wherein the robot arm comprises a base operatively coupled to the proximal end of the robot arm, and wherein the plurality of motors are disposed within the base.

12. The co-manipulation surgical system of claim 1, wherein the controller is configured to:

detect when the surgical instrument is coupled to distal end of the robot arm; and

cause a second vibration at the distal end of the robot arm, the second vibration indicative that the surgical instrument is successfully coupled to the distal end of the robot arm.

103

13. The co-manipulation surgical system of claim 12, wherein the distal end of the robot arm comprises a coupler interface, the co-manipulation surgical system further comprising:

a coupler body configured to be removably coupled to the coupler interface, the coupler body comprising a lumen sized and shaped to receive the elongated shaft of the surgical instrument therein.

14. The co-manipulation surgical system of claim 1, wherein the controller is configured to:

determine a level of fatigue of a user holding the handle of the surgical instrument;

compare the level of fatigue with a predetermined fatigue threshold; and

apply an impedance to the robot arm in the co-manipulation mode to reduce tremor of the user.

15. The co-manipulation surgical system of claim 14, wherein the controller is configured to determine the level of fatigue of the user holding the handle of the surgical instrument based on hand tremor of the user, distance traveled by the operating end of the surgical instrument, time to complete predefined steps of the surgery, and/or time to complete the surgery.

16. A method for assisting with surgery performed using a surgical instrument having a handle, an operating end, and an elongated shaft therebetween, the method comprising:

providing a robot arm comprising a proximal end, a distal end configured to be removably coupled to the surgical instrument, a plurality of links, and a plurality of joints;

causing, via a controller operatively coupled to the robot arm, the robot arm to automatically switch between a co-manipulation mode wherein the controller permits the robot arm to be freely moveable responsive to

104

movement at the handle of the surgical instrument for performing surgery using the surgical instrument, and a passive mode where the controller causes the robot arm to maintain a static position; and

causing a vibration at the distal end of the robot arm, the vibration indicative of the robot arm switching from the co-manipulation mode to the passive mode.

17. The method of claim 16, wherein causing the vibration at the distal end of the robot arm comprises causing a vibration that is perceivable by the user holding the handle of the surgical instrument, while causing negligible movement at the operating end of the surgical instrument.

18. The method of claim 16, wherein causing the vibration at the distal end of the robot arm comprises actuating at least one motor of a plurality of motors operatively coupled to at least some joints of the plurality of joints to cause the vibration at the distal end of the robot arm.

19. The method of claim 16, wherein causing the robot arm to automatically switch between the co-manipulation mode and the passive mode comprises causing the robot arm to switch to the passive mode responsive to determining that movement of the robot arm due to movement at the handle of the surgical instrument is less than a predetermined amount for at least a predetermined dwell time period.

20. The method of claim 16, wherein causing the robot arm to automatically switch between the co-manipulation mode and the passive mode comprises causing the robot arm to switch to the co-manipulation mode responsive to determining that force applied at the robot arm due to force applied at the handle of the surgical instrument exceeds a predetermined threshold.

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